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EFFECT OF WIND SPEED AND
DIRECTION ON THE PERFORMANCE
OF AIR-COOLED CONDENSERS

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PREFACE

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ABSTRACT

With competition for water resources, power producers are increasingly using air-cooled condensers (ACCs) for power plant cooling. It is well known that wind can degrade the performance of air cooled condensers, yet the mechanisms of this wind effect are not fully understood.

This project conducted field tests to determine the effect of wind speed and direction on air cooled condenser performance at five power plants. Continuous recordings of wind speed, wind direction, air cooled condenser inlet temperature, airflow to selected air cooled condenser cells, and plant operating variables including power output, steam flow, and turbine exhaust pressure made for several days at each plant.

Air cooled condenser performance was affected by both hot air recirculation and fan performance degradation. The reduction in airflow (estimated from measurements on selected cells) from degraded fan performance appears to be the more important mechanism. For certain wind directions at some locations, wind speeds above 15 to 20 miles per hour led to increases in turbine exhaust pressure of 1.5 to 2 inches of mercury (in Hga) above the level that would be predicted from air cooled condenser design curves for a given steam flow and ambient air temperature. For example, a rise in the turbine exhaust pressure to 7.5 in Hga results in a 13% reduction in steam turbine generation compared to operation at the design backpressure of 2.5 in Hga and does not include temperature related reductions.

Comparisons with flow modeling results support design recommendations for suppressing unfavorable airflow patterns under the air cooled condenser. Coordinated test and modeling efforts are recommended to further quantify and generalize the results and to advance development of wind-resistant air cooled condenser designs.

Keywords: Air-cooled condenser, wind effect, condenser airflow, power plant, fans

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Executive Summary

Introduction

Although wet cooling is still by far the most common system in use for power plant cooling, dry cooling systems with mechanical-draft air-cooled condensers are being selected with increasing frequency both in the United States and around the world. As the use of dry cooling becomes more widespread, the importance of ensuring adequate, predictable cooling performance becomes more critical to the efficient operation of the entire electric grid as well as to individual air-cooled plants.

It is well understood that air-cooled systems impose limitations on plant efficiency and output, particularly during the hottest hours of the year, which is precisely the time at which power demand is at its peak. This effect is exacerbated if winds, occurring concurrently with high temperatures, further degrade the performance of the air-cooled condenser and do so in an unexpected and unpredictable manner. Wind speeds above about 20 miles per hour can result in backpressures that are 1.5 to 2.0 inches of mercury absolute (in Hga) higher than those at the same ambient temperature and steam load at wind speeds below 4 miles per hour. Such backpressures correspond to a plant performance loss of 13 percent, on top of the temperature-related performance losses.

Moreover, wind gusts, if they occur when the air-cooled condenser is operating near the trip point of the turbine, can cause a brief excursion to pressures high enough to trip the turbine rapidly enough that operators cannot respond in time to keep the plant from coming off-line. A sudden trip—which may take several hours to recover from—can have a significant effect on plant revenue when occurring at a time when the plant is at full load, the network demand is at its peak, and the price for power is high.

Purpose

This research explores the mechanism(s) of wind effects on air-cooled condensers at power plants.

Objectives

The study was initiated to:

- Define and quantify the effect of wind on air-cooled condenser performance.
- Evaluate some specific existing attempts to mitigate the effects.
- Suggest further research to improve air-cooled condenser performance in high winds.

Specifically, the test program was intended to confirm or modify the generally accepted hypothesis that the negative effect of wind on air-cooled condenser performance is attributable to two major causes:

- Recirculation of heated plume air into the air-cooled condenser inlet air stream.

- Degradation of fan performance in some cells and, hence, reduced airflow to those cells.

These mechanisms were to be investigated through detailed monitoring of full-scale, operating air-cooled condenser under varying wind conditions.

Approach

Five sites were selected for testing:

- Bighorn power plant in Primm, Nevada
- El Dorado Energy Center in El Dorado, Nevada
- Apex generating station in Apex, Nevada
- Front Range in Fountain, Colorado
- Wygen plant in Gillette, Wyoming

As a group, these sites provided good coverage of the range of conditions of interest to the design, specification, and operation of an air-cooled condenser for power plant use, specifically:

- All sites normally experience high winds (>20 miles per hour) during the summer.
- Both a coal-fired steam plant and gas-fired combined-cycle plants were included.
- Air-cooled condensers from all of the (then) three major air cooled condenser vendors were included. This resulted in a variety of tube bundle designs (single row, multi-row; aluminum and galvanized tubes; plate and helical fins; etc.) and different fan designs from major vendors (varying blade shape, number of blades, etc.).
- A range of air-cooled condenser sizes (12 to 40 cells) in a variety of configurations (40 cells in a single 8 x 5 arrangement and two 4 x 5 clusters; 30 cells in both 5 x 6 and 6 x 5 arrangements; and 12 cells in a 2 x 6 arrangement).
- Air-cooled condensers with and without windscreens; windscreens at the three sites that had them differed in type and configuration.

Instrumentation was installed at each site to make those measurements required to:

- *Characterize the wind conditions at the air-cooled condenser.* Wind speed and direction were determined with a pole-mounted, vaned propeller anemometer at a height of about 15 feet. When available, comparable data were obtained from the plant's own meteorology station.
- *Determine the pattern and amount of recirculation.* The amount of recirculation was determined with inlet air temperature measurements in each cell. The probes were mounted on the fan bridge walkway about two-thirds of the way from the fan hub to the blade tip, corresponding roughly to the location of highest inlet velocity.

- *Determine the effect of wind on the performance of selected fans.* This was done by sensing any wind-induced flow reductions or fan stalling. A complete set of measurements, including inlet air velocity, fan static pressure rise, and fan power, was made at the first site (Bighorn) to determine a simpler, “surrogate” measurement that could be used at future sites to indicate any degradation in fan performance.

Tests at each plant lasted for eight or nine days. Measurements were obtained of:

- Air-cooled condenser operation
 - o Cell inlet air temperature
 - o Cell exit air temperature (Bighorn only)
 - o Cell inlet velocity
 - o Cell exit velocity (Bighorn only)
 - o Fan static pressure (Bighorn and El Dorado only)
 - o Fan motor current (Bighorn and El Dorado only)
 - o Fan operating status
- Plant operation
 - o Plant output
 - o Steam turbine output
 - o Steam (or condensate) flow to air-cooled condenser
 - o Turbine exhaust pressure
- Weather conditions
 - o Ambient air temperature
 - o Wind speed
 - o Wind direction

The monitoring of over forty data channels at one-minute intervals for over forty test days resulted in nearly 100 megabytes of data. While the complete data set cannot reasonably be included in this report, the full set is included on a CD available to interested parties from the California Energy Commission, the Electric Power Research Institute, or Maulbetsch Consulting.

Results

Average recirculation for most conditions at all sites was less than 3°F. Occasional excursions to higher values of 4° to 8°F, with very infrequent instances as high as 10°F, were noted. Individual cell recirculation of 10°F and above was not uncommon, primarily on edge and corner cells.

High recirculation was typically associated not with sustained high wind speeds, but rather with short-lived periods of variable wind direction. Occasional instances of apparent high recirculation were associated with nighttime periods of very still air when temperature inversions were established and cold air near the ground was entrained into one or two cells; this gave a spurious indication since the minimum-temperature cell was used as the recirculation baseline.

Fan performance results are less easy to quantify or generalize. Measurements were typically taken at four locations under each fan being monitored, and the average of the four velocities was used to estimate the total volume of airflow through the fan. Under low-wind conditions this was typically close to the design value. With increasing wind, cells near the edge of the air-cooled condenser would sometimes show a reduction in average flow, occasionally up to 60 or 70 percent. At other times no significant reduction was observed under what appeared to very similar if not identical conditions.

Air velocity across the inlet of a single fan usually, but not always, showed the greatest reduction in the upwind region. At times, a significant reduction was evident in one half of the fan inlet and little or no reduction in the other half. This is conceptually consistent with the idea that a separated flow region—beginning at the bottom of the windwall at the edge of the fan deck, extending in under the fan deck, and enveloping all or a portion of the inlet region of the nearest fan—is responsible for the reduced airflow through the fan. This differs from the conventional interpretation that a strong crosswind disrupts the preferred uniform parallel flow normal to the fan inlet plane—an assumption that is the basis for most fan design calculations.

Static pressure and motor current results were sufficiently erratic and inconsistent that any generalized interpretation of their significance is not attempted at this time.

Conclusions

The field test results led to the following conclusions:

- Wind speeds above 20 miles per hour can result in turbine backpressures 1.5 to 2 inches of mercury absolute higher than those which would be predicted from air-cooled condenser design curves at a given steam flow and ambient temperature.
- Recirculation is always present to some degree.
- It is highest at intermediate wind speeds and during times of variable wind direction.
- It occurs intermittently during times of variable winds, typically just after a shift in wind direction.
- Recirculation diminishes at the higher wind speeds, apparently as the hot exit air plume is blown clear of the air-cooled condenser inlet area.
- The effect, typically an average recirculation of less than 3°F, is not sufficient to account for the overall reduction in air-cooled condenser performance as inferred from the increase in turbine exhaust pressure in high-wind conditions.

- The major mechanism accounting for the observed effect of wind on air-cooled condenser performance appears to be a reduction in airflow through the condenser resulting from fan performance being degraded by wind.
- The patterns of reductions in inlet air velocity across fan inlet areas suggests the likelihood of a separated region from the edge of the air-cooled condenser enveloping the fan inlet.
- Precise correlation of wind conditions and measured air-cooled condenser conditions with turbine backpressure are difficult to establish. This is due in part to plant control actions related to megawatt (MW) control, steam flow control, and duct burning at combined-cycle plants which occur concurrently with variations in ambient wind conditions. The actions introduce variations in turbine backpressure, often with some time delay, confounding the interpretation of the instantaneous measurements made during the tests.

To develop more confident quantitative relationships between ambient wind conditions and air-cooled condenser performance, additional testing and analysis is required. This should include:

- Closely coordinated field tests and computational fluid dynamics modeling at a single well-described and completely instrumented operating air-cooled condenser.
- More detailed measurements of both far- and near-field velocity and temperature fields.
- More extensive measurements of inlet velocity and static pressure on a greater number of cells.
- More precise and extensive measurements of static pressure underneath the air-cooled condenser.
- More careful attention to pressure and velocity fields close to wind barriers under the air-cooled condenser.
- More detailed and precise inclusion of plant operation and control actions into the interpretation of the data in order to normalize the simple air-cooled condenser performance data of turbine exhaust pressure, steam flow, and ambient temperature against wind conditions.
- Careful calibration of detailed computational fluid dynamics models with field data, so the models can be confidently applied to assess design modifications that will mitigate wind effects on fan performance.

Benefits to California

The state's burgeoning population demands more water, even as global warming renders the Sierra snowpack less secure. Under these conditions, air-cooled condensers become an increasingly attractive choice for power plant cooling. By advancing knowledge of wind effects on air-cooled condenser performance—thereby enabling design modifications to mitigate those

effects—this research promotes wider use of air-cooled condensers. In turn, air cooling technology offers dramatic water savings over traditional wet cooling, along with greater freedom to build new power plants to meet growing demand and to site the new units closer to population centers.

CHAPTER 1:

Introduction

This report documents a study of the effect of wind on the performance of large air-cooled condensers at five power plants in the western United States.

1.1 Background

Although wet cooling is still by far the most common system in use for power plant cooling, dry cooling systems with mechanical-draft air-cooled condensers (ACCs) are being selected with increasing frequency both in the United States and around the world. Table 1-1 lists existing and “under construction” ACC installations in the United States. This total generation capacity exceeds 8,500 MW.¹ The first U.S. installation was at Neil Simpson I, a 20-MW coal-fired plant in Gillette, Wyoming, in 1968. Of the 73 existing units listed in Table 1-1, 29 have come on-line since 2000 and 57 since 1990.

Factors driving this increasing use of dry cooling include growing population and power demand in hot, arid areas such as Las Vegas and elsewhere in the southwest; a variety of environmental pressures including the reduction of cooling system water intake losses (“316(b)” regulations); airborne fine particulate rules and an aversion to visible plumes from wet cooling towers; and an increasing societal conviction that water resources are imperiled and must be conserved for “higher use.” Additionally, the choice of dry cooling can provide the siting flexibility to locate plants close to low-cost fuel, gas pipelines, or electric transmission access—desirable sites that may lack easy access to adequate water for wet cooling.

As dry cooling becomes more widespread, the importance of ensuring adequate, predictable cooling performance becomes more critical to the efficient operation of the entire electrical network as well as to the individual power plant. It is well understood that air-cooled systems impose limitations on plant efficiency and output, particularly during the hottest hours of the year, which is precisely the time at which power demand is at its peak. This performance limitation is exacerbated if winds, which often occur concurrently with high temperatures, further degrade the performance of the ACC and do so in an unexpected and unpredictable manner.

1. Includes only the output from the steam turbine side of any combined-cycle plants (see Table 1-1).

Table 1-1. Existing and planned ACC installations in the United States.

State	Vendor ¹	Customer	Project/Site	Plant Capacity ²	Steam Turbine Capacity ³		Service Date
				MWe	MWe	tons/hour	
NY	GEA	Astoria Energy, LLC	Astoria Enrgy II	~700	230	777	2008
CA	SPX	E&L Westcoast, LLC	Colusa	660	220		2008
NM	SPX	Colorado Energy Management	Hobbs	500	500		2008
CO	SPX	Foreruner Corporation	Lamar			44	2008
NV	SPX	Lockwood Green Constructors	Tracy	500			2008
VA	SPX	Dominion	Virginia City Hybrid Energy Center	585			2008
HI	GEA	Hawaii Electric & Light	Keahole ST-7	~115	38	70	2007
WY	GEA	Basin Electric	Dry Fork	425	425	1,000	2007
NM	GEA	Public Service New Mexico	Afton	148	148	270	2007
CO	GEA	Xcel	Comanche III	750	750	1530	2007
CA	SPX	Mirant	Gateway	530			2007
WY	SPX	Black Hills Generation	WyGen 3	100			2007
NY	SPX	Astoria Energy	Astoria	500	170	460	2006
NY	GEA	Caithness Long Island Energy	Long Island Energy Center	~300	100	297	2006
WY	GEA	Black Hills Generation	WyGen II/Unit 4	80	80		2005
CA	GEA	Noresco	32nd St. Naval Station	?	5		2004
UT	GEA	PacifiCorp	Current Creek	525	200		2004
MN	GEA	FibroWatt	Fibrominn Biomass	?	55		2004
AK	GEA	US Army	Ft. Wainwright CHP Plant	22.5	3 x 5		2004
AZ	GEA	Pinnacle West	Snowflake	?	3		2004
NY	Hamon	NYP&GE/Sargent & Lundy	Poletti	450	150		2004
NV	Hamon ¹	Genwest LLC/Silverhawk	Silverhawk	570	170		2004
NV	Hamon	Reliant/Sargent & Lundy	Big Horn	598	160		2003
MA	Hamon	Sithe/Raytheon	Fore River	750	250		2002
MA	Hamon	Sithe/Raytheon	Mystic	1600	500		2002
NY	Marley/BDT ¹	Key Span	Ravenswood	256	80	278	2002
MS	GEA	Reliant	Choctaw County	800	350		2001
NV	GEA	Nevada Power	Chuck Lenzie	1200	400		2001
CO	GEA	Front Range Power	Front Range	500	150		2001
PA	GEA	Reliant	Hunterstown	950	350		2001
CA	GEA	Calpine	Otay Mesa	500	277		2001
WY	GEA	Black Hill Generation	WyGen I/Unit 3	80	80		2001
CA	Hamon	Calpine/Bechtel	Sutter	500	170		2001
NV	Marley/BDT	Mirant	APEX	550	170	697	2001
MA	Marley/BDT	ABB	Bellingham	558	200	2 x 256	2001
WA	Marley/BDT	Parsons	Chehalis	520	170	490	2001
NY	GEA	PG&E Generating	Athens	1058	360		2000
WA	GEA	GoldendaleEnergy	Goldendale	250	110		2000
MA	Marley/BDT	ABB	Blackstone	570	180	2 x 256	2000
TX	Marley/BDT	ABB	Hays	1100	300	2 x 256	2000
CT	Marley/BDT	ABB	Lake Road	840	275	3 x 256	2000
TX	Marley/BDT	ABB	Midlothian	1650	500	4 x 255	2000
1990's							
ME	GEA	Rumford Power Associates	Rumford	115	80		1999
NV	GEA	Sempra Energy / Reliant Energy	El Dorado	500	150		1999
RI	GEA	Tiverton Power Associates	Tiverton	265	80		1999
MA	GEA	Energy Mangement Inc.	Dighton	169	60		1998
CA	Marley/BDT	Bechtel, Crockett	Crockett	240	80	275	1996
IL	GEA	Browning Ferris Gas Serv. Inc.	Mallard Lake Landfill	18	9		1996
MT	Marley/BDT	Billings Generation	Rosebud	42	42	210	1995
IA	GEA	Municipal Electric Utility	Cedar Falls/Streeter	50	40		1994
MA	GEA	Ogden Martin Sys. of Haverhill	Haverhill Extension	46.9	46.9		1994
MI	GEA	Browning Ferris Gas Serv. Inc.	Arbor Hill	9	9		1994
MN	GEA	Browning Ferris Gas Serv. Inc.	Pine Bend Landfill	15	6		1994
NY	GEA	MacArther Res. Rec. Agency	Islip	?	11		1994
NY	GEA	Dutchess County	Dutchess Co. Extension	10	10		1993
VA	GEA	Mission Energy	Gordonsville	240	100		1993
NY	GEA	Ogden Martin Sys.	Onondaga County	39	39		1992
NY	GEA	Falcon Seaboard	Saranac	240	80		1992
WY	GEA	Black Hills Power & Light	Neil Simpson II	80	80		1992
AK	GEA	University of Alaska	Fairbanks	?	10		1991
MA	Marley/BDT	CRS Sirrinc, Lowell	Lowell	27	27	73	1991
NJ	GEA	Ogden Martin Sys.	Union County	44	44		1991
NY	Hamon	Ogden/Marin/Huntington	Huntington	25	25		1991
VA	Hudson	FPL	Doswell	708	225		1991
NJ	GEA	Cogen Technologies Inc.	Linden	520	285		1990
NJ	GEA	Intercontinental Energy	Sayreville	292	100		1990
NY	Marley/BDT	Indeck Energy	Silver Springs	58	20	55	1990
PA	GEA	Falcon Seaboard	Norcon-Welsh	20	20		1990
MA	GEA	Intercontinental Energy	Bellingham	300	100		1990
WV	GEA	Energy America Southeast	North Branch	80	80		1990

Table 1-1. Existing and planned ACC installations in the United States (cont.)

State	Vendor ¹	Customer	Project/Site	Plant Capacity ²	Steam Turbine Capacity ³		Service Date
				MWe	MWe	ton/hour	
1980's							
CT	GEA	Oxford Energy	Exeter	30	30		1989
TX	Hudson	TXU	Sweetwater	250	80		1989
WA	GEA	Wheelabrator Environ. Sys.	Spokane	26	26		1989
NY	GEA	TBG Cogen	Grumman	?	13		1988
MA	GEA	Ogden Martin Systems	Haverhill	47	47		1987
PA	GEA	ABB	Hazleton	240	67.5		1987
IL	GEA	Chicago Northwest	Chicago	?	1		1986
MA	GEA	American Ref-Fuel	SEMASS	71	54		1986
ME	GEA	Wheelabrator Sherman Energy	Sherman	20	20		1985
CA	GEA	Pacific Ultrapower	Chinese Station	22.4	22.4		1984
CA	GEA	Pacific Gas & Electric	Gerber	?	3.7		1981
1970's							
AK	GEA	Chugach Electric	Beluga/Unit 8	55	55		1979
WY	GEA	Black Hills Power/Pacific Power	Wyodak	330	330		1977
CA	GEA	Exxon	Benicia	?	NA		1975
MA	GEA	Braintree Electric Light Dept.	Norton P. Potter	96	20		1975
1960's							
WY	GEA	Black Hills Power & Light Co.	Neil Simpson I	20	20		1968

¹ SPX currently owns Marley, BDT and Hamon; vendor designations indicate vendor name/design at time of purchase

² Data obtained from multiple sources

³ Values sometimes estimated from plant size and type

1.2 Wind Effects on ACC Performance

The importance of wind effects on the design and performance of ACCs derives from the following considerations. ACCs reject sensible heat to atmospheric air and thus are limited in the minimum condensing temperature they can achieve by the ambient (dry bulb) temperature. A typical ACC design might be sized to maintain a condensing steam temperature of 110°F—corresponding to a turbine exhaust pressure of 2.5 inches of mercury absolute (in Hga)—at full steam load and at the ambient temperature corresponding to an average summer temperature at the site of, perhaps, 70°F. However, the highest ambient temperatures at some sites might exceed 110°F resulting in a condensing temperature of nearly 150°F and a turbine exhaust pressure higher than 7.5 in Hga.

Two problems arise. First, for all times of the year when the ambient temperature exceeds 70°F, the turbine backpressure will exceed the design specification of 2.5 in Hga, and the turbine output, at constant firing rate, will decline. Figure 1-1 shows a typical temperature duration curve for a hot, arid site using meteorological data characteristic of desert areas in the southwestern United States. The ambient temperature exceeds 70°F for over 4,000 hours or nearly half the year. Figure 1-2 displays a typical performance curve for a turbine designed for use with dry cooling. A rise in the turbine exhaust pressure to 7.5 in Hga results in a 13% reduction in output compared to operation at the design backpressure of 2.5 in Hga.

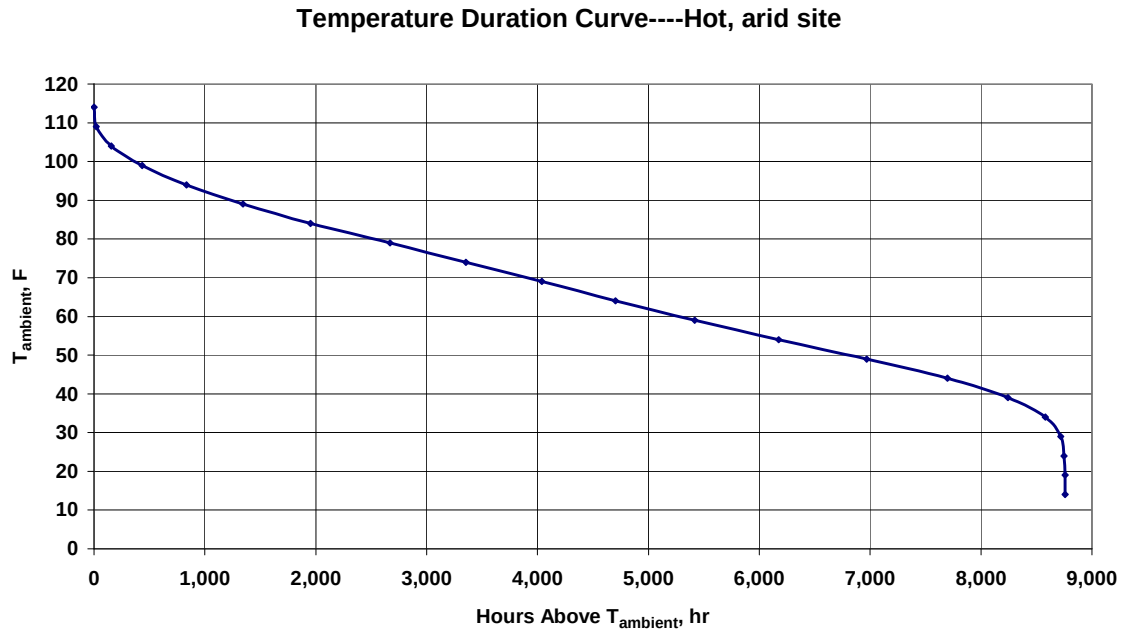


Figure 1-1. Ambient temperature duration, southwestern United States.

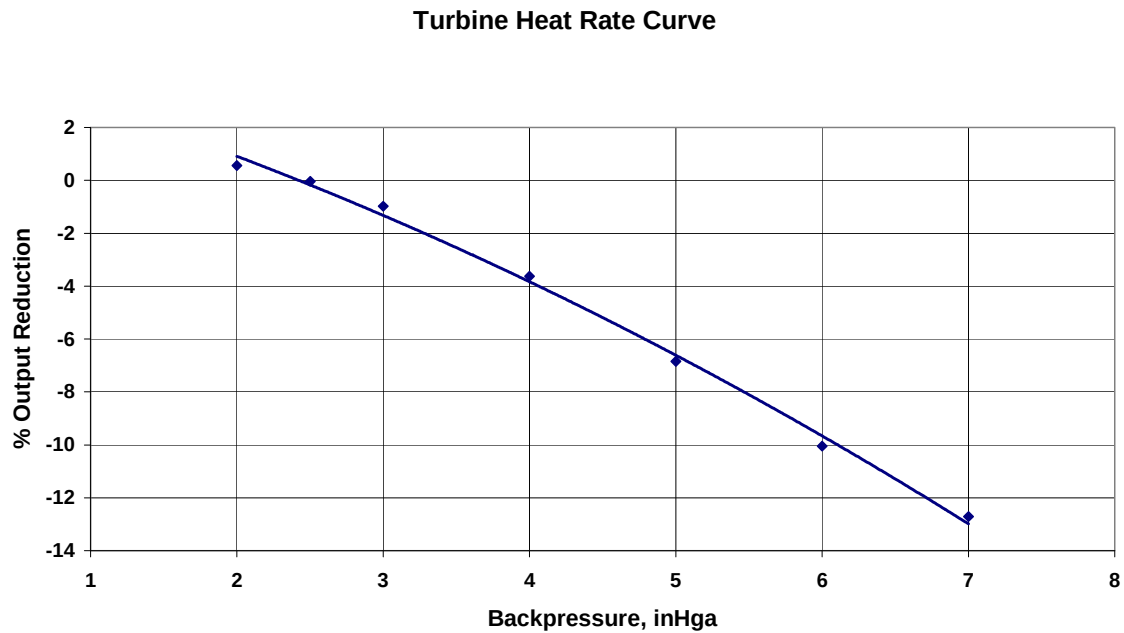


Figure 1-2. Steam turbine performance.

These performance limitations, while substantial, still assume that the ACC is performing at the full capability specified by its design performance curves. Such performance, however, assumes low-wind conditions. Acceptance tests, normally conducted to ensure that the ACC meets its

performance guarantees, do not provide assurance of any level of performance under windy conditions since test codes typically stipulate that valid tests must be conducted at wind speeds below 3 to 4 meters per second or 6.5 to 9.0 miles per hour.

Figure 1-3 illustrates the effect of wind on the performance of one ACC in the southwestern U.S. It is noteworthy that wind speeds above about 20 mph result in backpressure that are 1.5 to 2.0 in Hga higher than those at the same ambient temperature and steam load at wind speeds below 4 mph. In addition, backpressures approach the turbine alarm point of 7 in Hga at temperatures just above 100°F instead of 110°F as the ACC design performance curves would indicate. As shown in Figure 1-1, this condition, which requires the plant to consider reducing load to avoid the possibility of a turbine trip, would occur for nearly 500 hours per year. Furthermore, the wind speeds shown in Figure 1-3 are one-hour average values and not instantaneous. Wind gusts, if they occur when the ACC is operating near the turbine's trip point, can cause a brief excursion to pressures high enough to trip the turbine so rapidly that operators cannot respond in time to keep the plant from coming off-line. A sudden trip—which may take several hours to recover from—can have a significant effect on plant revenue when occurring at a time when the plant is at full load, network demand is at its peak, and the price for power is high.

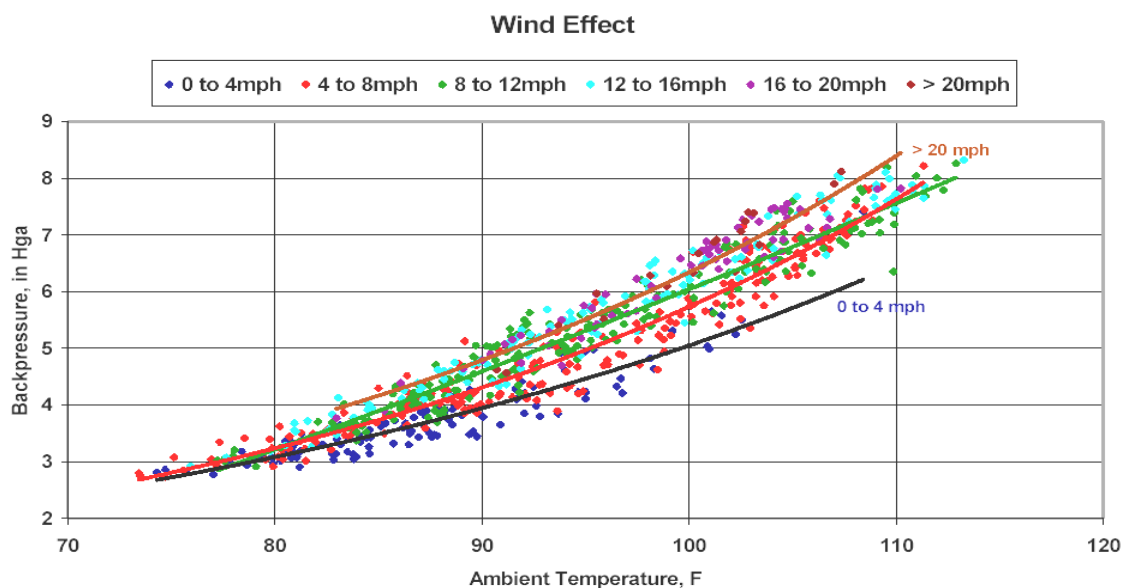


Figure 1-3. Effect of wind on ACC performance.

1.3 Awareness of the Problem

ACC performance test codes specify the maximum wind conditions under which tests are considered to be valid. The VGB code (VGB Technische Vereinigung der Grosskraftwerksbetreiber E. V., 1997), in worldwide use since 1997, states that the mean wind velocity at the upper edge of the air-cooled condenser must not exceed 3 meters per second (6.7 mph); peaks in

excess of 6 m/s (13.4 mph) must not occur more than 20 times per hour. Two codes currently in the final stages of preparation by the Cooling Technology Institute and the American Society of Mechanical Engineers specify similar limitations.

A recent publication of *Air-Cooled Condenser Design, Specification, and Operation Guidelines* (EPRI 2005) by the Electric Power Research Institute notes that, "The impact of ambient wind on ACC performance is not well understood by owner/operators or their representatives in the specification and bid/evaluation process," and that "this area of wind effects in total represents the major challenge associated with ACC specification, design, and performance."

1.4 Mechanisms of Wind Effect

The mechanisms causing the effect of wind on ACC performance will be discussed in detail throughout the report. However, it is noted here by way of introduction that the two commonly accepted mechanisms are *hot air recirculation* and *degraded fan performance*.

Recirculation occurs when the plume of heated air exiting the ACC, which normally rises vertically above the unit under quiescent conditions, is bent over by the wind to the point where a portion of it is entrained by the inlet air stream entering around the sides of the ACC below the fan deck. This results in the cooling air entering the condensing tube bundles being at a higher temperature than the far-field ambient air temperature. Recirculation usually has the greatest effect on the downwind cells.

Degraded fan performance occurs when wind passing underneath the fan deck distorts the inlet velocity field to the fans. ACC fans are typically large, low-speed fans with relatively low static pressure rise and are particularly susceptible to inlet losses. The effect, usually occurring on the upwind fans, can range from a modest decrease in inlet airflow to a stall condition on all or a portion of the fan blades, resulting in dramatically reduced airflow.

It is generally believed that fan performance degradation is the more important of the two effects. However, this can vary with the details of the site topography, the presence of nearby obstructions, ACC orientation relative to the prevailing winds, and any other factors influencing wind speed, direction, turbulence, and gustiness at the ACC inlet.

1.5 Earlier Wind Effects Studies

The adverse effects of wind on the performance of large ACCs have been recognized for many years. Kröger (1998), in a chapter entitled "Effect of Wind on Air-Cooled Heat Exchangers" states the situation succinctly: "In general, winds have a negative effect on the performance of mechanical-draft heat exchangers. Plume air recirculation tends to increase while fan performance is usually reduced during windy periods."

In the ensuing discussion, Kröger cites 16 references dating from 1958 dealing with recirculation in large cooling units (not all forced-draft air-cooled condensers) containing plant tests, physical model studies, and computational model analyses. The results of two studies—one field test at the Wyodak plant (Duvenhage, Du Toit, and Kroeger 1996; Kroger 1998) and one physical

model study of the Utrillas plant in Spain (Kröger 1998)—indicate the potential magnitude of the problem and are reproduced in Figures 1-4 and 1-5.

At a wind speed of 8.5 m/s (~20 mph) from the northeast, the turbine output was reduced by 12% to 14% at the Wyodak plant (Figure 1-4). Model studies at the Utrillas plant in Spain predicted output power reductions of 7% to 8% for southerly winds of 7 m/s (~15 mph) as shown in Figure 1-5.

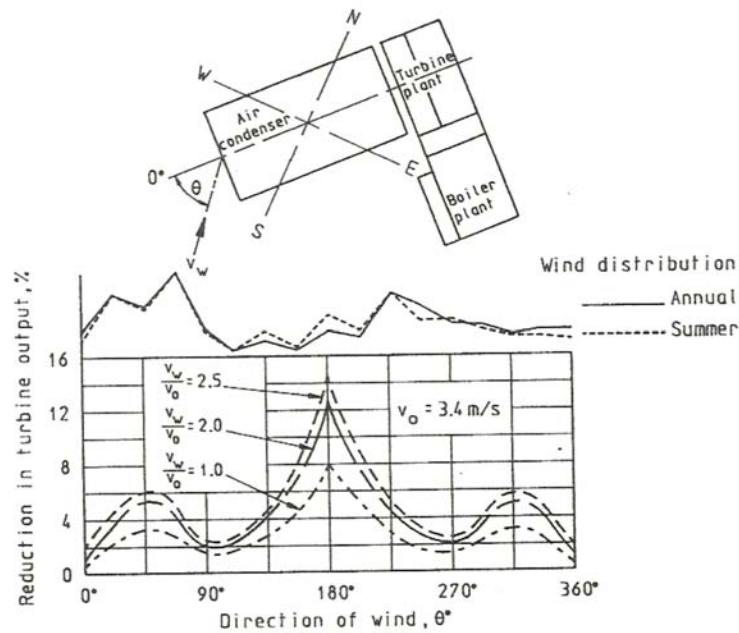


Figure 1-4. Effect of wind on turbine output at Wyodak (from Kröger 1998).

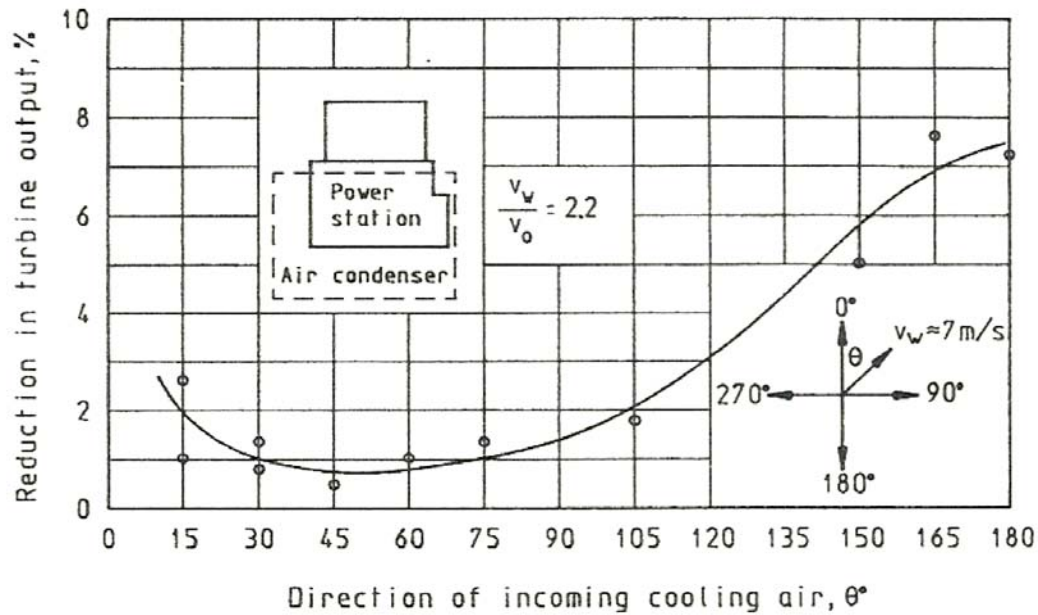


Figure 1-5. Effect of wind on turbine output at Utrillas (from Kröger 1998).

1.6 Approaches to Addressing the Problem

The authors are aware of no cases in which ACCs have been initially designed to mitigate high-wind effects. Presumably this would be done with additional heat transfer surface area, additional cells, additional fan power, or some combination of all three. This would, of course, lead to a higher unit cost—which many purchasers would be unwilling to accept to ameliorate conditions that may arise only occasionally. EPRI's ACC Guidelines (EPRI 2005) address this issue at some length from the point of view of both the purchaser and the vendor.

The usual situation is that attempts are made after the fact to retrofit the ACC with some sort of windscreen or barrier after operating experience has shown wind effects to be severe enough to warrant corrective action. Figures 1-6 and 1-7 show two examples. The first (Figure 1-6) is a porous windscreen in a cruciform shape under the ACC at the El Dorado Energy Center in El Dorado, Nevada. The second (Figure 1-7) shows a solid windwall added to ACCs on two units at the Black Hills Power and Light plant in Gillette, Wyoming. This barrier runs lengthwise between the two streets of the ACC and also includes a horizontal barrier (not shown in the photo) between the end of the ACC and the wall of the turbine building to prevent downdrafts from winds blowing over the ACC and hitting the building wall.

Both retrofits appear to have had beneficial effects, although other features of the units were changed at or about the same time, such as modified fans and additional cells, making a rigorous before-and-after comparison difficult in either case.



Figure 1-6. Windscreen at El Dorado.



Figure 1-7. Windwall at Black Hills.

1.7 Objectives

An understanding of the extent to which wind affects ACC performance, a determination of the mechanisms causing the adverse effects, and a possible approach to mitigating the effects would be valuable for making the best use of new and existing air-cooled condensers in the expanding national and worldwide power mix. In light of the importance and lack of understanding of the effect of wind on ACC performance, this study was initiated with the following objectives:

- Define and quantify the effects of wind on ACC performance.
- Evaluate some specific existing attempts to mitigate the effects.
- Suggest further research needed to better control wind effects for ACC performance improvement.

1.8 Organization of Report

The remainder of the report is organized as follows:

- Chapter 2 details the study approach.
- Chapter 3 describes the characteristics of the site, plant, and ACC at each of the five test sites.
- Chapter 4 describes the measurements made and the instrumentation used to make them.
- Chapter 5 summarizes the data obtained and describes the large database assembled from the measurements. Representative portions of the database are presented in Appendices to illustrate its content and format for use in future modeling work. The complete database is contained on a CD included with the report and can be acquired from the California Energy Commission.
- Chapters 6 and 7 presents a variety of analyses and correlation efforts intended to generate useful and general trends and to relate the observed effects to the physical behavior of the airflow and heat transfer mechanisms around the ACCs. Chapter 6 focuses on recirculation, while Chapter 7 discusses fan performance degradation.
- Chapter 8 summarizes conclusions and recommendations from the entire study.
- Chapter 9 lists references cited.
- Chapter 10 provides a brief glossary.

CHAPTER 2:

Study Approach

2.1 Study Approach

This study aimed to confirm or modify the generally accepted hypothesis that the negative effect of wind on ACC performance is attributable to two major causes:

- Recirculation of heated plume air into the ACC inlet air stream.
- Degradation of fan performance in some cells and, hence, reduced airflow to those cells.

These mechanisms were investigated through detailed monitoring of full-scale, operating ACCs under varying wind conditions.

2.2 Site Selection

Five sites were selected for testing. Detailed descriptions of the site, plant, and ACC characteristics of each site are given in Chapter 3. As a group, the sites provided good coverage of the range of conditions of interest to the design, specification, and operation of an ACC for power plant use, specifically:

- All sites normally experienced high (>20 mph) winds during the summer months.
- Both common plant types (a coal-fired steam plant and gas-fired combined-cycle plants) were included.
- ACCs from all of the (then) three major ACC vendors were included. This resulted in a variety of tube bundle designs (single row and multi-row, aluminum and galvanized tubes, plate and helical fins, etc.) and different fan designs from major vendors (blade shape, number of blades, etc.).
- The sites represented a range of ACC sizes (12 to 40 cells) in a variety of configurations (40 cells in a single 8 x 5 arrangement and two 4 x 5 clusters, 30 cells in both 5 x 6 and 6 x 5 arrangements, and 12 cells in a 2 x 6 arrangement).
- ACCs with and without windscreens were tested. The windscreens at the three sites which had them all differed in type and configuration.

2.3 Measurements

At each site, the project team erected a meteorological (met) tower to monitor wind speed and direction. The testing included extensive instrumentation on the ACCs for identifying separately the effects of recirculation and degraded fan performance.

2.3.1 Recirculation

Recirculation was determined by monitoring inlet air temperature in each cell. Inlet temperatures higher than the surrounding ambient temperature indicated hot plume

recirculation. The location of cells with elevated inlet temperatures could be interpreted as a recirculation pattern on the ACC.

2.3.2 Fan Performance

Fan performance was more difficult to determine. The approach chosen was to take extensive measurements of inlet air velocity static pressure rise and fan motor current on selected cells. This full suite of measurements was restricted to the first site with the intent of identifying a single, surrogate measurement that could be used to monitor fan performance at the remaining sites more economically. The selected measurement for the remaining sites was inlet air velocity. Not all cells could be monitored with available instrumentation. The cells chosen were those where high cross-wind velocity was expected.

2.3.3 ACC and Plant Performance

Plant data acquisition systems provided the data for turbine exhaust pressure, steam flow to the ACC, steam turbine/generator output, fan status (off/half-speed/full speed), and fan power (when available). In addition, plant weather data—ambient temperature, wind speed, and wind direction—were obtained to serve as a check on the project's own met station readings.

2.4 Data Analysis and Modeling

Data from project instrumentation were recorded at one-minute intervals. Plant data were provided at intervals varying from one minute to one hour, depending on the plant and on the metric being reported. (Plant met data were sometimes from a different system than the plant operating data.) Occasional gaps in both project and plant data records occurred as a result of power outages, computer malfunctions, or other interruptions.

Data organization, archiving, and assembling into spreadsheet format for analysis and plotting included:

- Creating a continuous, gap-free, one-minute time series and synchronizing all data from all sources.
- Identifying and removing any obviously erroneous points resulting from instrumentation or data acquisition system failure.
- Flagging all points for which not all fans were running at full speed and segregating them from the routine analysis.

2.4.1 Meteorological Data

Wind speed and direction data from the project met tower and, when available, from plant instrumentation were plotted for the duration of the test period and compared. Significant differences were examined with reference to site plans to determine if any structural or other sources to interference were responsible.

Ambient temperature data, normally available from plant instrumentation in at least one, and usually two, locations (typically on the administration building and under the ACC) were

plotted throughout the test period and compared to the minimum cell inlet temperature. Significant differences among the records were examined to identify any solar radiation or shielding effects or any possible sources of hot air in the vicinity of the ACC. Temperature data were plotted against time on the same plot with wind speed and direction and on plots of temperature vs. wind speed and wind direction to identify any obvious trends or relationships.

At one plant, for a limited period, numerous measurements were made both on the plant boundary fence and at three elevations at each of four locations on the perimeter of the ACC. Chapter 5 provides an extended discussion of the results and implications of those measurements on how recirculation is determined for the purposes of this study.

2.4.2 Recirculation Data

Average recirculation—defined as the average of all cell inlet temperatures minus the minimum cell inlet temperature—as well as recirculation to individual cells was calculated for all times when all fans were at full speed. These results were plotted several ways:

- Average and maximum recirculation vs. time (on the same plot as wind speed and/or direction).
- Recirculation vs. wind speed.
- Recirculation vs. wind direction.
- An array of all cell inlet temperatures at a given time arranged from minimum to maximum temperature with the points labeled with the cell identifier.

2.4.3 Fan Performance Data

Plots were made—in a similar manner and format as the recirculation plots described above—of fan inlet velocity (at four locations across the fan inlet areas of a few selected fans), fan static pressure rise, and fan motor current.

Fan inlet velocity measurements were examined both as averages of each of the four measurements on a single fan and for variation in velocity across the fan inlet. These variations were compared with wind speed and direction.

Details of the measurements and the subsequent analysis and results are given in Chapter 3 (measurements), Chapter 4 (data collection), and Chapters 5 and 6 (results).

CHAPTER 3:

Site, Plant, and ACC Information

Field tests of the effect of wind on ACC performance were conducted at five different sites, listed here in the order of testing:

- Bighorn power plant in Primm, Nevada
- El Dorado Energy Center in El Dorado, Nevada
- Apex generating station in Apex, Nevada
- Front Range in Fountain, Colorado
- Wygen plant in Gillette, Wyoming, owned by Black Hills Corporation; figures and tables in this report refer to this plant as Wygen or Black Hills interchangeably.

Summary information on the sites, the plants, and the ACCs is tabulated below. More detailed information on site information, plant information, and the ACCs is available from the authors.

This chapter provides information in photographic and sketch form on the layout of each site and the location and orientation of the ACC in relationship to other plant structures. This information is important for understanding the potential obstructions to and distortions of airflow to the ACC when the wind comes from different directions.

3.1 Site Information

All five sites were located in the western United States in relatively windy areas. Table 3-1 gives the location and elevation of each site along with summary information about the meteorology during the summer months.

Table 3-1. Wind study site information.

Site	Location	Lat/Long	Elevation (ft)	Summer Temperature (F)		Wind (Summer)	
				Average	Max	Speed Range (mph)	Prevailing Direction
Apex ⁽¹⁾	Apex, NV	36.33N/114.83W	2,563	87	112	8 - 40	S to SW
Bighorn ⁽¹⁾	Primm, NV	35.61N/115.39W	2,800				
El Dorado ⁽¹⁾	El Dorado, NV	35.78N/114.98W	1,820				
Front Range ⁽²⁾	Fountain, CO	38.63N/104.70W	5,464	66	95	5 - 35	N or S
Wygen ⁽³⁾	Gillette, WY	44.29N/105.50W	4,364	64	99	5 - 35	Variable (NW, S and SE)

(1) Met data from Las Vegas/McCarran Airport

(2) Met data from Colorado Springs/Peterson AFB

(3) Met data from Sheridan, Wyoming

3.1 Plant Information

Table 3-2 summarizes some general information about each plant including plant type, capacity, owner/operator, and the year the plant or unit came on line.

Table 3-2. Wind study plant information.

Plant	Type	Capacity (MW)	Owner	Date On-line
Apex	Gas-fired combined-cycle (2 x 1)	533	Mirant	2003
Bighorn	Gas-fired combined-cycle (2 x 1)	540	Reliant	2004
El Dorado	Gas-fired combined-cycle (2 x 1)	540	Reliant/Sempra	1998
Front Range	Gas-fired combined-cycle (2 x 1)	480	Front Range Power Company	2000
WyGen	Coal-fired steam	80	Black Hills Power and Light	2001

3.2 ACC Information

Tables 3-3 and 3-4 provide information about the ACC components and the design point at each site.

Table 3-3. Wind study ACC information.

Site	ACC Vendor	Cells		Fans			Windscreen	
		Number	Arrangement (Streets x Cells/Street)	Vendor	Number of blades	Diameter (ft)	Present	Arrangement
Apex	Marley (Balcke-Duerr)	30	6 x 5	BDT	5	34	No	
Bighorn	Marley (Hamon)	40	Two 4 x 5	Howden	7	34	Yes	Street direction
El Dorado	GEA	30	5 x 6	Cofimco	8	34	Yes	Cruciform
Front Range	GEA	40	8 x 5	Cofimco	6	32	No	
Black Hills	GEA	12	2 x 6				Yes	Cruciform

Table 3-4. ACC design points.

Site	ACC Vendor	Ambient Temp	Steam Flow	Exhaust Quality	Turbine Exhaust Pressure	Fan Power	Design ITD**
		F	lb/hr	(--)	in Hga	kW	F
Apex	Balcke-Duerr*	86	1,454,087	0.98	10	4,150	75.5
Bighorn	Hamon*	96	1,022,366	0.98	3.79	5,190	27.5
El Dorado	GEA	67	1,065,429	0.95	2.5	3,875	41.7
Front Range	GEA	80	1,266,468	0.97	3.57	4,950	41.3
WyGen	GEA	93	558,179	~ 0.95***	5.0	~ 1,500***	40.8

* Now part of SPX-Marley
 ** ITD = Initial Temperature Difference (Tcondensing - Tambient)
 *** Estimated

3.3 Site Layouts

ACCs consist of a number of individual cells arranged in rectangular arrays. Steam is ducted from the turbine exhaust to the cells through a main steam duct which feeds a number of parallel risers, each feeding a single horizontal duct which runs along the top of a number of cells. These groups of cells, each fed by a single steam duct, are referred to in this report as “streets.” The groups of cells aligned perpendicular to the steam ducts adjacent to one another are referred to in this report as “rows.” This terminology is common though not universal.

A typical arrangement will have five or six cells in each street. One of these cells, located near the middle of the street, is a “reflux” or “dephlegmator” cell, while the others are condensing cells (sometimes called K-cells and D-cells). The dephlegmator (D-cell) does not receive steam directly from the steam duct but takes steam and non-condensables from the condensate lines, returning the additional condensate to the condensate lines and removing the non-condensables through an air removal system at the top of the tube bundles.

The cells are usually, but not always, arranged in a single, rectangular cluster as is the case at El Dorado, Apex, Front Range, and Black Hills. The 40-cell ACC at Bighorn is arranged in two 20-cell clusters fed from a main steam duct, with risers centrally located between the two clusters

as can be seen in Figure 3-2. This arrangement may have been chosen to shorten or simplify the steam ducting.

3.3.1 Bighorn

Figure 3-1 shows an aerial view of the Bighorn Power Plant. The ACC is aligned essentially north-south with the evaporation pond on the west. The other plant structures are all to the east of the ACC. The HRSGs off the southeast wall of the ACC are the tallest obstructions. The combustion turbine buildings and other plant structures to the north and further to the east of the HRSGs are lower and provide less obstruction. Wind from the south, west, or north is unobstructed in its approach to the ACC. The low structure at the north end of the ACC is an air-cooled auxiliary heat exchanger with the potential for discharging hot air into the ACC inlet when winds come from the north.



Figure 3-1. Aerial view of Bighorn plant.



Figure 3-2. General ACC layout with steam duct and risers between north and south clusters.

Figure 3-2 shows the general layout of the Bighorn ACC as seen looking west from the catwalk near the top of the stack on HRSG #1 (closest to the ACC). It shows the steam duct and risers in the space between the north and south clusters.

Figure 3-3 shows the location of the HRSGs near the east wall of the south cluster of the ACC. The bulk of the HRSG buildings rises above the height of the ACC fan deck and can be expected to obstruct and distort airflow from southeasterly to easterly winds.



Figure 3-3. Obstruction of easterly wind flow by HRSGs.

Figures 3-4 through 3-8 show elements on and under the ACC that influence wind patterns. Figure 3-4 shows one of two “wings” on the northwest corner extending from the ground to the top of the windwall and approximately 30 feet to the north of the corner in line with the west wall. An identical wing is located at the southwest corner as well.



Figure 3-4. Wind wing at northwest corner by Cell 4A.

Figure 3-5 shows the windscreen running between Streets 2 and 3 in the north-south direction under both the north and south clusters. It extends from the ground to the fan deck and is mounted in three sections corresponding to the three vertical structural sections separated by horizontal beams. As indicated schematically in Figure 3-2, the lower section is moved to the north to run between Streets 3 and 4 underneath Cells C and H in order to avoid interference with the motor control center (MCC) buildings.



Figure 3-5. Windscreen under north cluster at Bighorn.

While both the screens and the corner “wings” exert important influence on the wind patterns, plant staff and vendor personnel both confirm that neither the screens nor the wings were originally intended to influence the wind patterns or to mitigate wind effects but rather were installed for aesthetic reasons.

Figures 3-6 indicates the extent of blockage from the steam ducts and risers at the north end of the south cluster and the south end of the north cluster.

Figures 3-7 and 3-8 indicate the extent of blockage underneath the ACC from the motor control centers under Rows C and H and the condensate systems (tanks, piping, and pumps) under Rows G and D.



Figure 3-6. Blockage from steam ducts and risers.



Figure 3-7. Blockage from motor control centers (MCCs).



Figure 3-8. Blockage from condensate return system.

3.3.2 El Dorado Energy Center

Figure 3-9 shows the El Dorado Energy Center as seen from the southeast. The ACC is located at the south end of the site with the steam ducts/streets aligned north and south. The south, west, and east sides of the ACC are completely open and unobstructed to incoming airflow.



Figure 3-9. El Dorado plant site.

Figure 3-10 shows the northwest corner of the ACC, the two steam risers to Streets 4 and 5 (see Figure 4-6) for ACC street, row, and cell designations) and the westerly combustion turbine and HRSG located to the north of the ACC. The HRSGs rise above the height of the ACC fan deck and can be expected to partially obstruct and distort winds from the northerly direction.



Figure 3-10. El Dorado blockage from risers and HRSGs at north end.



Figure 3-11. El Dorado windscreen.

Figure 3-11 illustrates the windscreen under the ACC. The screen is in a cruciform arrangement running north-south between Streets 3 and 4 and east-west between Rows 3 and 4. The screens extend from the ground to the fan deck and are divided into three vertical sections separated at the horizontal structural beams. The porosity of the screens varies from top to bottom with the top section being the most porous.

Figure 3-12 gives some indication of the blockage at the north end of the ACC from the steam risers on the north face, the MCC buildings under Row 2 and the condensate system under Row 1. Note that the picture was taken prior to the installation of the windscreens in 2003.



Figure 3-12. Blockage from steam risers, MCCs, and condensate system.

3.3.3 Apex Generating Station

Figure 3-13 shows the Apex Generating Station from the boundary at the northeast corner. The ACC is located in the south portion of the site. Figure 3-14 illustrates the layout of the site and its major structures more clearly.

Winds approaching the ACC are completely unobstructed from the west, south, and east with the exception of small structures in the water treatment area and the MCC buildings on the west side.



Figure 3-13. Apex plant from northeast corner of site.

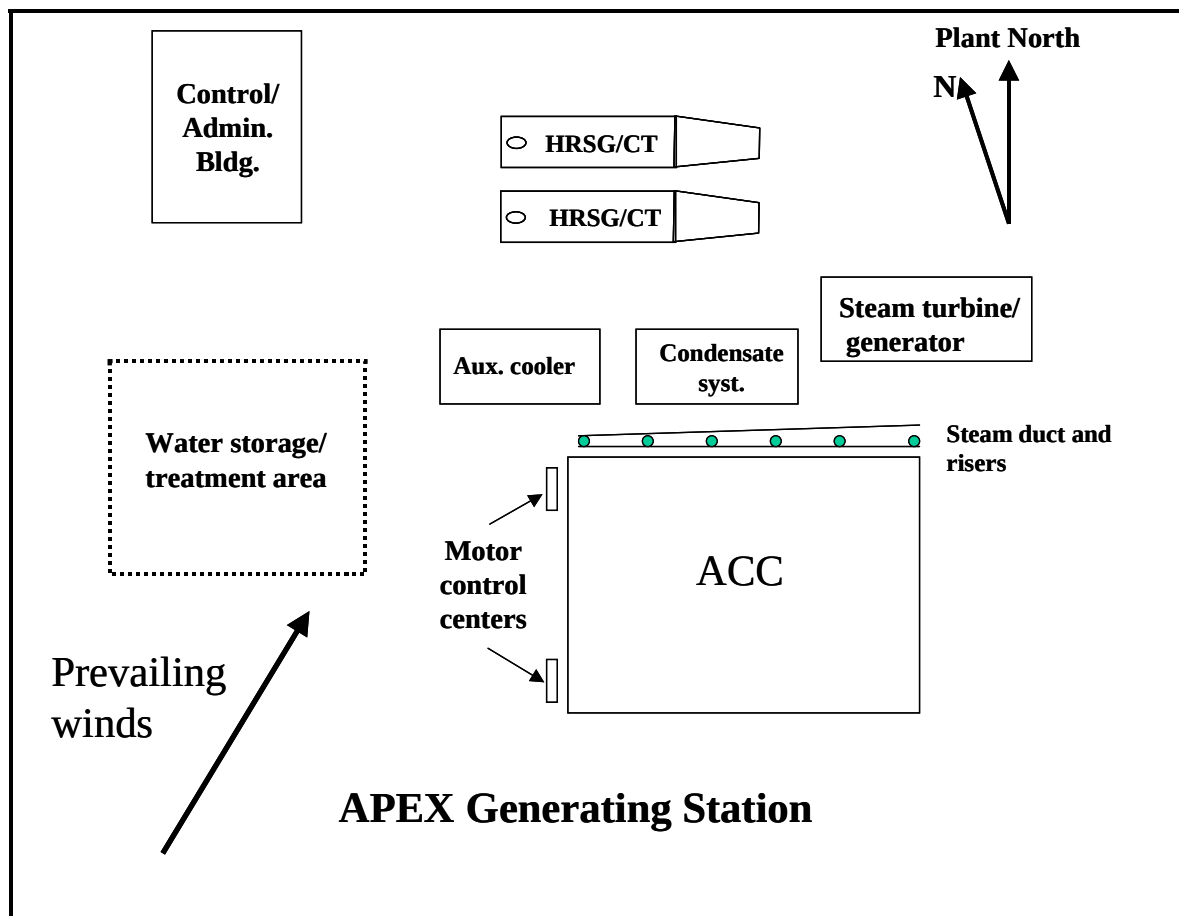


Figure 3-14. Schematic layout of Apex site.

Figure 3-15 shows the ACC and the supporting structure as seen from the ground at the southwest corner. The heavy obstruction on the north end can be seen through the structural members. The small building to the left is one of the motor control centers. Another is similarly located at the northwest corner.

Figure 3-16 shows the HRSGs and the steam turbine building located just to the north of the ACC. Both can be expected to influence the approaching patterns of winds coming from the north.

Figure 3-17 shows the blockage of the steam duct at ground level and the risers along the north face of the ACC and HRSG structures rising to the level of the fan deck further to the north.



Figure 3-15. Apex ACC from the southwest corner.



Figure 3-16. Apex HRSGs and steam turbine building to the north of the ACC.



Figure 3-17. Steam duct, risers, and plant buildings to north of ACC.

3.3.4 Front Range Generating Station

Figure 3-18 shows the Front Range ACC as seen from the HRSG stack platform (visible in Figure 3-19) off the southeast corner of the ACC. The ACC is located on the west side of the plant site and is open to unobstructed wind flow from the south, west, and north.



Figure 3-18. East side of the Front Range ACC from HRSG stack.

Figure 3-19 shows the plant structures as seen from the top of the caged ladder at the northeast corner of the ACC near the Street 1 steam duct (visible in Figure 3-18). The largest of the structures are the HRSG buildings located to the east of the south end of the ACC. The HRSG buildings rise above the level of the ACC fan deck.

Figure 3-20 shows the blockage at ground level on the east side of the ACC from the steam heater and risers and the proximity of the neighboring plant buildings.

Figure 3-21 shows the MCC buildings located under Street 7 of the ACC at about Row 3. (See Figure 4-8 for the street, row, and cell designations for Front Range.)

Figure 3-22 shows the blockage due to the presence of the condensate system under Row 2 at about Street 6.



Figure 3-19. Nearby HRSGs seen from northeast corner of ACC.



Figure 3-20. Blockage along east side from steam header, risers, and nearby HRSGs.



Figure 3-21. Motor control centers under the Front Range ACC.



Figure 3-22. Condensate system under the Front Range ACC.

3.3.5 Wygen I (Black Hills)

Figure 3-23 shows the top of the Wygen II ACC as seen from the roof of the boiler/turbine building. The view is toward the south. Wind flow approaching the ACC is unimpeded from the east, south and west.

The north end of the ACC is close to the south wall of the boiler/turbine building. The building is a large structure rising well above the ACC steam ducts and extending beyond the sides of the ACC to the east and west. It completely blocks and redirects wind from any direction in the northern half of the compass as shown in Figure 3-24.

Figures 3-25 shows the wind barrier installed to mitigate the effect of wind on ACC performance. Underneath the ACC it is in a cruciform arrangement running north-south between the two streets and east-west between Rows 3 and 4. In addition a panel closes off the northern opening beneath the ACC, effectively extending the windwall from the fan deck to the ground.



Figure 3-23. Wygen II ACC facing south from roof of boiler/turbine building.



Figure 3-24. ACC abutting boiler/turbine building.



Figure 3-25. Wygen windscreen arrangement.

The arrangement of the ACC is somewhat unusual in that the end cells in each street (K16 and K26) are separated from the other five cells in the street. They are helper cells which are used to provide additional cooling capability in the summer but are shutoff in the winter. These end cells are separated from the fifth row by a narrow passageway shown in Figure 3-26. The catwalk running between the two streets at the steam duct level provides access to the steam isolation valve shown in Figure 3-27.



Figure 3-26. Open region between Rows 5 (left) and 6 (right).



Figure 3-27. Catwalk and steam isolation valve for Cell K16.

CHAPTER 4:

Instrumentation and Measurements

Instrumentation was installed at each site to make those measurements required to:

- Characterize the wind conditions at the ACC.
- Determine the pattern and amount of recirculation.
- Determine the effect of wind on the performance of selected fans.

Figures 4-2 through 4-10 show the layout of each of ACCs, the location of the several test instruments, and other relevant features of the site and the ACC, such as windscreen location (if present).

4.1 Determination of Wind Conditions

Wind speed and direction were determined with a pole-mounted, vaned propeller anemometer at a height of about 15 feet. Figure 4-1 shows the project weather vane installed at Bighorn, secured to the plant boundary fence. The instrument was typically located 50 to 100 feet upwind of the ACC in the prevailing wind direction. The direction was set with a handheld compass.

When available, comparable data were obtained from the plant met station. In the case of Apex, where no met station was available, met data were obtained from the met station at Silverhawk, a neighboring plant located approximately upwind in the prevailing wind direction from the Apex ACC.



Figure 4-1. Project met tower at Bighorn.

4.2 Recirculation Measurements

The amount of recirculation was determined by measuring the inlet air temperature using one four-wire RTD (Resistance Temperature Detector) probe in each cell. The probes were mounted on the fan bridge walkway at a point about two-thirds of the way from the fan hub to the blade tip, corresponding roughly to the location of highest inlet velocity.

At one site (Bighorn), additional temperature probes were mounted on the cleaning ladders on the outer face of the finned tube bundles. This permitted measurement of the exit air temperature from both sides of the five cells in the south portion of Street #3, as shown in Figure 4-5.

4.3 Fan Performance

A number of measurements were made in an attempt to determine the effect of wind on fan performance by sensing any wind-induced flow reductions or fan stalling. The originally planned approach was to take a complete set of measurements, including inlet air velocity, fan static pressure rise, and fan power, at the first site (Bighorn) in the hopes of determining a simpler, “surrogate” measurement that could be used at future sites to indicate any degradation in fan performance. The location of the several instruments and the cells in which they were deployed are illustrated in Figures 4-3 through 4-6.

Table 4-1 lists the instruments installed at the five sites. All instruments were scanned at 10-second intervals with one-minute averages reported to the computer. Typical data logs are

shown in Chapter 5. All of the instrumentation and the data acquisition systems and computers were obtained by lease from Environmental Systems Corporation (ESC) in Knoxville, Tennessee. ESC personnel participated in the set-up and operation of the instrumentation and data acquisition systems at all five sites.

Table 4-1. Instrumentation.

Measurement	Instrument		Site				
	Type	Precision	Apex	Bighorn	El Dorado	Front Range	Wygen
Wind Conditions							
Wind speed	RM Young Prop Anemometer	± 0.6 mph	x	x	x		x
Wind direction	RM Yound Met Station	± 0.25%	x	x	x	x	x
Recirculation							
Inlet air temperatures	4-wire RTD probes	± 0.1 F	x	x	x	x	x
Ambient temperatures	4-wire RTD probes	± 0.1 F		x			
Ambient wet bulb	CTI Psychrometer (M133)	± 0.1 F		x			
Fan performance							
Inlet velocity	RM Young Prop Anemometer	± 2%	x	x	x	x	x
Static pressure	Rosemount Transmitter 3051 DP	± 0.075% F.S.	x	x	x		
Motor current	Yokogawa 2533 Power Meter	±0.04% RDG		x			
	Fluke 43B Power Meter	±1%+10 Counts	x		x		

4.4 ACC Layouts and Instrument Locations

The following sections identify—in the order of testing—the layout of each ACC, specifying its street, row, and cell designations. In addition, the type and location of each measurement is shown schematically for each site.

4.4.1 Bighorn

The 40-cell ACC at Bighorn is arranged in two 20-cell clusters. It is oriented in the north-south direction, and the prevailing winds are southerly. A porous fabric screen extends from a few feet above the ground to the fan deck which runs north and south under the ACC between Streets 2 and 3, as indicated in Figure 4-2. Under Row C in the north cluster and Row H in the south cluster, the lower third of the screen is placed between Streets 3 and 4 to avoid interference from the motor control center buildings. The screen was not intended to mitigate wind effects but rather was installed for aesthetic reasons to shield equipment from view from the neighboring casino resorts. Nonetheless, it clearly affects the wind patterns under the ACC for any wind not directly on the N-S axis.

Figure 4-3 identifies those cells in which inlet velocity and static pressure rise measurements were made. Three of these cells were located close to the south or windward end of the ACC where the major effect of wind on fan performance might be expected. The remaining instrumented cells were located progressively further from the windward edge to determine if the effect diminished with location.

Figure 4-4 identifies the location of the velocity-measuring anemometers. Four were placed within the inlet area of each fan being monitored. They were aligned with the prevailing wind direction to see if the effect varied with distance from the “leading edge” to oncoming wind.

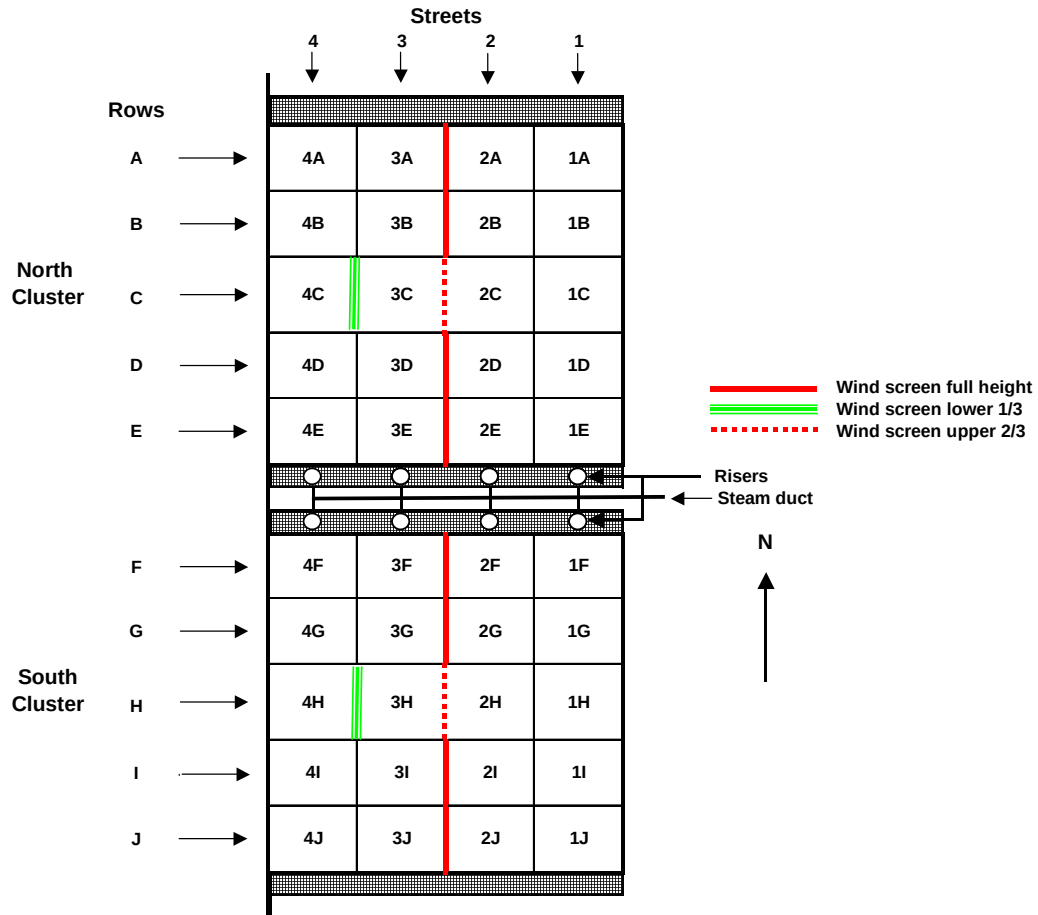


Figure 4-2. Bighorn ACC layout.

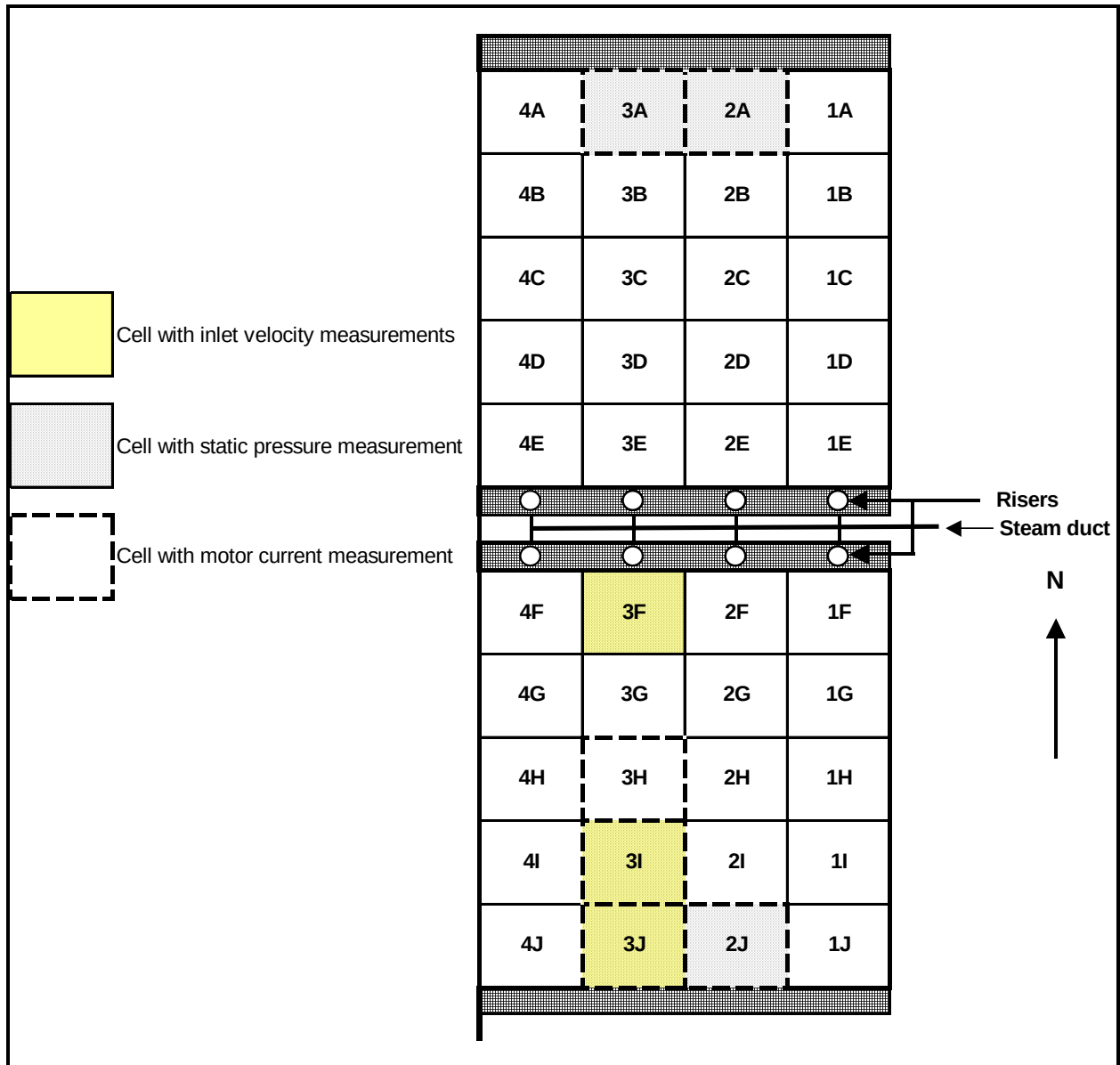


Figure 4-3. Bighorn ACC cells with additional instrumentation.

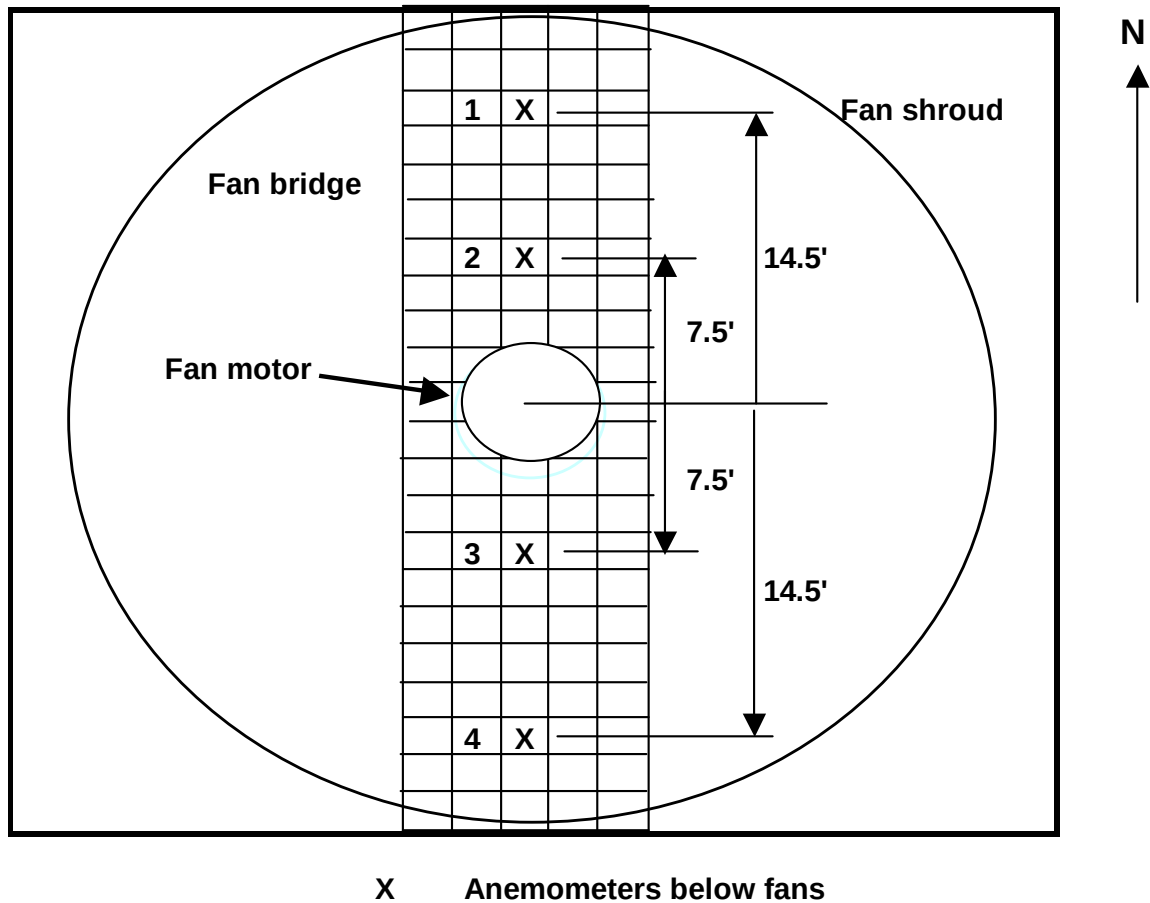


Figure 4-4. Bighorn inlet velocity probe locations (distances given in feet).

Figure 4-5 shows the placement of four anemometers and five temperature probes on the exit face of cell tube bundles. They were mounted on the movable cleaning ladders on the east and west sides of Street 3 in the south cluster. While the ladder could be moved the full length of the cluster from the north end of Cell 3F to south end of Cell 3J, the length of the instrument cables restricted the movement to Rows I and J.

Figure 4-6 shows the placement of 18 separate temperature measurement points at the south end of the south cluster. All were shielded in aspirating psychrometers. The three locations on the plant boundary fence had both wet and dry bulb probes at each location. The remaining 12 probes, all for shielded dry bulb measurements, were suspended at three levels in four locations on the east and west ends of Row I and at southeast and southwest locations on the south face of Row J. The location of the met tower (see also Figure 4-1) reporting wind speed and direction is shown as well.

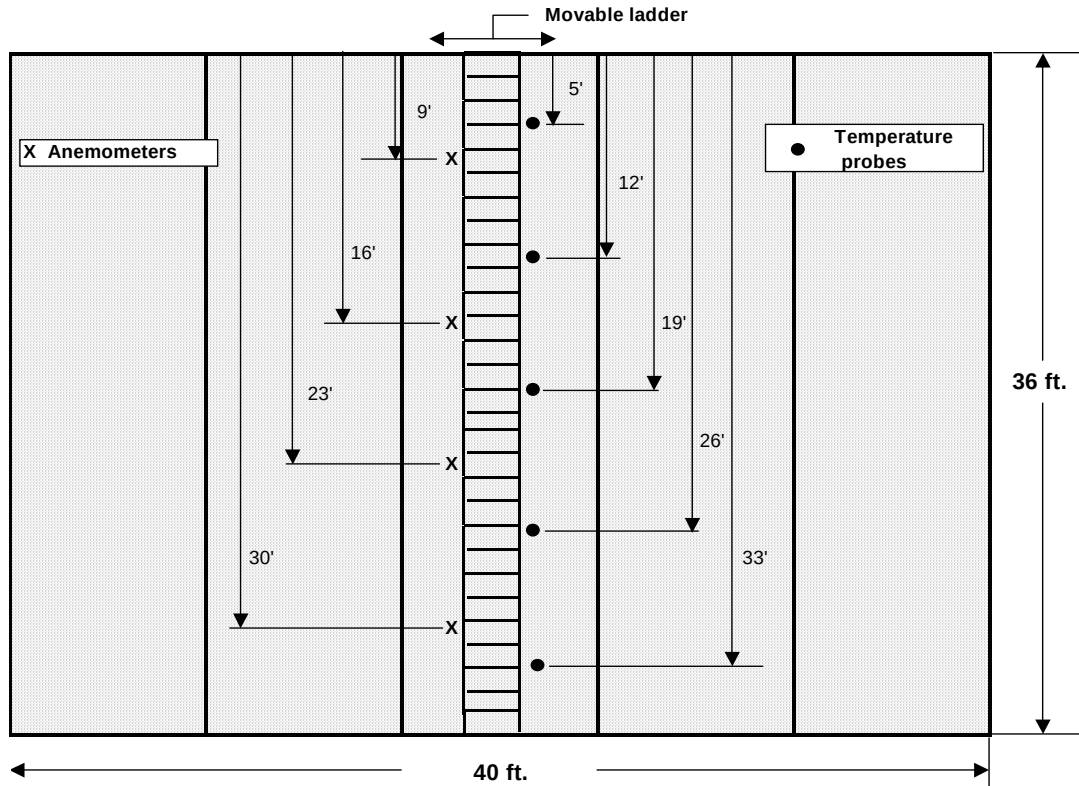


Figure 4-5. Bighorn exit velocity and temperature measurements.

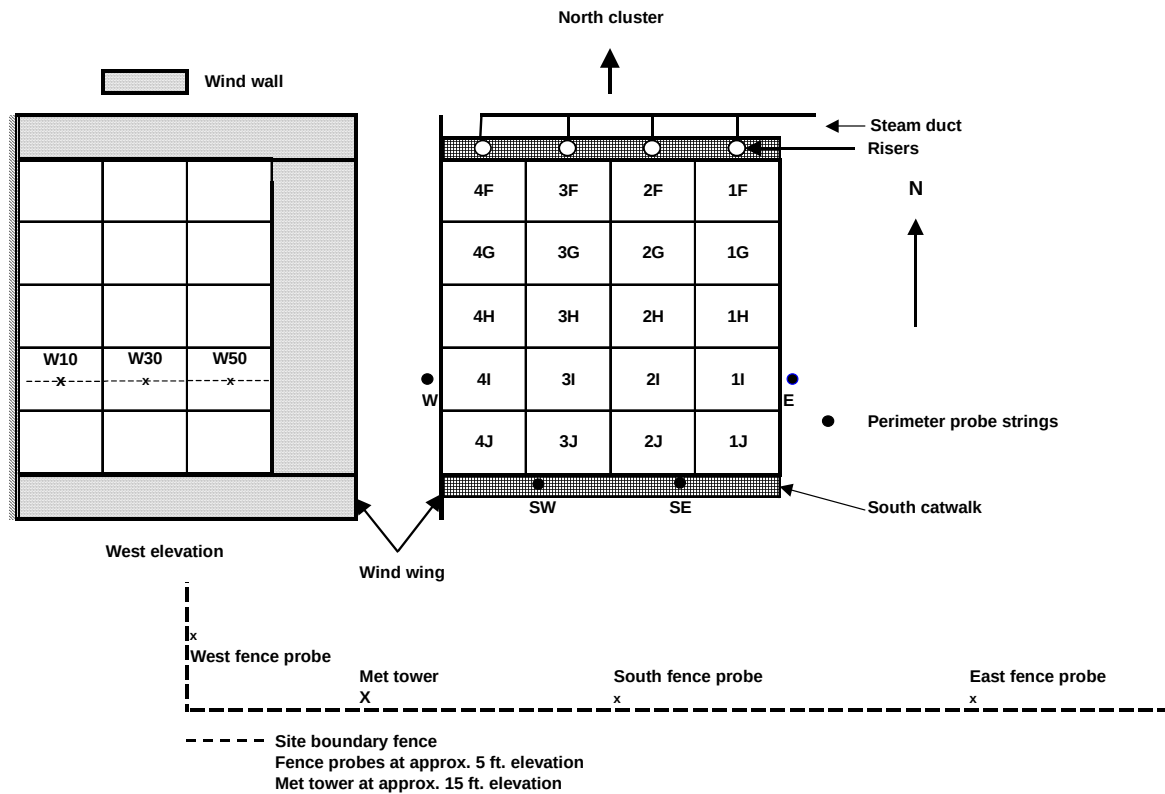


Figure 4-6. Bighorn additional perimeter and ambient measurements.

4.4.2 El Dorado

The ACC at El Dorado is a 30-cell unit arranged in a single 5 x 6 cluster aligned in the north-south direction in line with the prevailing summer winds. It was retrofitted in 2003 with a porous fabric windscreen after significant performance degradation was associated with high winds. It is arranged in a cruciform pattern running north-south between Streets 3 and 4 and east-west between Rows 3 and 4. It is reported to have minimized the effects of wind gusts on ACC performance.

Both inlet air velocity and static pressure rise were monitored in five cells (Cells 1, 19, 22, and 24) and static pressure rise alone in a fifth cell (Cell 15).

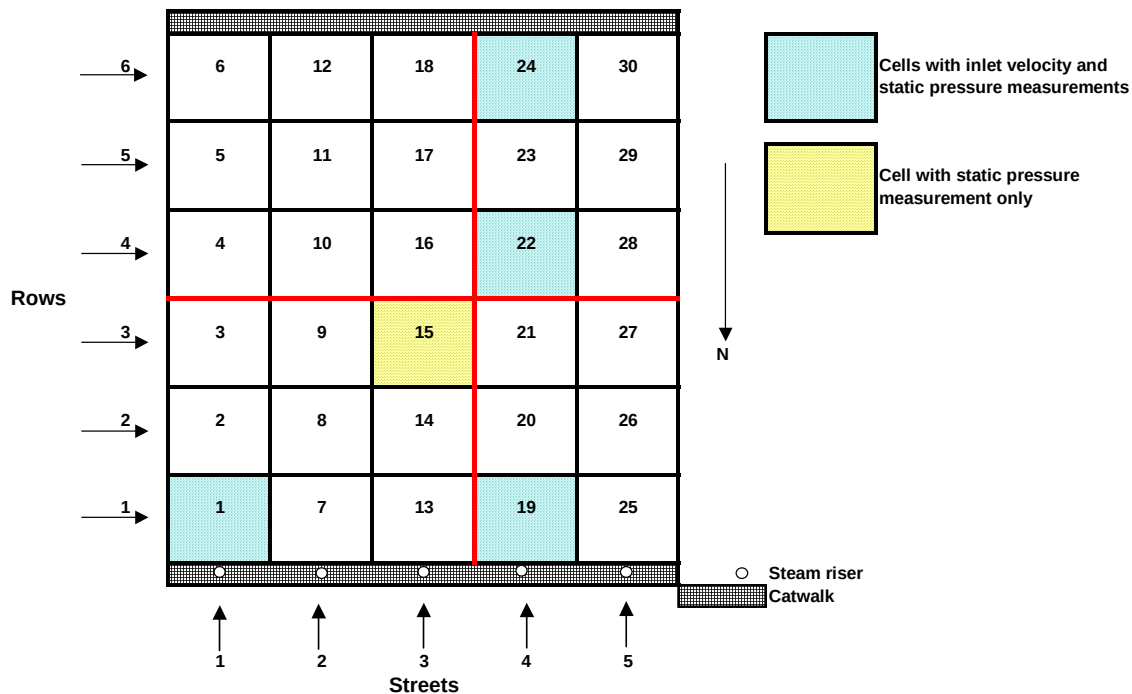


Figure 4-7. El Dorado ACC layout.

4.4.3 Apex

The Apex 30-cell ACC is arranged in a single 6 x 5 rectangular cluster, similar to El Dorado but with 6 streets and 5 rows as opposed to the 5 street/6 row arrangement at El Dorado. No windscreens are in place at Apex. In addition, as seen in Figures 3-14 and 3-15, the motor control center enclosures and the condensate return system are all located out from under the ACC, leaving the least obstructed region beneath the fan deck of any of the sites tested.

As shown in Figure 4-8, Cells K1D, K5D, and D4 were instrumented with anemometers to measure air inlet velocity. At Apex and at the two future sites, the anemometers were

suspended from the fan bridge above the fans for easier installation. Again, four anemometers were placed in each cell to determine the variation in velocity across the inlet area.

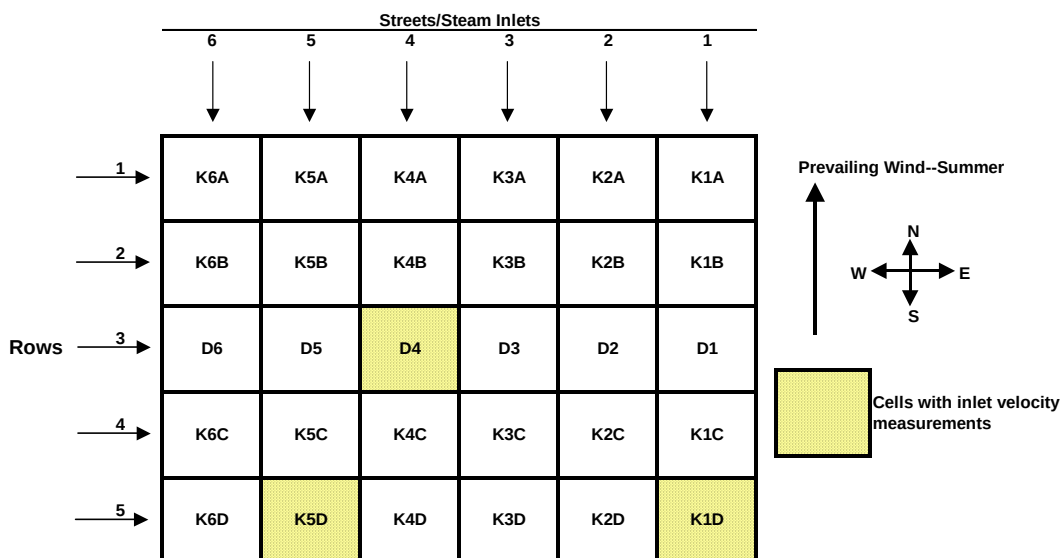


Figure 4-8. Apex ACC layout.

4.4.4 Front Range

The ACC at Front Range is a 40-cell unit arranged in an 8 x 5 rectangular cluster aligned in the north-south direction of the prevailing winds. At this site, unlike any of the other four, the prevailing winds blow across the rows, perpendicular to the street direction. It may be that this was done to locate fewer cells (five as opposed to eight) on the edge facing the prevailing winds than would the alternate arrangement. No windscreens were used on the Front Range unit. The condensate return system equipment and the MCC enclosures are located in the northwest area as shown in Figures 3-21 and 3-22.

Three cells (Cells K11, D43, and D83) were instrumented with anemometers suspended above the fans.

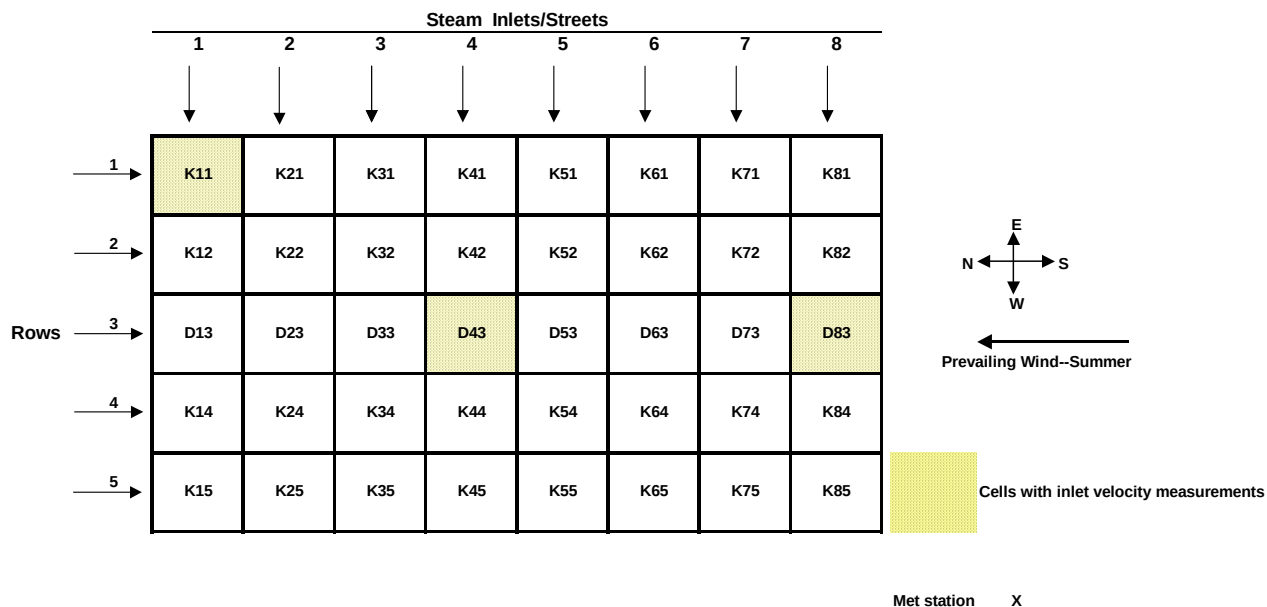


Figure 4-9. Front Range ACC layout.

4.4.5 Wygen II at Black Hills Power and Light

The ACC on Wygen II is a 12-cell unit arranged in two streets of six rows. The two end cells (K15 and K25) are “helper” cells and are used only in the summer. During colder periods the steam duct is valved off between Row 5 (Cells K14 and K24) and Row 6 (Cells K15 and K25). The unit is fitted with windwalls (solid walls as opposed to the porous fabric screening at Bighorn and El Dorado). The walls, in a cruciform arrangement, run beneath the fan deck the length of the ACC between Streets 1 and 2 and across the unit between the third (Cells D1 and D2) and fourth (Cells K13 and K23) rows.

The north end of the ACC is positioned close to the south wall of the Wygen II boiler house, which extends far above the top of the ACC (see Figures 3-23 and 3-24). Winds from the south blow the hot air exhaust plume from the ACC against the boiler house wall. When a portion of the wind encountering the obstruction turns downward, it would carry the hot plume into the air inlet area of Cells K11 and K21. To prevent this, a horizontal panel was installed at the level of the fan deck between the ACC and the boiler house wall.

Cells K11, K15, and K25 were instrumented with four anemometers in each cell suspended from the fan bridge above the fans. At this site, in order to explore the uniformity of the inlet air temperature and any variation with wind, four cells (K11, K15, K21, and K25) were instrumented with three additional inlet temperature probes positioned as shown in the inset in Figure 4-10.

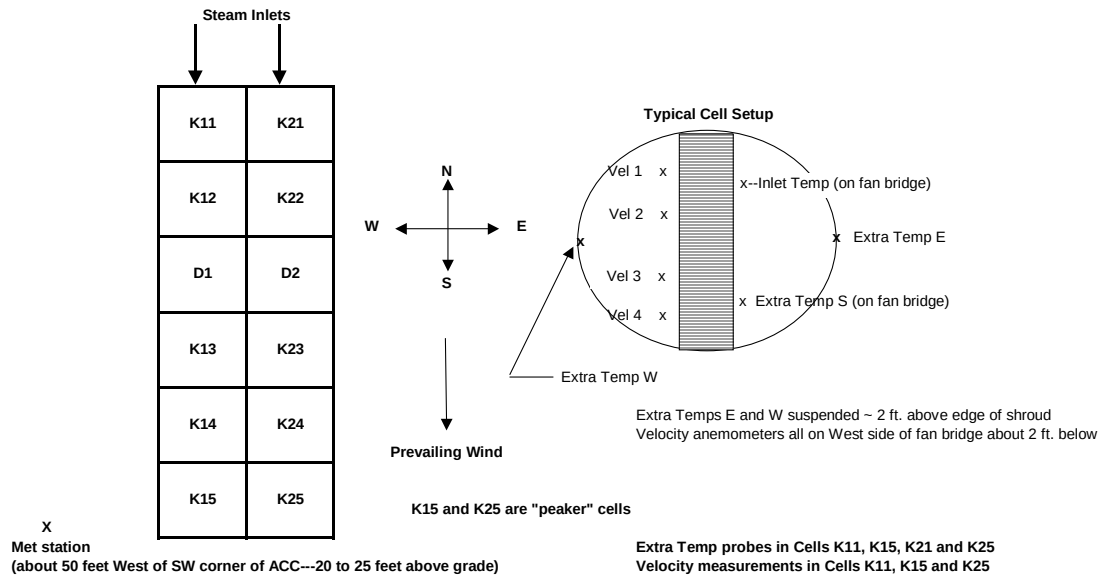


Figure 4-10. Wygen ACC layout.

CHAPTER 5:

Data Collection

At different periods during the summer of 2005, measurements were obtained of:

- ACC operation
 - o Cell inlet air temperature
 - o Cell exit air temperature (Bighorn only)
 - o Cell inlet velocity
 - o Cell exit velocity (Bighorn only)
 - o Fan static pressure (Bighorn and El Dorado only)
 - o Fan motor current (Bighorn and El Dorado only)
 - o Fan operating status
- Plant operation
 - o Plant output
 - o Steam turbine output
 - o Steam (or condensate) flow to ACC
 - o Turbine exhaust pressure
- Weather conditions
 - o Ambient air temperature
 - o Wind speed
 - o Wind direction

Table 5-1 gives the duration of the test periods at each site. Data were normally recorded at one-minute intervals continuously throughout the test period. Only occasional, intermittent gaps appear as a result of occasional instrumentation problems or plant shutdown.

Table 5-1. Test periods at five sites.

Plant	Data Acquisition Period	
	Start	End
Bighorn	June 10, 6:30 p.m.	June 18, 9:13 p.m.
El Dorado	June 28, 8:12 p.m.	July 7, 9:01 a.m.
Apex	July 13, 9:54 p.m.	July 21, 8:38 a.m.
Front Range	July 27, 11:42 a.m.	August 4, 6:43 a.m.
Wygen II	August 9, 10:16 a.m.	August 18, 9:38 a.m.

In addition, some supplementary data were taken at Bighorn at the end of the summer during another test program dedicated primarily to the study of spray enhancement of ACC performance. Data relevant to the wind study included vertical temperature profiles at four locations on the perimeter of the ACC and additional ambient temperature and wet bulb measurements at three locations on the plant boundary fence upwind of the ACC. The data were taken on selected days between August 18 and September 16 for only the few hours per day during which the spray tests were being conducted.

5.1 ACC Data

The monitoring of over forty data channels at one-minute intervals for over forty test days resulted in nearly 100 megabytes of data. While the complete data set cannot reasonably be included in this report, the full set is included on a CD available to interested parties from the California Energy Commission, Electric Power Research Institute, or Maulbetsch Consulting. However, in the interest of providing the reader with an understanding of the level of detail as well as the inherent uncertainty of the data upon which the results of this study are based, example data are presented in both tabular and graphical form and discussed briefly in the following section. The data in these examples are from Bighorn where the most comprehensive set of measurements was made.

5.1.1 Cell Inlet Air Temperature

Temperature measurements were taken at the inlet of each cell. Tables 5-2a and 5-2b give the cell inlet temperature at Bighorn for 1:00 to 1:20 p.m. on June 16, 2005. Some general observations include:

- Temperatures in Street 1 (Cells 1A to 1J) are generally higher, suggesting some recirculation into the cells on the east edge. (See Figure 5-2 for cell layout.)
- The southwest corner cell (Cell 4J) also shows evidence of recirculation.
- Minute-to-minute temperature variations in individual cells are small in most cells. Those cells with recirculation have greater minute-to-minute variability as would be expected for cells under greater influence of the wind.

5.1.2 Cell Exit Air Temperature and Velocity

Temperatures of the air leaving the finned tube bundles were measured at five vertically aligned positions on the east and west face of the cells in the south cluster of Street 3 (Cells 3F to 3J). Velocity measurements (four on each face) were similarly positioned. The probe numbering and locations are shown in Figure 4-5. The probes, mounted on the cleaning ladders, could be positioned at any location along the faces of Cells 3F to 3J. Measurements taken from 1:00 to 1:20 p.m. on June 16, 2005, are shown in Table 5-3 (exit temperature) and Table 5-4 (exit velocity). Velocity readings for June 16 are plotted in Figure 5-1. The ladders were located near the midpoint of Cell 3I for that period.

Table 5-2a. Bighorn cell inlet temperatures in Streets 1 and 2, June 16, 2005, 1:00–1:20 p.m.

Date/Time	Cell Number---Streets 1 and 2																			
	1A	1B	1C	1D	1E	1F	1G	1H	1I	1J	2A	2B	2C	2D	2E	2F	2G	2H	2I	2J
6/16/05 1:00 PM	92.09	93.44	93.36	91.74	90.83	92.15	93.19	92.57	91.23	97.53	90.44	89.41	88.71	88.62	89.18	90.44	90.16	88.41	88.22	88.51
6/16/05 1:01 PM	92.61	93.36	92.67	92.09	91.47	92.43	93.47	91.71	92.24	96.83	89.84	88.92	88.66	88.11	88.48	89.89	90.01	88.48	88.41	88.42
6/16/05 1:02 PM	91.45	93.80	92.90	91.33	90.57	91.39	92.24	91.65	90.71	94.57	89.86	89.31	88.79	88.46	88.83	89.92	89.62	88.81	88.70	88.83
6/16/05 1:03 PM	91.91	93.12	92.36	91.73	91.10	90.57	92.38	91.16	90.02	94.51	89.89	90.58	88.69	88.40	88.66	89.97	89.64	88.88	88.41	88.79
6/16/05 1:04 PM	91.88	92.42	92.34	91.91	91.24	90.70	92.78	91.56	90.22	95.09	90.02	89.74	88.98	88.42	88.45	90.11	89.83	88.69	88.24	88.57
6/16/05 1:05 PM	92.09	92.80	92.51	91.80	90.80	90.70	92.94	92.05	90.22	95.58	89.69	89.22	88.86	87.95	88.28	90.27	89.92	87.93	87.64	88.23
6/16/05 1:06 PM	92.47	93.03	91.53	92.33	91.13	91.87	93.98	91.74	90.69	97.34	89.75	90.04	88.15	87.98	88.34	89.87	89.87	88.14	87.79	88.04
6/16/05 1:07 PM	92.06	92.44	91.56	92.22	91.62	93.33	94.15	91.72	92.18	97.30	89.21	89.29	88.38	88.13	88.24	89.78	90.01	88.00	87.85	88.04
6/16/05 1:08 PM	91.35	91.10	90.82	92.25	91.09	93.39	93.80	91.39	92.19	96.46	88.59	88.71	87.93	88.35	88.16	89.36	89.78	88.26	88.11	88.46
6/16/05 1:09 PM	91.89	91.98	91.33	92.49	92.84	93.62	94.02	91.79	93.60	95.67	89.25	88.70	88.39	88.78	87.68	89.02	89.47	88.19	87.76	88.25
6/16/05 1:10 PM	92.90	92.49	91.44	93.33	92.25	93.29	93.51	92.29	92.80	96.21	90.04	89.17	88.65	89.46	88.87	89.18	90.01	88.70	88.15	88.63
6/16/05 1:11 PM	92.43	92.98	92.27	93.33	92.59	92.90	93.34	92.17	94.22	96.24	90.18	89.38	88.85	89.28	89.40	89.64	90.18	89.29	89.25	89.67
6/16/05 1:12 PM	92.90	93.78	93.36	92.66	91.88	92.99	93.59	92.94	94.35	96.10	90.19	89.78	89.76	89.37	89.59	90.33	90.68	89.22	89.41	89.45
6/16/05 1:13 PM	92.87	93.33	93.27	91.85	91.98	93.08	93.89	91.88	95.21	94.95	89.54	88.53	89.17	88.43	88.55	89.99	90.09	88.54	88.64	89.45
6/16/05 1:14 PM	92.42	93.97	92.97	91.70	92.06	92.97	93.49	91.40	92.99	93.87	89.65	89.76	88.99	88.68	88.98	89.98	90.22	89.34	88.83	88.97
6/16/05 1:15 PM	91.71	92.90	92.55	91.49	91.66	91.23	91.70	90.87	90.63	92.65	90.72	90.12	89.69	89.49	89.67	90.35	90.21	89.49	89.42	89.36
6/16/05 1:16 PM	92.01	92.58	92.31	92.24	91.77	90.93	92.04	91.14	90.53	92.81	89.64	89.41	89.16	88.70	89.23	90.58	89.99	89.10	88.91	89.36
6/16/05 1:17 PM	92.08	92.86	92.31	92.44	92.10	91.12	93.43	91.77	90.82	94.71	90.08	89.31	88.84	88.55	89.37	90.65	90.53	89.47	89.10	89.49
6/16/05 1:18 PM	92.50	93.05	91.95	91.71	91.13	91.47	93.96	92.53	91.02	95.77	89.99	89.82	89.47	88.44	89.15	90.83	90.14	88.82	88.25	88.74
6/16/05 1:19 PM	91.62	93.30	93.07	91.02	90.69	92.22	93.80	92.48	91.54	96.59	89.67	89.11	88.87	88.36	88.99	90.75	90.41	88.67	88.35	88.68
6/16/05 1:20 PM	92.11	93.83	92.56	91.11	90.93	92.96	94.08	92.72	91.63	96.66	90.36	90.15	88.95	88.60	88.93	91.11	90.92	88.82	88.50	89.02

Table 5-2b. Bighorn cell inlet temperature measurements in Streets 3 and 4, June 16, 2005, 1:00–1:20 p.m.

Date/Time	Cell Number--Streets 3 and 4																			
	3A	3B	3C	3D	3E	3F	3G	3H	3I	3J	4A	4B	4C	4D	4E	4F	4G	4H	4I	4J
6/16/05 1:00 PM	87.92	87.80	87.66	87.64	88.64	88.56	88.58	88.29	88.32	90.09	88.59	88.23	88.12	88.18	89.23	88.27	88.36	88.17	88.54	93.06
6/16/05 1:01 PM	87.59	87.42	87.32	87.28	87.79	88.15	88.39	88.25	88.06	88.45	88.51	88.02	87.86	88.10	89.84	88.11	88.23	88.10	88.35	91.69
6/16/05 1:02 PM	87.90	87.89	87.83	87.68	88.38	88.53	88.64	88.46	88.38	89.48	88.69	88.47	88.31	88.47	89.65	88.20	88.26	88.09	88.40	92.95
6/16/05 1:03 PM	87.41	87.70	87.69	87.74	87.97	88.58	88.69	88.41	88.26	90.08	88.45	87.99	88.09	88.27	89.59	88.19	88.30	88.14	88.36	93.71
6/16/05 1:04 PM	88.06	87.99	87.81	87.69	88.22	88.49	88.71	88.32	88.10	90.02	88.73	88.24	88.19	88.31	89.77	88.15	88.31	88.21	88.33	93.47
6/16/05 1:05 PM	87.78	87.81	87.63	87.40	88.53	88.06	88.16	87.95	87.65	88.97	88.10	87.92	87.85	87.87	88.95	87.77	87.91	87.73	87.93	92.63
6/16/05 1:06 PM	87.41	87.52	87.45	87.29	88.44	88.03	88.12	87.88	87.77	88.47	88.01	87.71	87.64	87.78	89.02	87.73	87.86	87.71	87.91	91.50
6/16/05 1:07 PM	87.75	87.32	87.19	87.14	87.63	87.80	87.92	87.79	87.59	88.12	88.33	87.75	87.60	87.83	89.36	87.58	87.76	87.62	87.88	91.17
6/16/05 1:08 PM	88.18	86.88	86.83	86.87	87.05	87.80	87.89	87.91	88.01	88.29	87.97	87.27	87.29	87.54	89.29	87.50	87.80	87.77	88.16	91.10
6/16/05 1:09 PM	87.20	86.90	86.75	86.82	86.99	87.76	87.87	87.78	87.64	87.50	88.03	87.43	87.37	87.53	89.50	87.50	87.69	87.60	87.92	90.56
6/16/05 1:10 PM	87.51	87.47	87.41	87.43	87.81	88.46	88.31	87.98	88.20	88.25	88.76	87.99	87.96	88.13	89.71	87.87	87.99	87.83	88.06	90.70
6/16/05 1:11 PM	87.83	87.84	87.85	87.89	88.75	88.77	88.94	88.55	88.70	88.85	88.54	88.42	88.39	88.71	90.47	88.41	88.54	88.42	88.62	90.96
6/16/05 1:12 PM	88.50	88.58	88.49	88.53	89.47	88.79	89.02	88.80	88.68	88.88	88.93	88.95	88.98	89.40	91.23	88.81	88.90	88.77	89.00	91.57
6/16/05 1:13 PM	87.75	87.65	87.57	87.66	87.90	88.33	88.58	88.52	88.28	88.42	88.56	88.39	88.32	88.66	91.00	88.56	88.64	88.54	88.86	90.80
6/16/05 1:14 PM	88.15	88.14	87.98	88.07	88.67	88.95	89.07	88.79	88.57	88.72	89.45	88.69	88.58	88.83	90.64	88.97	88.97	88.64	88.84	90.43
6/16/05 1:15 PM	88.83	88.73	88.53	88.65	89.70	89.51	89.61	89.18	89.27	89.48	89.17	88.90	88.81	89.07	90.05	89.12	89.25	89.04	89.25	91.07
6/16/05 1:16 PM	88.03	87.92	87.88	88.00	89.06	88.92	89.32	89.06	88.91	90.91	88.13	88.22	88.19	88.48	90.00	88.77	89.01	88.78	89.04	93.70
6/16/05 1:17 PM	88.23	88.16	88.18	88.22	89.56	89.22	89.44	89.27	89.13	91.38	88.65	88.47	88.56	88.91	90.38	88.97	89.25	89.02	89.25	94.28
6/16/05 1:18 PM	88.12	88.01	87.97	87.93	89.71	88.20	88.65	88.67	88.20	90.48	88.47	88.42	88.45	88.92	90.66	88.70	88.82	88.69	89.02	93.95
6/16/05 1:19 PM	87.73	87.71	87.75	87.83	89.17	88.20	88.60	88.46	88.22	89.92	88.28	88.42	88.41	88.80	90.88	88.49	88.66	88.50	88.80	93.28
6/16/05 1:20 PM	87.87	87.93	87.89	87.93	88.93	88.59	88.79	88.61	88.41	90.33	88.36	88.56	88.43	88.76	90.34	88.57	88.74	88.60	88.86	93.51

Table 5-3. Bighorn cell exit temperature measurements in Cell 3I June 16, 2005, 1:00–1:20 p.m.

Date/Time	West Face					East Face				
	Top	Upper Mid	Middle	Lower Mid	Bottom	Top	Upper Mid	Middle	Lower Mid	Bottom
6/16/05 1:00 PM	116.78	116.22	117.96	120.77	118.38	114.83	115.57	114.68	115.64	80.92
6/16/05 1:01 PM	116.72	116.14	118.02	120.71	119.03	114.56	115.75	114.53	115.57	105.04
6/16/05 1:02 PM	117.04	116.45	118.33	120.91	119.19	114.51	116.14	114.67	115.99	96.33
6/16/05 1:03 PM	116.87	116.27	118.13	120.76	118.85	114.35	116.05	114.67	115.70	94.24
6/16/05 1:04 PM	116.16	115.53	117.20	119.94	118.13	114.16	115.51	114.49	114.90	94.09
6/16/05 1:05 PM	115.42	114.83	116.25	118.99	117.38	113.94	114.94	114.17	114.12	92.31
6/16/05 1:06 PM	114.87	114.37	115.60	118.27	117.04	113.77	114.52	113.96	113.60	101.60
6/16/05 1:07 PM	114.69	114.24	115.45	117.99	117.21	113.80	114.42	114.01	113.57	94.93
6/16/05 1:08 PM	114.76	114.33	115.56	117.99	117.44	113.94	114.53	114.21	113.77	94.60
6/16/05 1:09 PM	114.92	114.50	115.84	118.19	117.90	114.14	114.74	114.45	114.01	94.69
6/16/05 1:10 PM	115.37	114.98	116.35	118.72	118.67	114.67	115.27	115.04	114.64	96.32
6/16/05 1:11 PM	115.98	115.57	117.22	119.56	119.29	115.35	115.94	115.75	115.37	94.37
6/16/05 1:12 PM	116.09	115.56	117.71	119.86	118.90	115.56	116.07	115.86	115.42	88.57
6/16/05 1:13 PM	115.93	115.39	117.63	119.87	119.02	115.54	116.03	115.80	115.23	94.46
6/16/05 1:14 PM	116.23	115.73	117.67	119.72	119.53	115.87	116.39	116.23	115.76	97.27
6/16/05 1:15 PM	117.12	116.74	118.39	119.70	119.77	116.61	117.10	117.02	116.71	96.95
6/16/05 1:16 PM	117.43	116.91	118.57	120.42	119.62	116.59	117.48	117.31	116.88	92.56
6/16/05 1:17 PM	117.67	117.08	118.89	121.10	119.80	116.07	117.84	117.62	117.07	88.25
6/16/05 1:18 PM	117.53	116.81	119.24	121.47	119.51	115.23	117.83	117.54	116.93	81.44
6/16/05 1:19 PM	117.19	116.47	119.07	121.32	119.07	114.55	117.38	117.20	116.48	83.97
6/16/05 1:20 PM	117.04	116.32	118.66	121.10	118.90	114.23	116.88	117.04	116.29	66.48

Table 5-4. Bighorn cell exit velocity measurements in Cell 3I June 16, 2005, 1:00–1:20 p.m.

Date/Time	West Face				East Face			
	Top	Upper Mid	Lower Mid	Bottom	Top	Upper Mid	Lower Mid	Bottom
6/16/05 1:00 PM	477.5	442.0	397.1	229.2	311.0	440.7	393.2	515.5
6/16/05 1:01 PM	485.3	447.2	391.3	240.2	349.1	435.9	391.7	504.3
6/16/05 1:02 PM	493.3	443.6	383.2	230.4	319.3	449.2	404.5	540.9
6/16/05 1:03 PM	484.2	439.1	415.3	231.4	318.4	439.6	391.7	533.2
6/16/05 1:04 PM	489.2	455.2	417.3	245.3	337.7	446.4	401.8	524.8
6/16/05 1:05 PM	479.2	439.5	414.4	247.7	328.4	433.0	385.6	514.8
6/16/05 1:06 PM	502.2	448.6	396.0	217.3	329.7	439.7	387.3	517.1
6/16/05 1:07 PM	476.3	440.4	408.7	253.4	313.7	433.3	383.2	507.3
6/16/05 1:08 PM	476.0	445.7	403.0	235.4	318.3	439.4	381.9	519.2
6/16/05 1:09 PM	475.7	445.5	405.6	236.7	318.4	429.3	384.8	504.7
6/16/05 1:10 PM	488.6	450.8	390.9	220.4	283.6	440.5	388.4	519.2
6/16/05 1:11 PM	499.2	456.3	376.7	206.7	325.5	451.9	401.9	541.1
6/16/05 1:12 PM	517.6	482.1	378.7	209.4	342.6	462.5	405.5	545.3
6/16/05 1:13 PM	509.8	468.7	415.3	238.1	368.9	454.7	400.4	542.4
6/16/05 1:14 PM	450.4	405.1	384.0	326.6	268.3	422.0	371.0	474.3
6/16/05 1:15 PM	455.2	392.5	399.4	288.4	288.5	431.1	384.2	508.9
6/16/05 1:16 PM	475.0	444.4	399.8	222.8	331.2	446.2	398.3	534.1
6/16/05 1:17 PM	502.7	467.6	391.8	220.8	326.0	456.4	419.9	532.6
6/16/05 1:18 PM	528.2	473.6	350.6	203.3	335.7	474.2	425.0	563.6
6/16/05 1:19 PM	474.6	460.3	391.7	214.4	309.1	453.1	401.7	543.5
6/16/05 1:20 PM	492.2	469.1	360.2	209.8	389.6	463.7	413.8	538.2

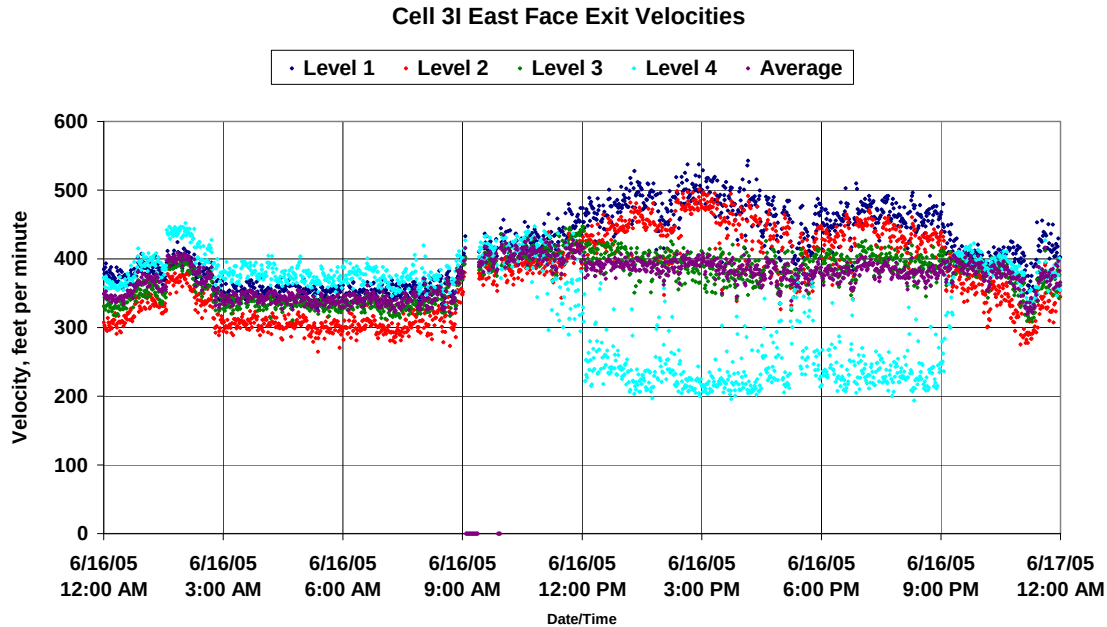


Figure 5-1. Bighorn east face, Cell 3I exit velocity, June 16, 2005.

5.1.3 Cell Inlet Velocity

Measurements were made of inlet air velocity in several selected cells using propeller anemometers mounted at four positions along the fan diameter in the prevailing wind direction, as shown in Figure 5-4. At two sites (Bighorn and El Dorado) the anemometers were hung from the inlet screens below the fans. At the other three sites, the anemometers were positioned above the fans by mounting them on the side rail of the fan bridge at a level of about 3 feet below the bridge.

Table 5-5 shows the inlet velocities in three of the cells at Bighorn from 1:00 to 1:20 p.m. on June 16, 2005. Cell 3J has only three anemometers due to a shortage of instrumentation. All locations show considerable difference from one anemometer to another and minute-to-minute variability. Figure 5-2 shows the individual measurements as well as the average of the four measurement points on Cell 3I (the same cell for which the exit velocities are tabulated in Table 5-4) for all of June 16.

Table 5-5. Bighorn cell inlet velocity measurements, June 16, 2005, 1:00–1:20 p.m.

Cell Inlet Velocity, ft./min.											
Date/Time	Cell 3J			Cell 3I				Cell 3F			
	3J1	3J2	3J3	3I1	3I2	3I3	3I4	3F1	3F2	3F3	3F4
6/16/05 1:00 PM	1,586	1,159	1,043	1,323	1,295	1,364	912	1,580	1,401	957	1,161
6/16/05 1:01 PM	1,365	1,297	1,251	1,337	1,413	1,178	941	1,534	1,463	959	1,134
6/16/05 1:02 PM	1,237	1,188	991	1,558	1,472	1,206	958	1,529	1,494	897	1,154
6/16/05 1:03 PM	1,508	1,274	1,037	1,483	1,352	1,288	906	1,609	1,420	963	1,135
6/16/05 1:04 PM	1,422	1,324	852	1,580	1,476	1,152	891	1,698	1,441	1,013	1,134
6/16/05 1:05 PM	1,365	1,374	1,038	1,381	1,219	1,260	878	1,624	1,338	959	1,142
6/16/05 1:06 PM	1,499	1,246	1,099	1,360	1,236	1,378	952	1,544	1,397	967	1,113
6/16/05 1:07 PM	1,416	1,164	1,039	1,253	1,199	1,386	929	1,399	1,259	1,150	1,278
6/16/05 1:08 PM	1,331	1,167	1,204	1,245	1,158	1,512	977	1,481	1,304	988	1,265
6/16/05 1:09 PM	1,285	1,166	1,150	1,290	1,159	1,361	1,030	1,381	1,320	1,129	1,262
6/16/05 1:10 PM	1,247	1,167	1,144	1,230	1,186	1,490	995	1,439	1,231	1,089	1,138
6/16/05 1:11 PM	1,398	1,205	1,094	1,456	1,356	1,340	1,003	1,736	1,565	914	1,193
6/16/05 1:12 PM	1,366	1,336	1,195	1,319	1,318	1,562	1,007	1,582	1,435	1,051	1,193
6/16/05 1:13 PM	1,444	1,170	1,062	1,321	1,235	1,526	1,101	1,387	1,305	950	1,277
6/16/05 1:14 PM	1,214	1,521	951	1,475	1,375	1,189	895	1,689	1,657	796	1,022
6/16/05 1:15 PM	1,403	1,188	962	1,561	1,493	1,183	875	1,564	1,455	964	1,108
6/16/05 1:16 PM	1,255	1,059	1,087	1,548	1,436	1,122	873	1,526	1,450	975	1,109
6/16/05 1:17 PM	987	1,170	815	1,521	1,492	1,190	916	1,615	1,648	902	1,039
6/16/05 1:18 PM	1,156	1,139	1,112	1,482	1,552	1,252	907	1,675	1,626	861	1,033
6/16/05 1:19 PM	1,188	1,155	1,026	1,486	1,483	1,149	890	1,577	1,489	873	1,079
6/16/05 1:20 PM	1,288	1,198	984	1,573	1,570	1,163	934	1,717	1,557	972	1,115

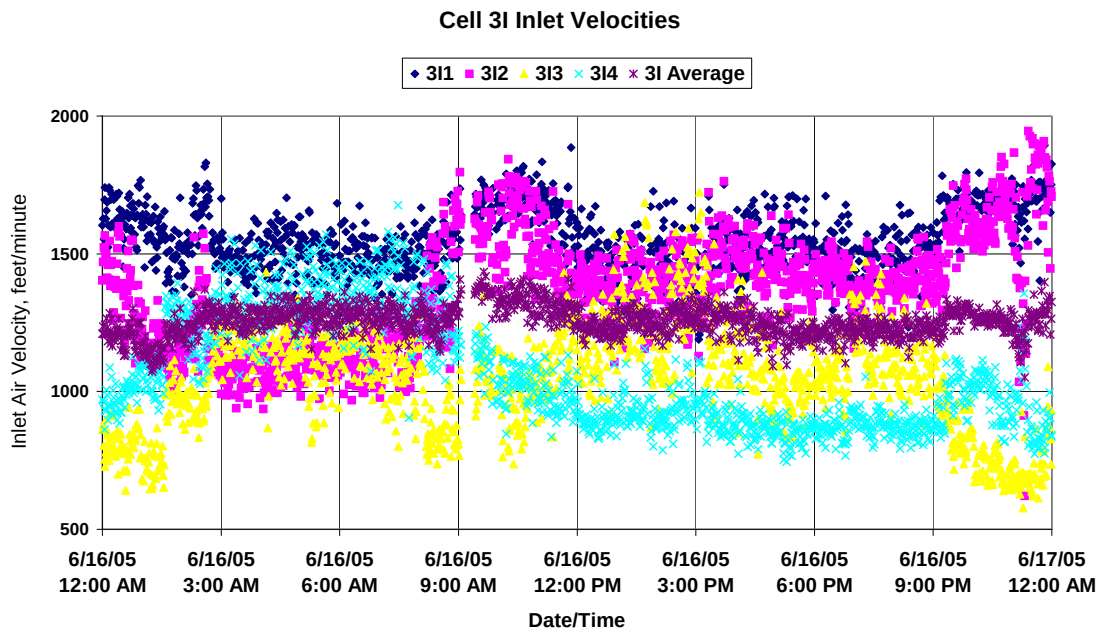


Figure 5-2. Cell 3I inlet velocity measurements.

5.1.4 Fan Static Pressure

Fan static pressure was monitored in several cells at Bighorn and El Dorado. The location of the selected cells at Bighorn (2A, 2J, 3A, 3F, 3I, and 3J) is shown in Figure 4-3. Table 5-6 shows the values recorded from 1:00 to 1:20 p.m. on June 16, 2005. Figure 5-3 plots the readings for the full day on June 16.

The measurement in Cell 3F erroneously reads negative for several hours during the day before the malfunction was noted and corrected. The measurements show reasonable agreement from cell to cell at levels consistent with the expected performance of the fans. Cell 3A values are significantly higher, suggesting the influence of wind on the cell. All cells exhibit some amount of minute-to-minute variability.

Table 5-6. Static pressure in selected Bighorn cells, June 16, 2005, 1:00–1:20 p.m.

Date/Time	Static Pressure---in H2O					
	Cell 2A	Cell 2J	Cell 3A	Cell 3F	Cell 3I	Cell 3J
6/16/05 1:00 PM	0.26	0.26	0.66	-0.48	0.24	0.31
6/16/05 1:01 PM	0.28	0.26	0.71	-0.48	0.24	0.28
6/16/05 1:02 PM	0.26	0.23	0.69	-0.48	0.22	0.28
6/16/05 1:03 PM	0.24	0.29	0.68	-0.48	0.21	0.29
6/16/05 1:04 PM	0.25	0.22	0.68	-0.48	0.20	0.22
6/16/05 1:05 PM	0.24	0.26	0.66	-0.48	0.23	0.28
6/16/05 1:06 PM	0.26	0.28	0.70	-0.48	0.25	0.30
6/16/05 1:07 PM	0.24	0.27	0.68	-0.48	0.21	0.28
6/16/05 1:08 PM	0.27	0.31	0.70	-0.48	0.26	0.34
6/16/05 1:09 PM	0.27	0.28	0.68	-0.48	0.21	0.27
6/16/05 1:10 PM	0.29	0.23	0.70	-0.48	0.24	0.31
6/16/05 1:11 PM	0.27	0.33	0.71	-0.48	0.28	0.35
6/16/05 1:12 PM	0.26	0.25	0.72	-0.48	0.23	0.31
6/16/05 1:13 PM	0.28	0.24	0.71	-0.48	0.25	0.30
6/16/05 1:14 PM	0.26	0.28	0.71	-0.48	0.25	0.28
6/16/05 1:15 PM	0.25	0.27	0.67	-0.48	0.17	0.22
6/16/05 1:16 PM	0.24	0.31	0.66	-0.48	0.19	0.25
6/16/05 1:17 PM	0.24	0.30	0.70	-0.48	0.17	0.28
6/16/05 1:18 PM	0.25	0.33	0.73	-0.48	0.25	0.31
6/16/05 1:19 PM	0.24	0.31	0.69	-0.48	0.22	0.29
6/16/05 1:20 PM	0.25	0.32	0.71	-0.48	0.23	0.31

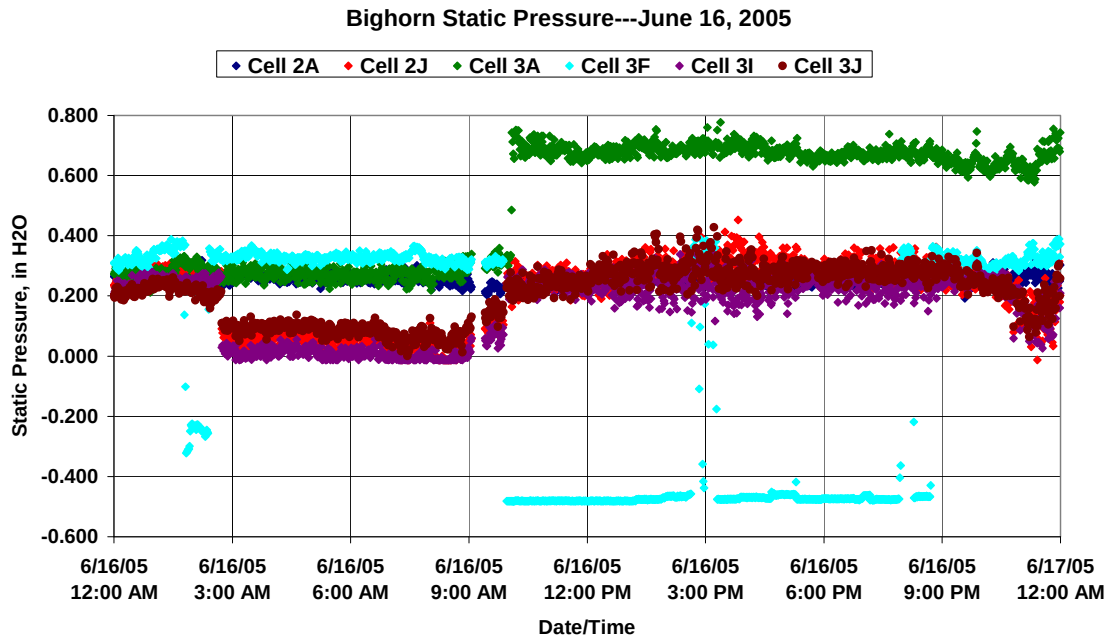


Figure 5-3. Bighorn static pressure, Cells 2A, 2J, 3A, 3F, 3I, and 3J.

5.1.5 Fan Motor Current

In a further attempt to determine the effect of wind on fan performance, the fan motor current was monitored in selected cells. In five cases, these were chosen to be the same cells in which fan static pressure was also monitored. The configuration of the current-monitoring instrument and the location of the individual motor controllers in the motor control centers made it necessary to select one cell in which static pressure was not also monitored.

Table 5-7 gives the data for the period of 1:00 to 1:20 p.m. on June 16, 2005. These data are recorded at 7½ minute intervals. With the exception of Cell 2J, there is reasonable agreement from cell to cell and only modest temporal variability. Figure 5-4 plots the readings for the full day of June 16. For a seven-hour period during the early to mid-morning, there was an interruption in the data acquisition. The fan in Cell 3H was turned off from about 1:30 a.m. until noon.

Table 5-7. Fan motor current in selected Bighorn cells, June 16, 2005, 1:00–1:20 p.m.

Date/Time	Fan Motor Current---amps					
	Cell 2A	Cell 2J	Cell 3A	Cell 3H	Cell 3I	Cell 3J
6/16/05 12:58 PM	155.56	144.29	159.83	161.85	160.96	156.95
6/16/05 1:06 PM	155.00	143.67	159.35	161.79	161.51	156.24
6/16/05 1:13 PM	158.87	144.98	160.56	163.26	158.37	163.95
6/16/05 1:21 PM	152.38	142.82	159.09	159.08	161.60	150.61
6/16/05 1:28 PM	153.06	142.09	158.37	159.18	161.95	153.87
6/16/05 1:36 PM	152.57	142.79	158.80	160.27	162.35	154.97
6/16/05 1:43 PM	153.33	143.35	157.92	160.98	158.54	158.17
6/16/05 1:51 PM	153.76	142.86	158.72	161.37	157.93	164.18
6/16/05 1:58 PM	153.22	141.82	158.83	160.02	158.99	160.40

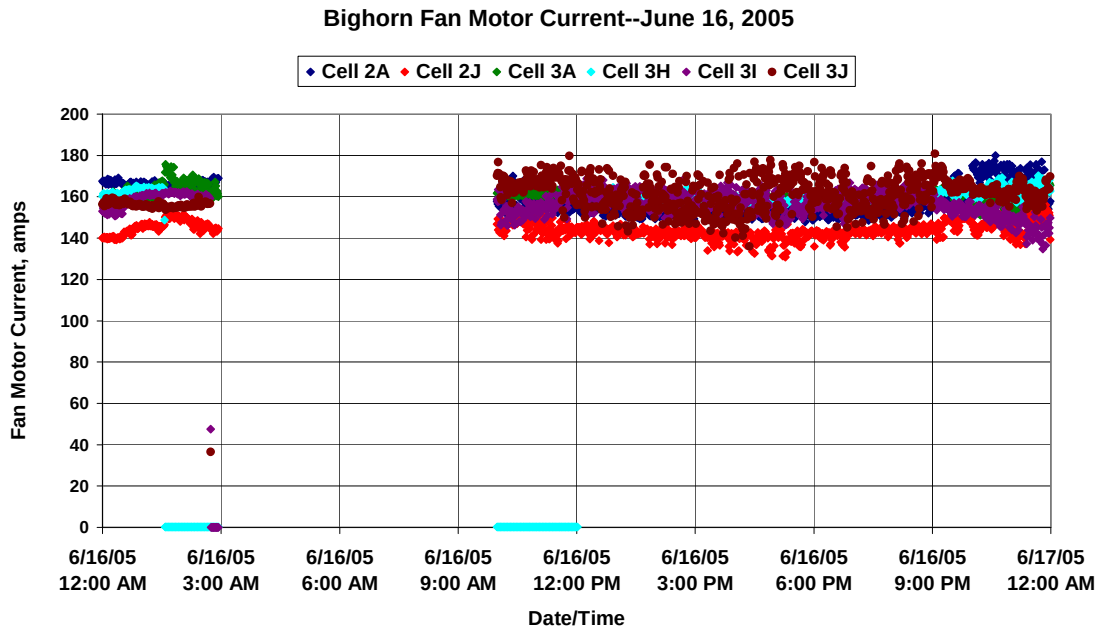


Figure 5-4. Bighorn fan motor current in Cells 2A, 2J, 3A, 3H, 3I, and 3J, June 16, 2005.

5.1.6 Fan Status

Fan operation at Bighorn is monitored continuously by the plant. At one-minute intervals, each of the 40 fans is recorded as “0” (fan off), “2” (fan on at half speed), or “4” (fan on at full speed). These readings can be summed into a collective “fan status” number where “160” indicates all 40 fans operating at full speed. Wind effects on ACC performance were analyzed using only data for time periods during which all fans were at full speed.

Figure 5-5 shows the collective fan status for June 16, 2005. During the period from about 1:30 a.m. to noon, when the plant was at partial load and the ambient temperature was low, some of the fans were shut off or operated at half speed.

Individual fan status records confirm that Fan 3H was shut off from 1:30 a.m. to noon as indicated by the motor current data shown in Figure 5-4. Fan status data were unavailable after 4:30 p.m. for reasons related to the plant data acquisition system.

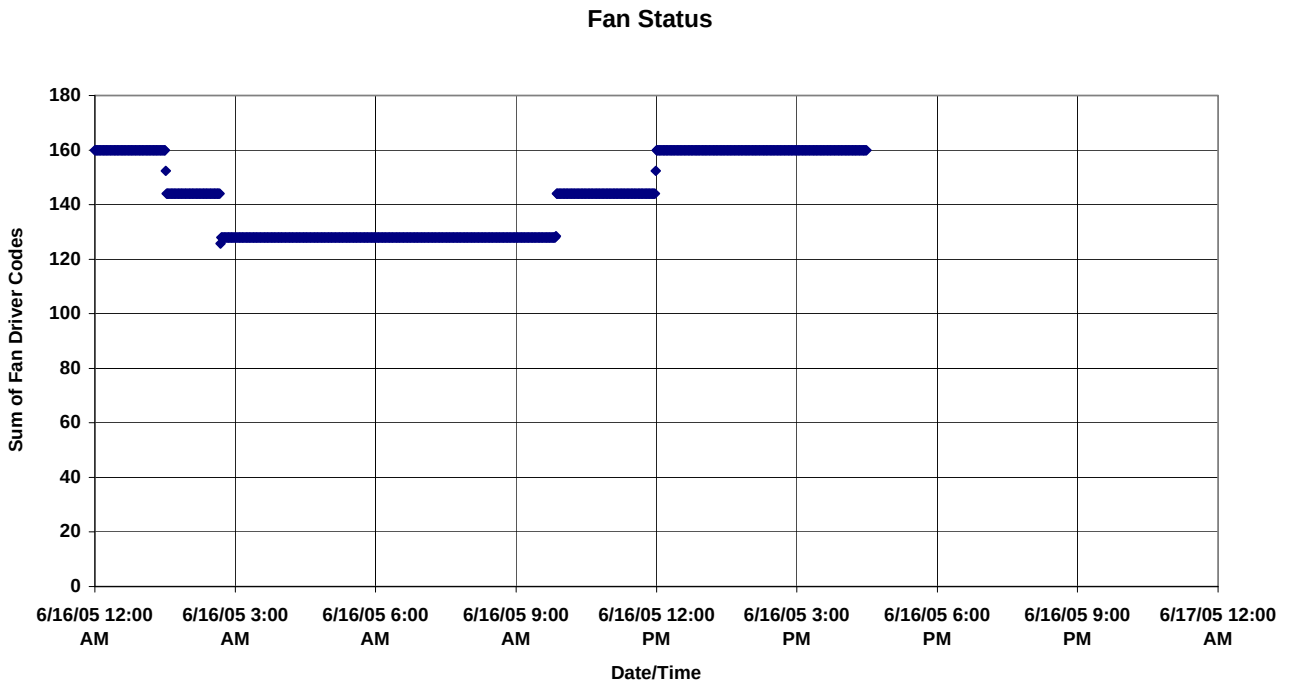


Figure 5-5. Fan status at Bighorn, June 16, 2005.
 (Fan status code: Off = 0; ½ speed = 2; Full speed = 4—Summed over 40 fans).

5.2 Plant Operating Data

Data on plant operations relevant to ACC performance were provided by the plant. Table 5-8 shows data for net plant output, gross steam turbine output, steam flow to the ACC, and turbine exhaust pressure for the period from 1:00 to 1:20 p.m. on June 16, 2005. The readings are typical of normal plant operation with little minute-to-minute variability.

Table 5-8. Bighorn Plant operating data, June 16, 2005, 1:00–1:20 p.m.

Date/Time	Plant Output (Net)	Steam Turbine Output (Gross)	Steam Flow	Turbine Exhaust Pressure
	MW	MW	lb/hr	in Hga
6/16/05 1:00 PM	455.43	163.27	1167.30	3.45
6/16/05 1:01 PM	455.56	163.25	1167.57	3.46
6/16/05 1:02 PM	455.69	163.24	1167.56	3.46
6/16/05 1:03 PM	455.68	163.23	1167.21	3.46
6/16/05 1:04 PM	455.65	163.22	1167.02	3.46
6/16/05 1:05 PM	455.62	163.20	1167.69	3.47
6/16/05 1:06 PM	455.59	163.18	1168.42	3.47
6/16/05 1:07 PM	455.56	163.17	1168.74	3.47
6/16/05 1:08 PM	455.53	163.15	1168.46	3.47
6/16/05 1:09 PM	455.50	163.14	1168.00	3.48
6/16/05 1:10 PM	455.47	163.12	1168.42	3.48
6/16/05 1:11 PM	455.44	163.11	1168.89	3.48
6/16/05 1:12 PM	455.42	163.09	1168.89	3.49
6/16/05 1:13 PM	455.40	163.07	1168.55	3.49
6/16/05 1:14 PM	455.38	163.06	1168.34	3.49
6/16/05 1:15 PM	455.36	163.04	1168.01	3.49
6/16/05 1:16 PM	455.34	163.03	1167.43	3.50
6/16/05 1:17 PM	455.32	163.01	1167.16	3.50
6/16/05 1:18 PM	455.31	163.00	1168.21	3.50
6/16/05 1:19 PM	455.29	162.98	1168.51	3.50
6/16/05 1:20 PM	455.27	162.97	1167.97	3.51

5.3 Weather Conditions

Meteorological data during test periods was obtained from both the plant and a project met tower. Specific data available from the plant differed from site to site as shown in Table 5-9.

Table 5-9. Plant meteorological measurements.

Site	Ambient Temperature		Wind Conditions	
	Admin. Building	Under ACC	Speed	Direction
Apex	Yes	No	No*	No*
Bighorn	Yes	Yes	Yes	Yes
El Dorado	Yes	Yes	Yes	Yes
Front Range	Yes	Yes	Yes	Yes
Wygen II	Yes	No	No	No

* Obtained from Silverhawk (neighboring plant)

Project data consisted of wind speed and direction measured with a met tower typically 20 feet high and located a hundred feet or so upwind of the ACC for the prevailing summer wind patterns. Project data, with the exception of those taken during testing periods in late August to mid-September as part of the spray enhancement tests at Bighorn, did not provide an independent measurement of ambient air temperature away from the ACC. The inlet air temperature was assumed to be well represented by the lowest of the inlet air temperatures measured at each of the cells on the basis that no plausible mechanism existed for cooling the air as it entered the ACC.

5.3.1 Ambient Temperature

Project data, with the exception of those taken during testing periods in late August to mid-September as part of the spray enhancement tests at Bighorn, did not provide an independent measurement of ambient air temperature away from the ACC. The inlet air temperature was assumed to be well represented by the lowest of the inlet air temperatures measured at each of the cells on the basis that no plausible mechanism existed for cooling the air as it entered the ACC.

5.3.2 Plant Measurements

Plant measurements at Bighorn (and all others with the exception of Apex) were monitored at two locations: one on top of a plant building (usually the administration/control building); the other at a location underneath the ACC, usually up near the fan deck. Figure 5-6 shows the two ambient temperature measurements at Bighorn for June 16, 2005. Differences between the two measurements can be quite large as shown in the figure. In this case, they are most likely attributable to differences in solar radiation since they are small to vanishing during nighttime hours. At other sites, the differences may be attributed to other factors or else the agreement is much better.

Figure 5-6 includes a plot of the minimum inlet air temperature for purposes of comparison. The agreement with the plant's "Ambient at ACC" measurements is quite good with the exception of brief periods in the early morning and late afternoon. These periods correspond to sunrise and sunset; times at which the plant sensor mounted under the ACC could be exposed to direct solar radiation. The cell inlet temperature measured inside the cells above the fan are never affected by solar radiation.

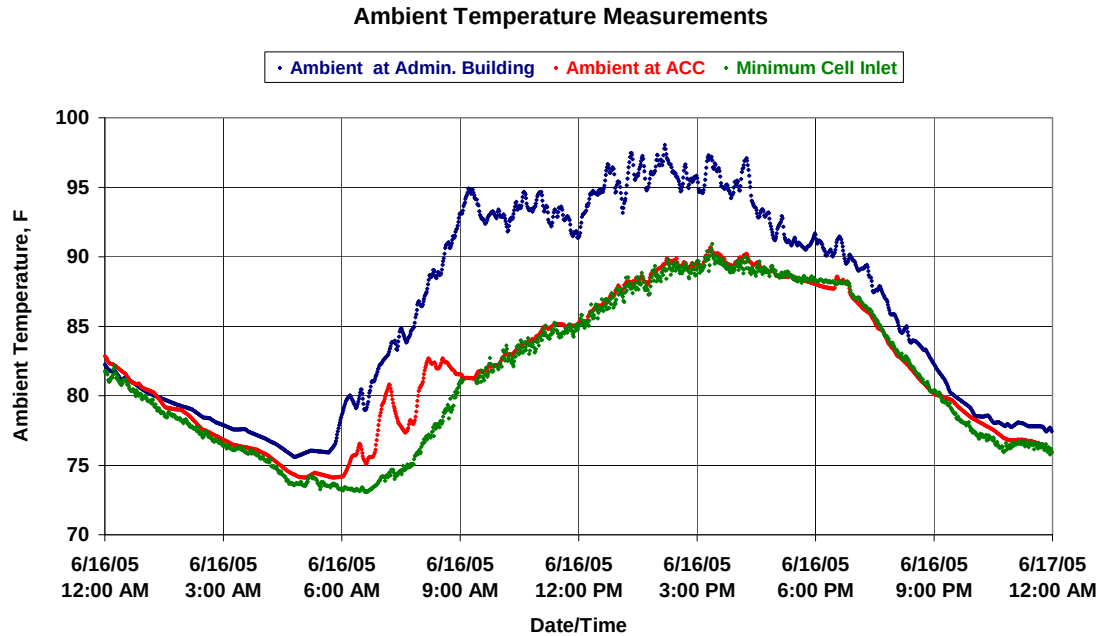


Figure 5-6. Bighorn ambient air temperature—comparison of measurements.

5.3.3 Additional Perimeter Measurements

Some additional information is available from tests of inlet air spray cooling conducted at Bighorn in late August through mid-September 2005. During these tests, air temperature measurements were made several locations:

- The plant boundary fence (three locations).
- The ACC periphery (three heights at each of four locations).
- The cell inlets above the fans (the H, I, and J cells in Streets 1, 2, and 3).

The probe locations are shown in Figure 4-5. In addition, plant ambient temperature measurements at two locations (on the administration building and underneath the ACC) were provided for these test days.

Figures 5-7 through 5-12 compare the various measurements taken on September 14, 2005.

Figure 5-7 shows the temperatures at the three boundary fence locations and their average value. The probes were tied to the fence at about 4 feet above the ground. They were mounted in aspirating psychrometers to eliminate or minimize the effect of solar radiation. The measurements differ by less than 2 to 3 degrees with the east and west probes showing essential agreement and the south probe consistently lower.

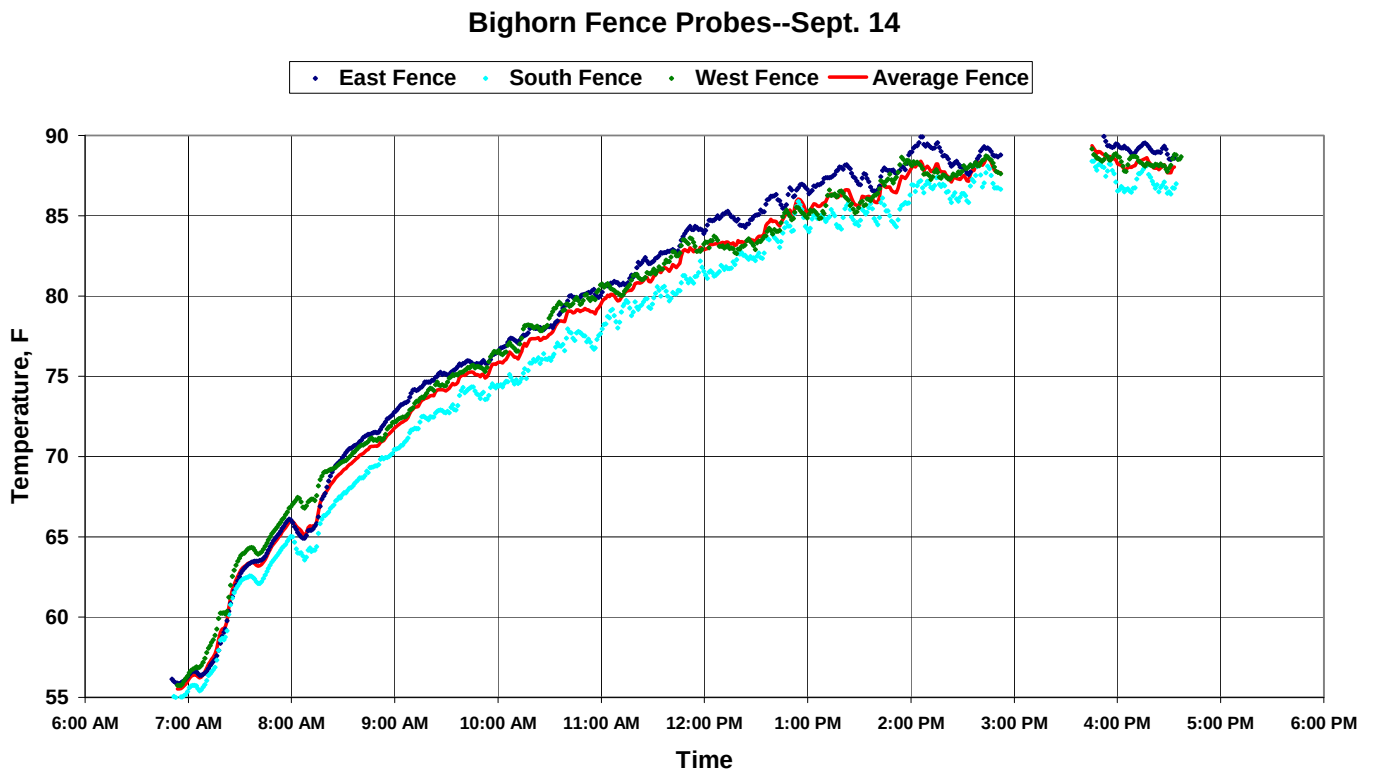


Figure 5-7. Ambient air temperature—plant boundary fence south of ACC.

Figures 5-8a through 5-8d show the values from the 12 probes on the ACC perimeter. As indicated in Figure 4-6, they are located on the east face underneath Cell 1I, on the south face under Cells 2J and 3J, and on the west face under Cell 4I. At each location there was one probe at each of three levels—10, 30, and 50 feet above grade. Figure 5-9 compares the average values of each of the three levels at each of the four locations; Figure 5-10 compares the average of the fence probes with the average of the peripheral probes at the lowest (10 foot) level; and Figure 5-11 compares the average values for each of the four locations at each of the three levels.

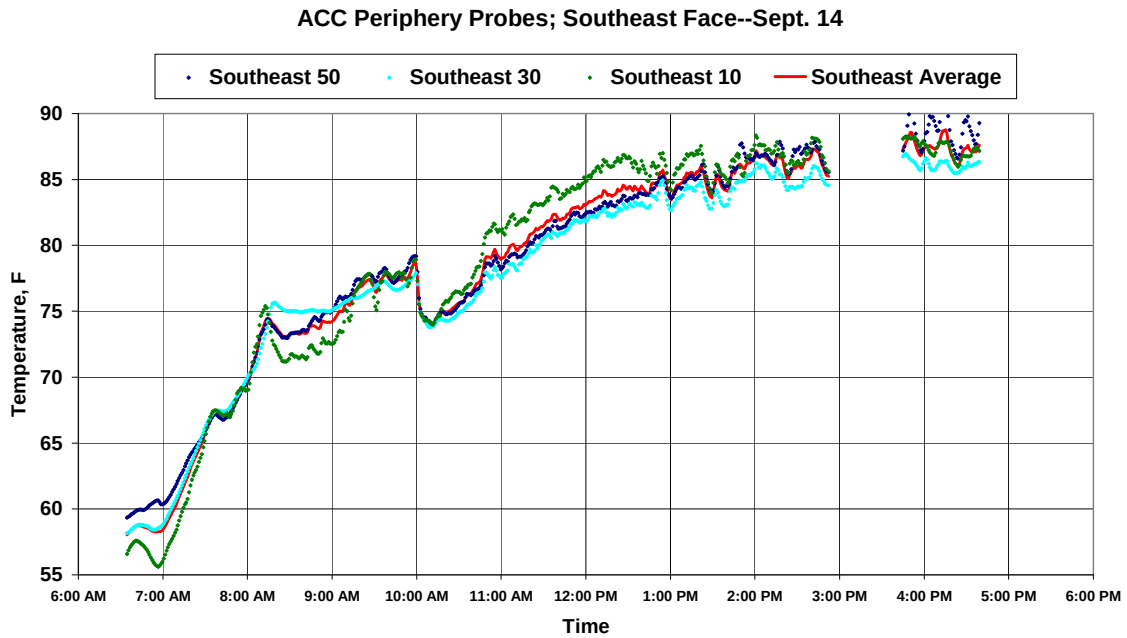


Figure 5-8a. Bighorn ambient air temperature—ACC periphery, east face.

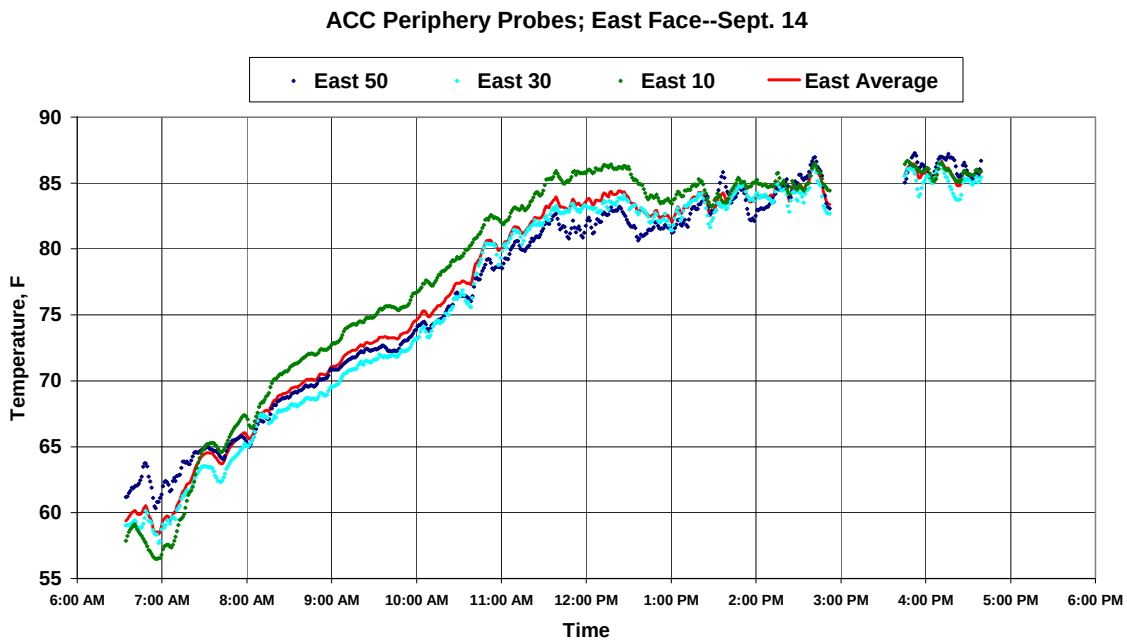


Figure 5-8b. Bighorn ambient air temperature—ACC periphery, southeast face.

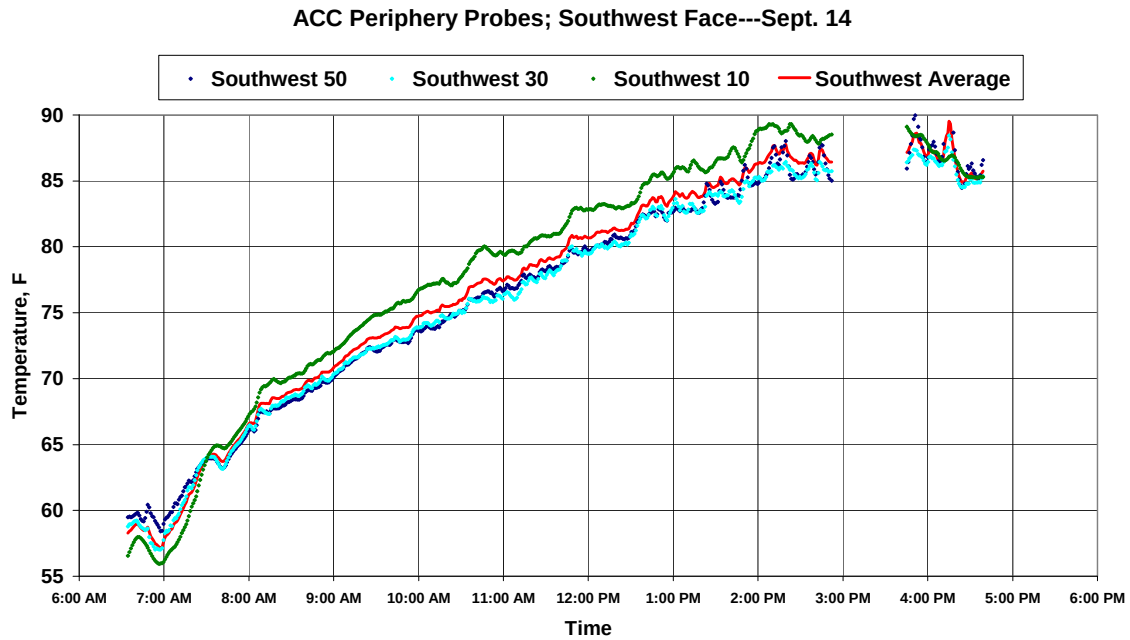


Figure 5-8c. Bighorn ambient air temperature—ACC periphery, southwest face.

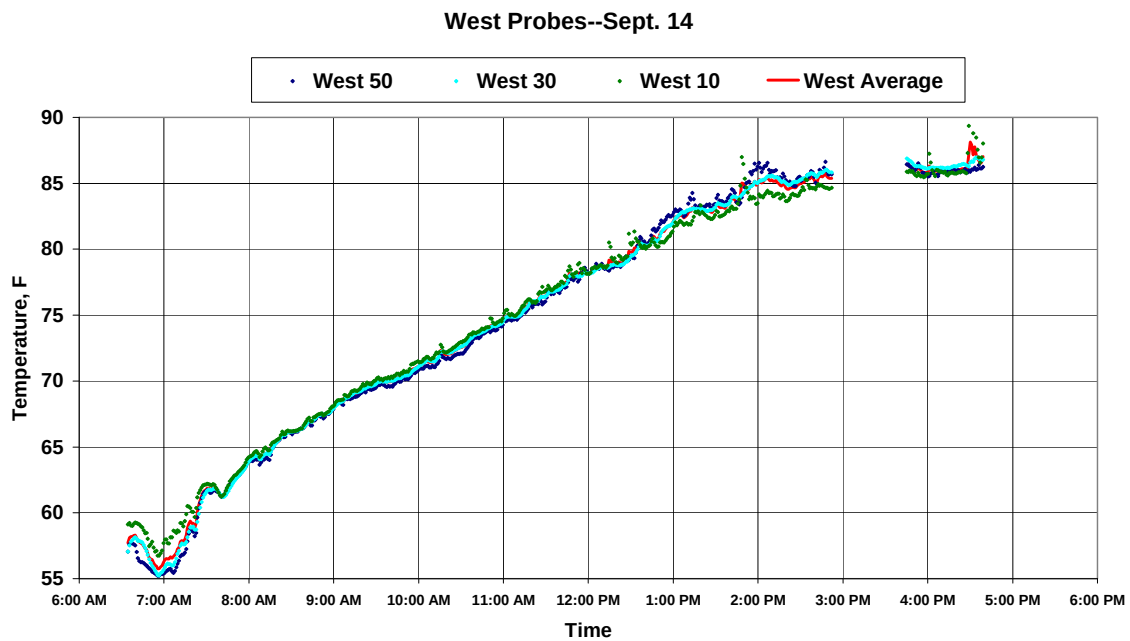


Figure 5-8d. Bighorn ambient air temperature—ACC periphery, west face.

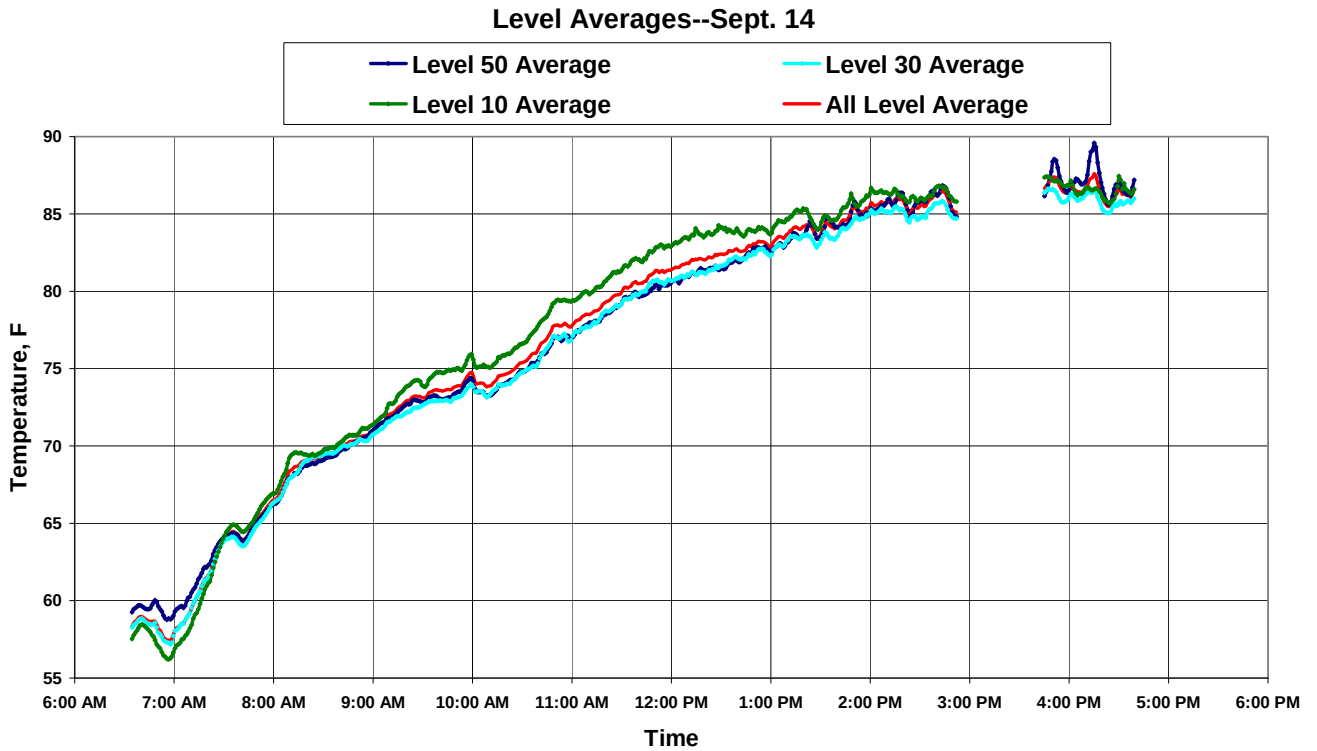


Figure 5-9. Bighorn ambient air temperature—average temperatures at each level (10, 30, and 50 feet above grade).

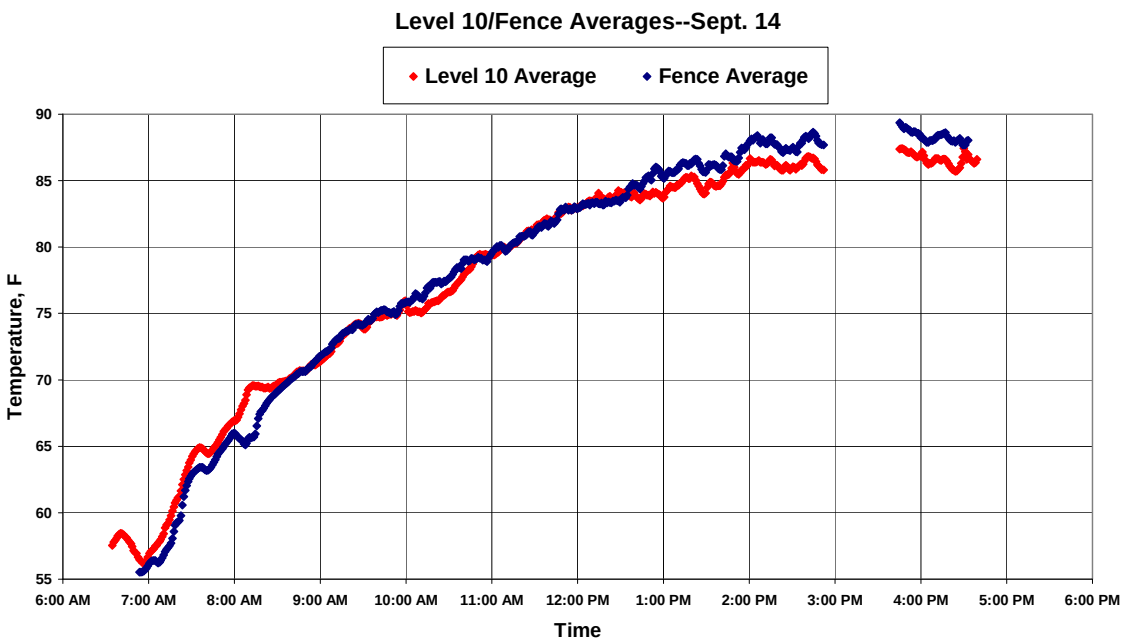


Figure 5-10. Bighorn ambient air temperature—comparison of fence and Level 10 averages.

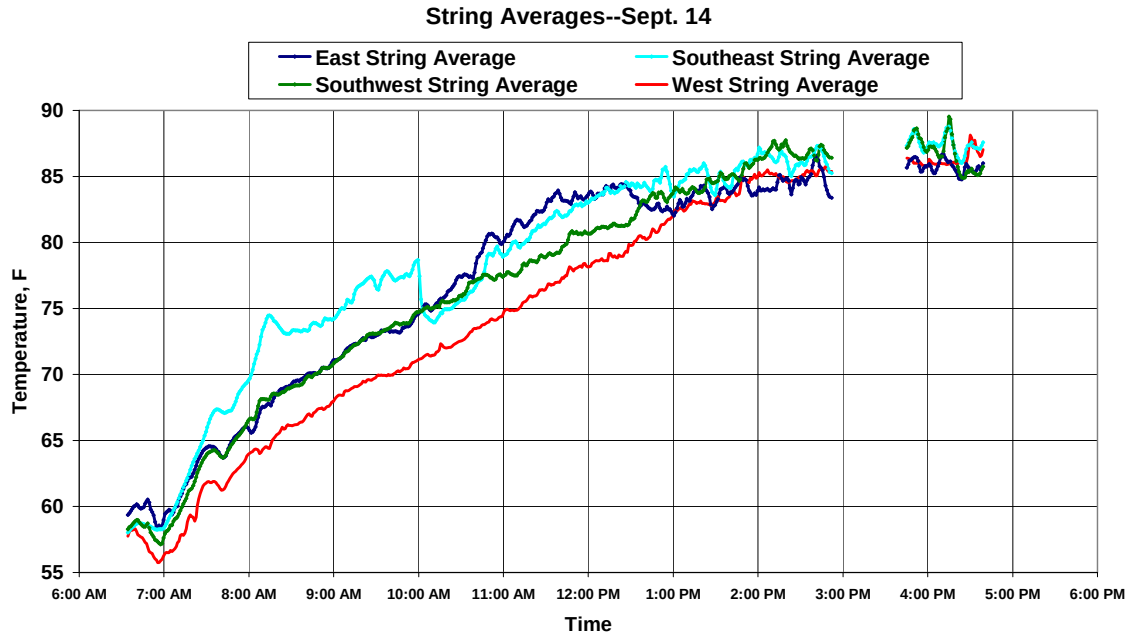


Figure 5-11. Bighorn ambient air temperature—average temperatures at each face.

All of the peripheral probes exhibit similar behavior with a few noteworthy exceptions:

- On all but the west face, the lowest probe (10 feet above grade) is typically 2° to 3°F hotter than the higher probes, as shown in the comparison of “Level Averages” in Figure 5-9. The low probes compare reasonably well with the average of the measurements taken on the boundary fence as shown in Figure 5-10. Higher air temperatures at the lower levels are consistent with the significant heating of the ground, especially in the morning hours.
- The variation in “string” averages from location to location (displayed in Figure 5-11) is also consistent with the expected influence of solar heating. The probes on the west string, as well as the ground beneath the string, are shaded throughout the morning and read consistently lower until early afternoon when all probes are more nearly consistent.
- The east string and the southwest string are partially shaded by the HRSGs (east string) and the ACC itself (southwest string) during the morning hours. They read close to one another and are lower than the southeast string, which is less shaded, and lower than the west string, which is more shaded.
- The behavior of the southeast string with a rapid rise from 7:30 a.m. to about 8:15 a.m. and a sudden drop at 10:00 a.m. is difficult to explain and may be due to aberrant wind currents.

Figure 5-12 compares the minimum cell inlet temperature, the average of all four peripheral strings at the 30 and 50 foot levels, and the average of the west string. Here again, the comparisons are consistent with the expected influence of solar radiation:

- The cell inlet temperature probes are located inside the cells above the fans and are completely shielded from solar radiation at all times.
- They agree well with the west string average for the hours during which the west string is also shaded.
- Most of the air entering the cells comes from the upper portion of the open areas on the periphery of the ACC. The average air temperature at the 30/50 foot level tracks the cell inlet temperatures well and is consistently 1° to 2°F higher as the different strings are warmed by solar radiation.

Although plant measurements of the ambient air temperature are not available for September 14, the data from June 16, 2005, plotted in Figure 5-6 indicates excellent agreement between the plant measurements taken under the ACC and the minimum cell inlet temperatures with the exception of the period from 6:00 a.m. to 9:00 a.m. when the plant probe may be exposed to heating from the low-lying sun.

Therefore, for all of the following analysis and comparisons in this study, the “ambient” air temperature will be taken to be the lowest of the set of cell inlet temperatures.

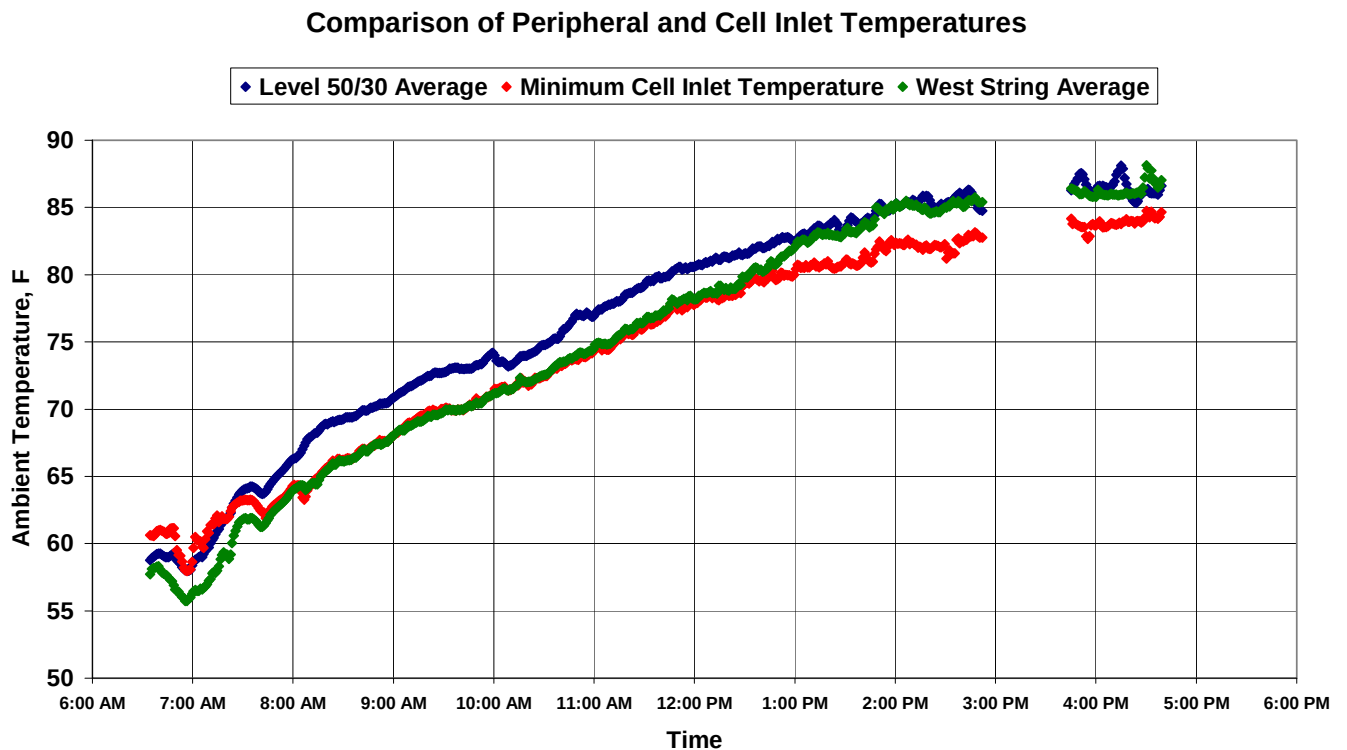


Figure 5-12. Bighorn ambient air temperature—comparison of peripheral and cell inlet temperatures.

5.3.4 Wind Conditions

Primary wind information was taken from the project met tower since it was located, on all sites, in a position close to and upwind (for prevailing summer winds) of the ACC with no intervening obstructions.

Table 5-10 shows the wind speed and direction for the period of 1:00 to 1:20 p.m. on June 16, 2005. Similarly, Figure 5-13 plots the wind conditions at one-minute intervals for the day of June 16. The obvious observation is that there is a high degree of variability on the minute-to-minute timescale for both speed and direction. On the directional scale, southerly winds are plotted at 180°. Northerly winds are represented by both 0° and 360° so winds varying around the northerly direction appear as points at both the bottom of the chart (if slightly east of north) or the top of the chart (if slightly west of north), lending some confusion to the visual interpretation.

A comparison of these measurements with plant data is made in Figures 5-14 and 5-15 simply to confirm the general reliability of the project met data. However, the plant data are felt to be less representative of conditions near the ACC.

Table 5-10. Bighorn wind conditions, 1:00–1:20 p.m., June 16, 2005.

Date/Time	Weather Conditions	
	Wind Speed	Wind Direction
	mph	degrees
6/16/05 13:00	16.5	210
6/16/05 13:01	13.0	83
6/16/05 13:02	13.2	97
6/16/05 13:03	16.5	90
6/16/05 13:04	14.1	91
6/16/05 13:05	16.1	200
6/16/05 13:06	15.6	201
6/16/05 13:07	13.5	189
6/16/05 13:08	16.6	208
6/16/05 13:09	18.0	319
6/16/05 13:10	18.3	200
6/16/05 13:11	17.4	331
6/16/05 13:12	21.9	200
6/16/05 13:13	15.8	312
6/16/05 13:14	14.8	195
6/16/05 13:15	15.7	313
6/16/05 13:16	16.7	180
6/16/05 13:17	16.4	84
6/16/05 13:18	18.0	322
6/16/05 13:19	16.8	186
6/16/05 13:20	14.9	313

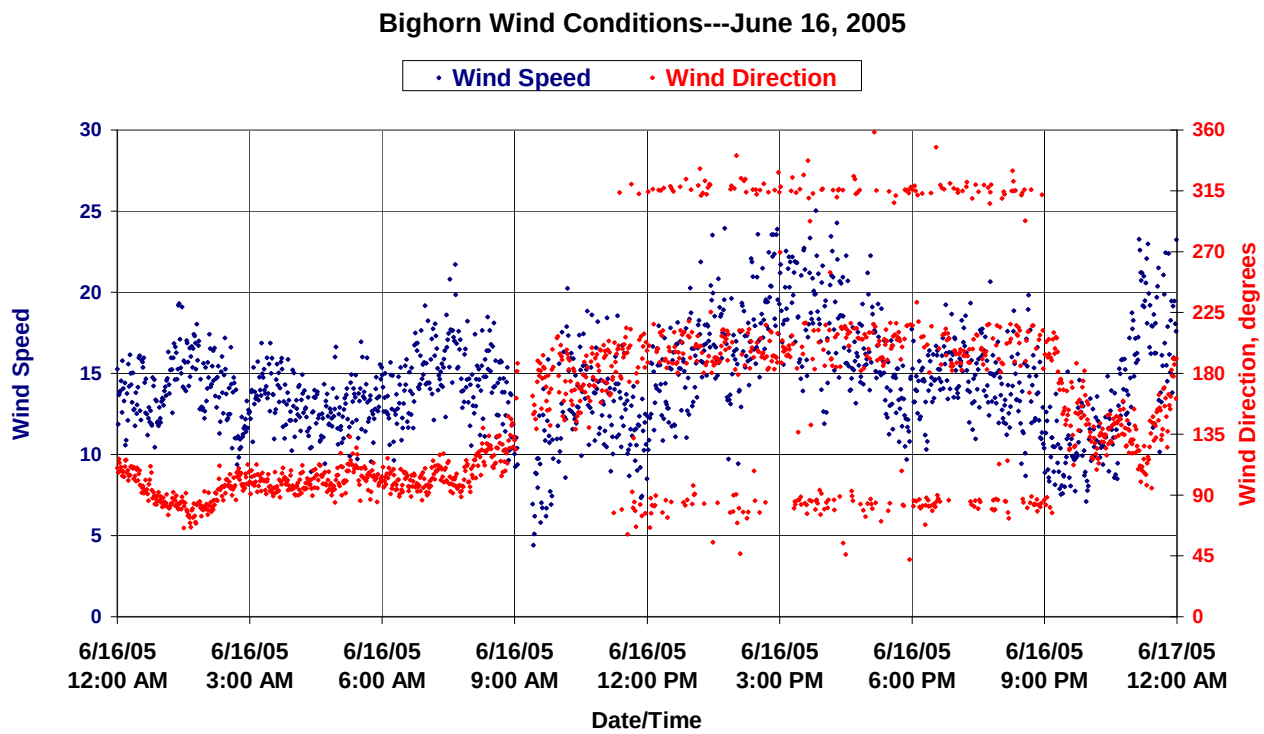


Figure 5-13. Bighorn wind conditions, June 16, 2005.

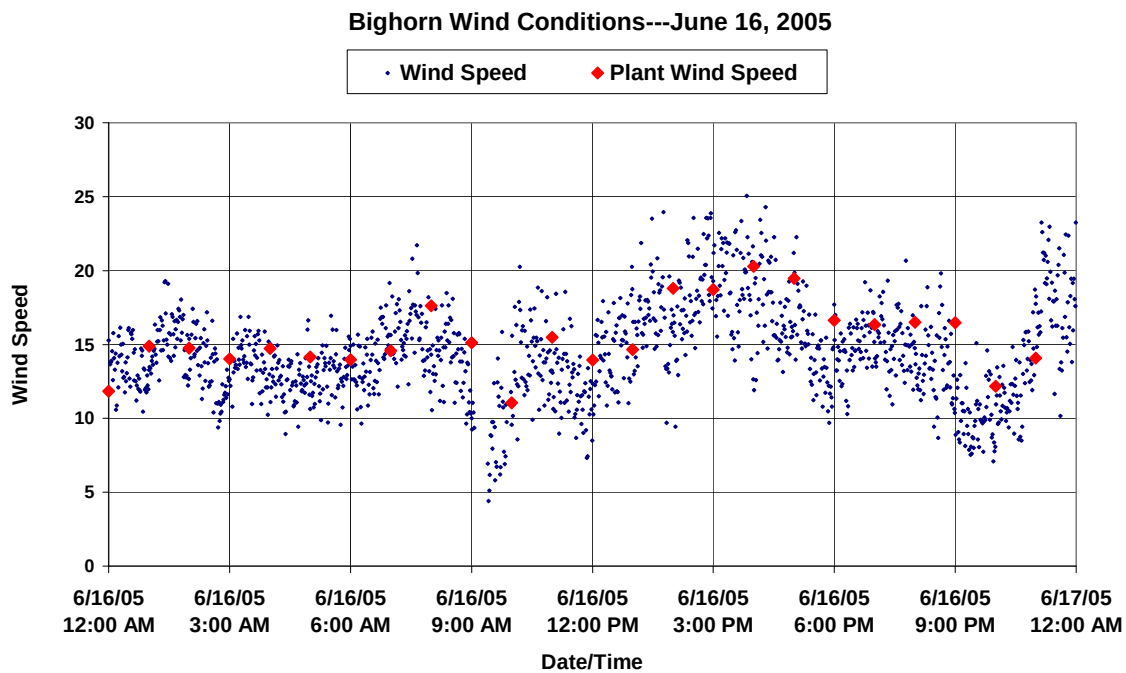


Figure 5-14. Comparison of plant and project wind speed measurements, June 16, 2005.

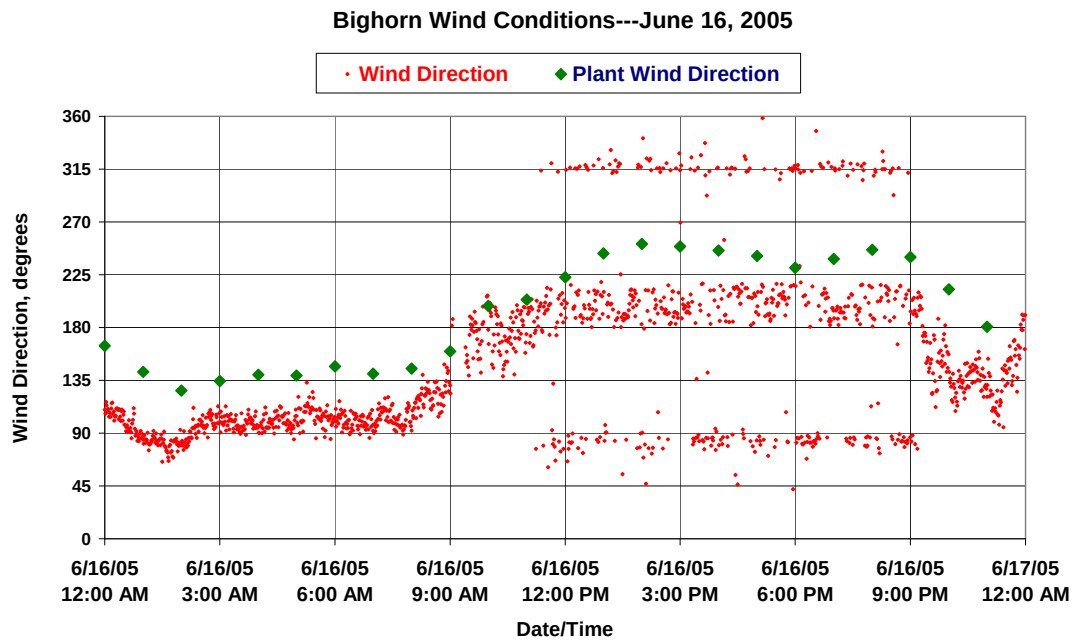


Figure 5-15. Comparison of plant and project wind direction measurements, June 16, 2005.

CHAPTER 6:

Analysis of Results: Recirculation

Wind is believed to effect the performance of ACCs in two ways:

- Recirculation—caused by the entrainment of a portion of the hot exhaust air back into the inlet air stream of some of the cells, resulting in an elevated inlet temperature into these cells.
- Fan performance degradation—caused by distortion of the inlet airflow to some of the fans, resulting in reduced airflow to the affected cells.

The data taken at the five test sites permit these two mechanisms to be examined and evaluated separately. This chapter analyzes the recirculation results; fan performance degradation is analyzed in Chapter 7. The first part of this chapter presents the average recirculation data for each site (Figures 6-1 through 6-5) along with general observations and estimates of the potential effect on plant performance from the measured recirculation. The chapter then examines each site's recirculation measurements in greater detail.

6.1 Average Recirculation

6.1.1 Definition of "Average Recirculation"

Recirculation is properly defined as the difference between the average of all cell inlet temperatures and an ambient temperature some distance from the ACC. In ideal, "zero-recirculation" conditions where no air from the warmed exhaust plume is entrained into the inlet air stream to the ACC fans, the average inlet temperature should equal the far-field ambient temperature. The difficulties of determining this "ambient" temperature with a high degree of confidence will be discussed in Section 6.2.1. For this analysis, the baseline temperature for defining average recirculation will be taken to be the minimum measured cell inlet temperature. The adequacy of this choice will be examined for each site in following sections.

6.1.2 Average Recirculation at Each Site

Figures 6-1 through 6-5 display the average ACC recirculation at each of the five sites for all times during the test periods when *all fans were operating at full speed*.

A consistent observation from the results at the five sites is that the average recirculation—based on the average inlet temperature of all the cells minus the lowest inlet temperature at any of the cells—is almost always less than 3°F:

- In the case of Bighorn (Figure 6-1), nearly all the points lie between 0.5° and 2.5°F with only three excursions above 3°F but less than 4°F and only one above 4°F.
- Similar results are seen at El Dorado (Figure 6-2) with nearly all between 1.0 °F and 2.5°F, seven excursions above 5°F, and two above 6°F.

- Apex (Figure 6-3) operated most of the time with an average recirculation less than 2°F, a few periods between 2° and 4°F, and seven brief excursions above 4°F.
- Front Range (Figure 6-4) showed a slightly higher degree of recirculation with most points between 1 and 4 °F and with three excursions over 6°F and one as high as 12 °F.
- At Black Hills (Figure 6-5) nearly all points lie between 0.25° and 2.5°F, with only three excursions above 3°F and none above 4°F.

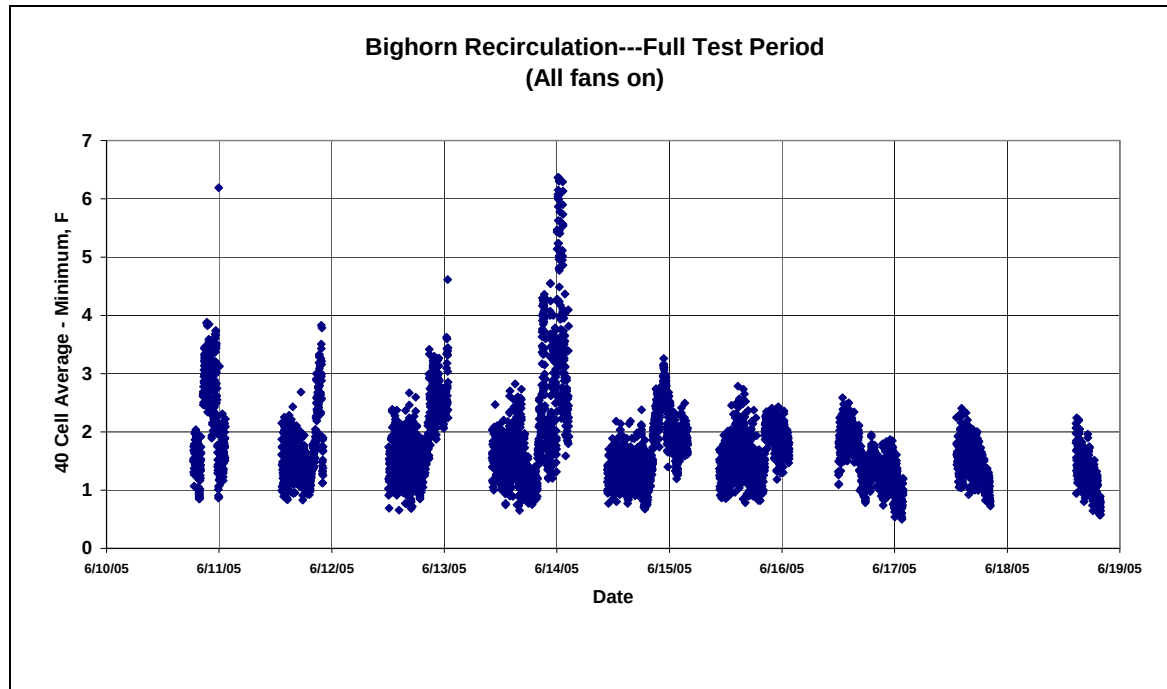


Figure 6-1. Average recirculation at Bighorn.

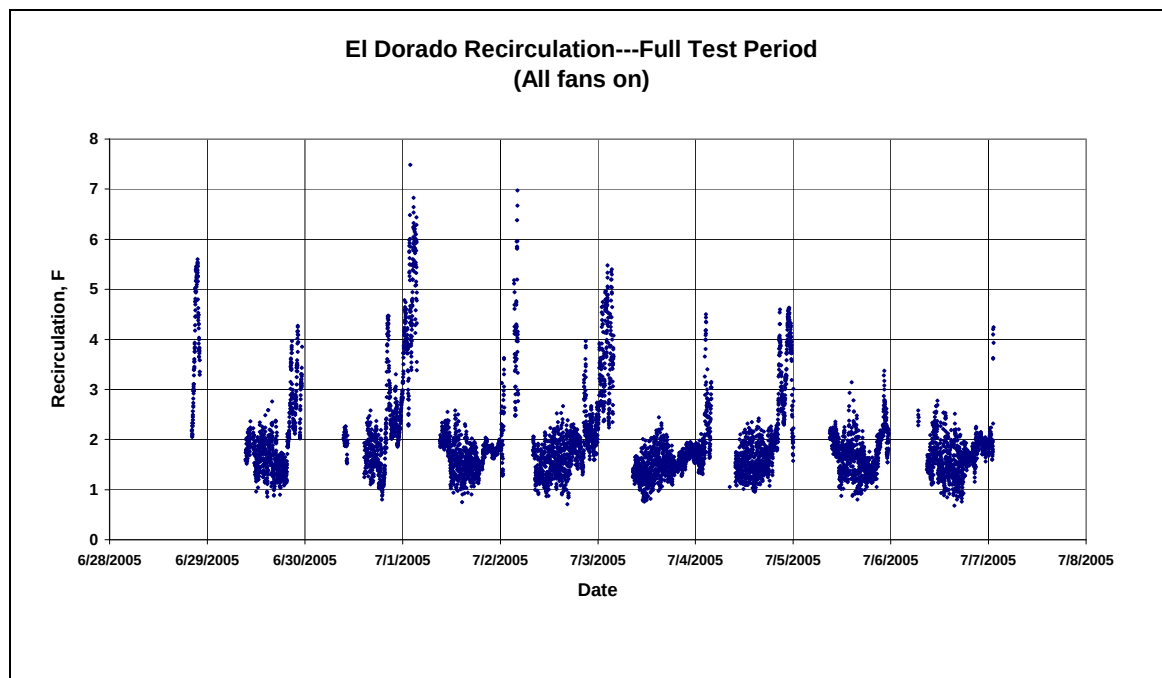


Figure 6-2. Average recirculation at El Dorado.

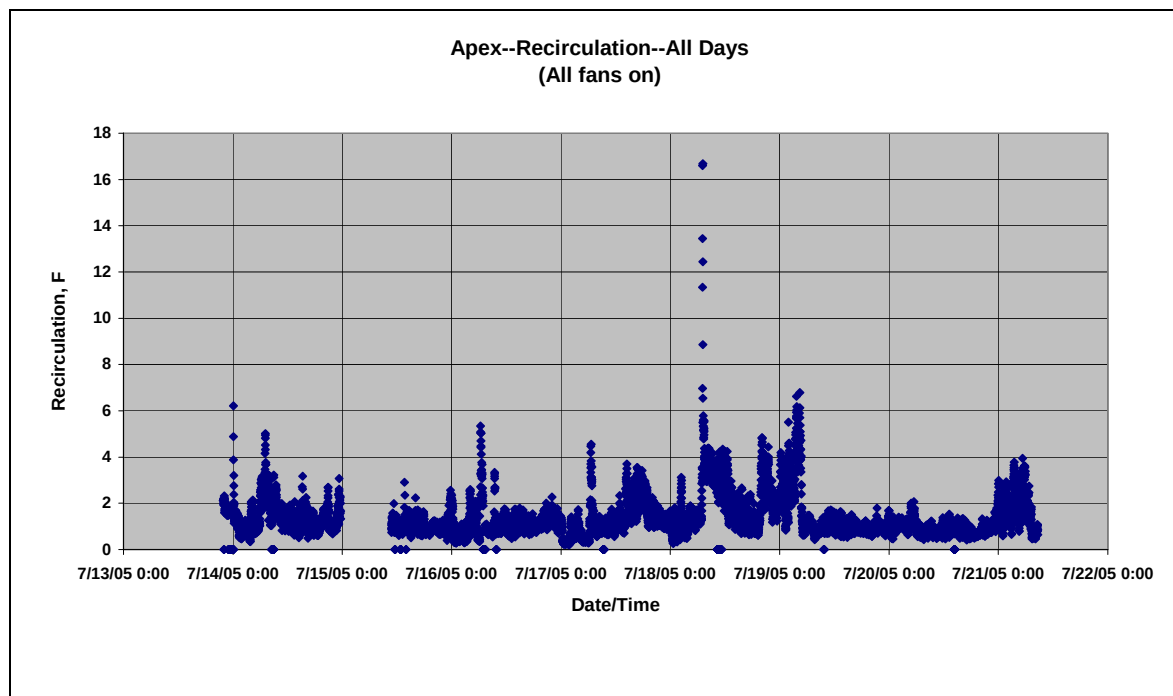


Figure 6-3. Average recirculation at Apex.

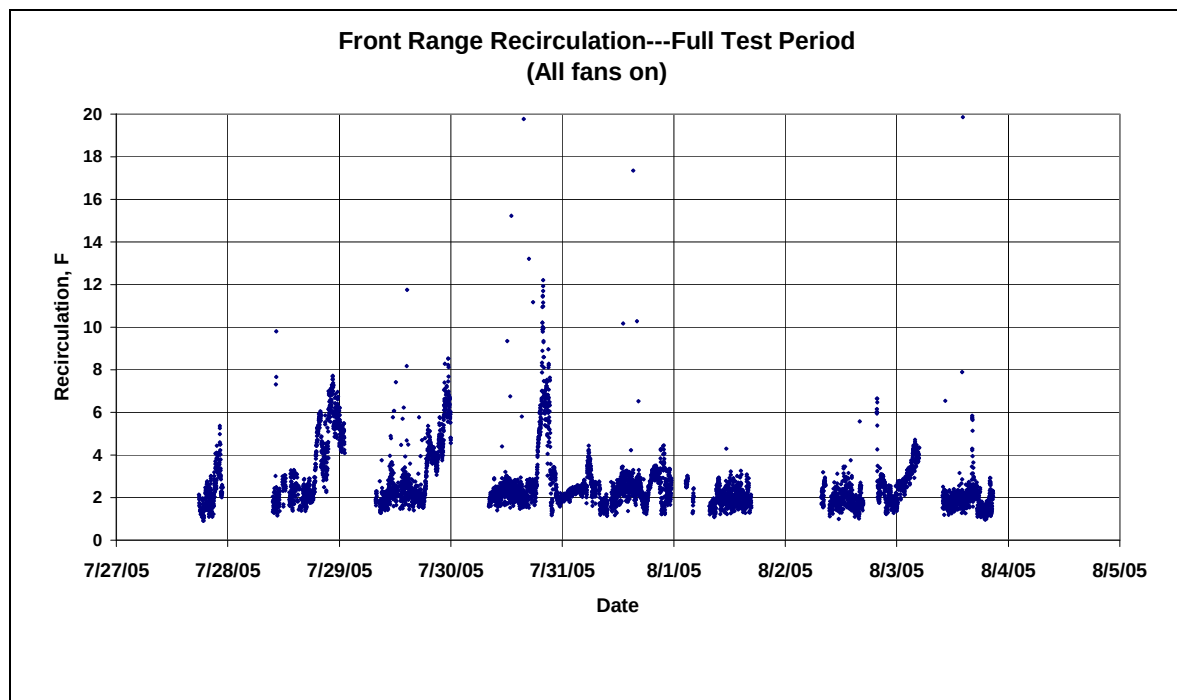


Figure 6-4. Average recirculation at Front Range.

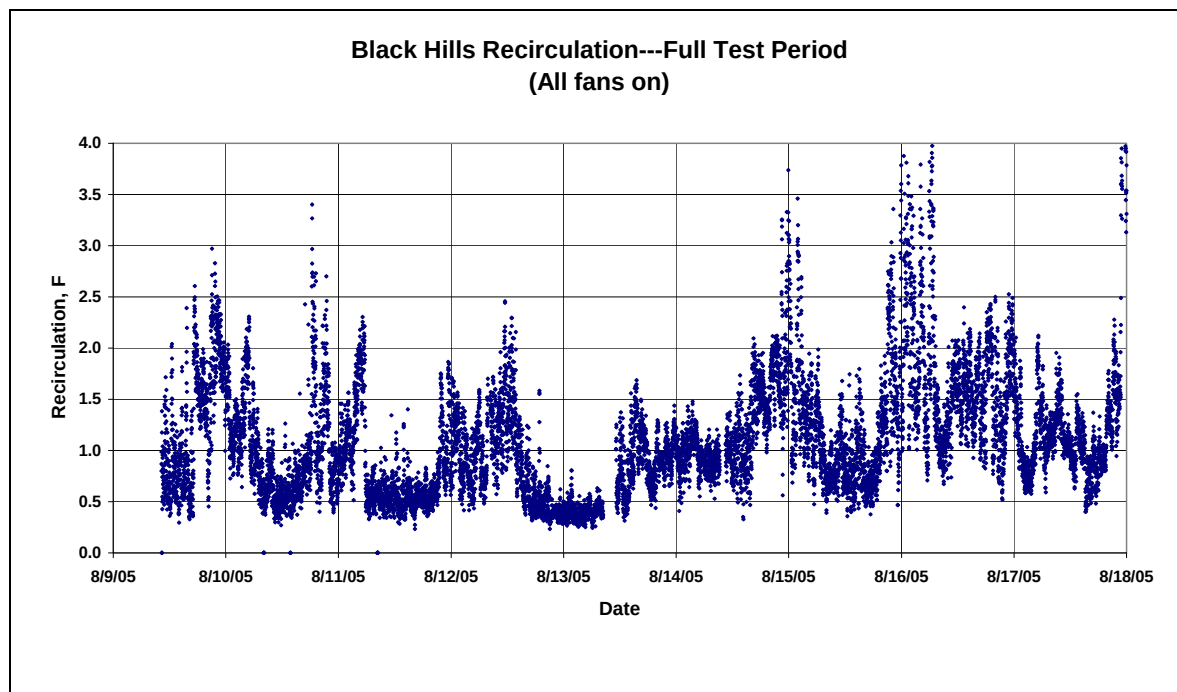


Figure 6-5. Average recirculation at Black Hills.

Several points are noteworthy. First, at essentially no time, is the average recirculation ever zero; it ranges, at a minimum, from 0.25°F to slightly above 1°F. As will be seen in the more detailed examination of the data, there is also always one or a few cells for which the inlet temperature exceeds the minimum cell inlet temperature by at least one and in most cases several degrees. For example, Figure 6-6 displays the *maximum* recirculation experienced by any cell (defined as maximum cell inlet temperature minus minimum cell inlet temperature) on the same plot as the average recirculation for the test period at Bighorn (as in Figure 6-1).

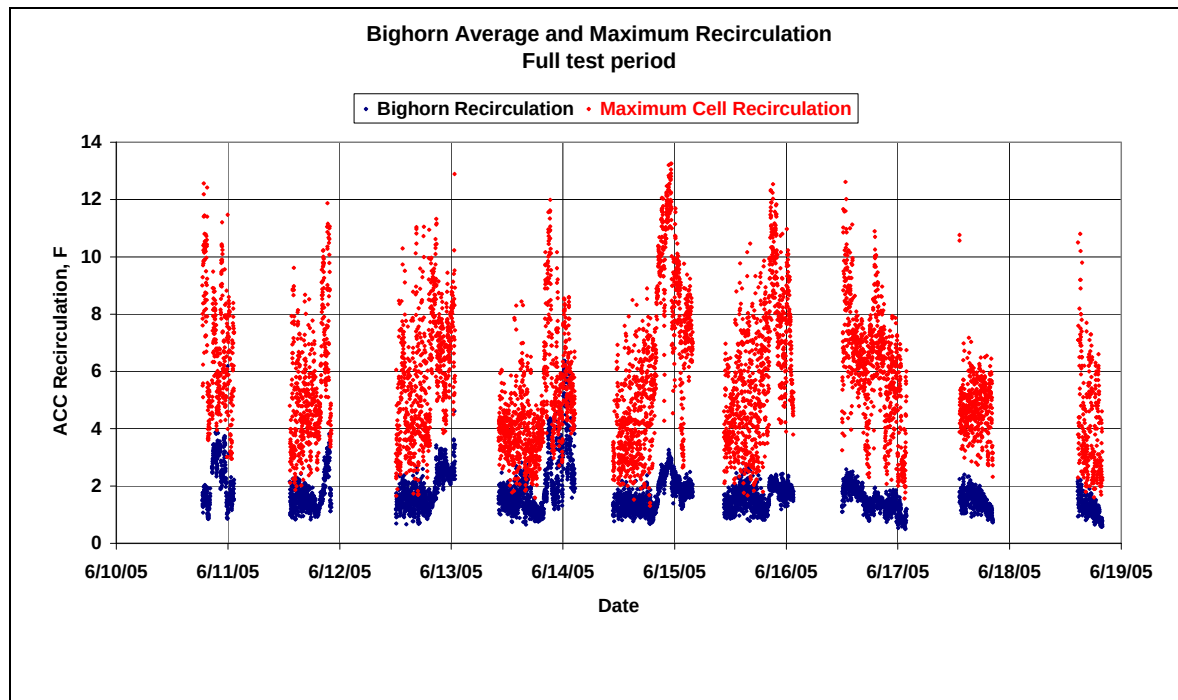


Figure 6-6. Average and maximum recirculation at Bighorn,

Second, the average recirculation seldom exceeds 3°F at any time at any of the sites. Figure 6-7 displays typical operating curves for an ACC based on vendor-supplied information. The curves plot condenser pressure vs. steam flow with ambient temperature as a parameter. Interpolation between curves indicates that the expected effect of an average recirculation of 3°F would be no more than 0.5 in Hga.

Figure 6-8, however, displays operating data from the El Dorado Energy Center, taken over a period of several months. The data are plotted as condenser pressure vs. ambient air temperature with wind speed as a parameter. All the points are for approximately the same steam flow. No distinction is made for wind direction. Best-fit lines have been plotted on the graph using the Excel curve fit function. The difference in backpressure, at temperatures above 90°F, between operating points at low (0 to 4 mph) wind speeds and at high (>20 mph) wind speeds is about 1.5 to 2.0 in Hga.

The basic conclusion drawn from this examination of average recirculation data is that recirculation is not a sufficiently large effect to account for the increase in backpressure of 1 inHga or more that is observed at higher wind speeds. It remains to examine some details of the recirculation mechanism.

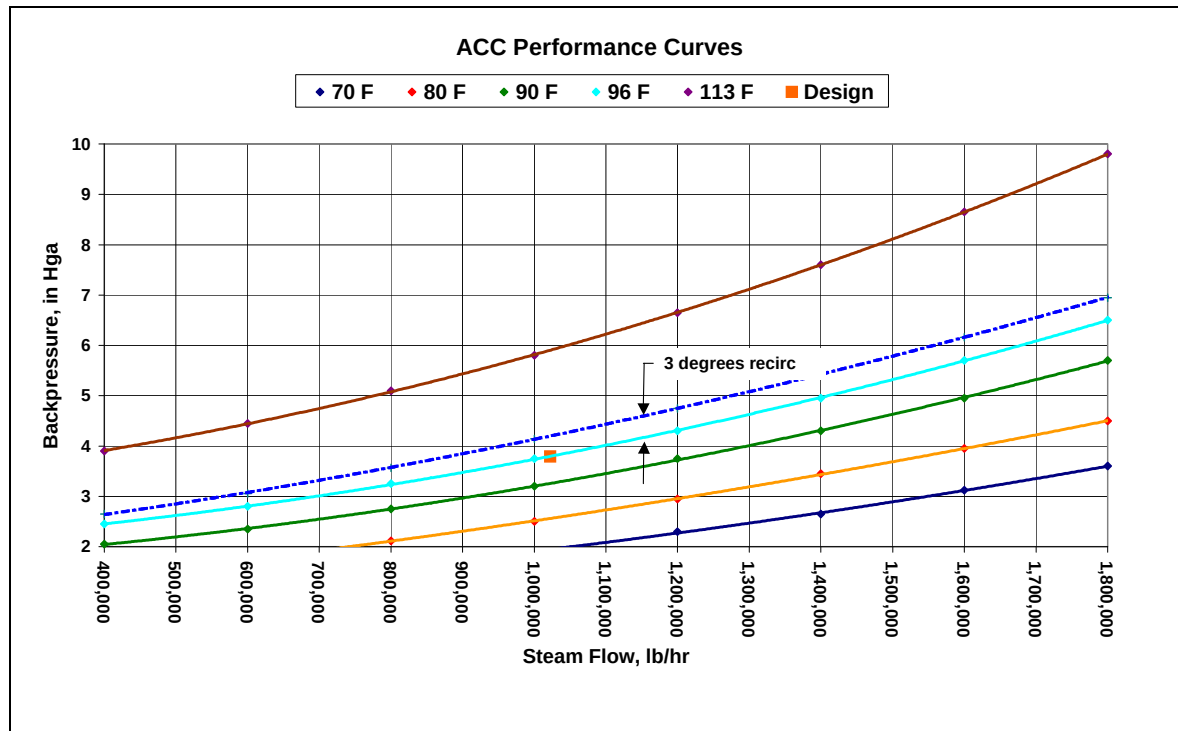


Figure 6-7. Expected effect of 3°F recirculation.

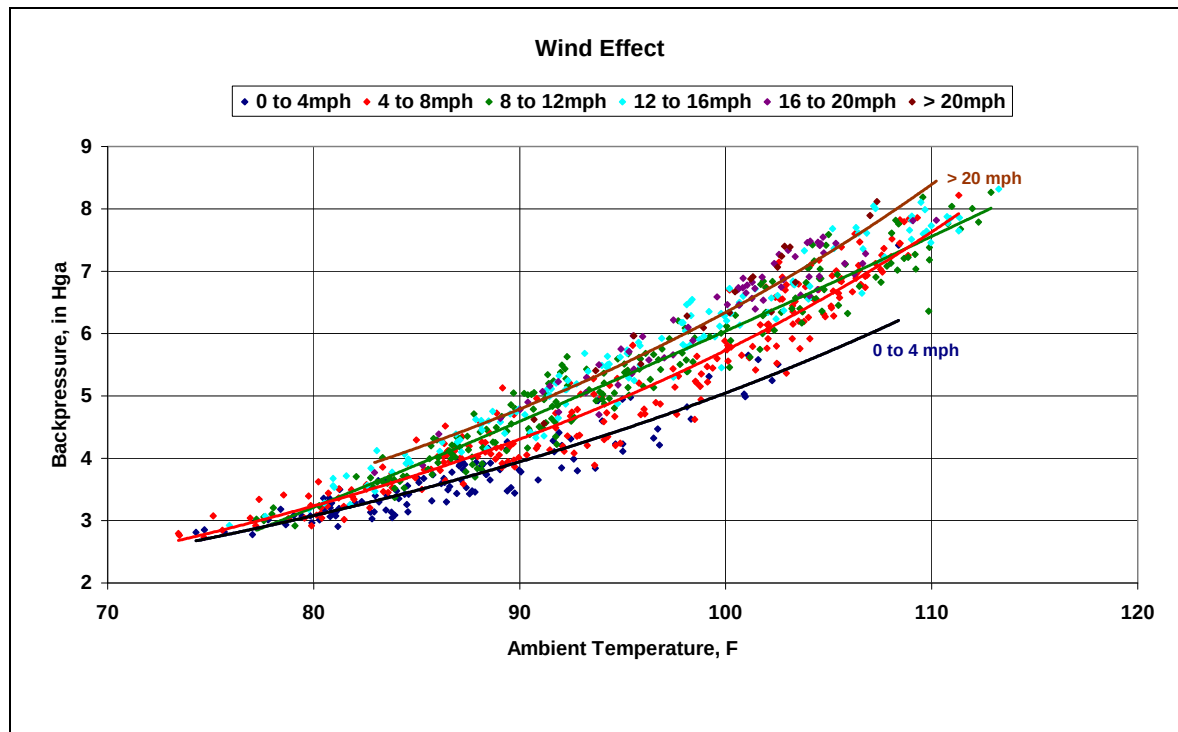


Figure 6-8. Wind effect on turbine backpressure.

6.2 Recirculation Tests—Bighorn Power Plant

The most extensive testing with the most complete set of instrumentation was conducted at Bighorn. Some of the conclusions drawn there are applied to the analysis of data from the other sites.

6.2.1 Determination of Baseline (“Zero-Recirculation”) Inlet Temperature

The average recirculation displayed in Figures 6-1 through 6-5 is calculated on the basis that the appropriate ambient temperature is reasonably represented by the minimum cell inlet temperature. This is based on the assumption that there is no plausible cooling mechanism affecting the air stream as it enters the ACC. However, Figures 6-9 through 6-12 indicate that different measurement points can give temperature readings which differ from the minimum cell inlet temperature, sometimes by a significant amount.

Plant instrumentation at Bighorn monitors ambient conditions in three locations: on the administration building, under the ACC, and at a met tower located several hundred yards to the west of the plant.

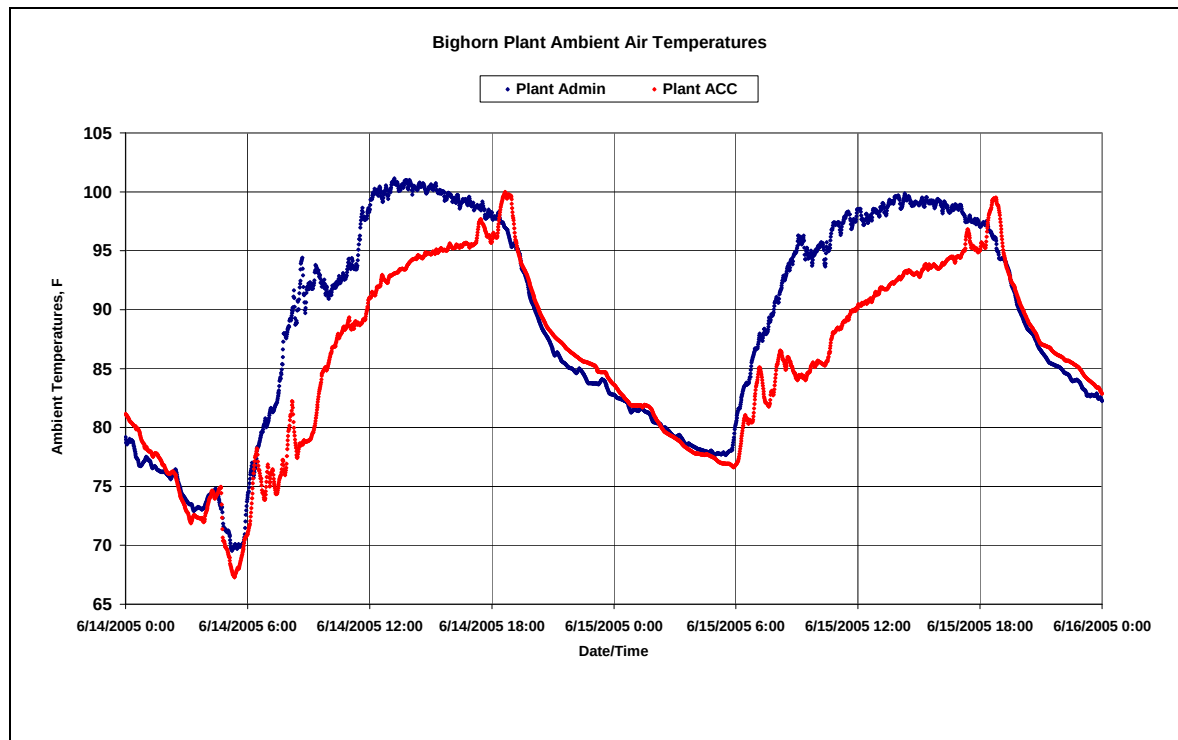


Figure 6-9. Plant ambient temperature readings.

The plant readings from the locations under the ACC and on top of the administration building (Figure 6-9) show essentially perfect agreement between the hours of 7 p.m. and 7 a.m. For the rest of the day, during sunlit periods, the ACC readings from a temperature sensor shaded from the sun by the ACC gives significantly lower readings. Two daily “spikes” in the ACC measurements are seen in the early morning (about 8:30 a.m.) and again in the late afternoon (about 6:30 p.m.), times when the sun is low in the sky and briefly illuminates the point under the ACC where the temperature probe is located. Therefore, the ACC readings are considered to be the more reliable.

Figure 6-10 compares the minimum inlet cell temperature with the plant ACC reading.

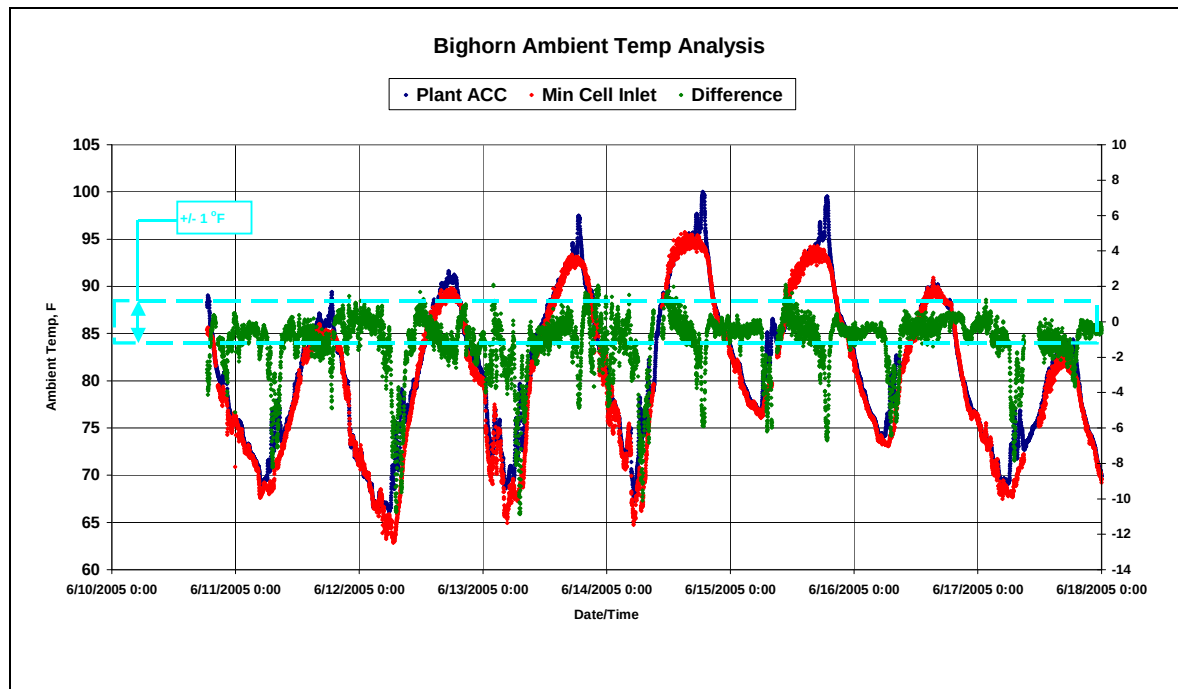


Figure 6-10. Comparative inlet temperature measurements—plant ACC and minimum cell.

Most points during daylight hours lie within $\pm 1^{\circ}\text{F}$. Points with negative differences are for instances where the plant ACC reading is higher than the minimum cell inlet temperature. Most of these are associated with early morning and late afternoon periods during which the probe beneath the ACC is affected by solar radiation. The cell inlet temperature probes, located inside the ACC cells, are always shielded from the sun. Other periods with significant negative differences occur occasionally around midnight; see, for example, midnight on June 12 and June 13. These differences may be attributable to temperature inversions as will be discussed further in Section 6.2.3. Periods with positive differences greater than $+1^{\circ}\text{F}$ may indicate some degree of recirculation affecting even the lowest-temperature cell. These occur around midday on June 14 and June 15.

Later in the summer of 2005, additional alternative measurements were made. A full description of these tests is found in a final report entitled *Inlet Air Spray Cooling Enhancement for Air-Cooled Condensers* (Maulbetsch and DiFillipo, in press). Figure 6-11 displays the difference between ambient temperature readings at the 10-meter level on the plant met tower, located several hundred yards west of the plant boundary in an open area, and those taken on the roof of the administration building. From 9:00 a.m. to 6:00 p.m. over a three-day period in September, 2005, the readings on the administration building were consistently higher, the difference increasing throughout the day. This suggests a stronger solar radiation effect on the administration building probe compared to the more adequately shielded probe on the met tower.

Similarly, Figure 6-12 shows the differences between the met tower and the probe under the ACC, ranging from -2°F to $+2.5^{\circ}\text{F}$. The cooler readings under the ACC may be due to the fact

that the ground under the ACC is always shaded and presumably cooler than the ground in the open field where the met tower is located, with the likely result that the air temperature in the lower layers (less than 10 meters) is also lower.

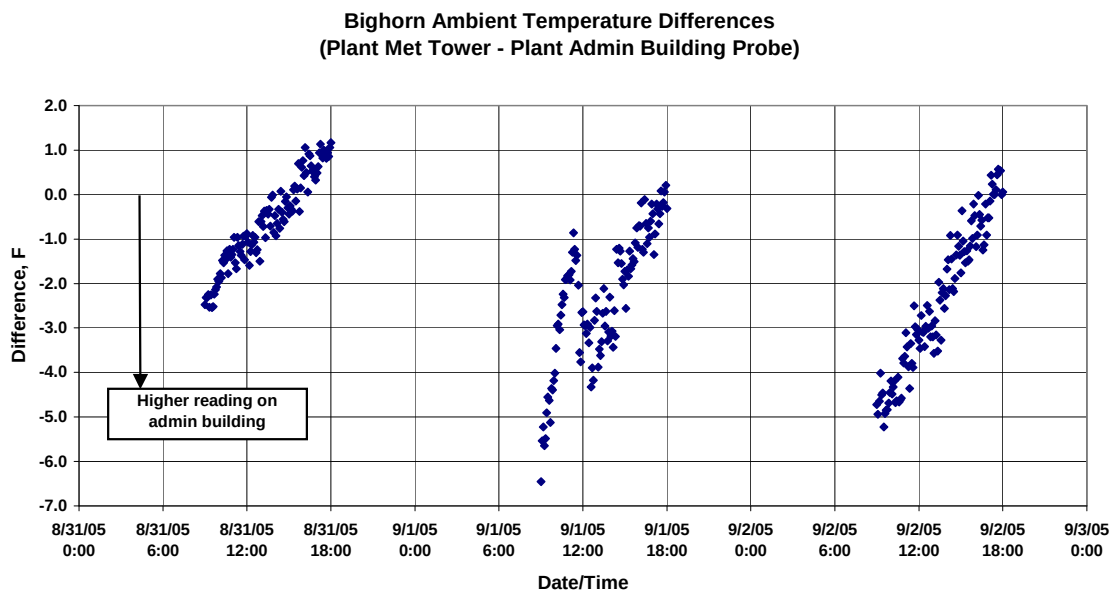


Figure 6-11. Ambient temperatures—plant met tower minus plant admin building.

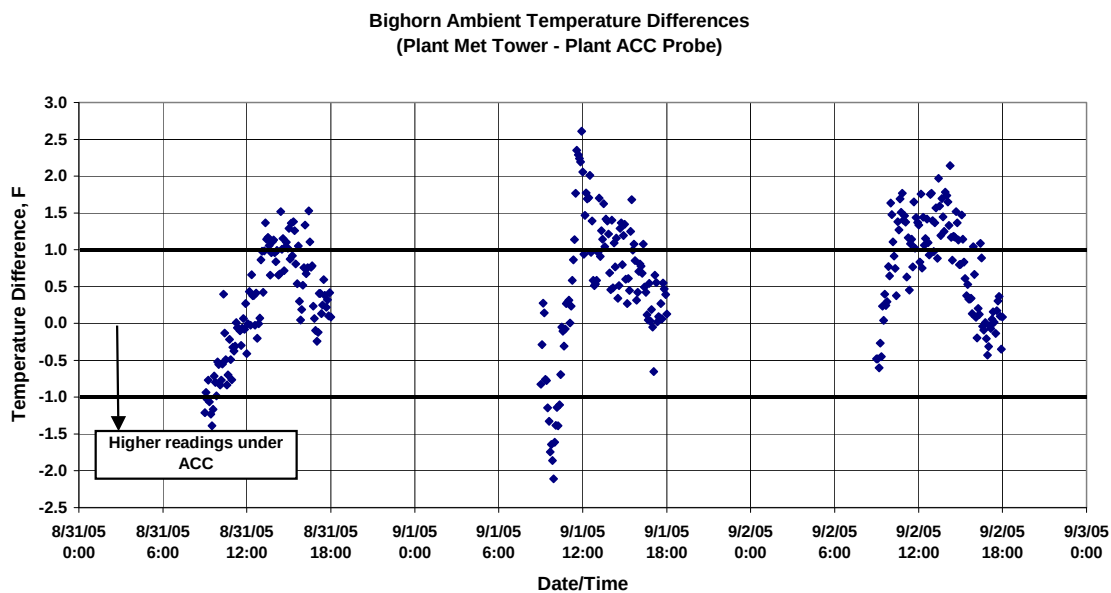


Figure 6-12. Ambient temperatures—plant met tower minus plant ACC.

In addition, measurements were taken with shielded aspirating probes at three locations on the plant boundary fence (approximately 4 to 5 feet above grade), at three levels (10, 30, and 50 feet above grade) at each of four locations on the periphery of the ACC, and at several places on the fan bridge inside Cell 2I (with both shielded and unshielded probes). Again the conclusion is that hot ground surface effects can be significant and that defining a representative “ambient” temperature within 1° to 2°F is problematic.

The minimum cell inlet temperature, measured with an unshielded probe inside the ACC cells above the fan is, therefore, chosen on the basis that it is reasonably consistent with other shaded or shielded readings, it is always shaded from solar radiation, and is taken in a consistent manner with all of the other cell inlet temperatures.

6.2.2 Effect of Wind on Recirculation

Figure 6-1 presented the average recirculation as it varied with time during the test period at Bighorn. Figures 6-13a and 6-13b display the wind behavior over the same test period. The wind behavior changed from the early portion of the test period, during which it was consistently low (usually below 10–15 mph) and primarily from the northwest, to the latter portion when it was generally higher (seldom less than 10 mph) and primarily from the east-southeast to the south-southwest.

Figure 6-13b shows the relationship between wind speed and direction and indicates that the highest wind speeds (above 20 mph) are almost exclusively from the south. Low and intermediate winds have almost no preferred direction with the exception that winds from the northeast quadrant are rare. Winds from the northwest quadrant are typically lower speed than those from the more southerly directions.

Figure 6-14 shows the average recirculation as a function of wind speed. The results exhibit several interesting features. First, the highest recirculation is indicated at the lowest wind speeds (<5 mph). Second, at the higher wind speeds (~13 to 30 mph) there is no evident increase in recirculation with wind speed. Finally, there appears to be a small cluster of moderately high (~3° to 4.5°F) recirculation indicated at winds around 10 mph. With few exceptions, the average recirculation is less than 3°F, and usually less than 2°F, for the entire range of wind speeds.

Figure 6-13a. Wind speed and direction during test period at Bighorn.

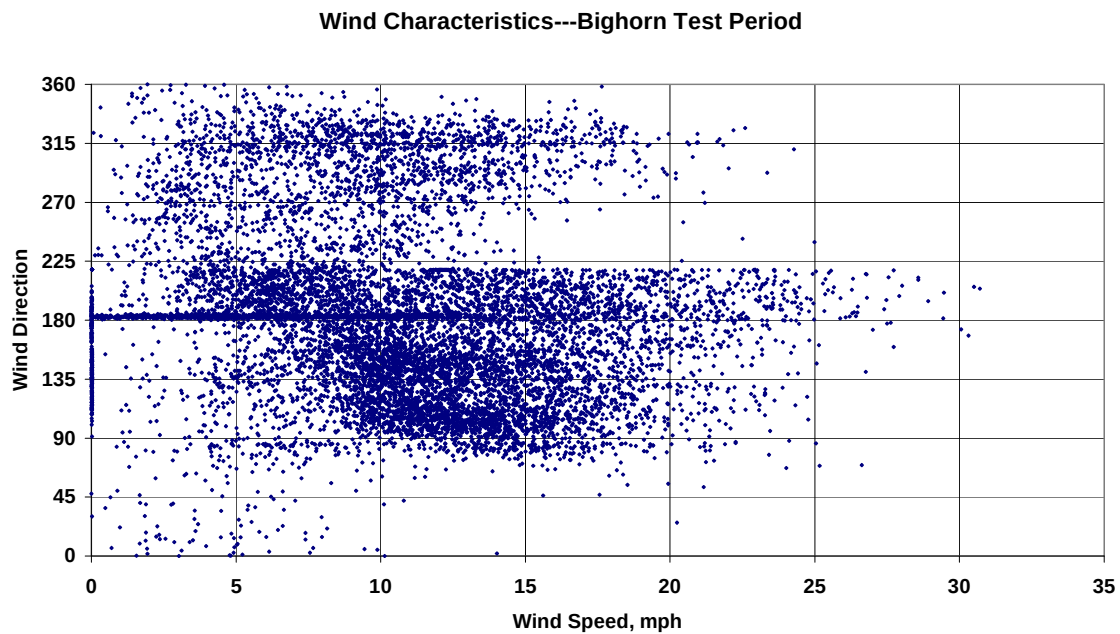
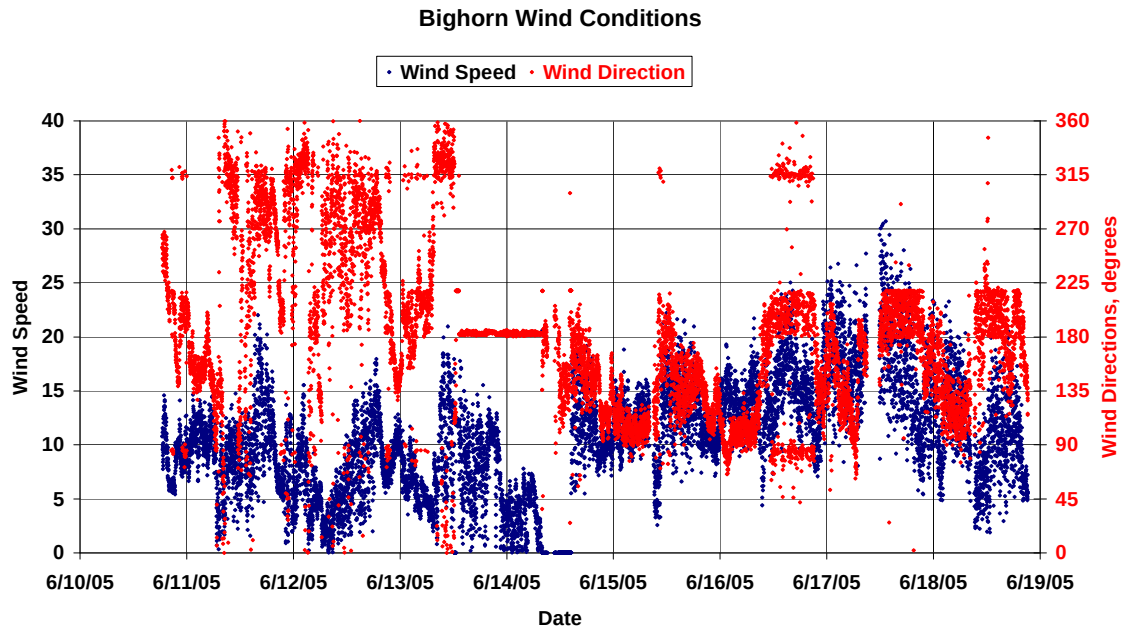


Figure 6-13b. Wind speed and direction relationship at Bighorn.

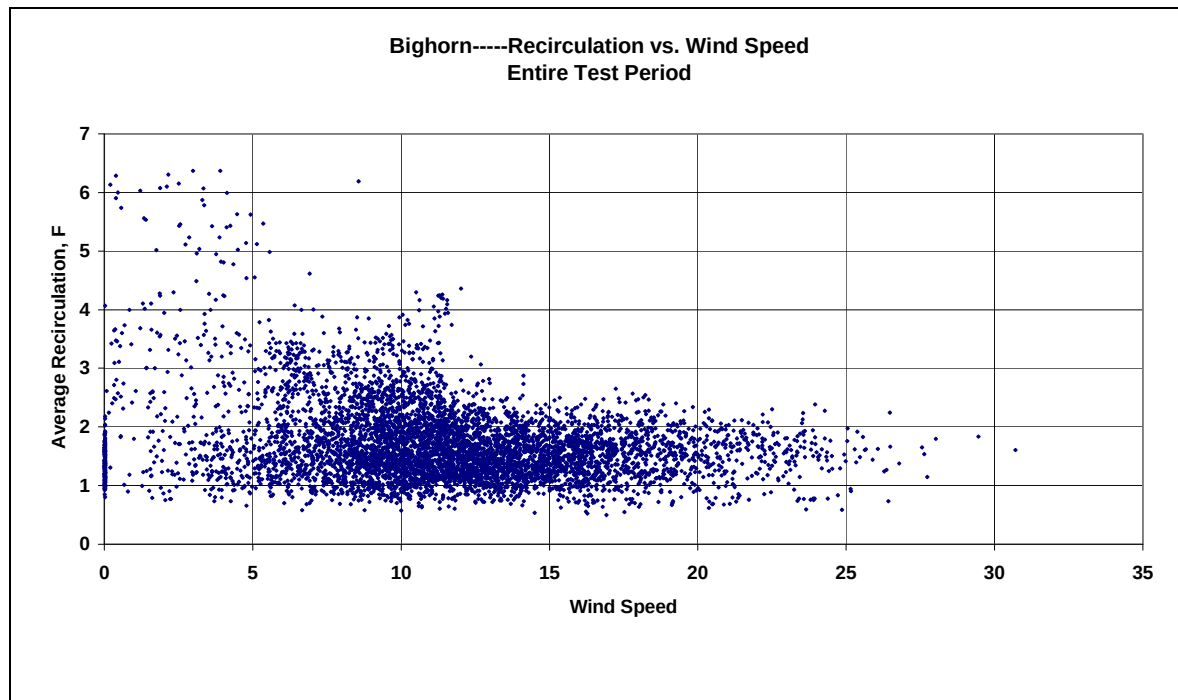


Figure 6-14. Recirculation vs. wind speed at Bighorn.

6.2.3 High Recirculation at Low Wind Speed

The occurrence of high indicated recirculation at the lowest speeds was seen most dramatically around midnight on June 13 (between ~10 p.m. on June 13 and 1 a.m. on June 14); see Figure 6-1. Figure 13a shows this to be a period of sustained low wind, coming almost directly from the south. Other periods of higher-than-usual recirculation are seen on other days, all at or near midnight. This period on June 11 is also associated with very low winds.

These periods are anomalous situations during which the definition of “average recirculation” used in Figures 6-1 and 6-5 and discussed at length in Section 6.2.1 gives spurious results. This appears to be the result of a temperature inversion, where the air temperature at and near the ground is several degrees cooler than the air temperature at the fan deck level. Analysis of this phenomenon is presented in some detail by Kröger (1998). Figure 6-15, excerpted from this reference, shows temperature inversions in data obtained at a site with similar climate and topology to Bighorn from midnight to ~5 a.m. where the temperature at 20 meters—the approximate height of the fan deck at Bighorn—ranges from 7° to 9°C (13° to 16°F) above the temperature at ground level (1 meter). Such inversions exist during periods of clear sky and low wind and are suppressed or disrupted by cloud cover or wind.

Figure 6-16 shows the inlet temperatures in each the 40 cells at Bighorn at 1:17 a.m. on June 14, 2005, arranged in order of increasing temperature and labeled with the cell number. At that time, the wind was directly from the south at or below 1 mph. The average ACC inlet temperature was 80.5°F. Note, however, that two cells (Cells 2I and 2H) have inlet temperatures which are significantly lower than any of the others and deviate from any expected pattern. The

minimum cell temperature (Cell 2I at 74.9°F) gives a “recirculation” of 5.6°F ($80.5^{\circ} - 74.9^{\circ} = 5.6^{\circ}$). If the minimum cell temperature had been more consistent with the general pattern (as for example, Cell 2G), the recirculation would have been between 2° and 3°F and within the normal range.

The two low-temperature cells (2I and 2H) are located directly above the motor control center building, which presents a 12- to 15-foot-high obstruction to air coming in under the ACC from the south. Cold air from low levels could be diverted upward, bypassing the J-row cells but being entrained by the fans in Rows I and H. This postulated phenomenon gives a consistent explanation for the anomalously high recirculation readings at low wind conditions in the middle of the night and early, pre-dawn periods observed at several of the sites.

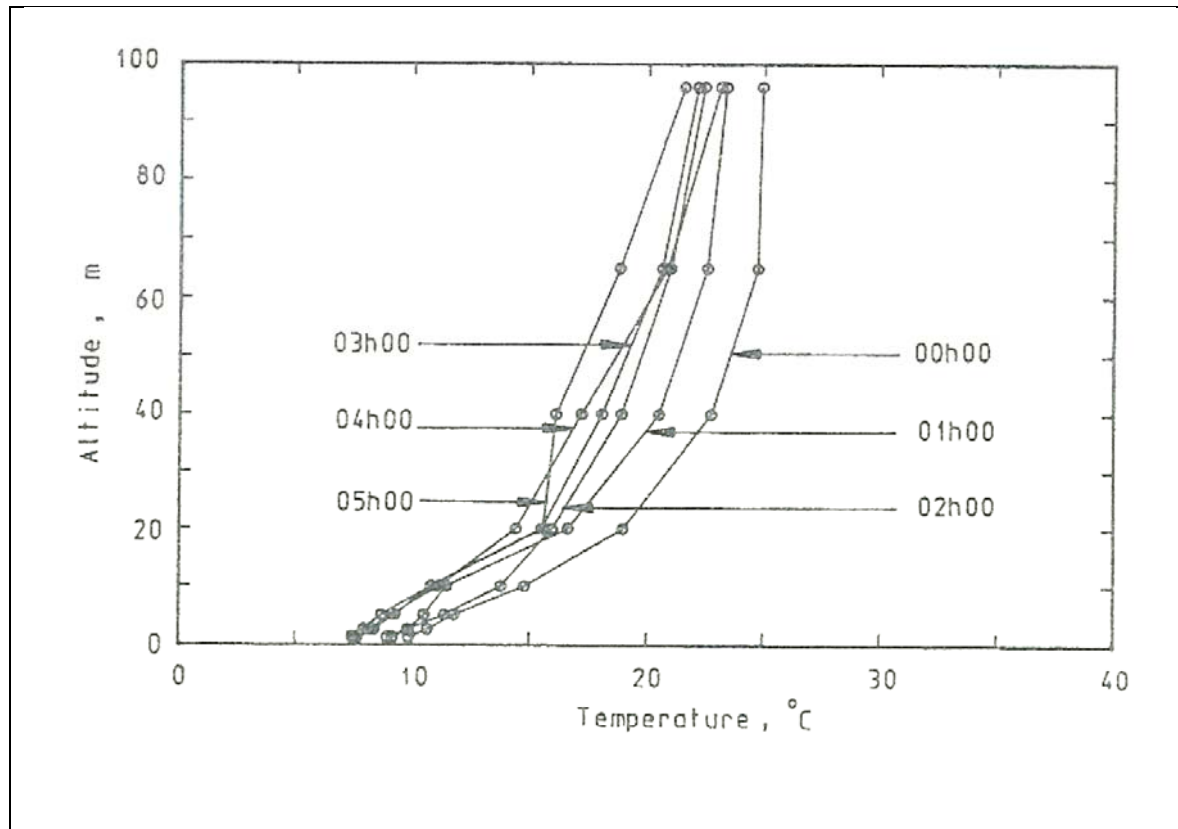


Figure 6-15. Temperature inversion conditions (excerpted from Kröger 1998).

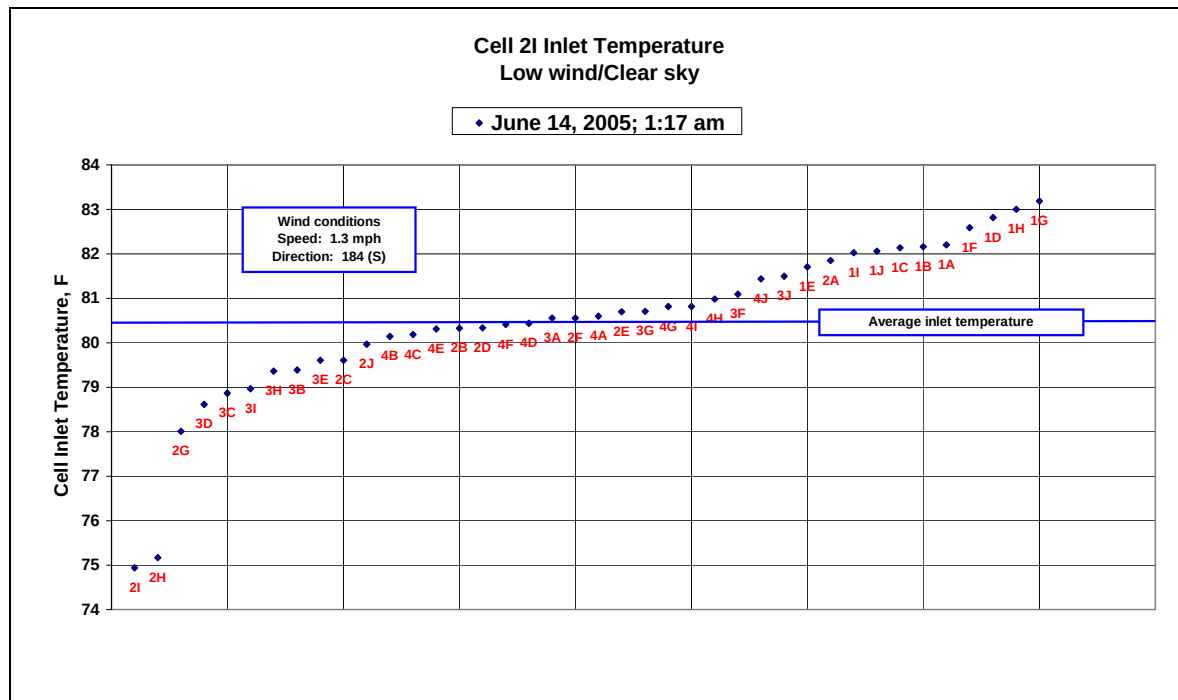


Figure 6-16. Cell inlet temperature distribution—temperature inversion conditions.

6.2.4 Recirculation at High Wind Speeds

Average recirculation data from Bighorn plotted in Figure 6-1 shows a cluster of moderately high results ($>2.5^{\circ}\text{F}$) occurring during the afternoon of June 15. Figure 6-17 plots the recirculation data from 1 p.m. to 4:30 p.m. on that day. Figures 6-18a and 6-18b show the wind speed and direction for the same period. The time of the highest single recirculation reading of 2.8°F is 2:36 p.m. This time is identified with a vertical bar in each of the figures.

There is nothing extraordinary about the magnitude or trend of either the wind speed or direction at that time. The higher recirculation readings are somewhat more frequent after 2:00 p.m. The wind speed readings show a modest increase in the number of higher-speed points beginning around 2:15 p.m. and persisting throughout the period. The distribution of lower and intermediate speed winds remains essentially unchanged. The wind direction shows a small shift from generally SSE between 1:00 and 1:30 p.m. to SE after 2:00 p.m.

At the time of highest recirculation, the cell inlet temperature distribution, shown in Figure 6-19, is consistent with expectations that recirculation would occur predominantly in the downwind cells. In this instance, winds from the southeast produce the highest inlet temperature in the northwest cluster of cells (Cells 4A to 4E). A plot of recirculation in the cells with highest inlet temperatures is shown in Figure 6-20.

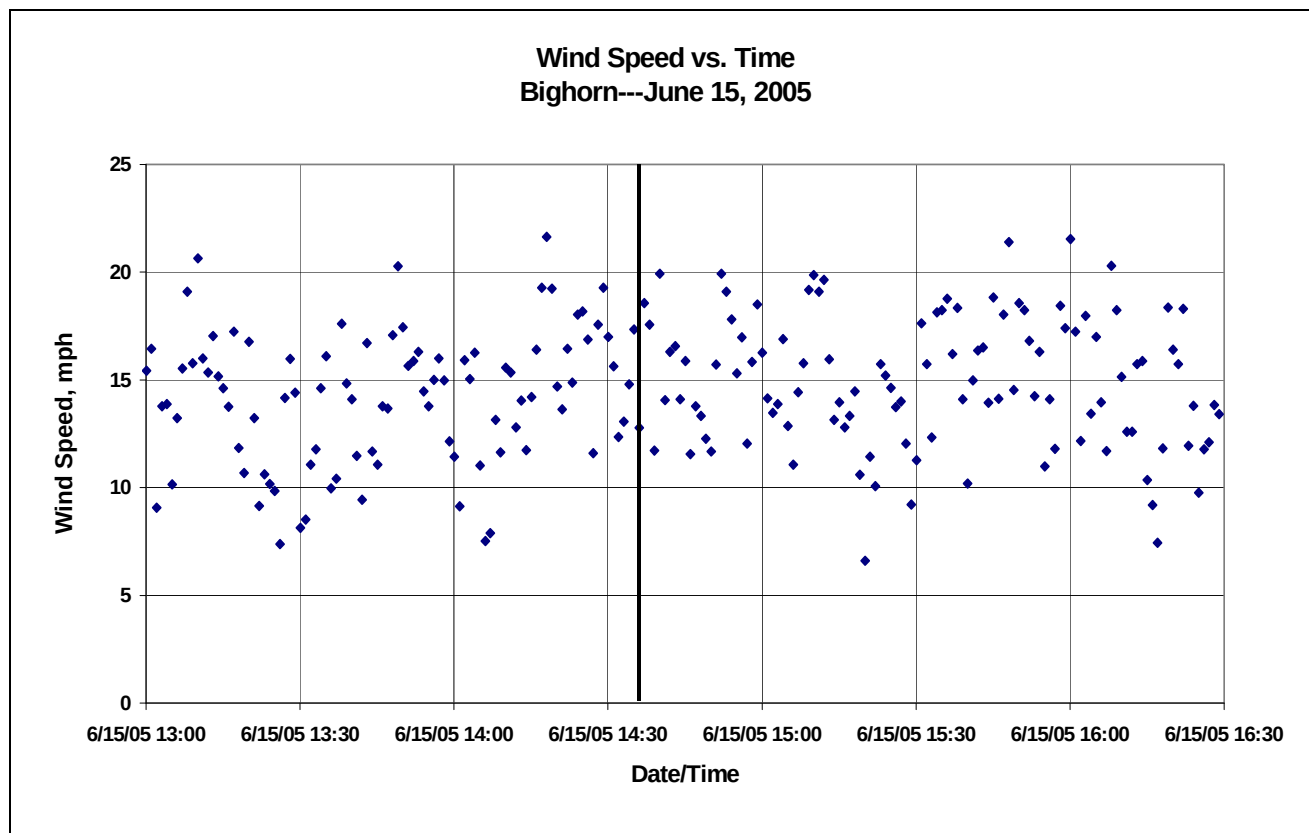


Figure 6-18a. Wind speed at Bighorn, mid-afternoon, June 15, 2005.

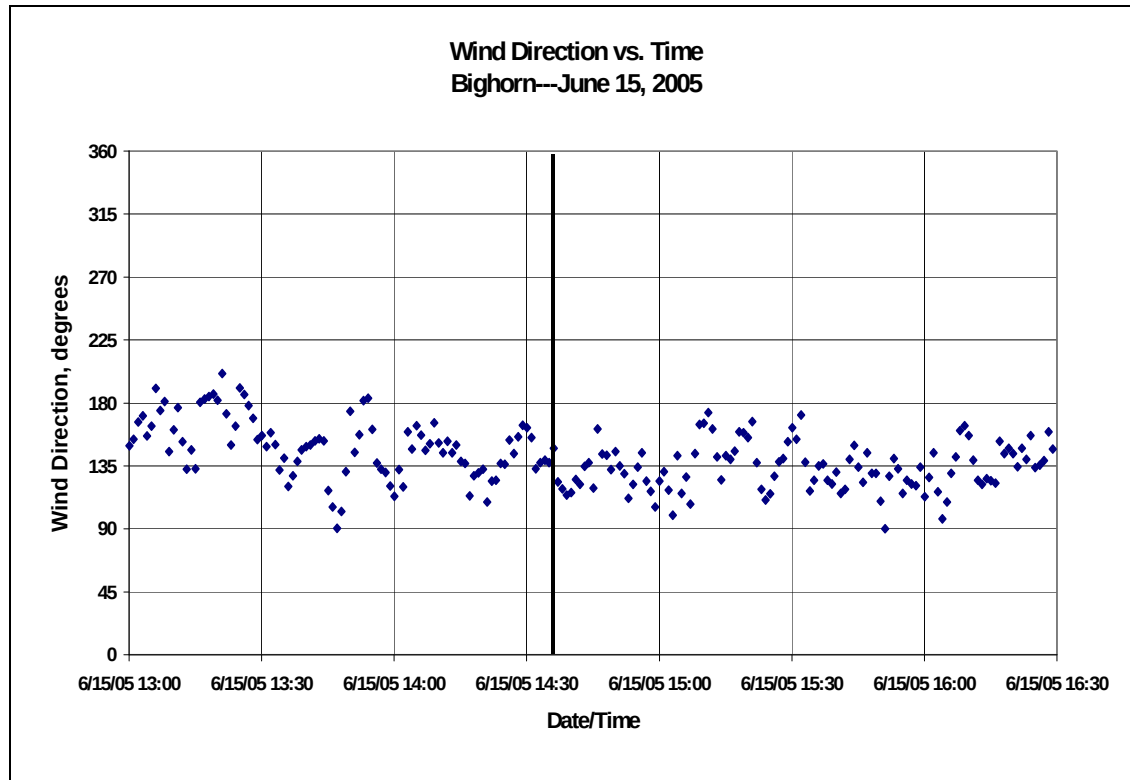


Figure 6-18b. Wind direction at Bighorn, mid-afternoon, June 15, 2005.

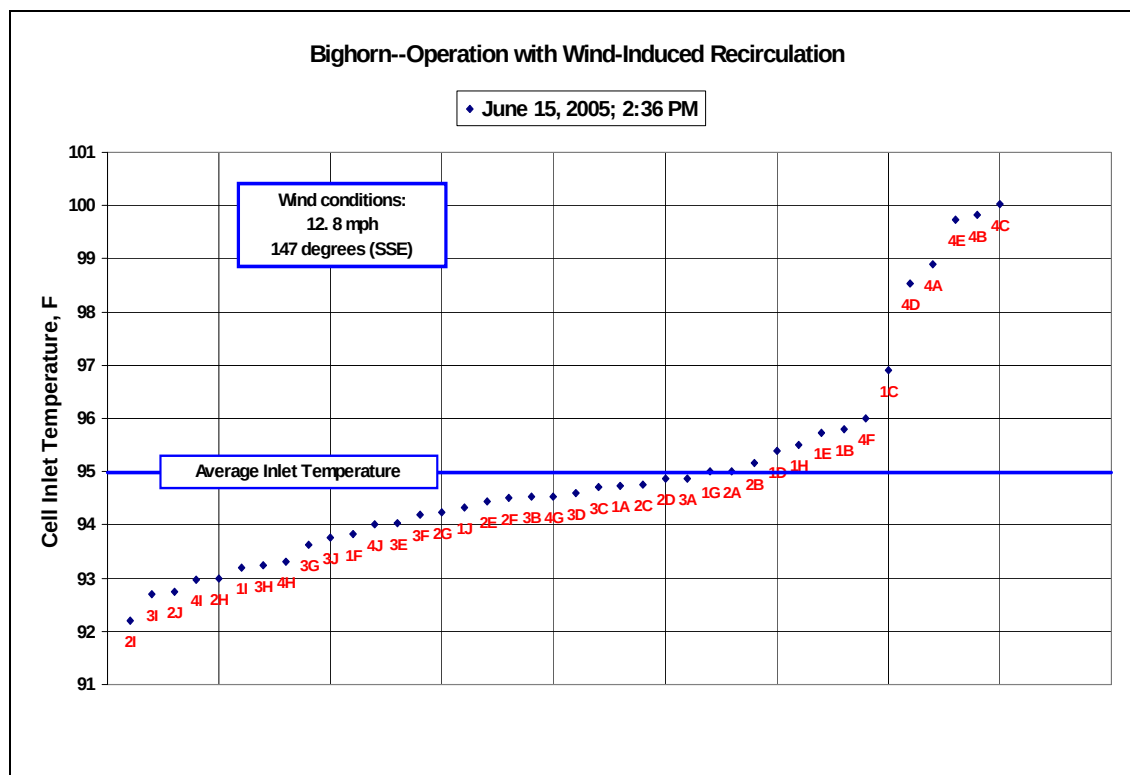


Figure 6-19. Cell inlet temperature distribution at Bighorn, mid-afternoon, June 15, 2005.

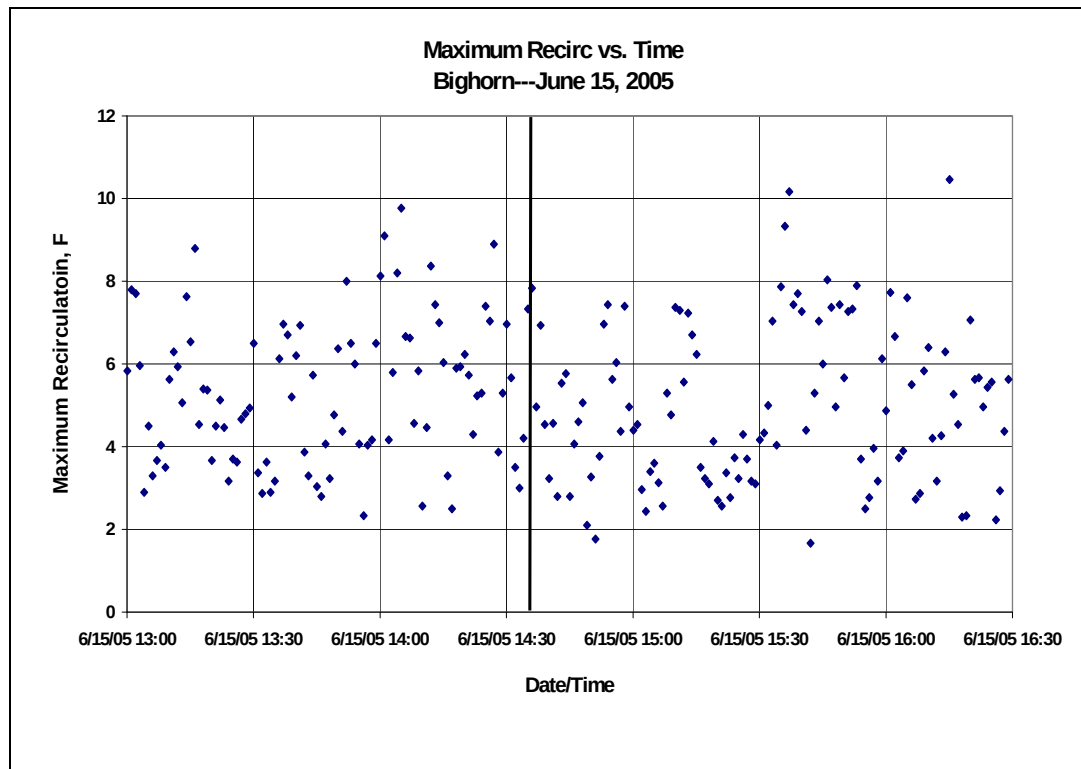


Figure 6-20. Maximum recirculation at Bighorn, mid-afternoon, June 15, 2005.

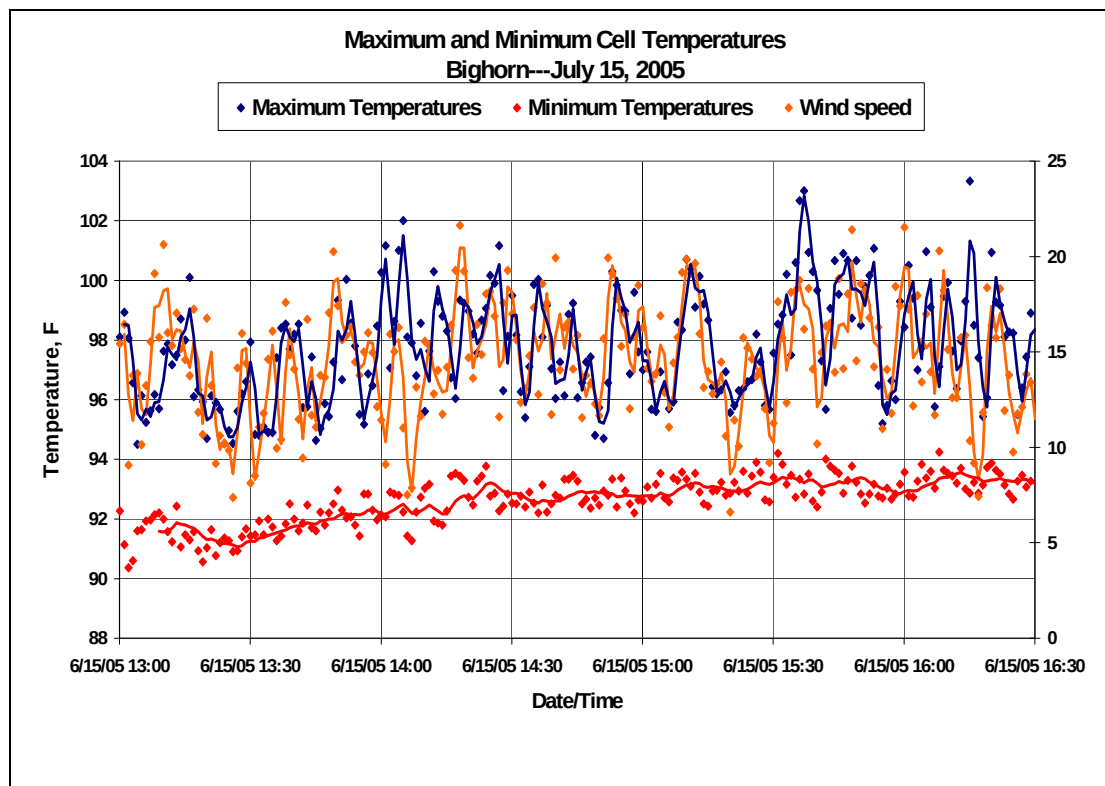


Figure 6-21. Comparison of variability of maximum temperatures (°F) and wind speed (mph).

Figure 6-21 plots the minimum and maximum inlet temperatures and the wind speed for the mid-afternoon. The minimum temperatures fluctuate very little ($\pm 1^\circ\text{F}$) from minute to minute and show a modest increase (from 90°F to 94°F) throughout the afternoon. The maximum temperature shows significant variability of nearly $\pm 4^\circ\text{F}$. Similarly, the wind speed is highly variable from <10 mph to >20 mph. It is noteworthy, however, that the 10-minute rolling average lines of maximum temperature and wind speed exhibit very similar variation over nearly the entire period.

6.2.5 Additional Observations at Bighorn

Observations during the late afternoon (4:00 to 6:00 p.m.) of June 11, 2005, illustrate some behavior contrary to expectations. Figures 6-22a, -22b and -22c show the distribution of cell inlet temperatures for 5:15, 5:25, and 5:30 p.m., respectively. All show a characteristic pattern of recirculation. Winds were consistently moderate (~ 10 to 14 mph) from the west to west-northwest. At 5:15 and 5:30 p.m. the recirculation occurred, as expected, in the cells on the eastern (downwind) side of the ACC. However, at 5:25 p.m. the pattern switched to recirculation in the western (upwind) cells. Figure 6-23 shows the eight hottest cells for each of the three observing times illustrating this shift in behavior. Figure 6-24 shows the wind conditions during the period and provides no obvious explanation for the changing recirculation pattern.

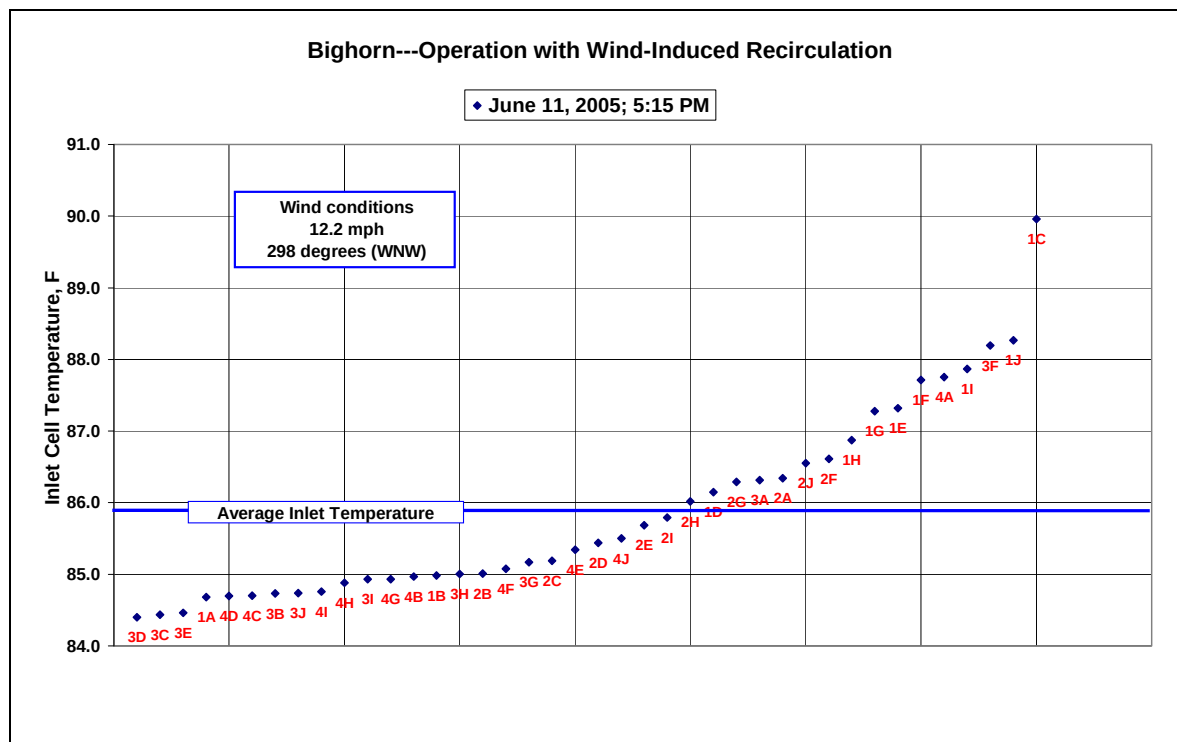


Figure 6-22a. Cell inlet temperature distribution at Bighorn, June 11, 2005, 5:15 p.m.

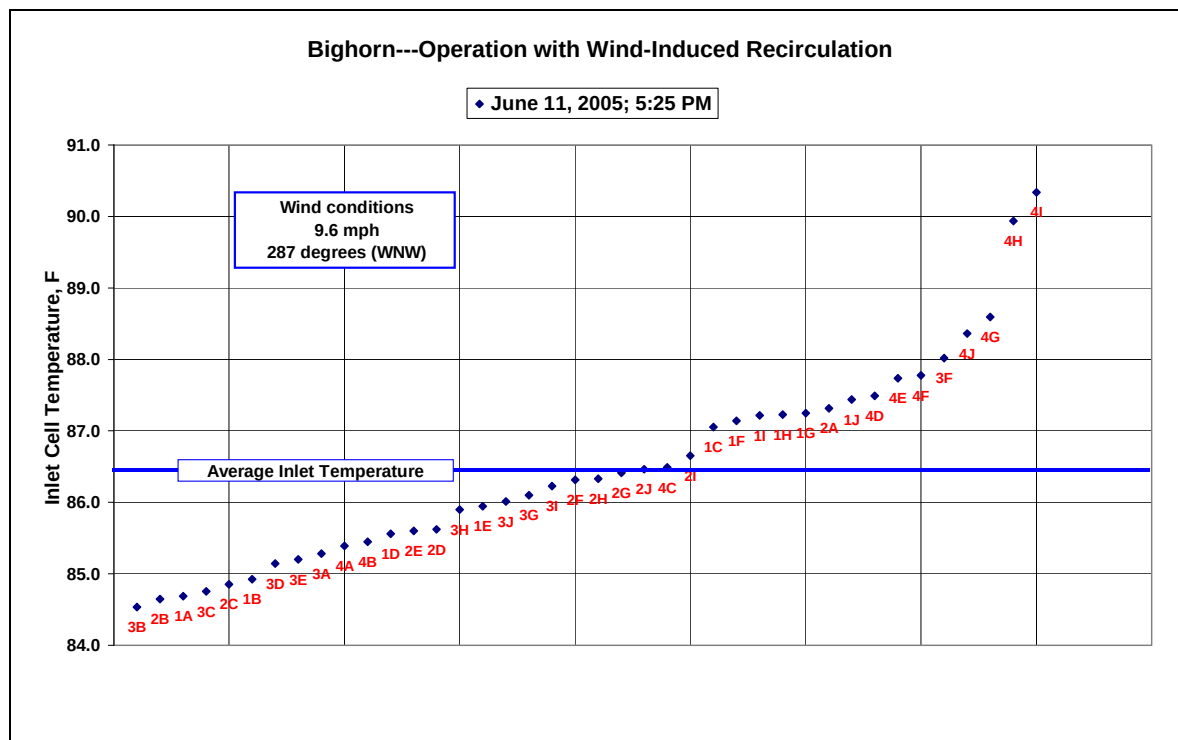


Figure 6-22b. Cell inlet temperature distribution at Bighorn, June 11, 2005, 5:25 p.m.

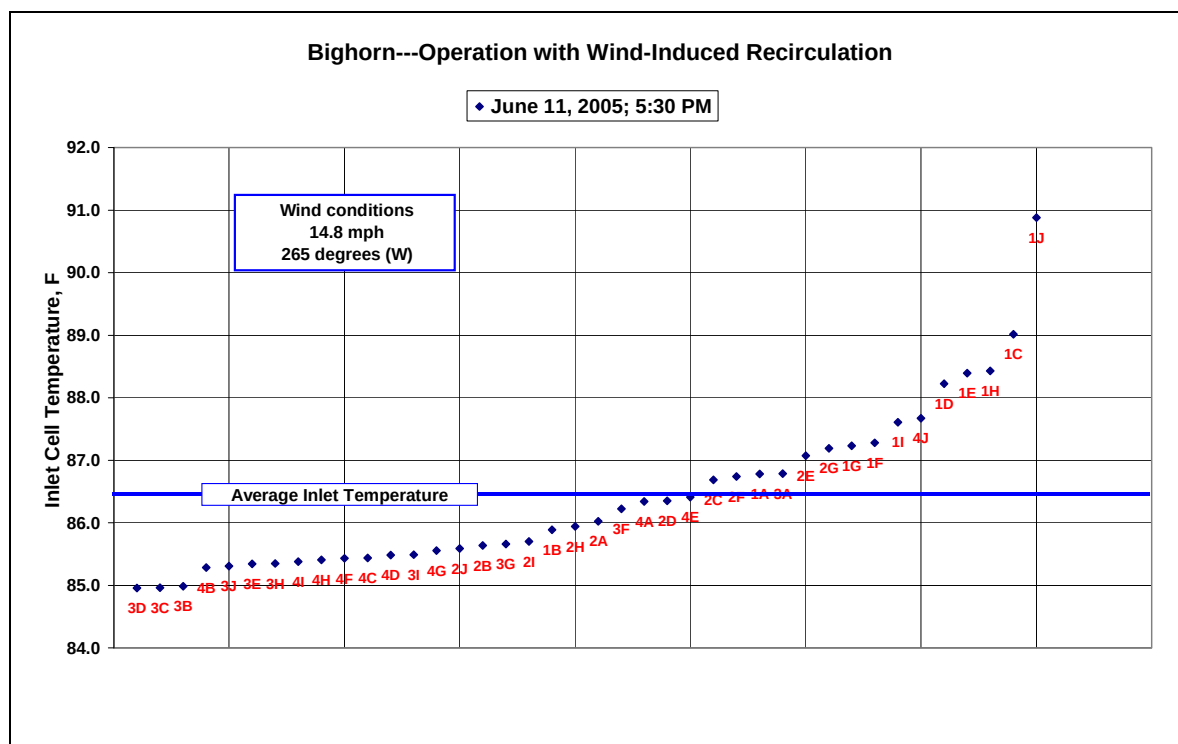


Figure 6-22c. Cell inlet temperature distribution at Bighorn, June 11, 2005, 5:30 p.m.

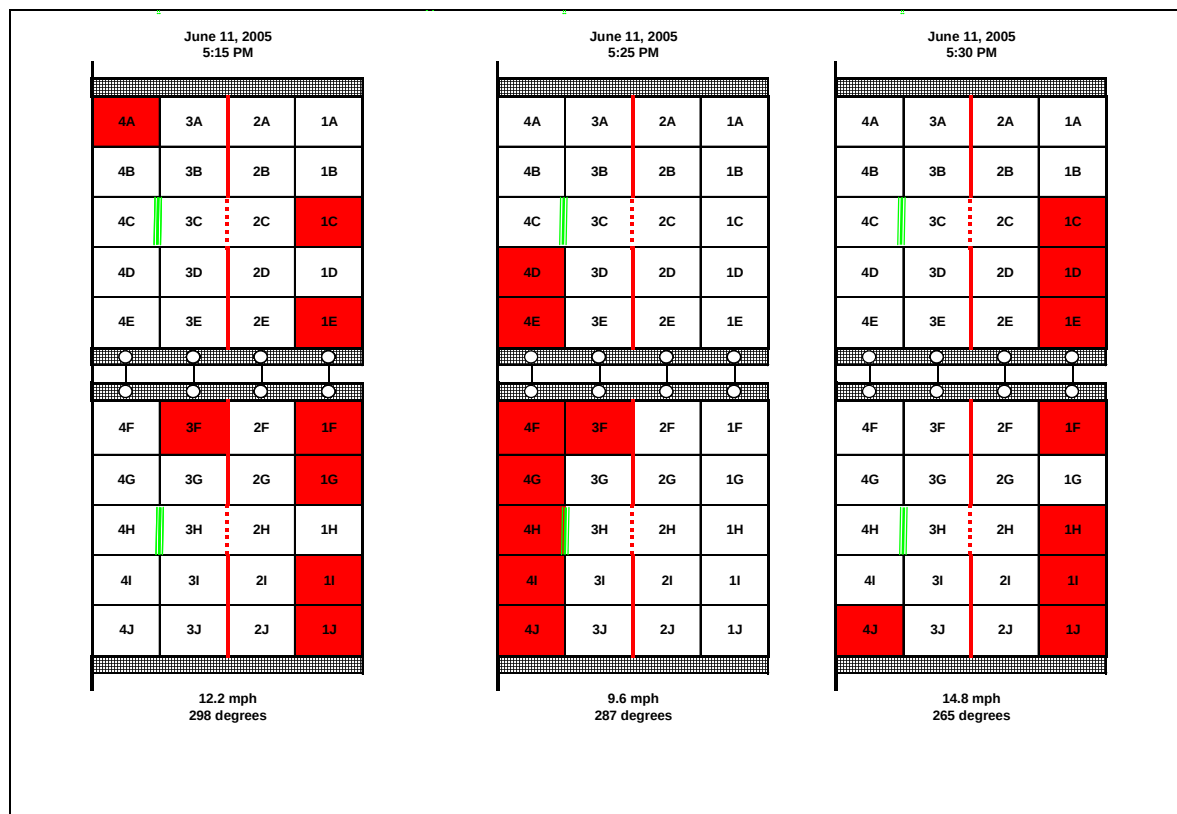


Figure 6-23. Location of cells with highest recirculation.

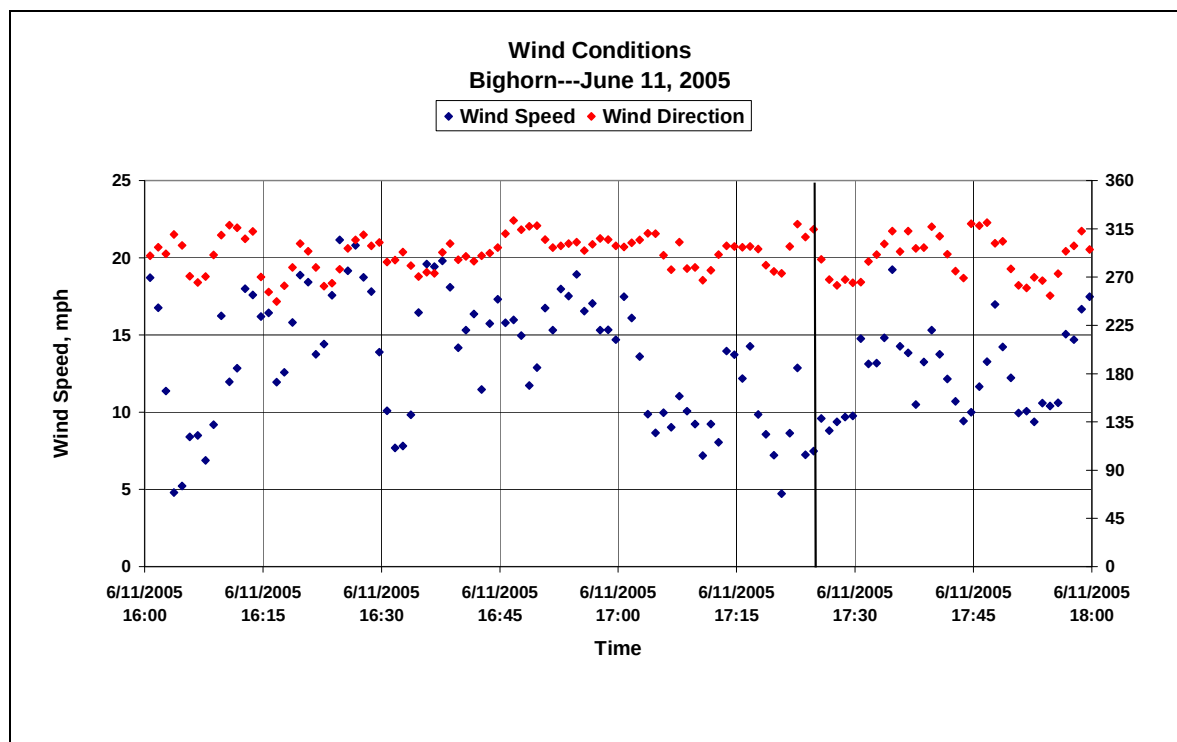


Figure 6-24. Wind conditions at Bighorn, July 11, 2005.

Figure 6-25 plots the instantaneous values of the maximum cell inlet temperature and the corresponding wind speed vs. time. Figure 6-26 is a similar plot of average recirculation and wind speed. Figure 6-27 shows the same data of average recirculation and wind speed vs. time plotted as 3-minute rolling averages. Figure 6-28 shows the same data (instantaneous points) with the average recirculation points shifted 5 minutes to the left (back in time) to provide an visualization of recirculation compared to wind conditions from 5 minutes earlier. This might be more illuminating if the recirculating plume takes some time to respond to changing wind forces. While both manipulations appear to improve the apparent correspondence of recirculation to wind speed, there is no evident consistent physical interpretation of the results at this time.

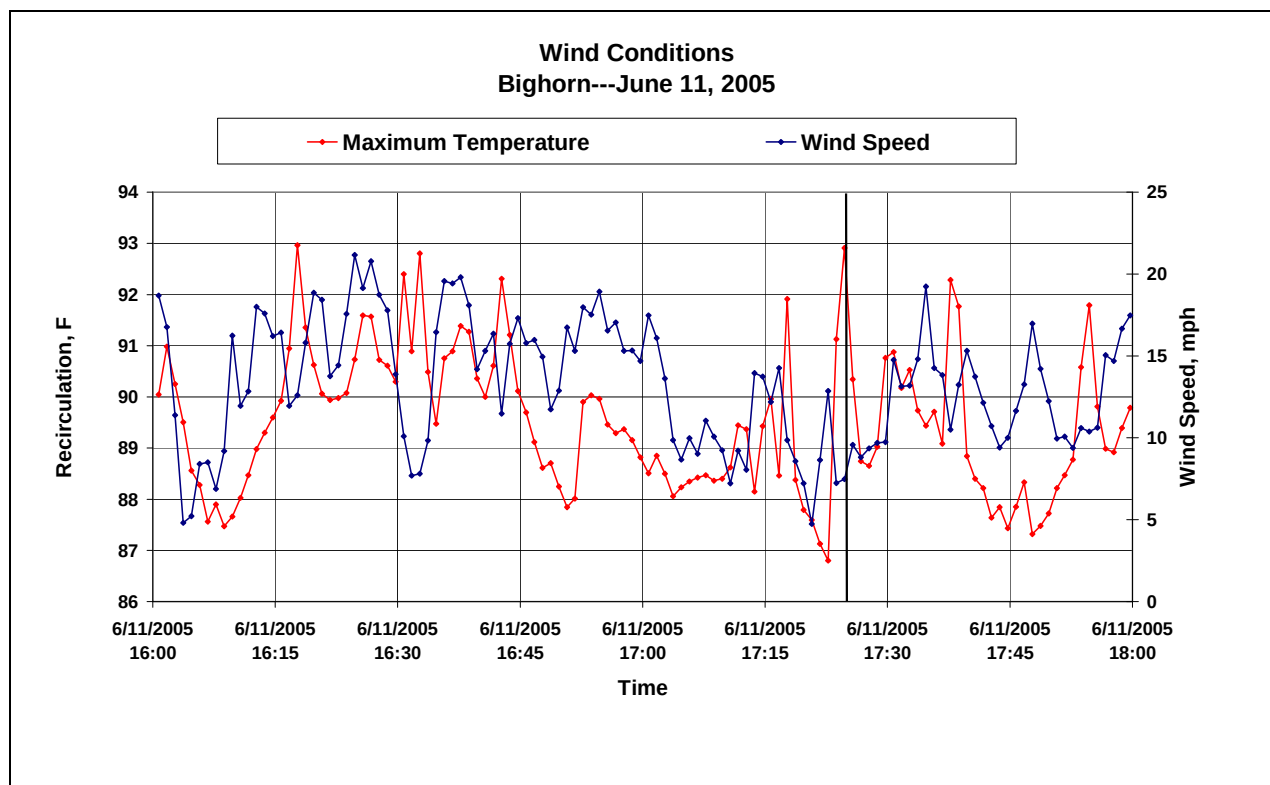


Figure 6-25. Maximum cell inlet temperature and wind speed at Bighorn, June 11, 2005.

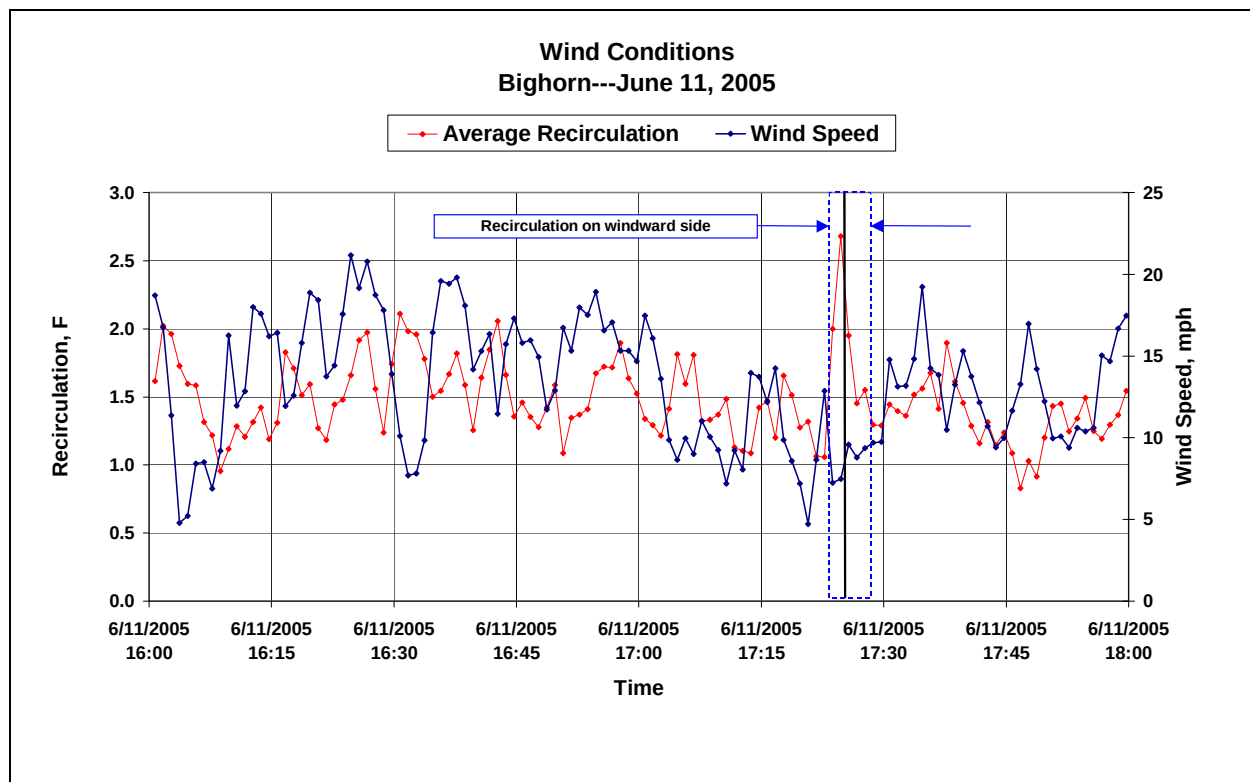


Figure 6-26. Average recirculation and wind speed at Bighorn, June 11, 2005.

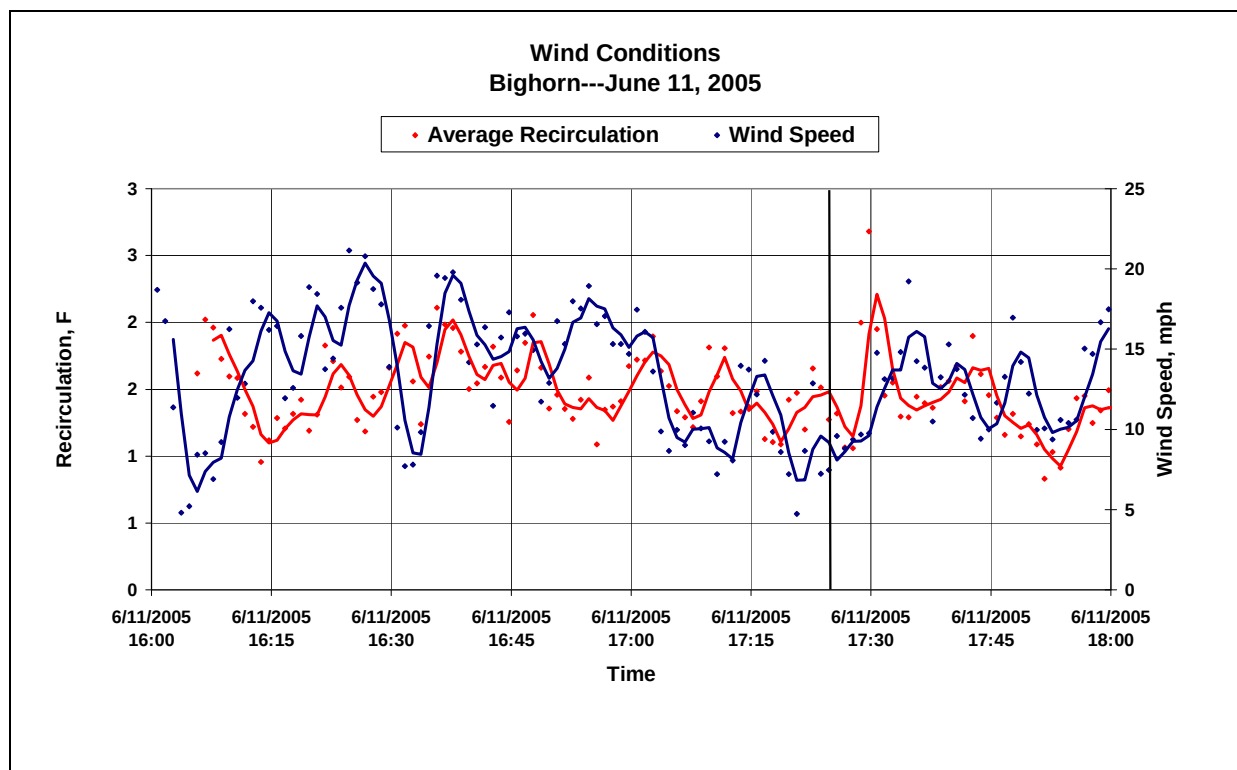


Figure 6-27. Average recirculation and wind speed—3-minute rolling averages (Bighorn, June 11, 2005).

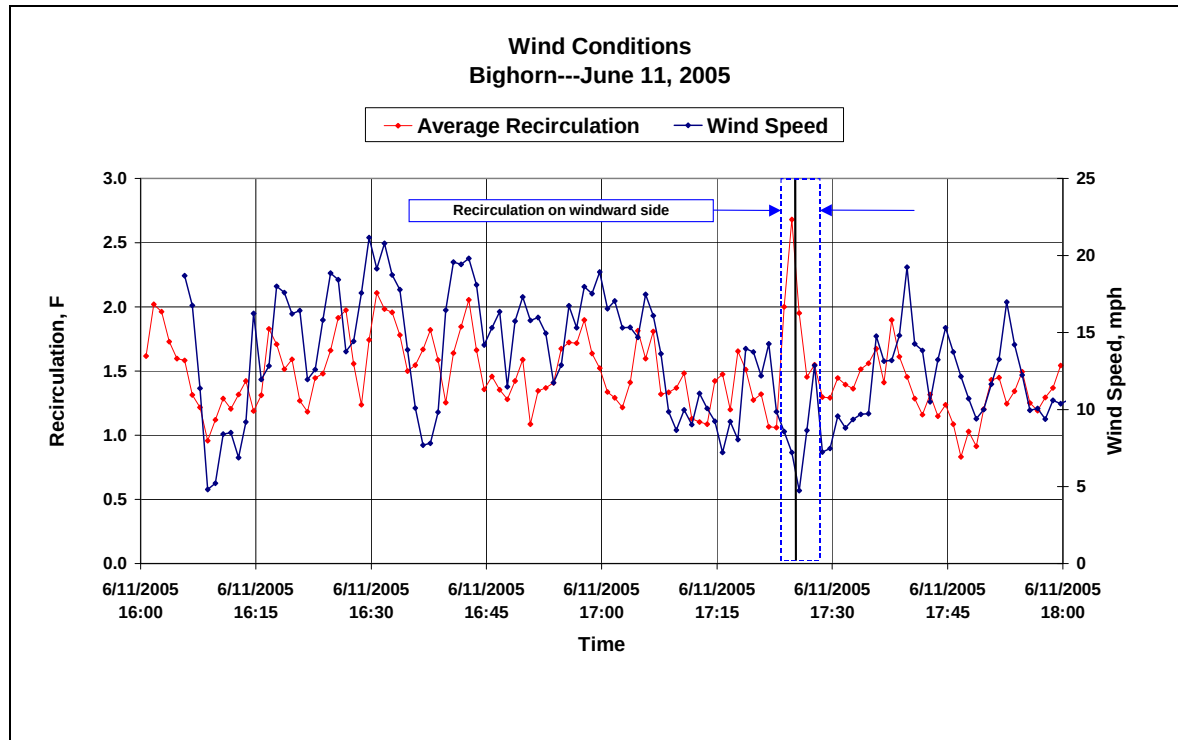


Figure 6-28. Average recirculation and wind speed—5-minute shift in time scale (Bighorn, June 11, 2005).

6.3 Recirculation Tests—El Dorado Energy Center

The test period at El Dorado ran from 8:12 p.m. on June 28, 2005, until 1:12 a.m. on July 7, 2005. Figure 6-2 displays the average recirculation for all times during the test period when all fans were running at full speed. Figure 6-29 gives the wind speed and direction for the duration of the test period exhibiting the usual desert pattern of high winds during the day and especially the afternoon and lower speed during the night. The prevailing direction for the high afternoon winds is from the south/southwest.

Plant instrumentation records the ambient temperature at one location on top of the administration building and another under the ACC. Figure 6-30 compares the two readings, indicating that the more exposed instrument on the administration building shows some influence of solar radiation—with higher readings during the sunlight hours and lower readings during the night compared to the shaded probe under the ACC. Figure 6-31 compares the readings from the plant ACC probe with the minimum inlet cell temperature. The agreement is generally excellent, within $\pm 2^{\circ}\text{F}$ with the exception of rapid warming periods in the morning where the plant probe appears to lag the “in-cell” probe by as much as 6° to 8°F . Given that the in-cell probe is exposed to high-velocity air at the fan exit, a quicker response is to be expected. On the basis of these comparisons, the minimum cell inlet temperature will be used as the baseline for determining the amount of recirculation to the ACC, as was discussed in Section 6.2 in the context of the Bighorn measurements.

Figure 6-32 plots all data points as average recirculation vs. wind speed. As at Bighorn, the instances of higher recirculation occur at the lower wind speeds while the higher wind speeds above 20 mph result in little recirculation.

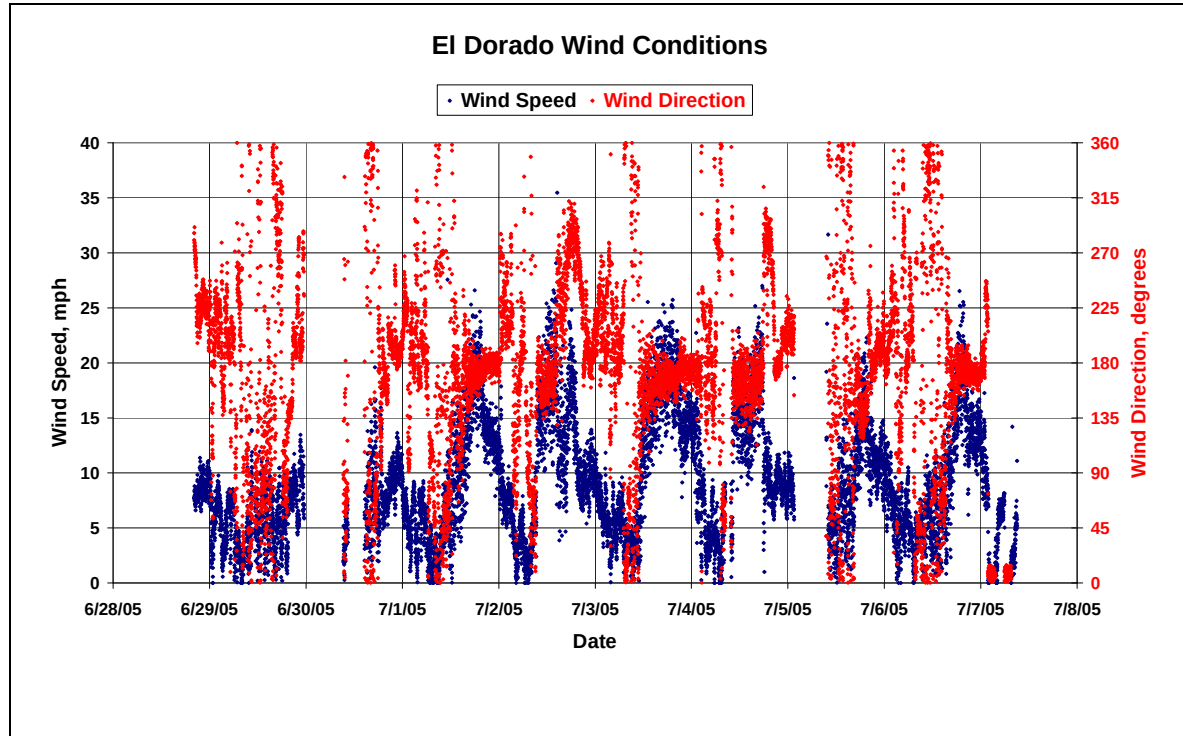


Figure 6-29. El Dorado wind speed and direction during test period.

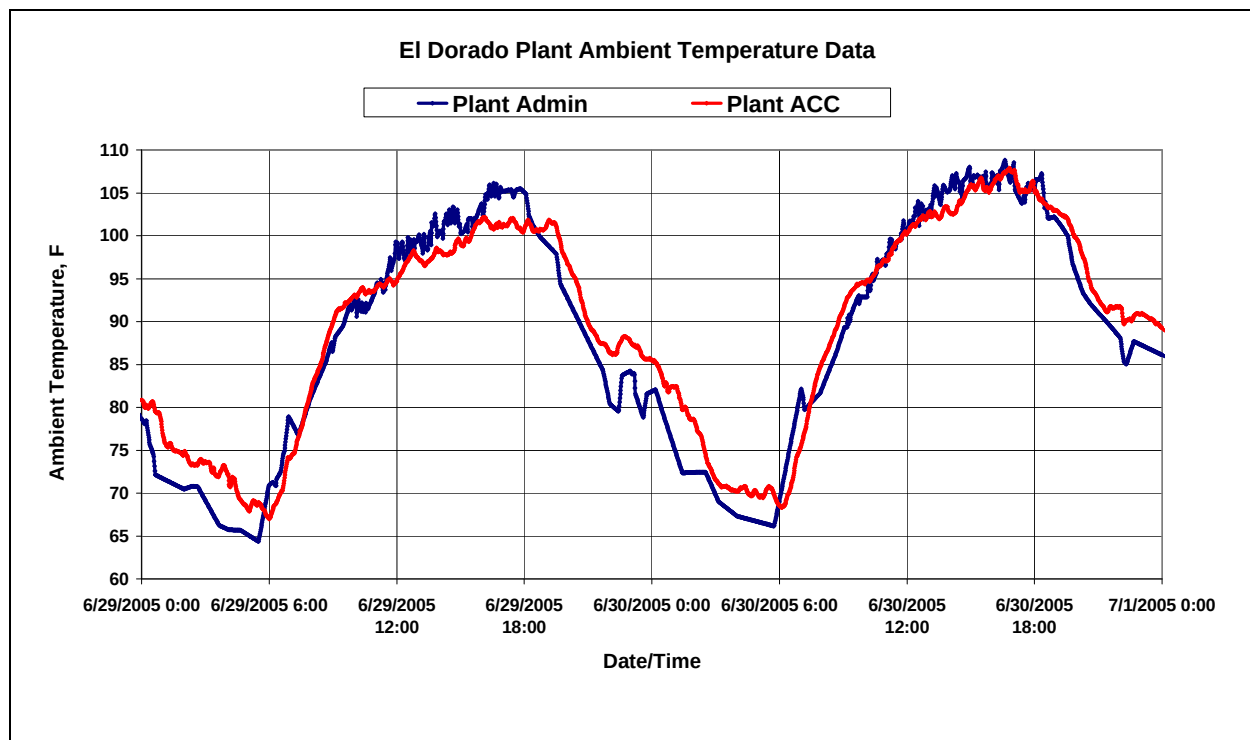


Figure 6-30. Comparison of plant ambient air temperature readings.

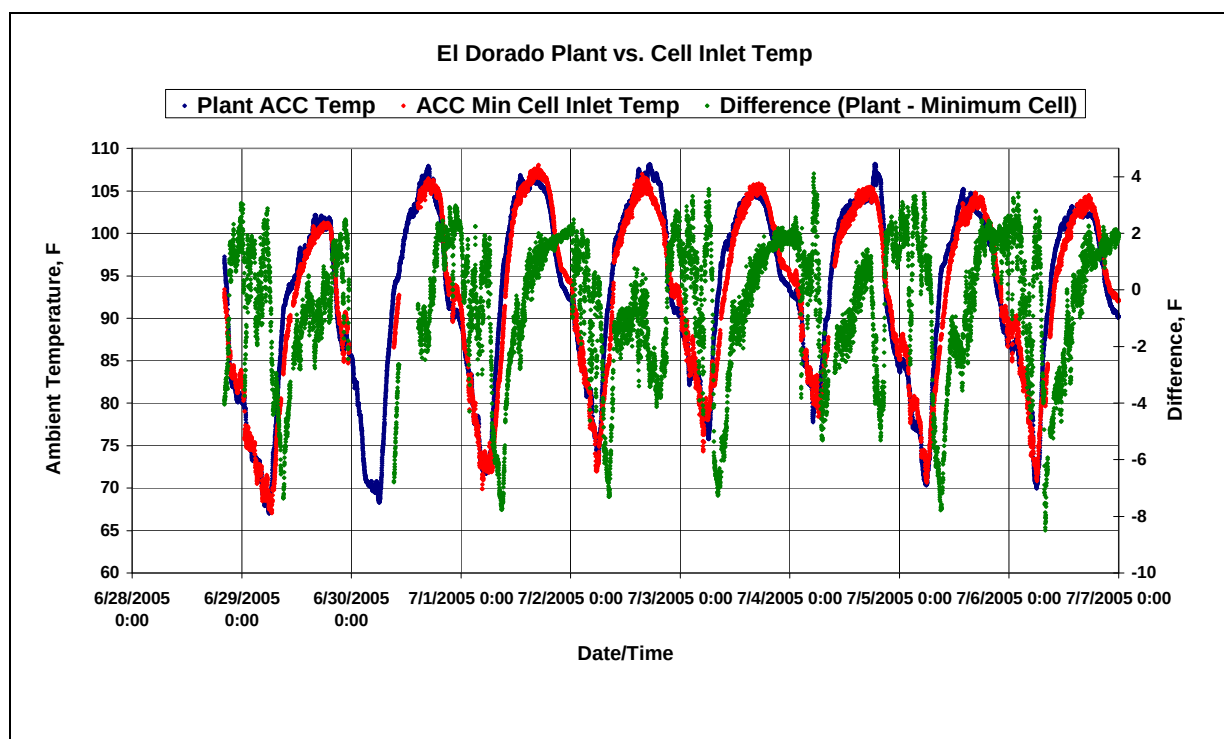


Figure 6-31. Comparison of plant (under ACC) and minimum cell inlet temperature.

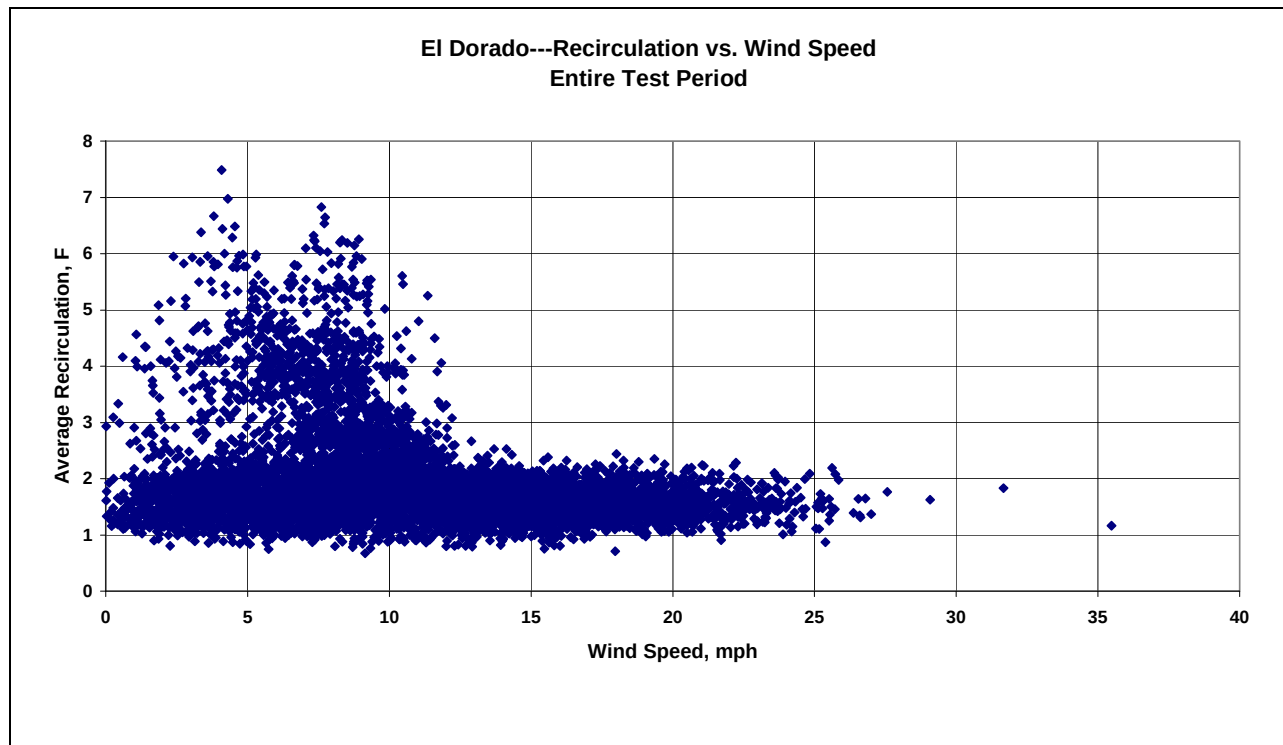


Figure 6-32. El Dorado recirculation vs. wind speed.

Three periods will be examined in more detail:

- June 30, 6:00 p.m. to July 1, 3:30 a.m.: Period of highest recirculation (as high as 7°F); low wind speeds from 1 to 10 mph; wind direction variable but primarily from the S/SW.
- July 3, 3:00 to 6:00 p.m.: Period of typical recirculation (1° to 2°F with a few times exceeding 2°F); wind speed high with many occurrences between 20 and 25 mph; wind direction steady from the S/SE.
- July 5, 1:00 to 5:00 p.m.: Period of highest daytime recirculation approaching 3°F on a few occasions; wind speed moderate to high varying from 10 to over 20 mph; direction is highly variable from all directions.

6.3.1 June 30, 6:00 p.m. to July 1, 3:30 a.m.

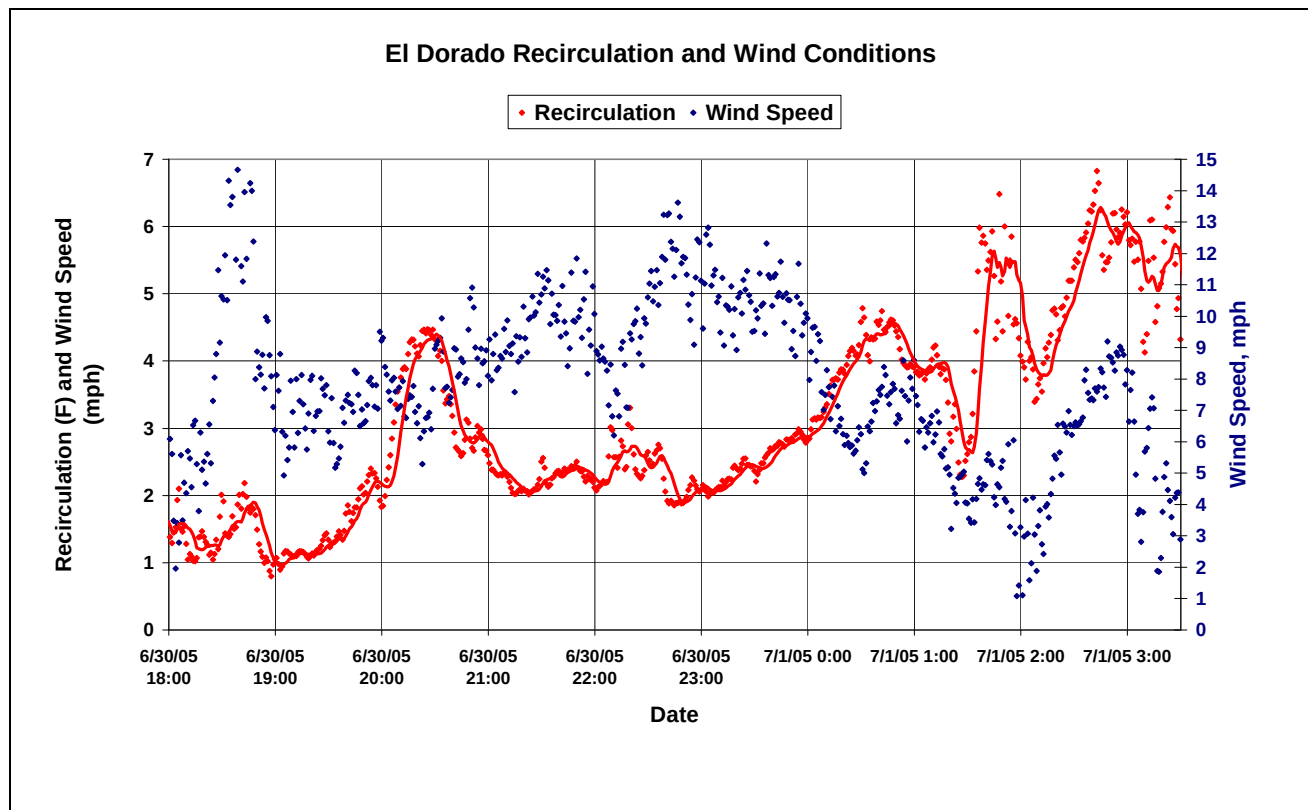
Figures 6-33a and 6-33b show average recirculation along with the corresponding wind speed (6-33a) and wind direction (6-33b) from 6 p.m. on June 30 until 3:30 a.m. on July 1. Peaks in recirculation occur around 6:45 p.m., 8:25 p.m., 10:20 p.m., 12:45 a.m., 1:45 a.m. and 2:45 a.m. Rapid decreases in recirculation occur following each peak.

There is no compelling correlation of recirculation with wind speed. The following observations are noted:

- An abrupt increase in wind speed followed by an equally abrupt decrease occurs at 6:45 p.m. but the corresponding increase in recirculation is modest. The increase in recirculation at 2:45 a.m. is accompanied by a corresponding increase in wind speed.
- Other increases in recirculation (8:25 p.m., 10:20 p.m., 12:45 a.m., and 1:45 a.m.) are not associated with a high-wind-speed event.
- The general trend of increasing wind speed from 7:15 p.m. to midnight is not reflected in any corresponding increase in recirculation.
- The period from 11:00 p.m. to 2:00 a.m. is characterized by a general *decrease* in wind speed and a general *increase* in recirculation.
- From 2:00 a.m. to 3:30 a.m., there is a reasonable correspondence in the general shapes of the recirculation and wind speed curves.

Similarly, wind direction has only a sporadic effect:

- The wind direction for the entire period is generally southerly.
- Two occasions, at 6:45 p.m. and 1:45 a.m., when the wind shifts suddenly and temporarily to easterly, the recirculation increases.
- Other recirculation events do not appear to be associated with a wind shift.



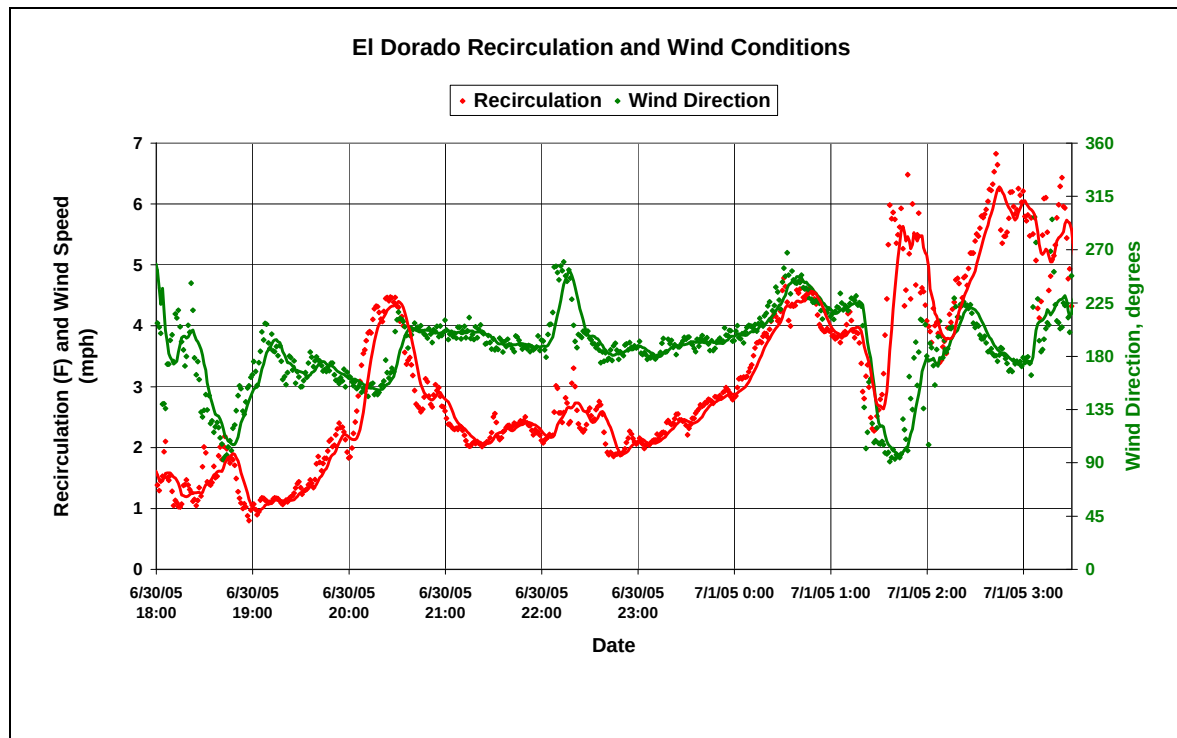


Figure 6-33b. Recirculation and wind direction (low to moderate wind speed).

The following figures present a detailed look at the distribution of cell inlet temperatures at the beginning, peak, and end of two of the excursions in recirculation. Figures 6-34 (8:00 p.m.), 6-35 (8:40 p.m.) and 6-36 (9:20 p.m.) surround the 8:45 p.m. peak on June 30, and Figures 6-37 (1:15 a.m.), 6-38 (1:40 a.m.) and 6-39 (2:10 a.m.) surround the 1:45 a.m. peak on July 1. The location on the ACC of the various cells referred to in the following discussion can be found in Figure 4-7 in Section 4.4.2.

Figures 6-34 and 6-36 show similar patterns with only a slight variation in cell inlet temperature over most of the cells, but significant recirculation in a limited number of downwind and corner cells. Figure 6-35 shows the distribution at the point of highest recirculation with a much greater variation in cell inlet temperature over the entire ACC. The downwind and corner cells (C1, C2, and C6) still show higher-than-average recirculation, but the variation across the remaining 27 cells is over 5°F (from ~95.5°F to ~100.5°F) as opposed to only slightly over 2°F (~102°F to 104+°F) prior to the start of the recirculation peak. This variation occurred at an essentially constant wind speed of 7 to 9 mph. The shift in wind direction from SE to SW may have pushed the previously established plume back into the ACC, resulting in a temporary infusion of warm air under the entire structure, but this is only speculative at this time.

Figures 6-37, 6-38 and 6-39 exhibit a similar progression. In this instance at the lower wind speeds, particularly at the later time, the falloff in the lowest inlet temperatures *may* indicate the beginning of a temperature inversion associated with the lower wind speeds. These two cells (16 and 22), located at the generally upwind corners of the intersection of the two windscreens, may be subject to the upflow of low-level air as a result of the windscreen barrier.

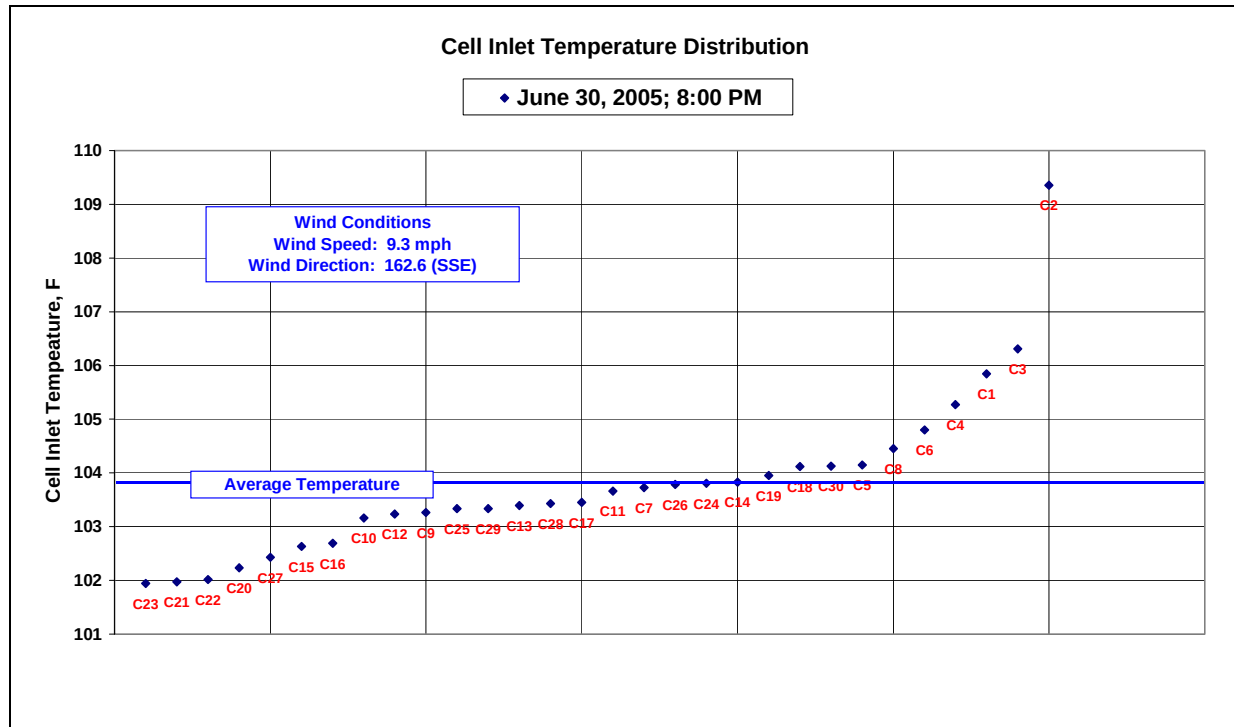


Figure 6-34. Cell inlet temperature distribution, June 30, 2005, 8:00 p.m.

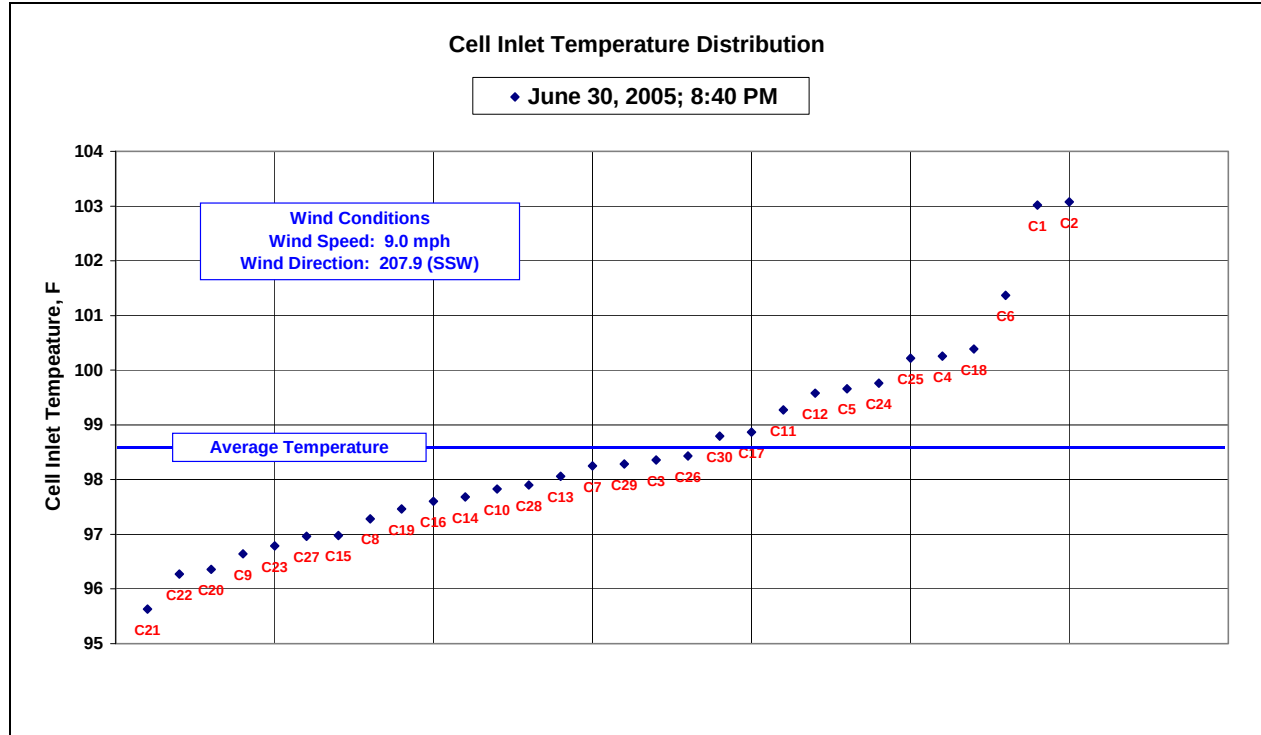


Figure 6-35. Cell inlet temperature distribution, June 30, 2005, 8:40 p.m.

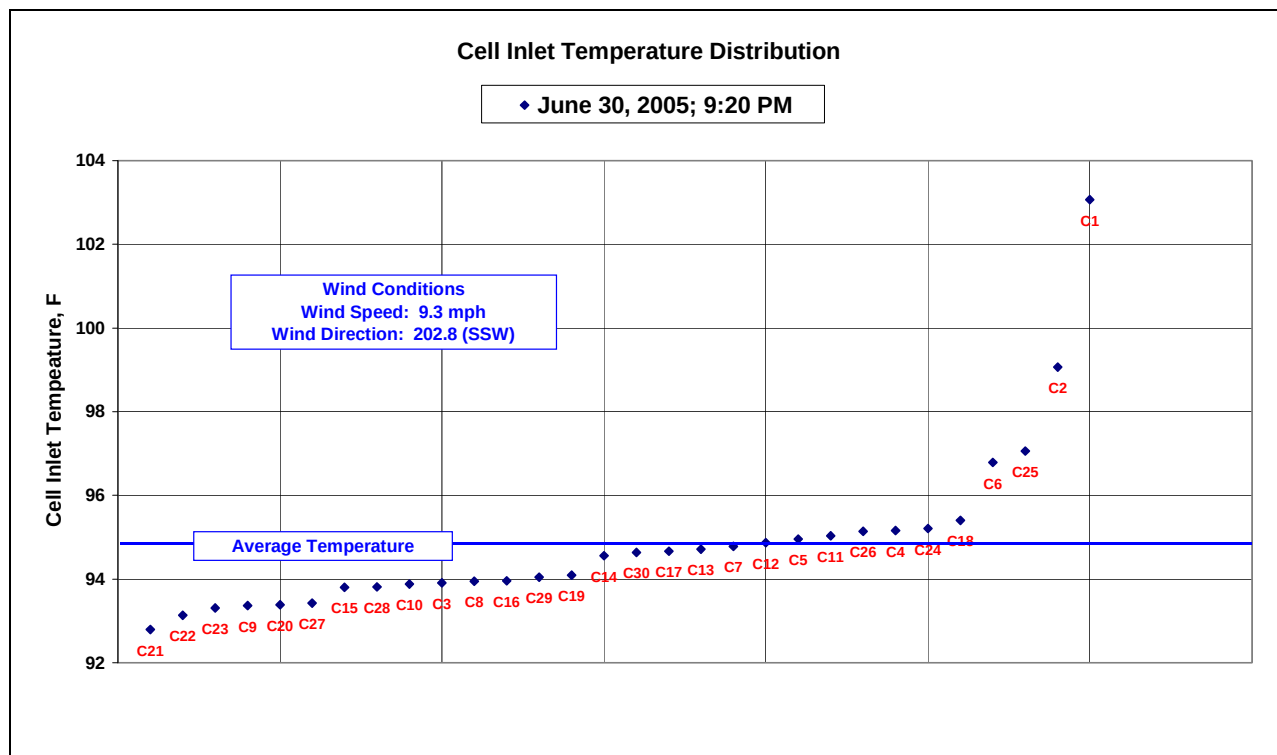


Figure 6-36. Cell inlet temperature distribution, June 30, 2005, 9:20 p.m.

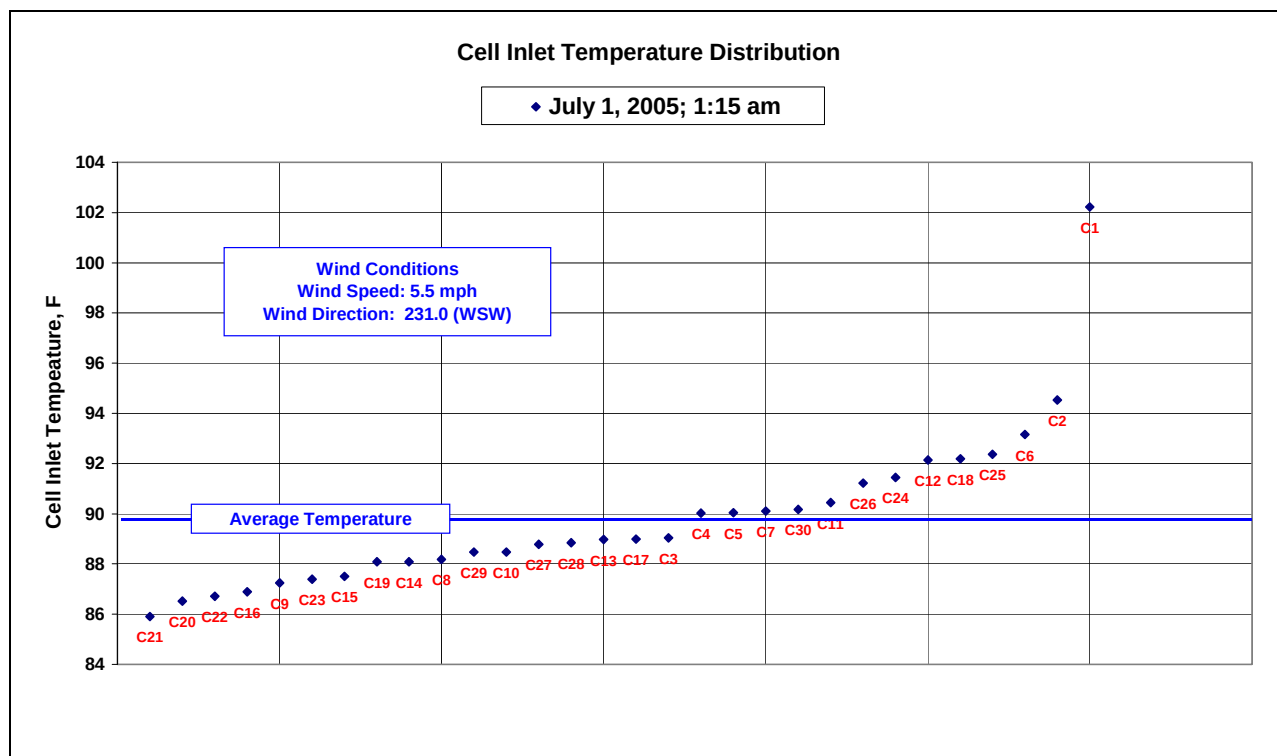


Figure 6-37. Cell inlet temperature distribution, July 1, 2005, 1:15 a.m.

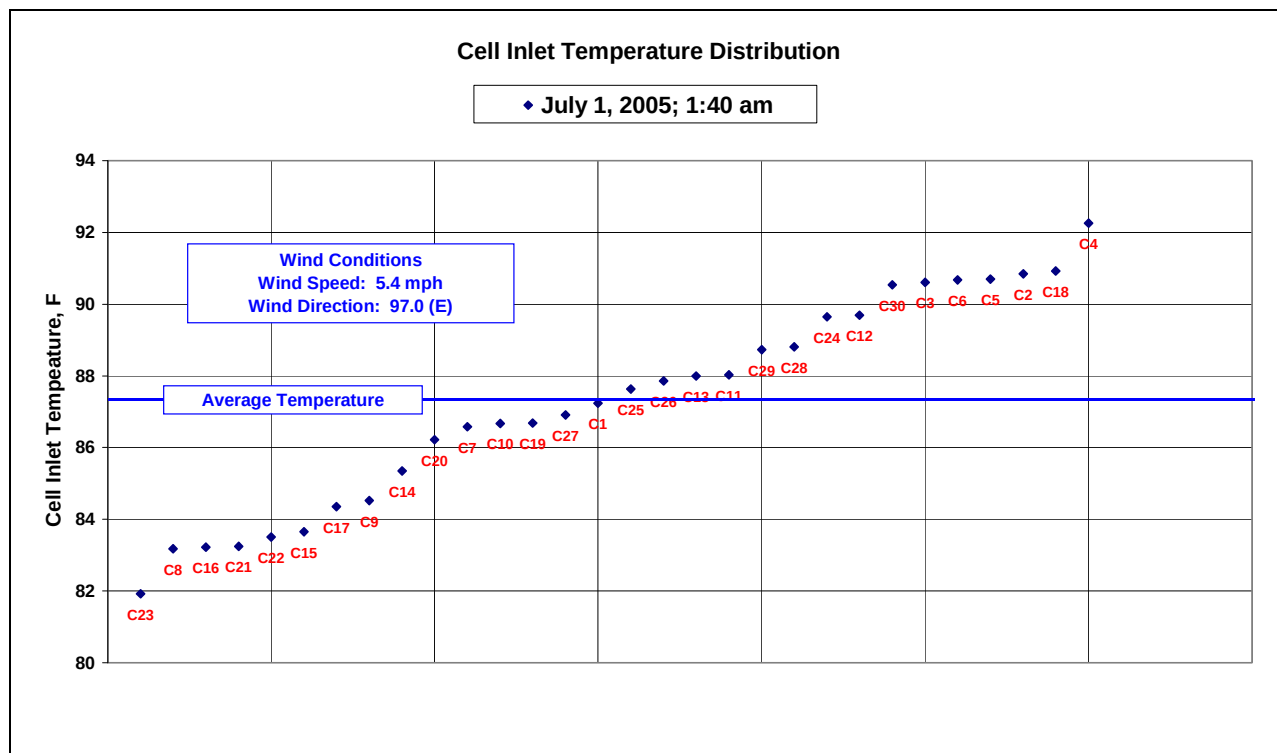


Figure 6-38. Cell inlet temperature distribution, July 1, 2005, 1:40 a.m.

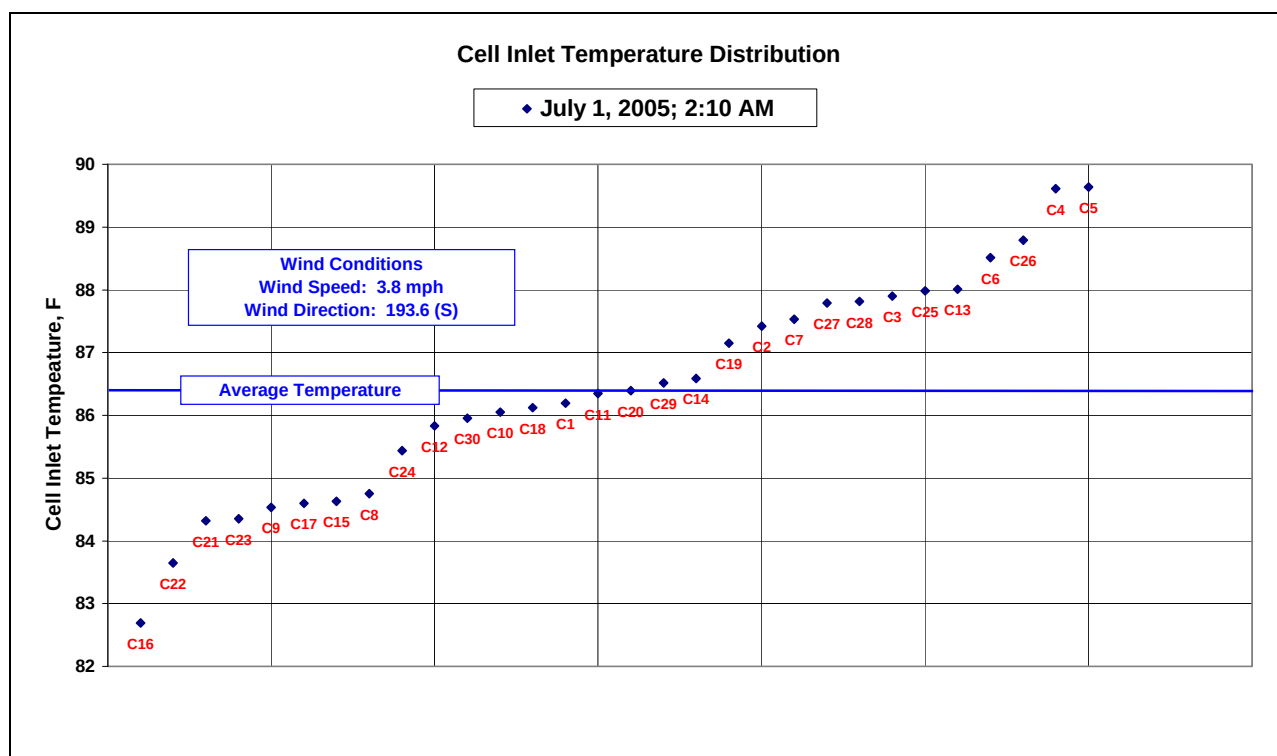


Figure 6-39. Cell inlet temperature distribution, July 1, 2005, 2:10 a.m.

6.3.2 July 3, 3:00 p.m. to 6:00 p.m.

The afternoon of July 3, 2005, exhibits typical summer afternoon conditions at El Dorado. The temperature is well over 100°F; the wind is strong, ranging consistently from 15 to 25 mph (Figure 6-40) from the south to slightly south-southeast (Figure 6-41).

Cells with the most recirculation (Figure 6-42) are at the north end of the ACC, primarily in the northwest corner (Cells 25, 26, and 19) downwind of a south-southeast wind. Some significant recirculation is also seen in the northeast corner cells (Cells 1 and 2). Nonetheless, the average recirculation is still consistently less than 1.5°F with only occasional excursions to 2°F.

Figure 6-40 shows a reasonable correlation between average recirculation and wind speed. Figure 6-41 shows essentially no effect of wind direction, but since the variation in wind direction is very small, no obvious correlation would be expected.

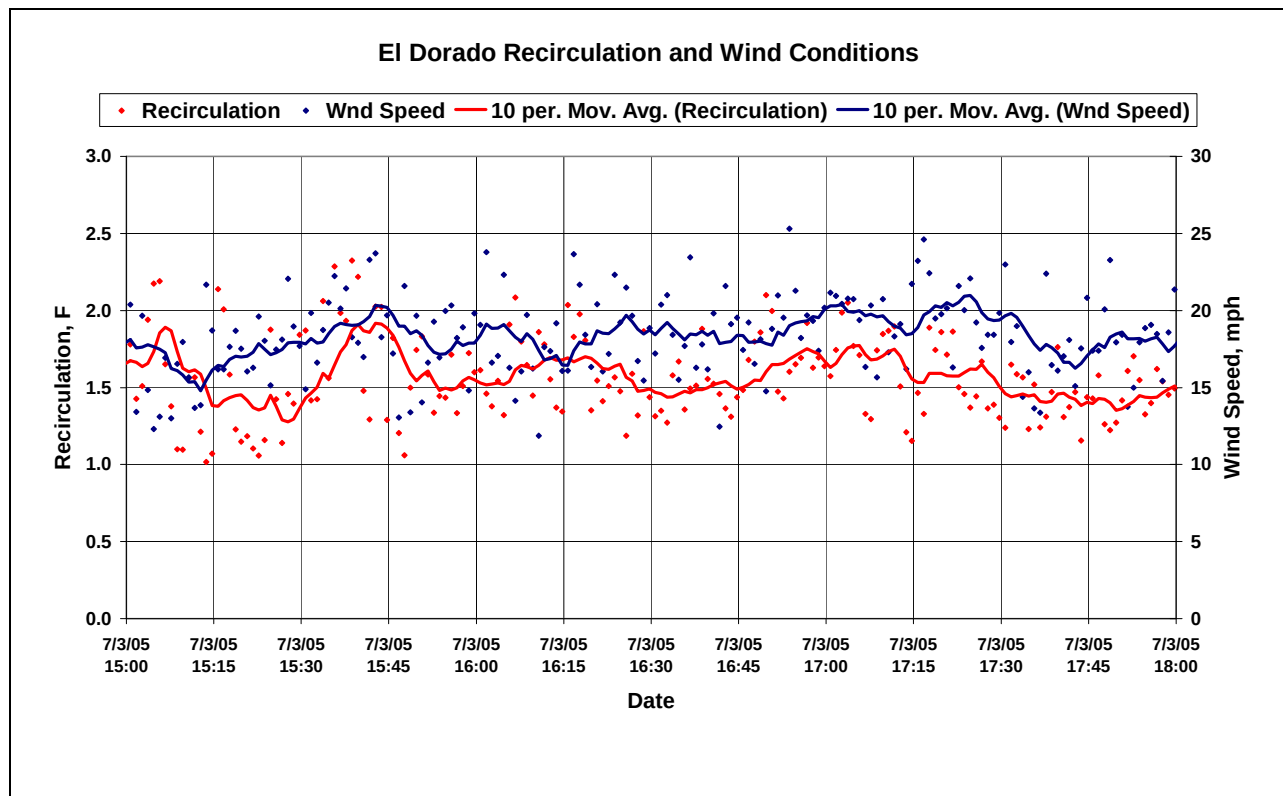


Figure 6-40. Recirculation and wind speed, July 3, 2005.

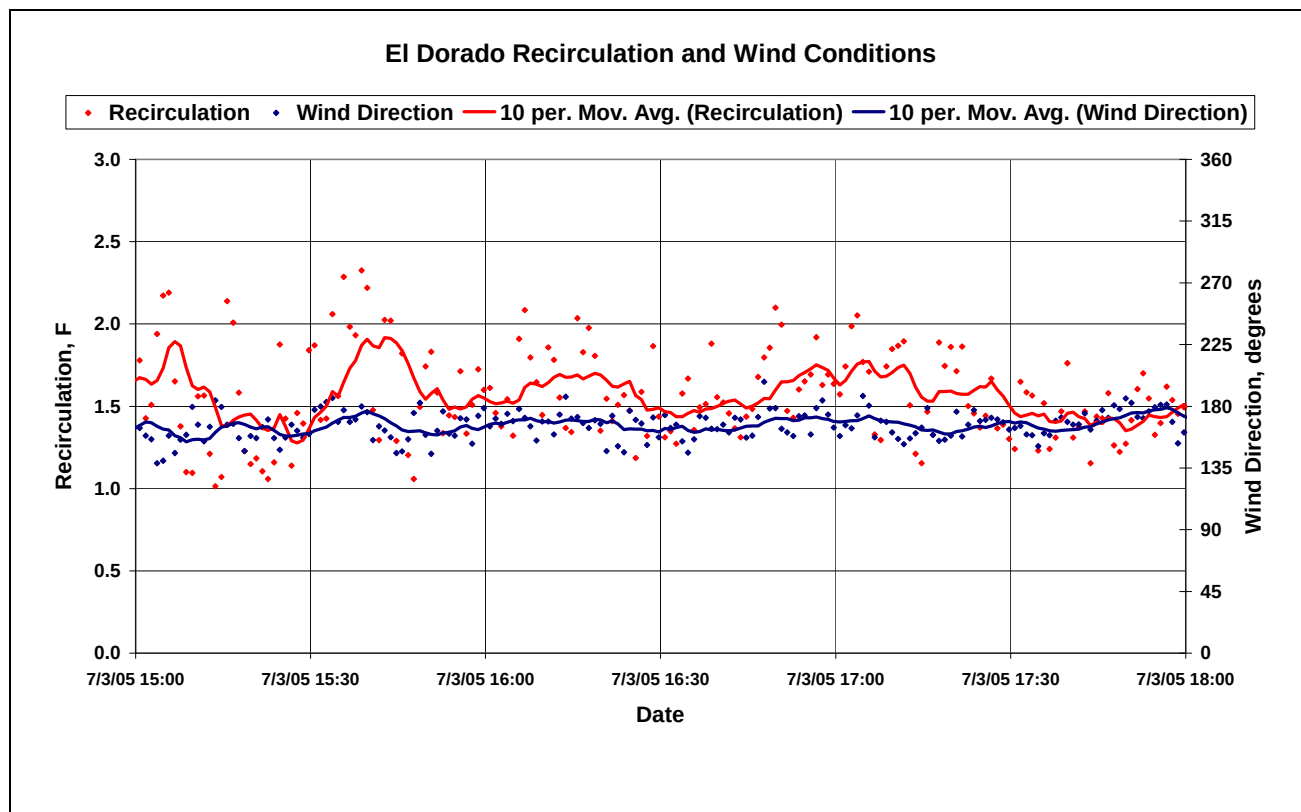


Figure 6-41. Recirculation and wind direction, July 3, 2005.

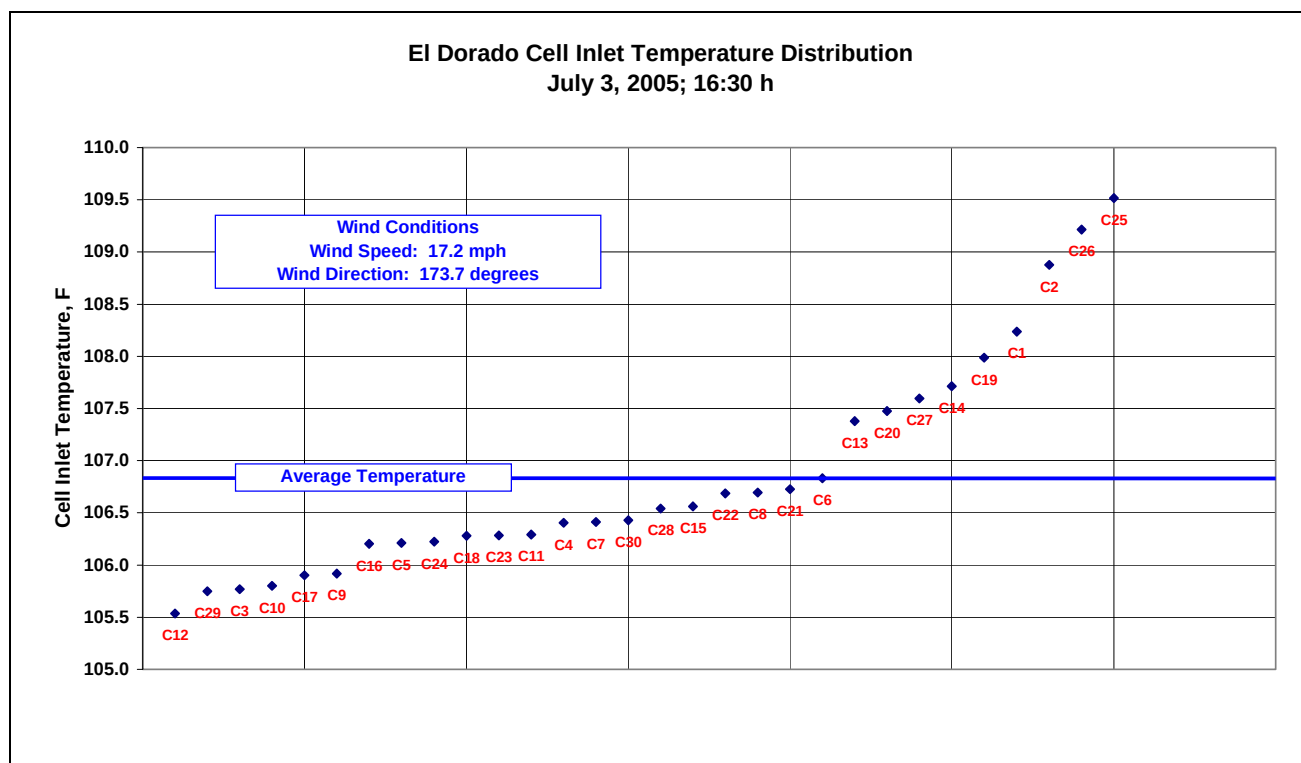


Figure 6-42. Cell inlet temperature distribution, July 3, 2005, 4:30 p.m.

6.3.3 July 5, 1:00 p.m. to 5:00 p.m.

The afternoon of July 5, 2005, exhibits somewhat different conditions from those of July 3. The temperature still exceeds 100°F, but the winds are more moderate, ranging from under 5 mph to just over 15 mph (Figure 6-43). The wind direction is highly variable and, while averaging generally southerly, there are frequent shifts to both westerly and easterly with occasional periods of northerly winds (Figure 6-44). This variability in direction shows little correlation with any variability in wind speed.

Little correlation of average recirculation with wind speed can be discerned in Figure 6-43. Figure 6-44 is difficult to interpret, but exhibits no evident correlation of average recirculation with particular wind directions. It does appear, however, that occasional periods of increased average recirculation such as observed between 1:30 p.m. and 2:00 p.m., between 2:15 p.m. and 2:30 p.m., between 2:45 and 3:15 p.m., and between 3:20 and 3:45 p.m. are reasonably coincident with fairly rapid *shifts* in wind direction.

This may suggest, as hypothesized in Section 6.3.1, that a warm air plume established under the influence of wind from one direction may be pushed back onto or under the ACC by winds that abruptly shift to another direction. In this argument, the particular wind directions prior to and after the shift are of less importance than the occurrence of the shift itself.

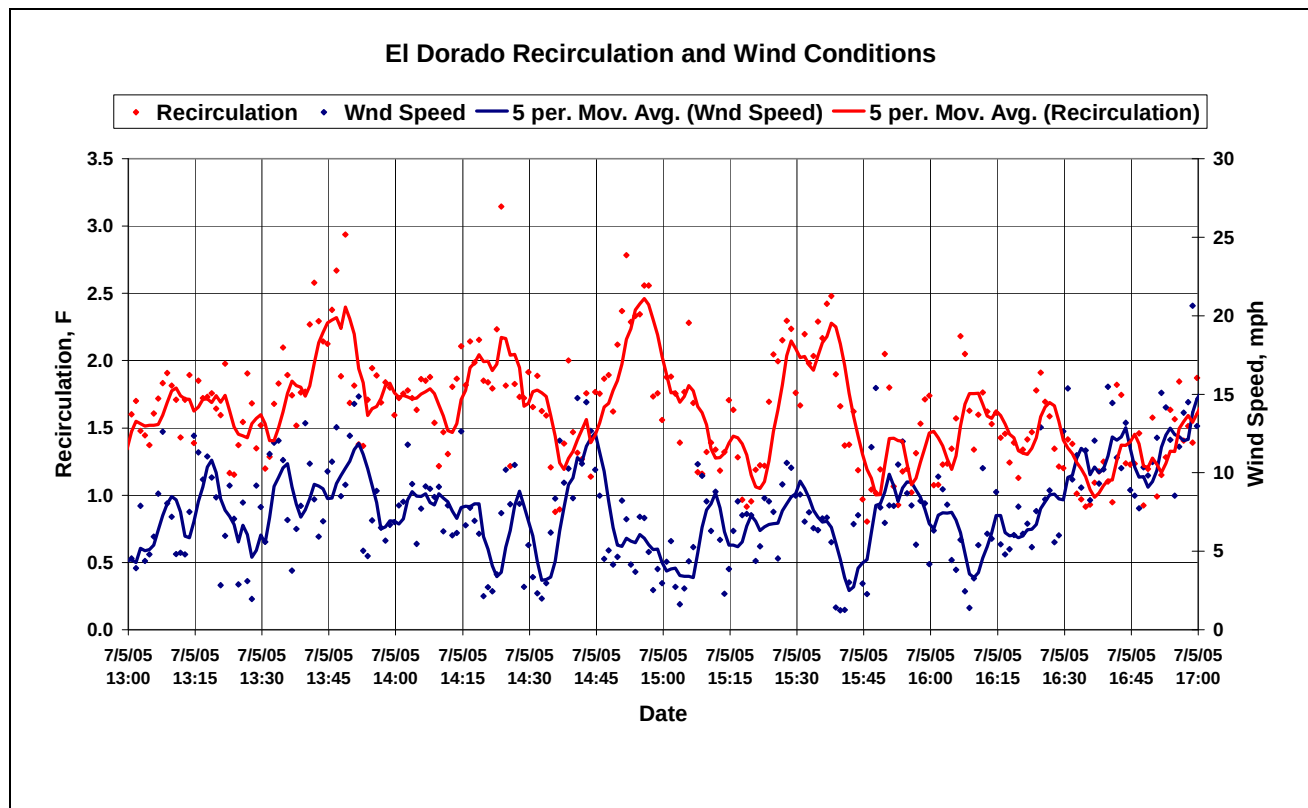


Figure 6-43. Recirculation and speed, July 5, 2005.

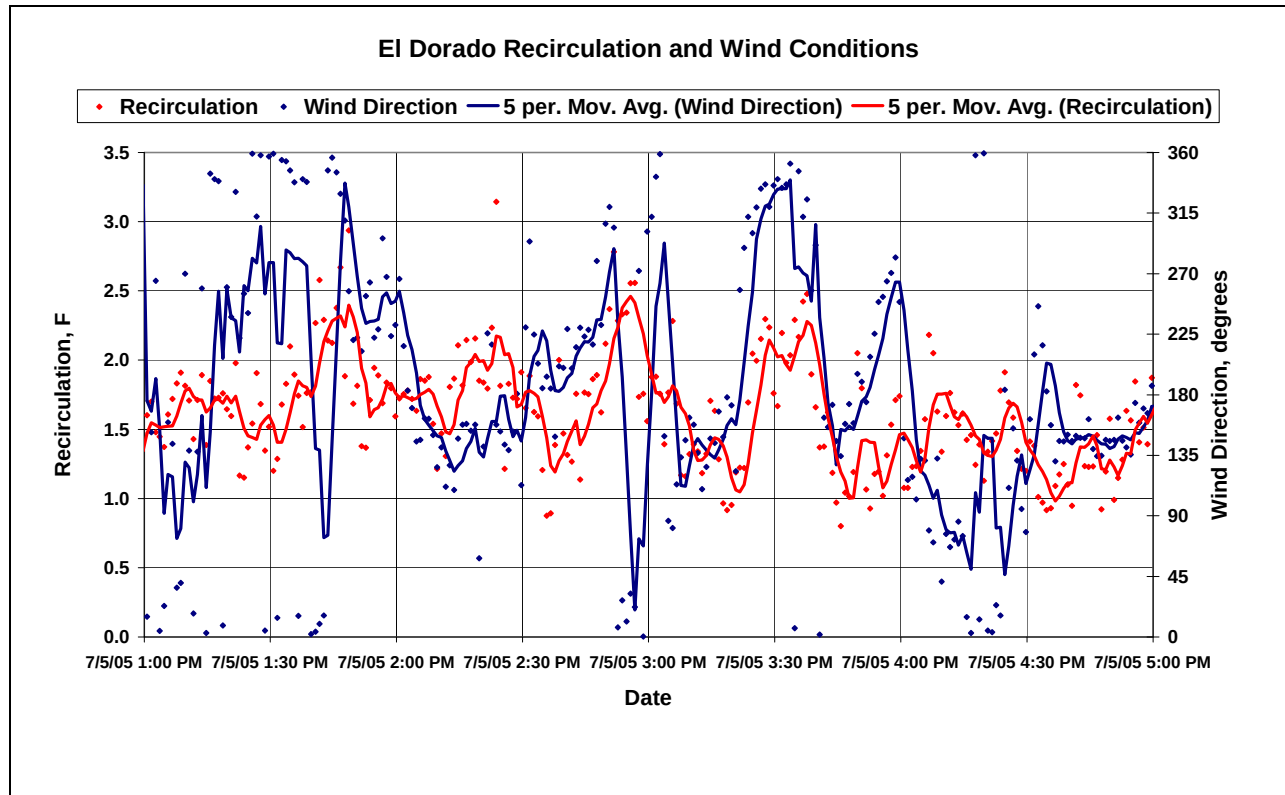


Figure 6-44. Average recirculation and wind direction, July 5, 2005.

Inlet cell distributions are plotted before (Figure 6-45), during (Figure 6-46), and after (Figure 6-47) the transient increase in recirculation observed between 2:45 p.m. and 3:15 p.m. The recirculation patterns at the beginning and end of the period exhibit a classic recirculation shape with relatively uniform inlet cell temperatures (especially in Figure 6-47) and selected edge and corner cells showing high recirculation effects. In the middle of the period, when the wind had shifted from primarily southerly to northerly, the cell temperatures show a very different pattern—a uniform variation from high to low across the ACC with the higher temperatures cells mostly at the north end. This suggests the explanation that the hot air plume, blown toward the north by the previously southerly winds, may have been forced back onto the ACC resulting in a high average inlet temperature without the usual pattern of recirculation.

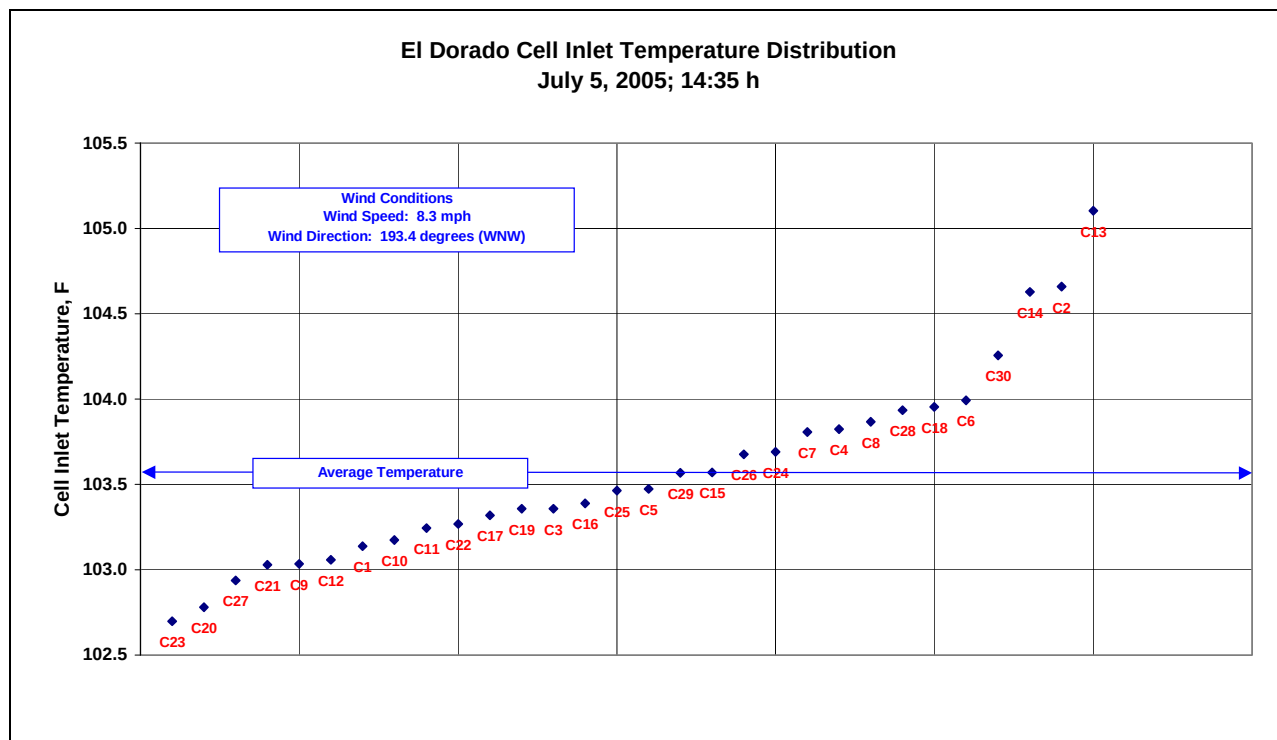


Figure 6-45. Cell inlet temperature distribution, July 5, 2005, 2:35 p.m.

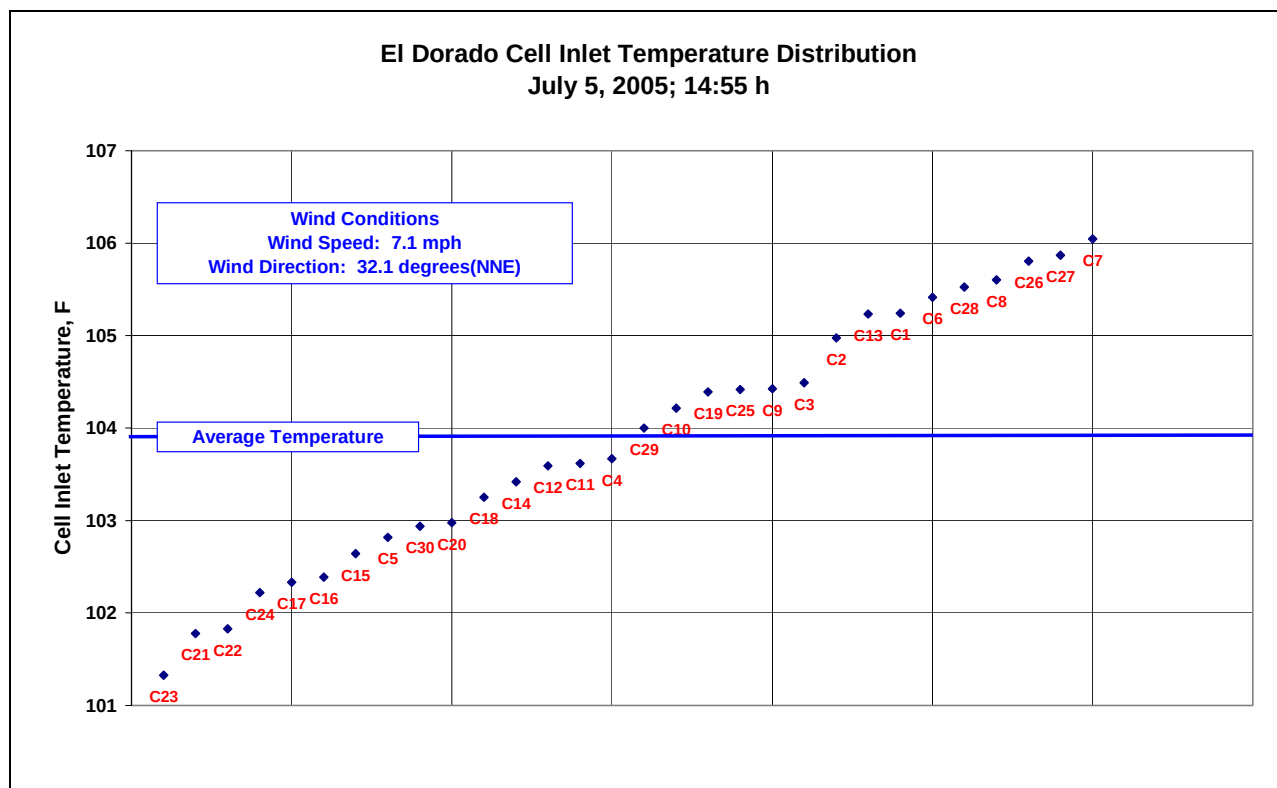


Figure 6-46. Cell inlet temperature distribution, July 5, 2005, 2:55 p.m.

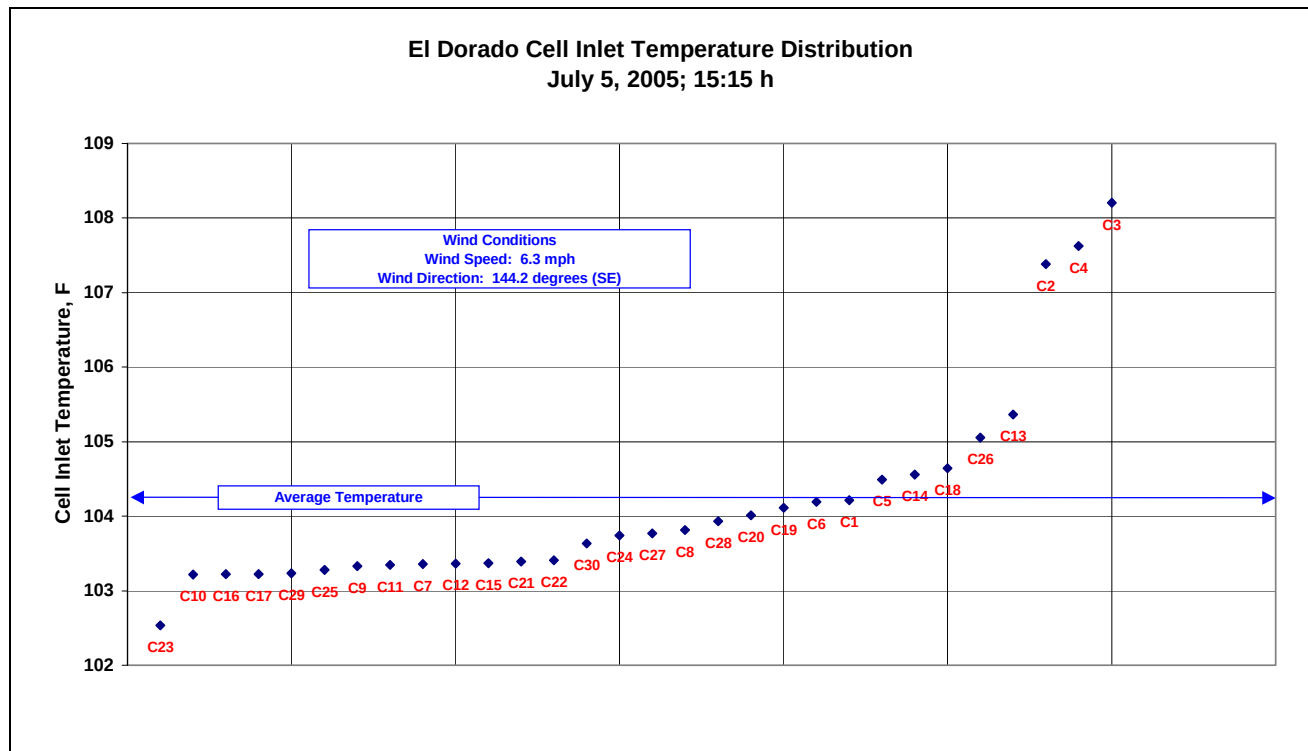


Figure 6-47. Cell inlet temperature distribution, July 5, 2005, 3:15 p.m.

6.4 Recirculation Tests—Apex Power Plant

The test period at Apex ran from 9:54 p.m. on July 13, 2005, until 1:40 p.m. on July 22, 2005. Figure 6-48 displays the average recirculation for selected times during the test period. The plant came off-line most nights around midnight and came back on the following morning at varying times. It was difficult to determine when all fans were running at full speed. Therefore, only data taken between 10 a.m. and 10 p.m. is used in this analysis since full-power operation with all fans on full is assured during those periods.

Figure 6-49 gives the wind speed and direction for the entire duration of the test period with the exception of July 14 when the wind instrumentation malfunctioned. Wind speeds throughout the test period are generally moderate, usually between 5.0 and 15.0 mph with a few periods between 15.0 and 20.0 mph and rare occurrences above 20.0 mph. As shown in Figure 6-50, the prevailing wind direction during the middle of the day when the ambient temperature and wind speed were the highest was southerly. Winds at night were frequently from the northern quadrant, more commonly from the northwest. Data for ambient temperature, wind speed, and wind direction were obtained from the met station at Silverhawk, a neighboring plant. The agreement with project test data for ambient temperature and wind speed was generally good. Silverhawk wind direction data, however, were reported to be essentially constant for the entire test period varying only from 95 to 110 degrees. This was contrary to both the project measurements and our personal observations; therefore, the wind direction data from Silverhawk were disregarded.

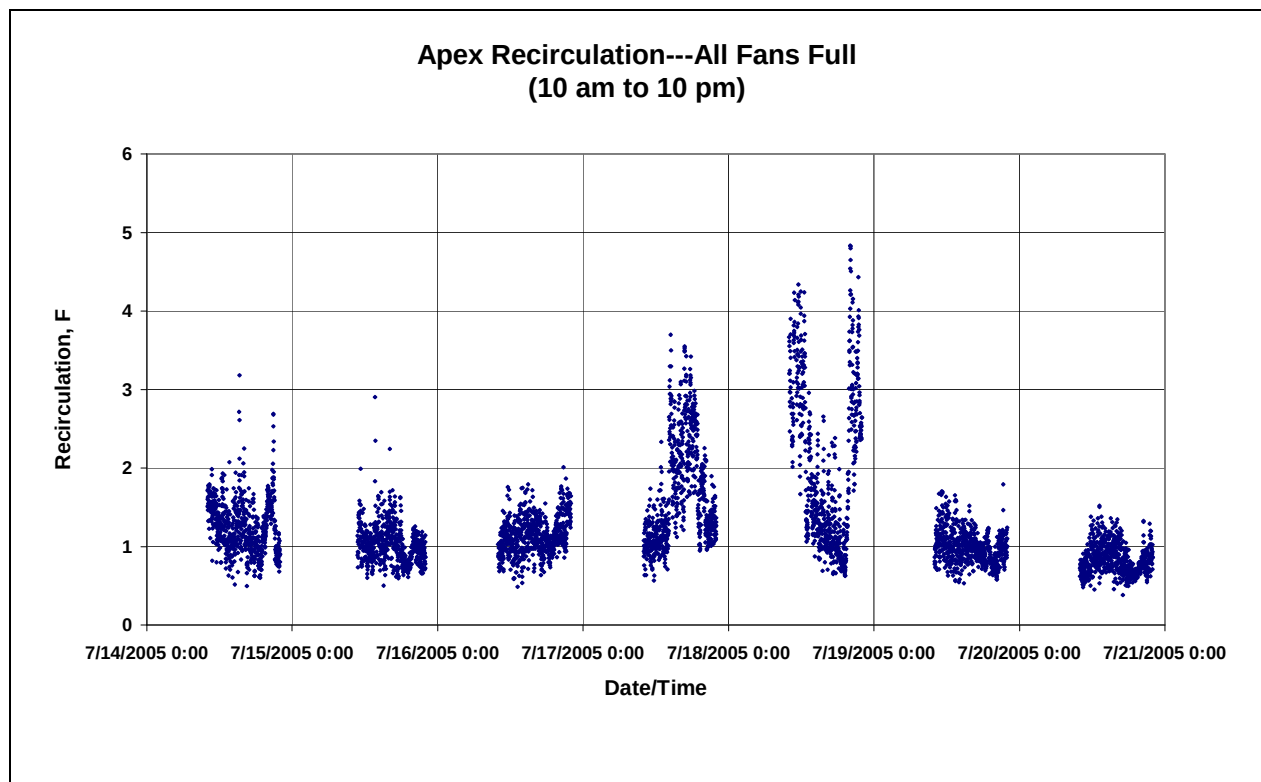


Figure 6-48. Average recirculation for selected periods at Apex.

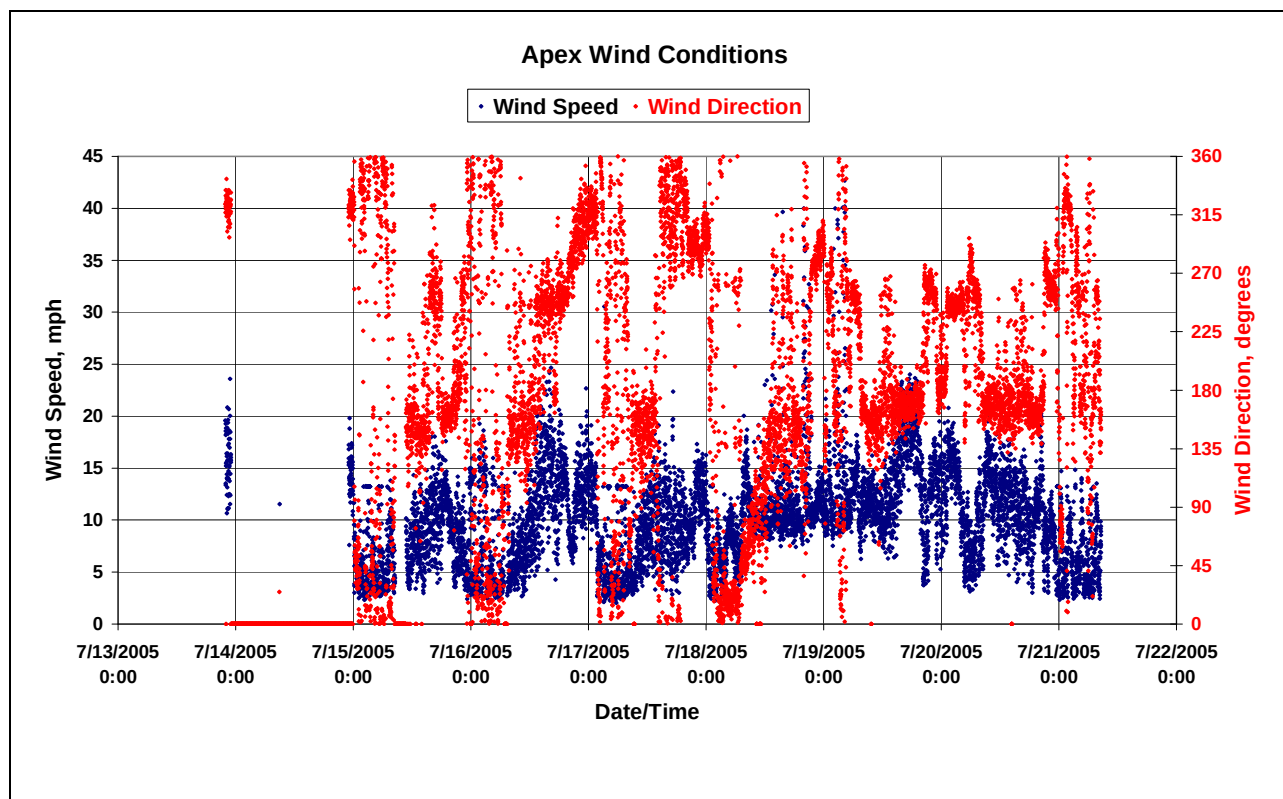


Figure 6-49. Apex wind speed and direction during test period.

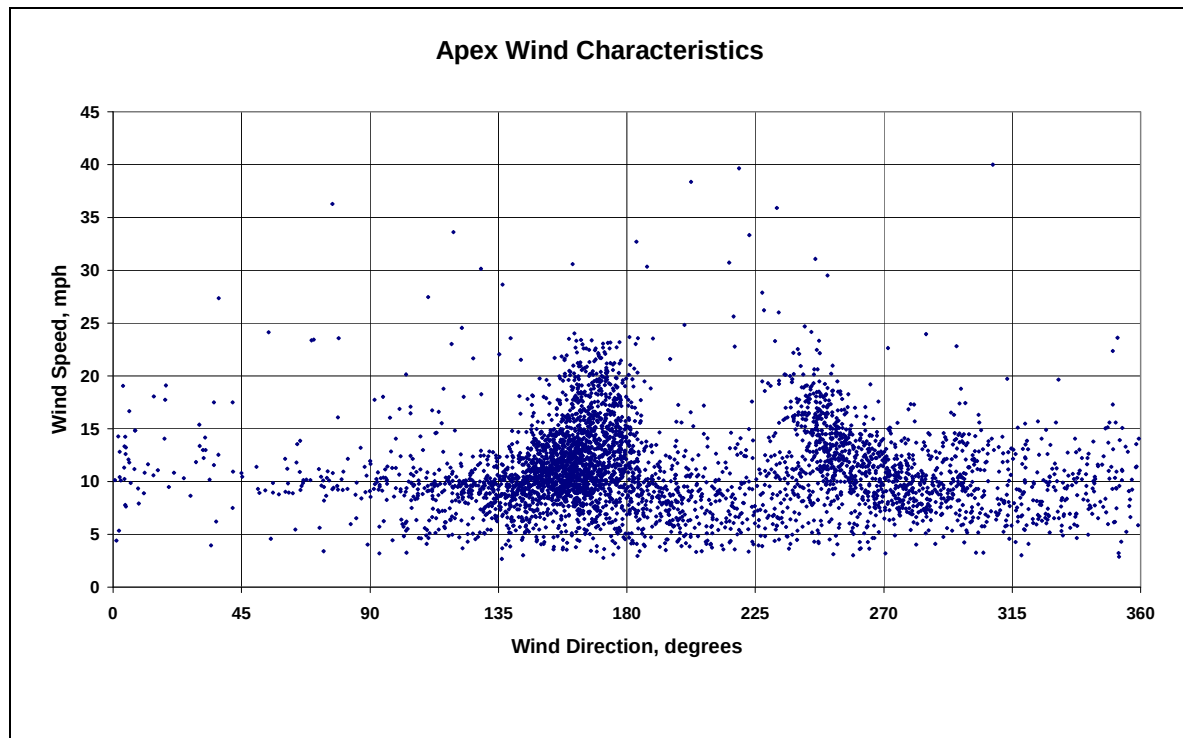


Figure 6-50. Wind characteristics at Apex.

Apex plant instrumentation reports the ambient temperature from an instrument on the administration building at hourly intervals. The agreement with minimum cell inlet temperature is shown to be excellent in Figure 6-51.

Figure 6-52 plots the average recirculation against wind speed. Higher levels of recirculation above 3°F occur at all wind speeds, but there is no apparent trend of increasing recirculation with increasing wind speeds.

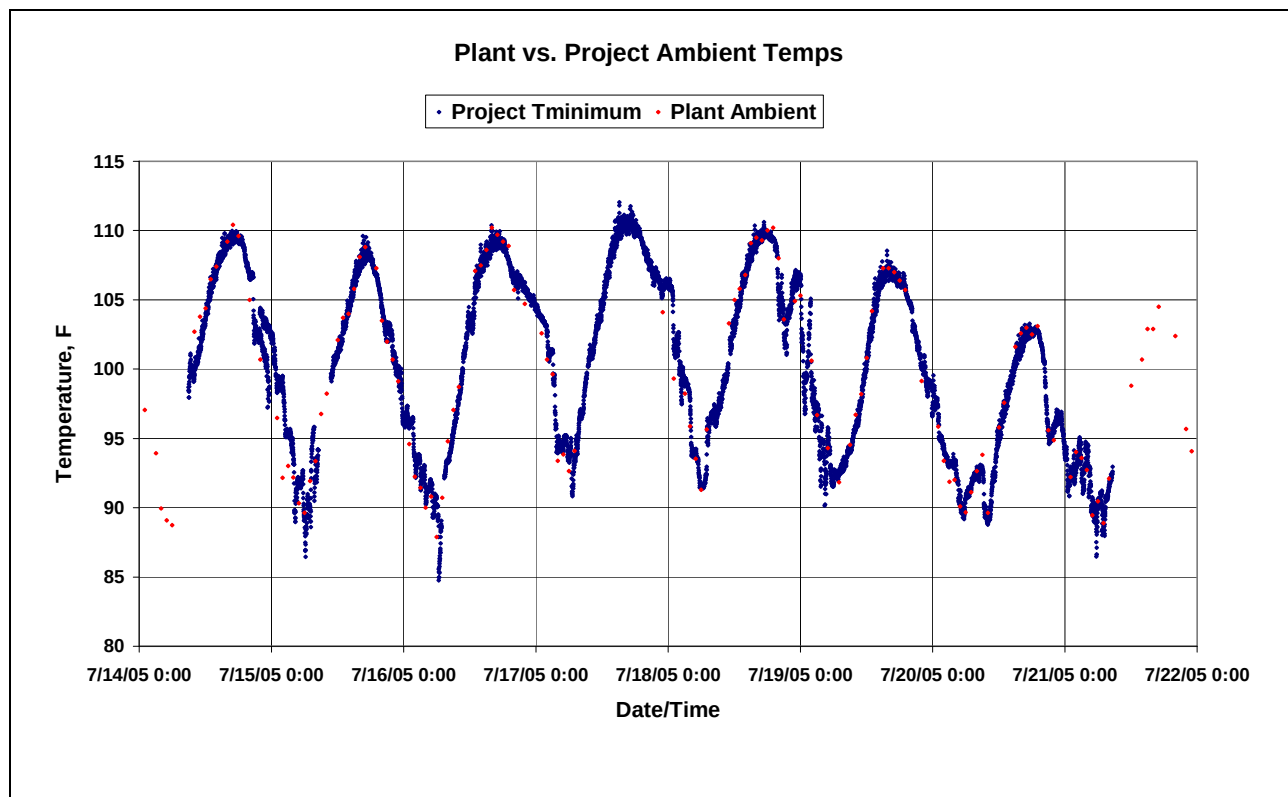


Figure 6-51. Apex ambient temperature comparison.

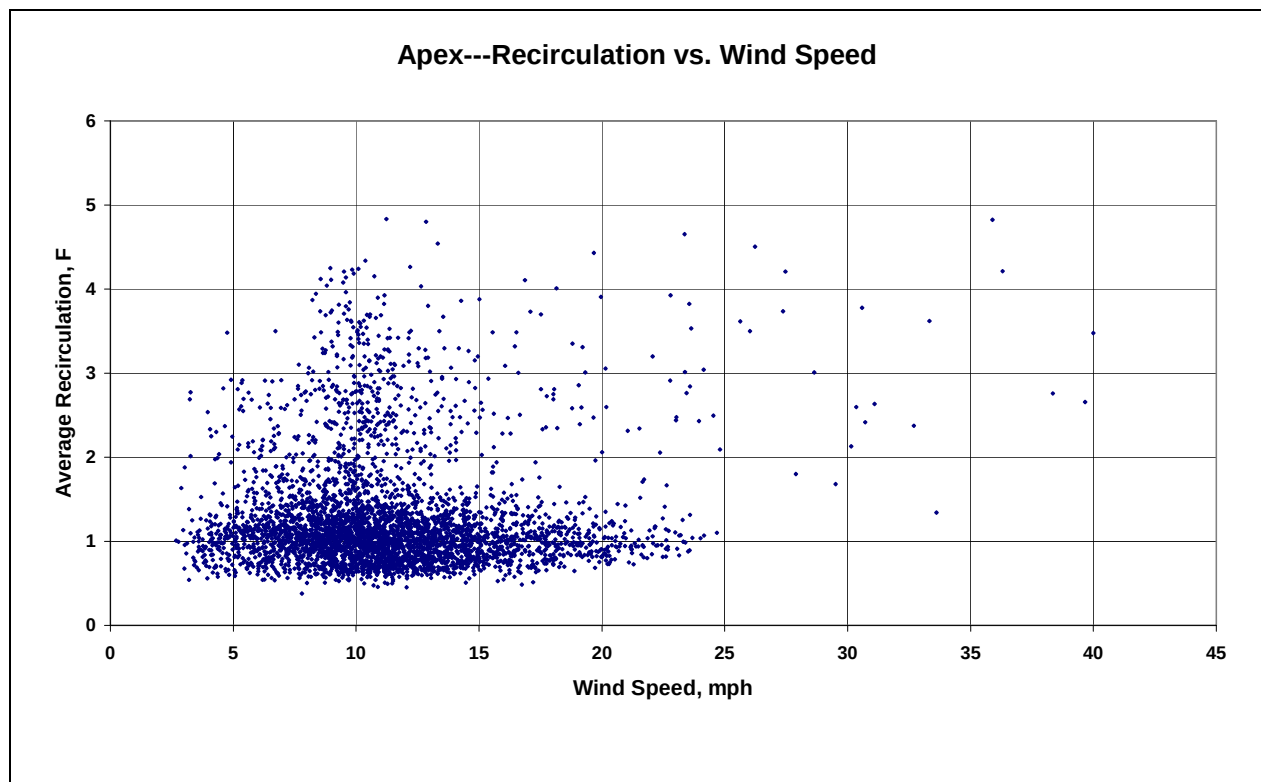


Figure 6-52. Recirculation vs. wind speed at Apex.

6.4.1 July 17, 2005

Recirculation behavior will be examined in more detail for July 17 and 18, which had high levels of average recirculation as shown in Figure 6-48. Figures 6-53 and 6-54 display the recirculation on the same plot as wind speed and direction for July 17.

Recirculation on July 17 averages from 1.0 to 1.5°F at the beginning of the day from 10:00 a.m. to 2:00 p.m. and again from 7:00 p.m. to 10:00 p.m. During the intervening hours, from 2:00 p.m. to 7:00 p.m., there is an abrupt increase to between 2.0 and 3.0°F which continues for five hours and then abruptly ends. There is nothing about the wind speed pattern that would appear to cause this behavior.

The wind direction, on the other hand, is steady from the southeast with only mild variability up until 2:00 p.m. At that time, abrupt, significant shifts occur from the southeast to the southwest, back to the northeast, and finally to the northwest where it remains, although with significant variability, until about 6:00 p.m. Between 6:00 and 7:00 p.m., the wind shifts from north-northwest to southeast and back abruptly and then remains from the northwest with only mild variability until 10:00 p.m.

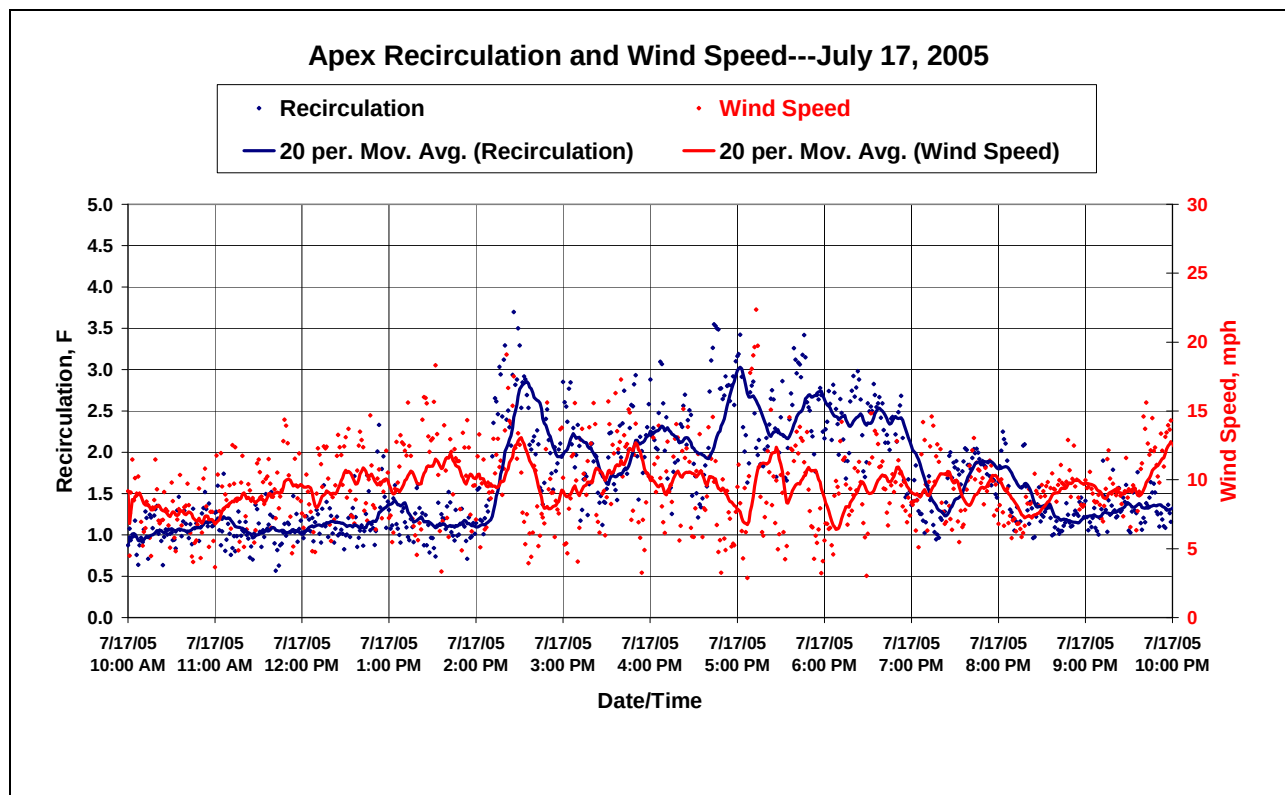


Figure 6-53. Recirculation and wind speed, July 17, 2005.

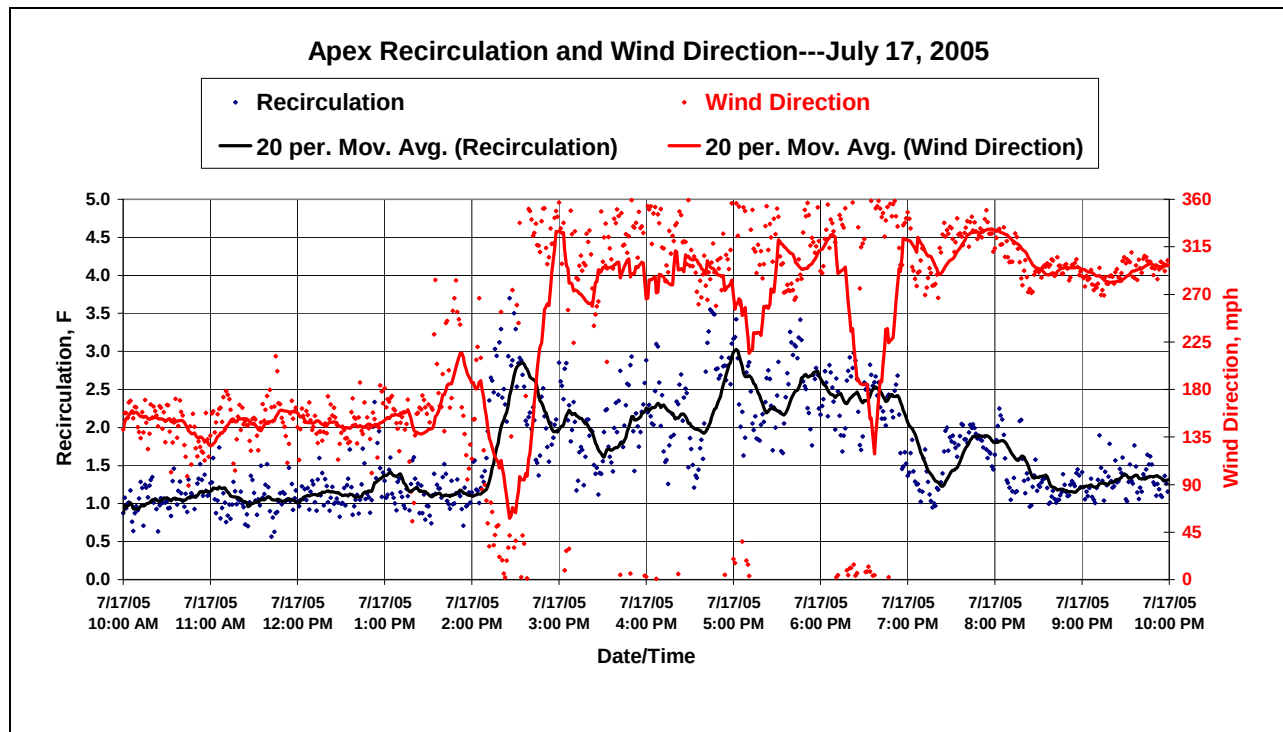


Figure 6-54. Recirculation and wind direction, July 17, 2005.

Figures 6-55 and 6-56 examine the period of 1:00 to 5:00 p.m. more closely. The finer-structure look at the recirculation superimposed with wind speed and direction gives a picture that is inconclusive. While there are occasional events in which an increase in wind speed is coincident with an increase in recirculation (see, for example, 2:15 to 2:45 p.m. on Figure 6-55), much of the time no relationship can be observed. The effect of wind direction is similarly problematic. It does appear, however, as has been suggested previously, that *changes* in direction result in changes in recirculation, perhaps through the disruption and reestablishment of plume patterns.

Figures 6-57 through 6-60 provide a closer look at the cell inlet temperature distributions for the four times between 1:30 and 3:00 p.m. on July 17 identified on Figure 6-56. The location of the various cells referred to in the following discussion can be found in Figure 4-8 in Section 4.4.3.

At 1:30 p.m., wind direction and recirculation have been steady for several hours (see Figure 6-54). Figure 6-57 shows nearly uniform cell inlet temperatures across the ACC with only the slightest hint of recirculation in the northeast corner (Cells K1A and K1B) and in one south cell (K2D).

At 1:52 p.m. the wind has shifted from southeasterly to westerly with little or no change in the average recirculation, but a distinct recirculation pattern is emerging (see Figure 6-58) with high inlet temperatures appearing in the southwest corner as well as the northeast corner.

At 2:35 p.m., the wind direction has shifted back all the way to easterly and the average recirculation has increased to almost 3°F. Figure 6-59 indicates that the increase in cell inlet

temperature has spread across nearly the entire ACC, with the hottest cells still in the northeast corner. It appears that the previously established recirculating plume has been pushed back under the ACC.

It is noteworthy that from 1:30 to 2:35 p.m. the inlet temperatures of the lowest-temperature cells have remained nearly constant between 108.4°F to 109°F. At 2:58 p.m. (see Figure 6-60), however, while the hottest cells are still at about 115°F, the same temperature as they were at 2:35 p.m., the lower inlet temperatures have increased to 111°F. This may indicate that the recirculated plume has been carried completely under the ACC and is affecting all of the cells.

This hypothesis, if correct, points out a methodological problem with using the lowest cell inlet temperature as the base point for calculating the average recirculation. The condition at 2:58 p.m. may in fact entail *more* recirculation than at 2:35 p.m. However, the increase in the minimum cell inlet temperature results in a *lower* calculated average recirculation.

While this may cloud and even confuse the interpretation of short-term variability of recirculation with wind conditions, the authors believe that the overall approach is appropriate and the conclusion that average recirculation is modest under essentially all wind conditions is valid.

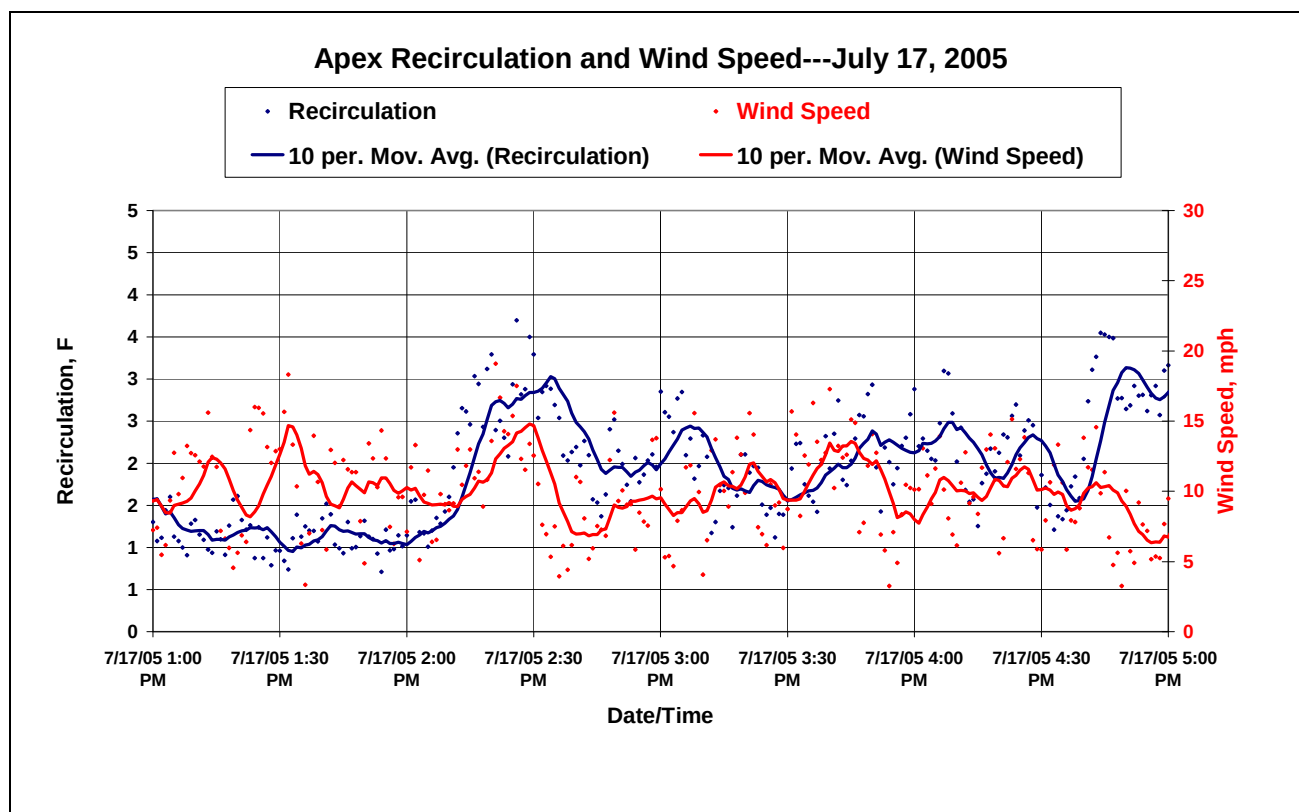


Figure 6-55. Recirculation and wind speed, July 17, 2005, 1:00 to 5:00 p.m.

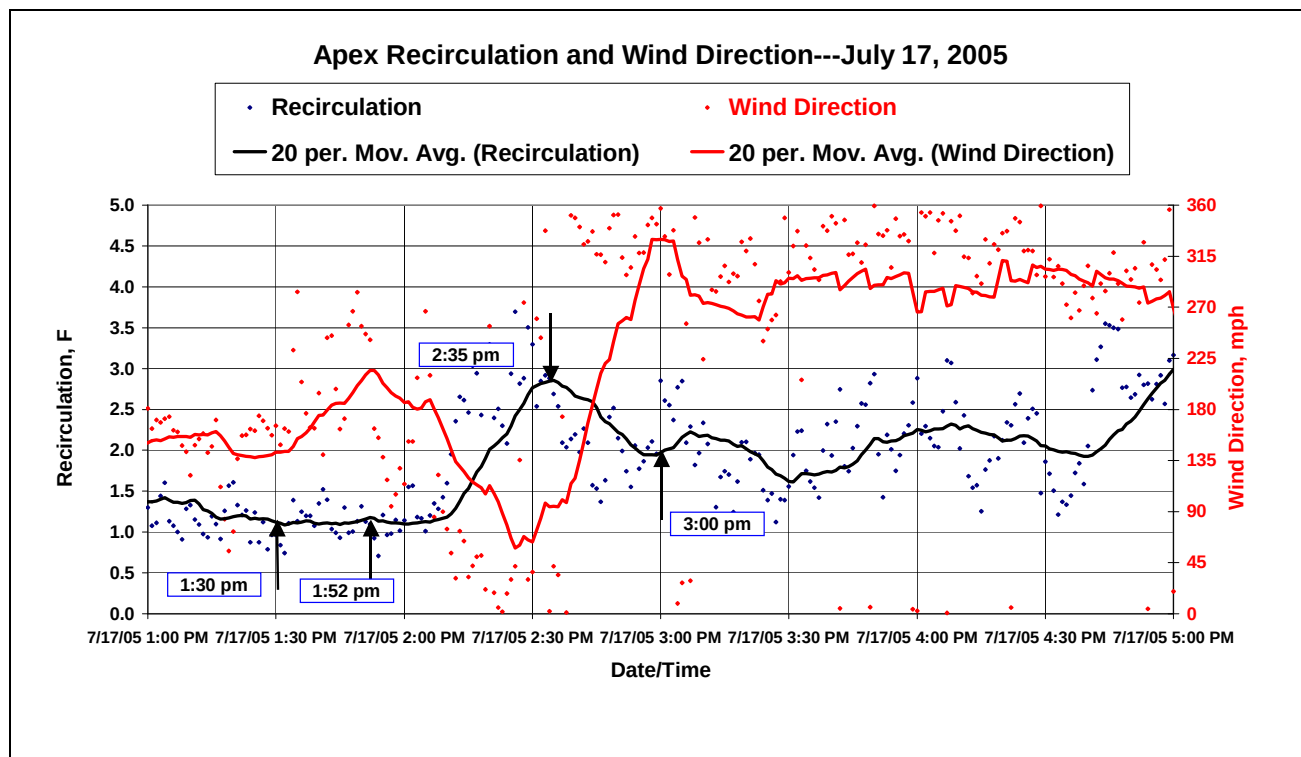


Figure 6-56. Recirculation and wind direction, July 17, 2005, 1:00 to 5:00 p.m.

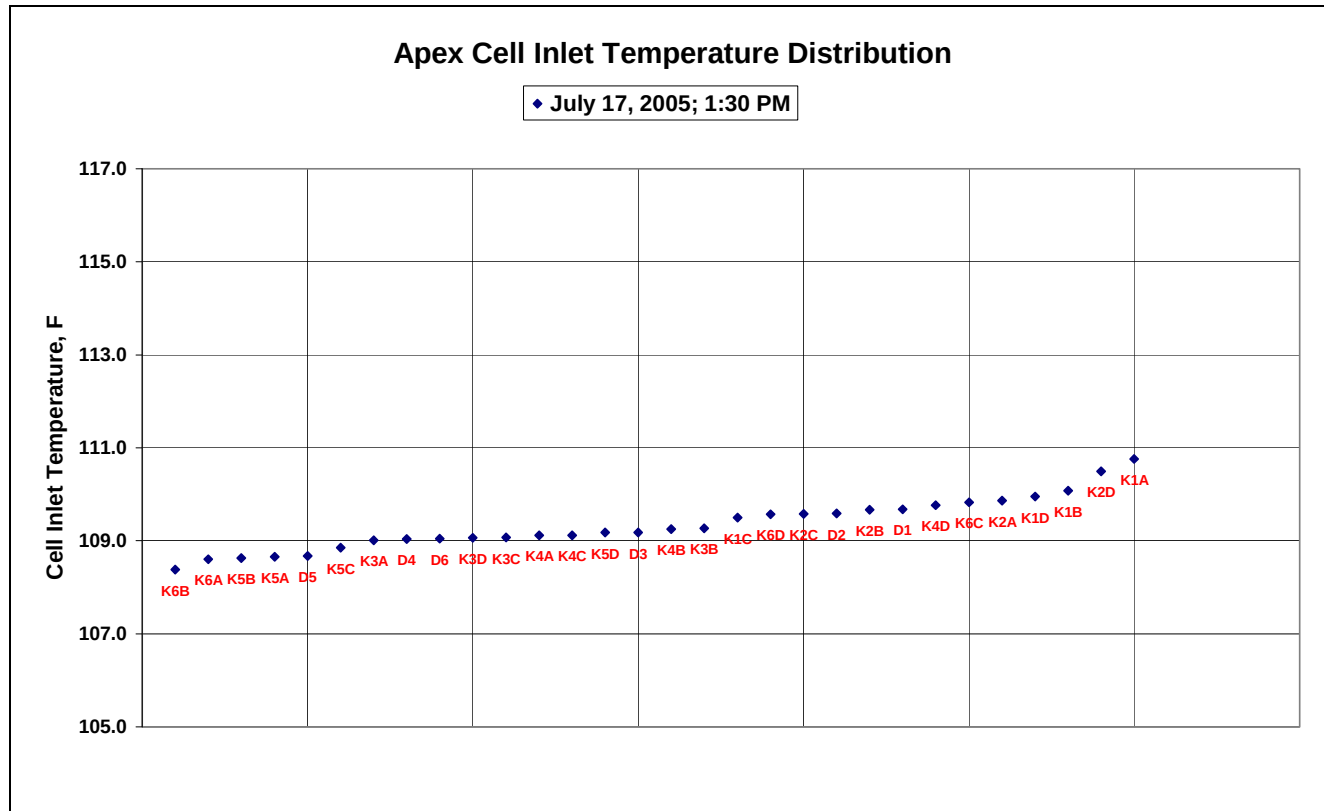


Figure 6-57. Cell inlet temperature distribution, July 17, 2005, 1:30 p.m.

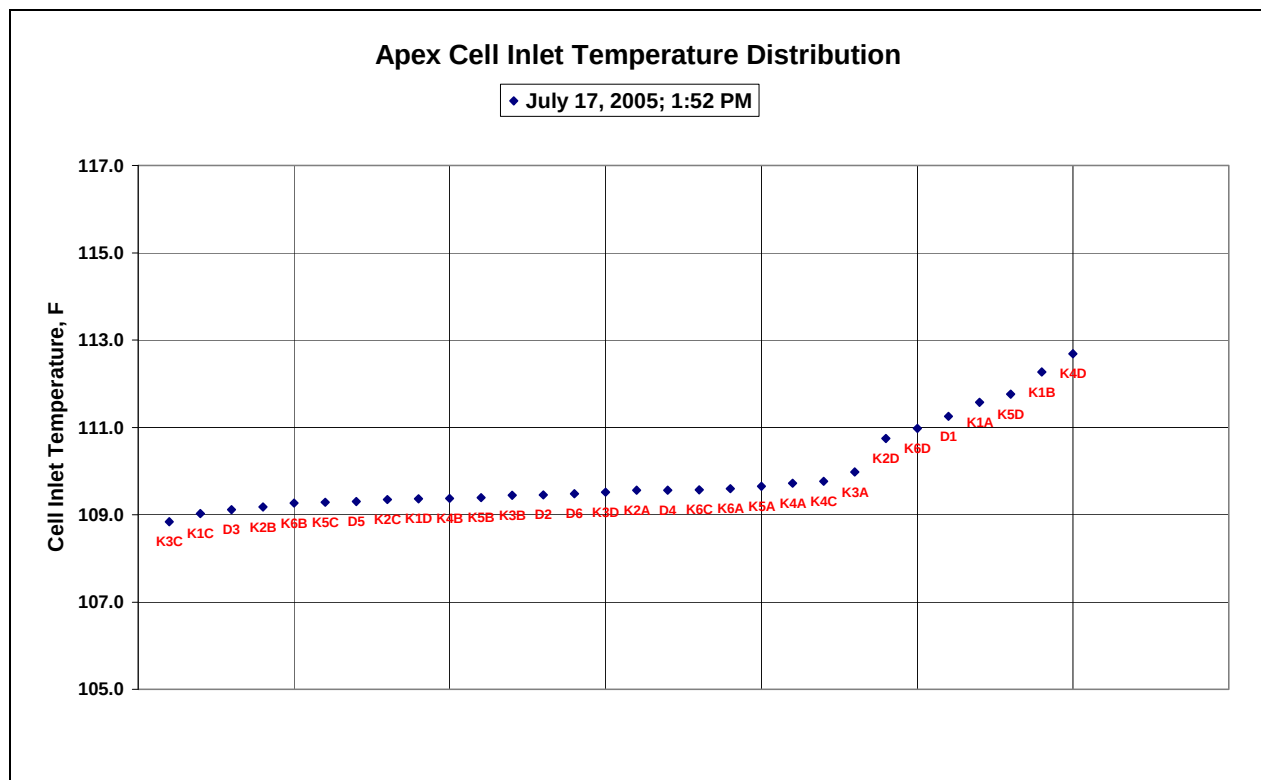


Figure 6-58. Cell inlet temperature distribution, July 17, 2005, 1:52 p.m.

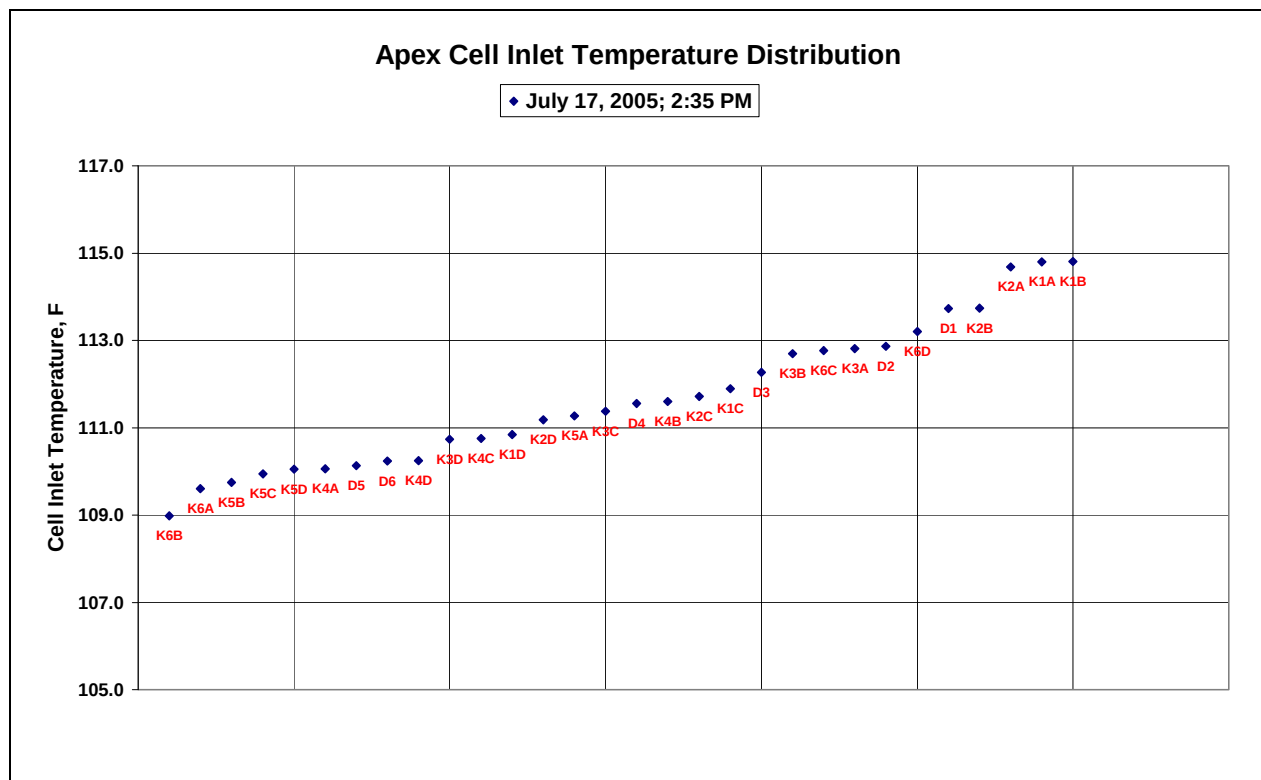


Figure 6-59. Cell inlet temperature distribution, July 17, 2005, 2:35 p.m.

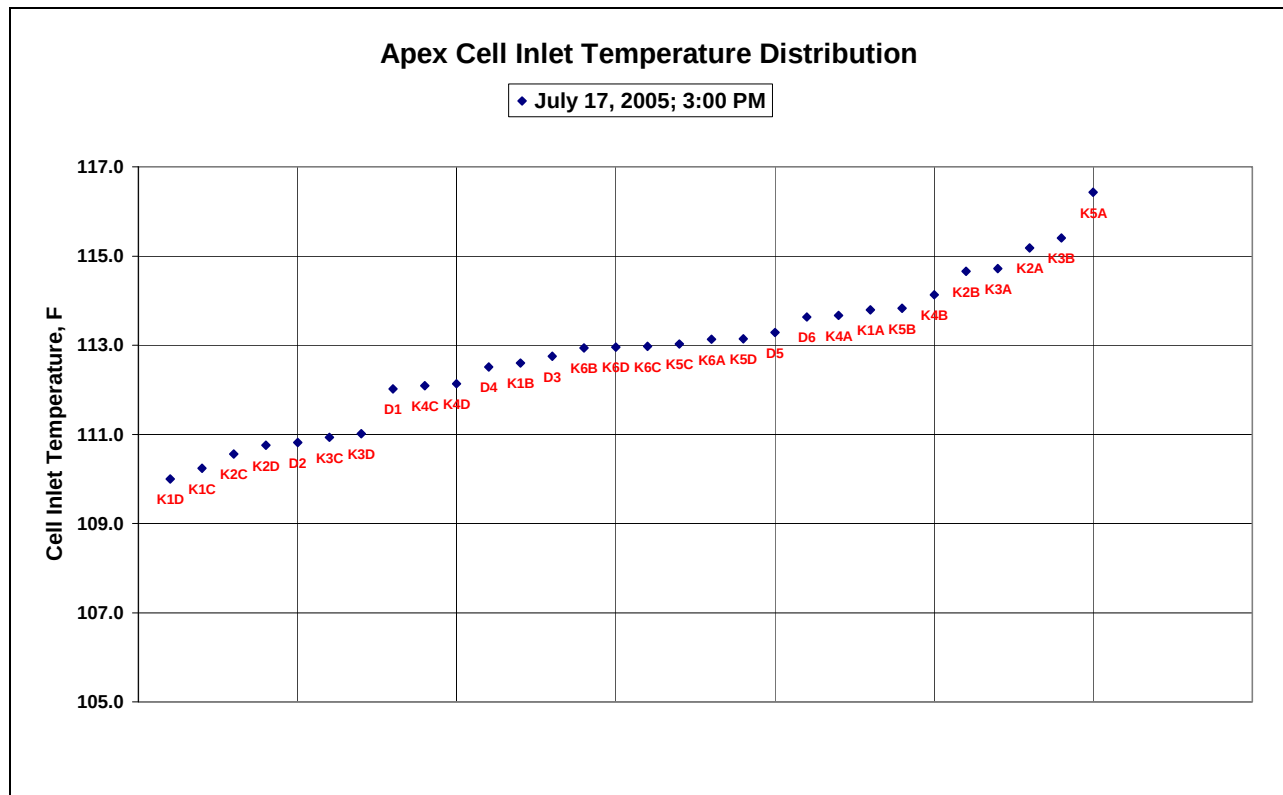


Figure 6-60. Cell inlet temperature distribution, July 17, 2005, 3:00 p.m.

6.4.2 July 18, 2005

Recirculation on July 18 is displayed along with coincident wind speed and direction in Figures 6-61 and 6-62. The recirculation is moderate to low for most of the day, declining from around 2.0° to 2.5°F in the mid- to late morning to around 0.5°F just after 7:00 p.m. During that period the wind is moderate at about 10.0 mph with little variability. The wind direction, while primarily from the southeast, exhibits repeated, abrupt, significant variability from 1:30 to 5:30 p.m. with five major shifts from southeast to westerly and back again. The recirculation was not noticeably affected by these shifts.

Just after 7:00 p.m., the recirculation increases rapidly from 0.6°F to about 3.5°F in about an hour, followed by significant variability with the recirculation going as high as 4.0°F and as low as 1.3°F in a period of about two hours. Figures 6-63 and 6-64 display the behavior during that period from 7:00 to 10:00 p.m. These figures suggest a stronger influence of wind direction on recirculation than of wind speed. Figure 6-64 identifies sometimes at which the cell inlet temperature distribution is examined in an attempt to understand the relationship between wind direction and recirculation during this period. Figures 6-65 through 6-71 shows the cell inlet temperature distributions at the times indicated on Figure 6-64.

At 7:15 p.m. (Figure 6-65), the recirculation and wind speed have been steady for several hours (see Figure 6-61) and the wind direction has been reasonably steady from the southwest for nearly an hour and a half. Figure 6-65 shows a nearly uniform cell inlet temperature

distribution with a small amount of recirculation on the south edge (Cells K2D and K4D) and in the northeast corner (Cells K1A and K2A).

At 8:11 p.m. (Figure 6-66), wind conditions have changed significantly. The wind speed has increased from 10 mph to over 25 mph, the direction has shifted from southwesterly to westerly or even northwesterly, and the ambient temperature has dropped about 4°F. The recirculation has increased from 0.6°F to 3.0°F. The increased recirculation is not due to an increased recirculation of hot plume air, but rather to a significant decrease in the temperature of the coolest cell (K3C) and some neighboring cells—mostly of interior cells in the center and northwest corner of the ACC. The reason for this is not understood.

At 8:25 p.m. (Figure 6-67), the wind speed has declined, the wind has shifted to the east, and the inlet cell temperature distribution is nearly uniform with no evidence of hot air recirculation. Two cells (K2B and D3) are distinctly cooler than the others with the result that the calculated recirculation is somewhat higher than it was prior to 7:15 p.m. At the prevailing wind speeds, temperature inversion effects are unlikely. There is no obvious explanation for the cooler cells.

At 8:30 p.m. (Figure 6-68), following a 180° shift in the wind direction from easterly to westerly, the situation completely changes to one of significant recirculation. The hottest cells are on the east edge and in the southwest corner. The recirculation on the east edge is consistent with a westerly wind. The hotter cells on the south edge may result from the wind's pushing a pre-established hot air plume under the ACC as it shifts from easterly to westerly through the southern quadrant.

By 8:45 p.m. (Figure 6-69), the wind had diminished slightly and shifted back to the southeast. The recirculation has decreased with the inlet temperature to the hottest cells dropping from 112°F to 109°F.

At 9:30 p.m. (Figure 6-70), the wind has shifted steadily to the southwest and the ambient temperature has decreased slightly. The cell inlet temperatures vary uniformly across the ACC from cooler cells in the northwest corner to hotter cells around the eastern and southern edges.

By 10:00 p.m. (Figure 6-71), with the wind steadily from the west, modest recirculation is seen on the northeast and south edge cells with a cooler cluster of cells in the northeast corner. The increase in the coolest cell inlet temperatures from 101°F at 9:30 p.m. to 104.°F at 10:00 p.m. is likely related to the shift in wind direction.

It is difficult to infer a consistent understanding of the effect of wind on recirculation from these observations beyond the observation that frequent changes in wind conditions prevent the formation of a well-established hot air discharge plume. As the wind direction changes, this leads to the carry-back of hot air into the ACC inlet and shifting cell inlet temperature distributions that are not consistent with the classical pattern of downwind recirculation.

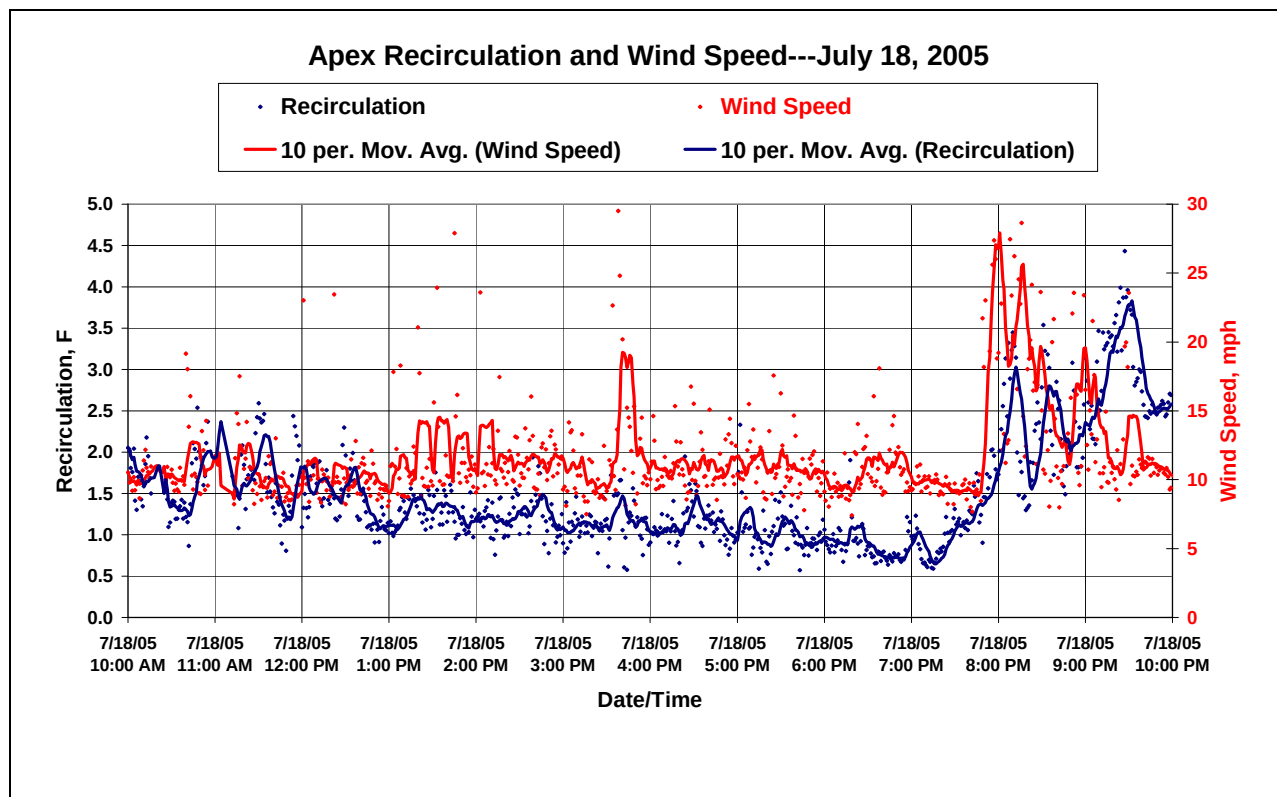


Figure 6-61. Recirculation and wind speed, July 18, 2005.

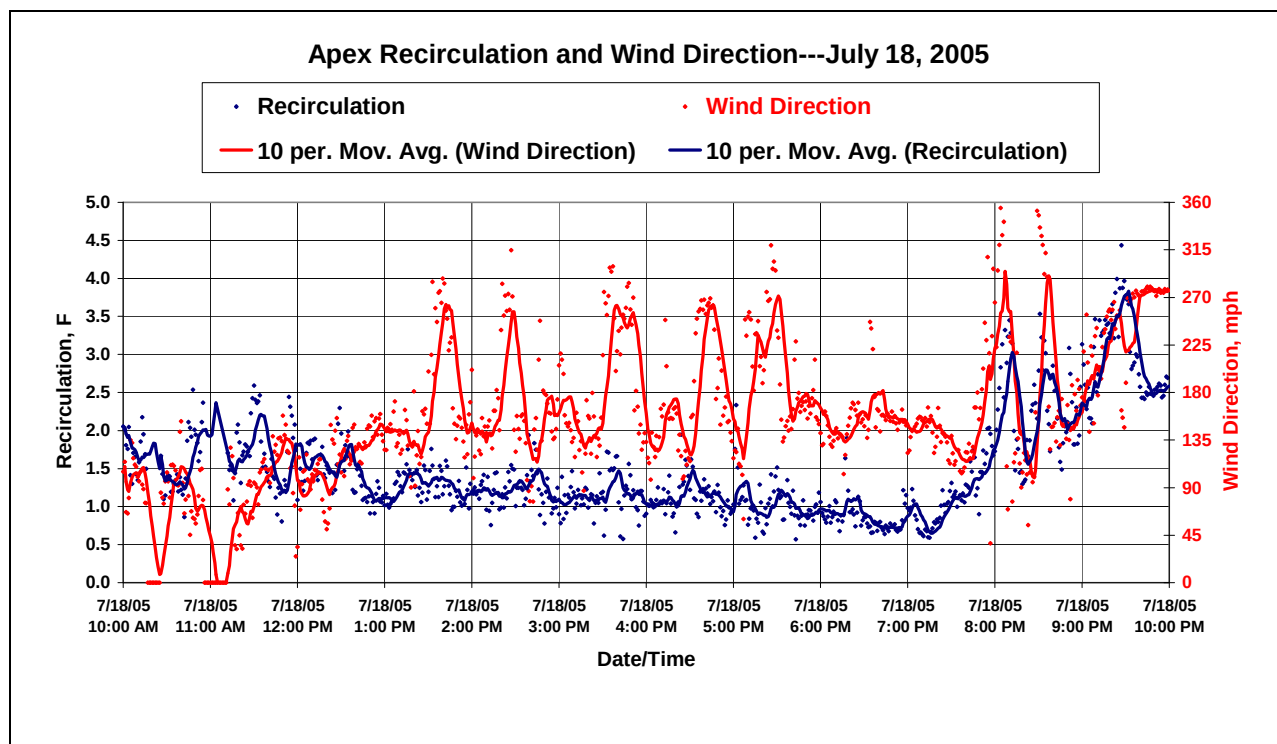


Figure 6-62. Recirculation and wind direction, July 18, 2005.

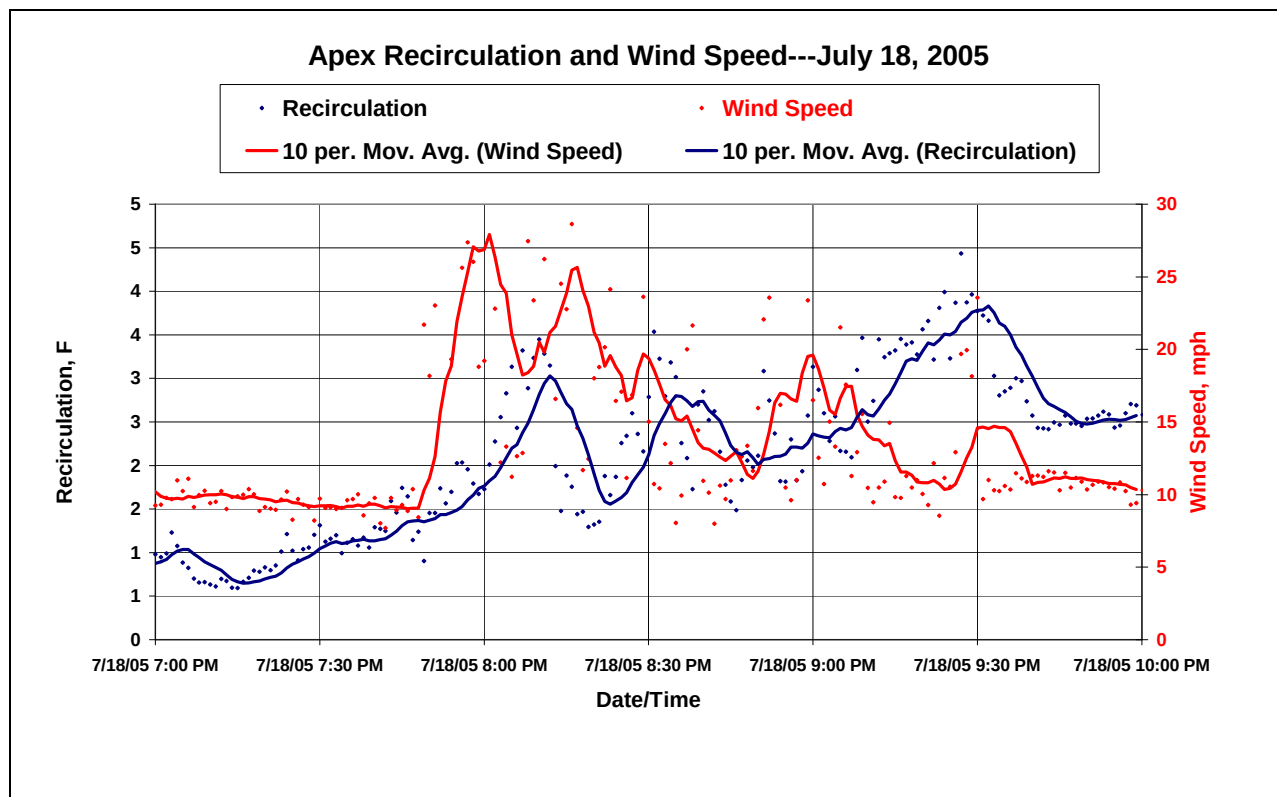


Figure 6-63. Recirculation and wind speed, July 18, 2005, 7:00 to 10:00 p.m.

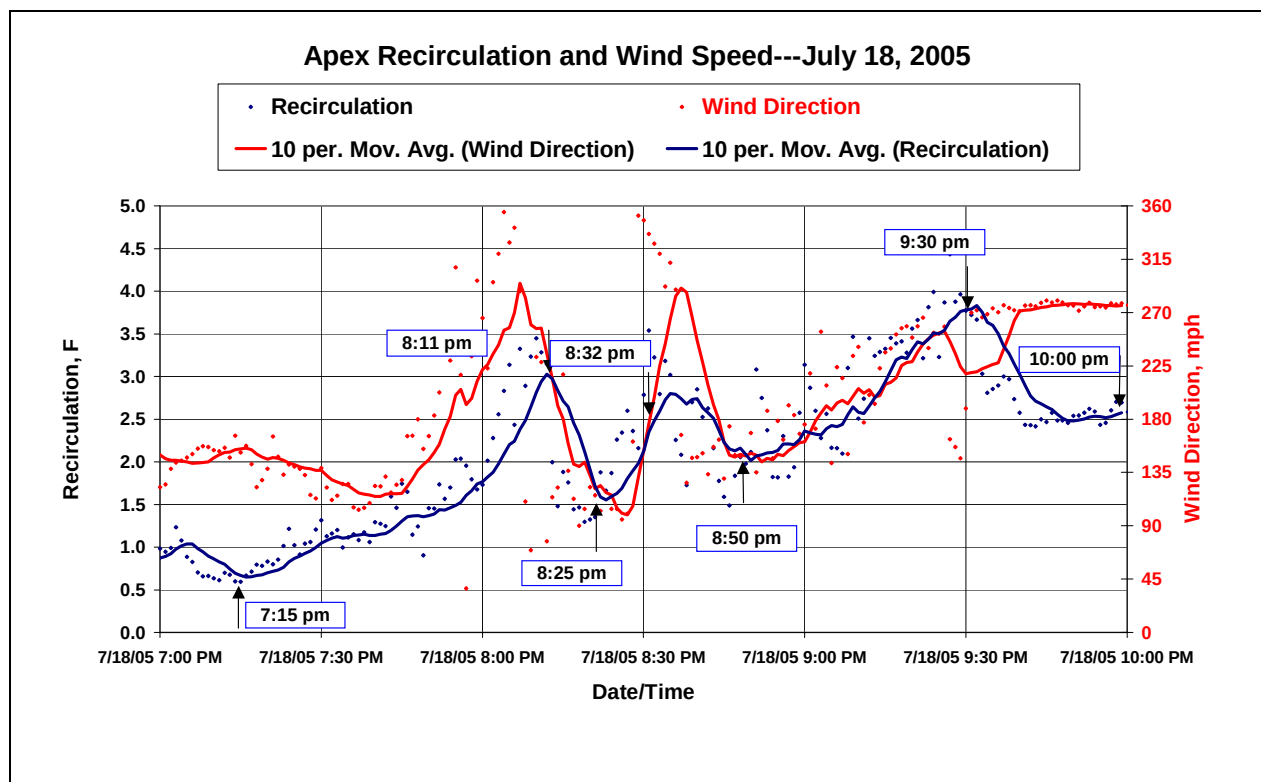


Figure 6-64. Recirculation and wind direction, July 18, 2005.

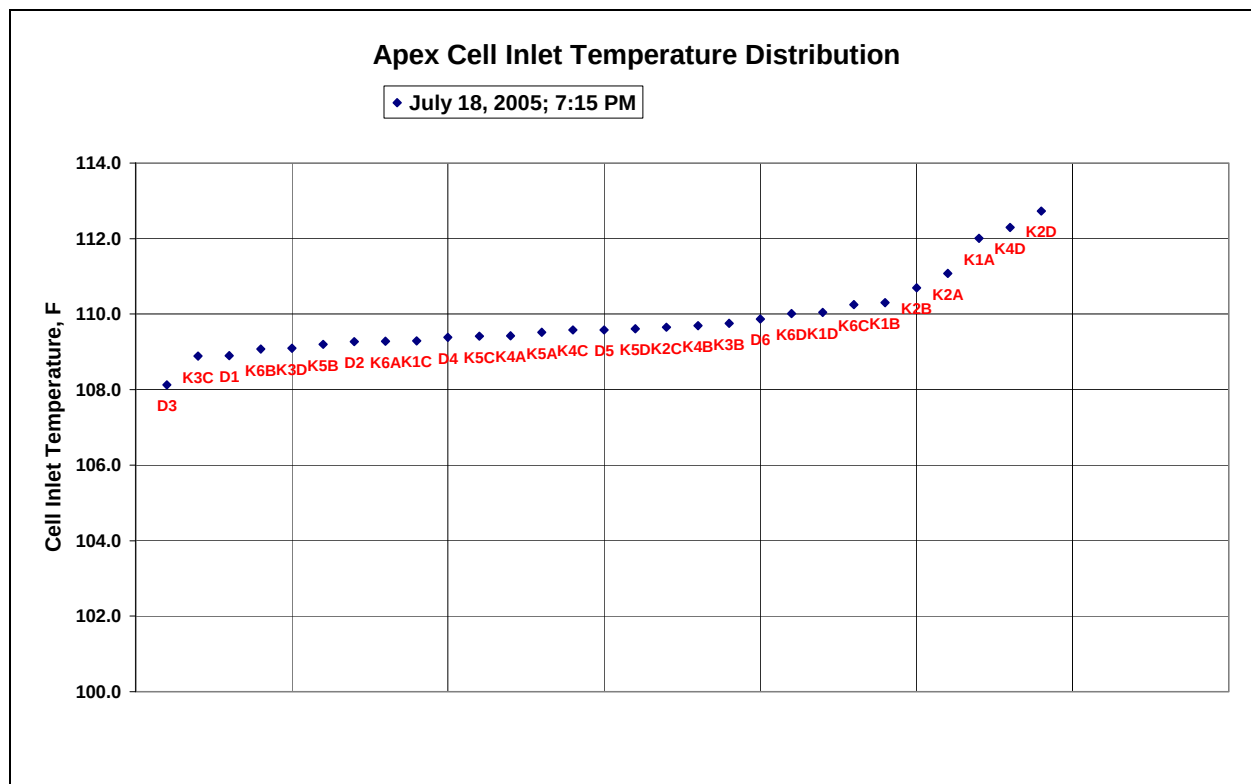


Figure 6-65. Cell inlet temperature distribution, July 18, 2005, 7:15 p.m.

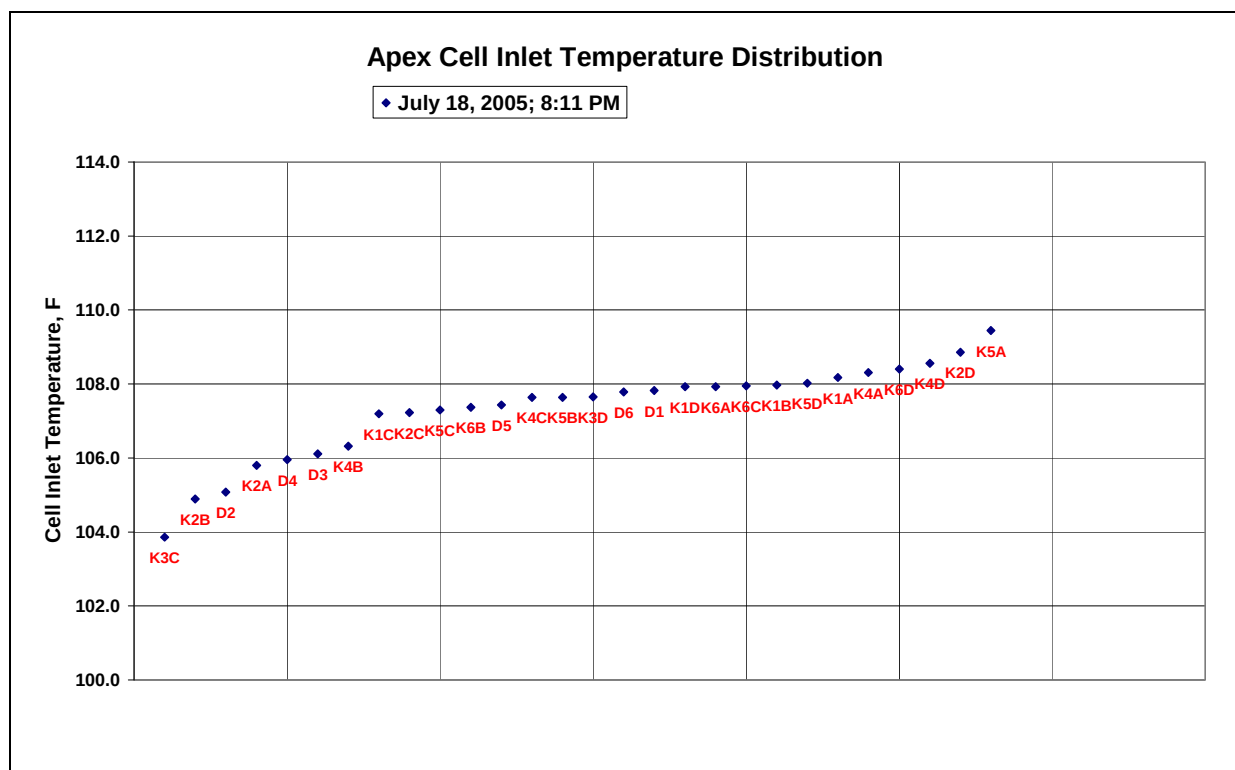


Figure 6-66. Cell inlet temperature distribution, July 18, 2005, 8:11 p.m.

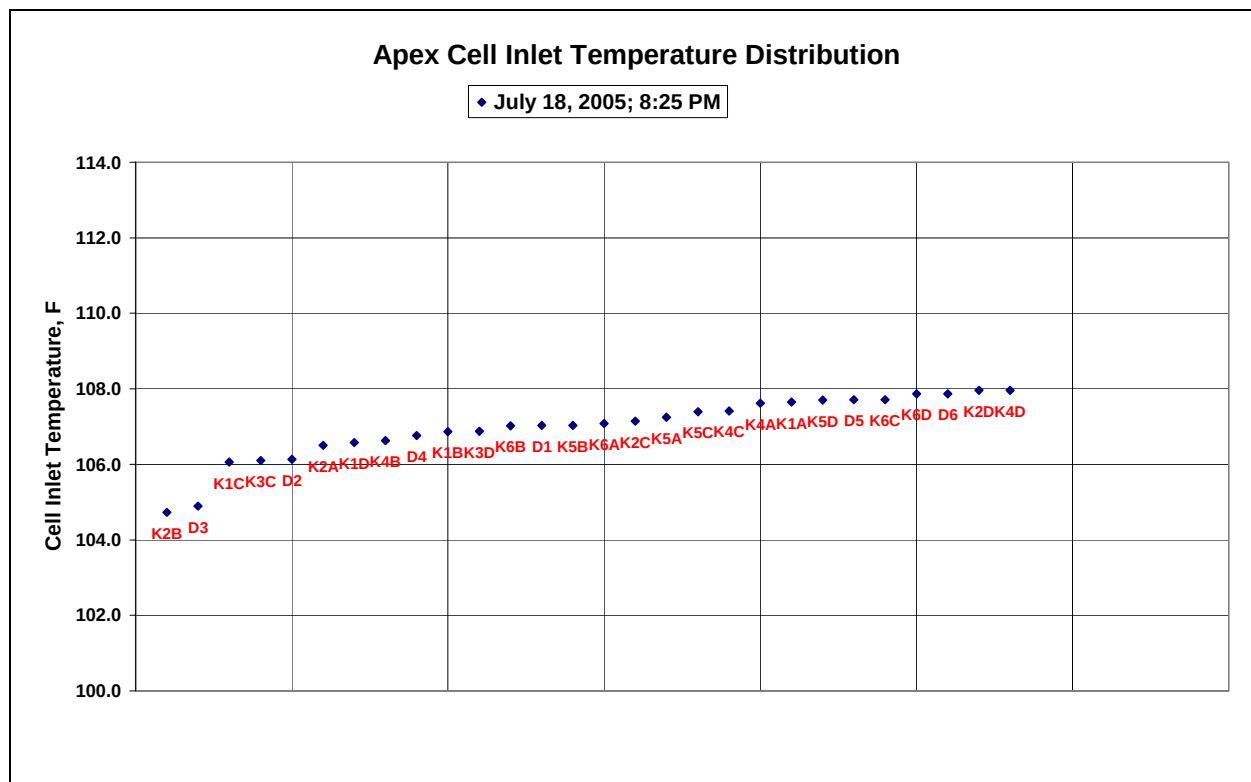


Figure 6-67. Cell inlet temperature distribution, July 18, 2005, 8:25 p.m.

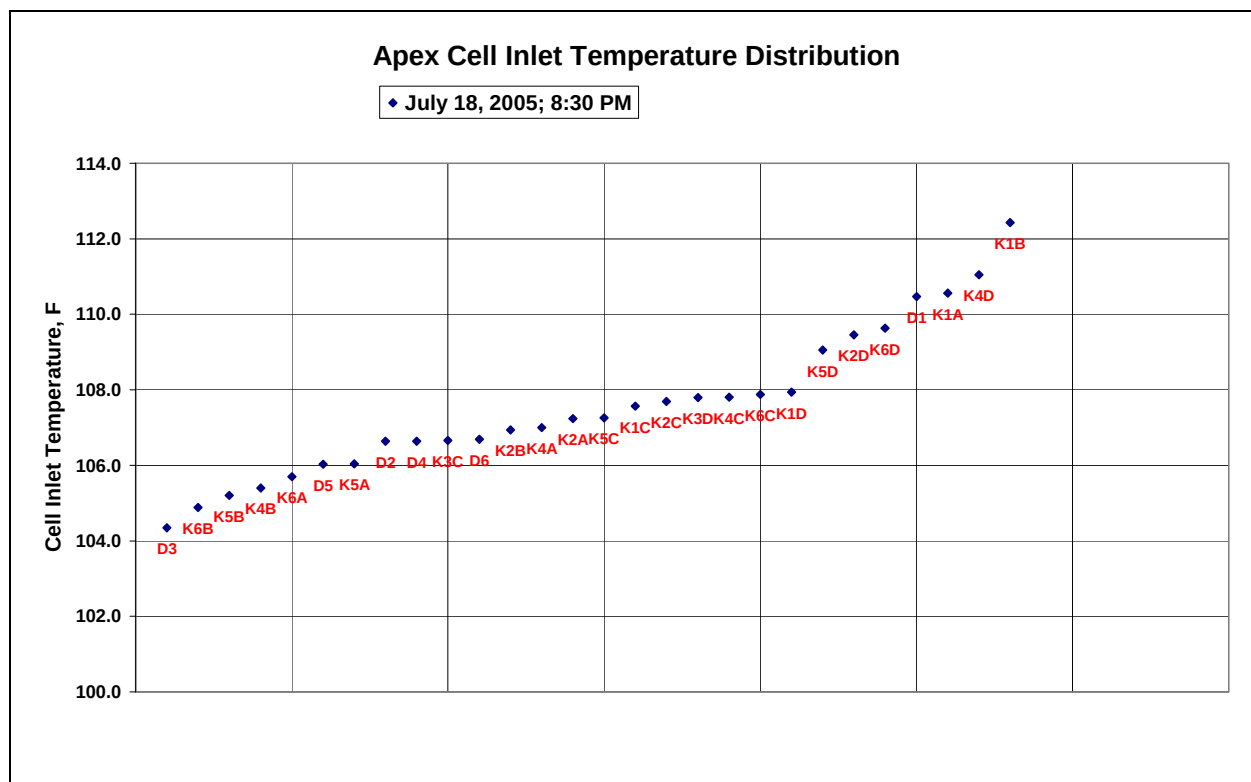


Figure 6-68. Cell inlet temperature distribution, July 18, 2005, 8:32 p.m.

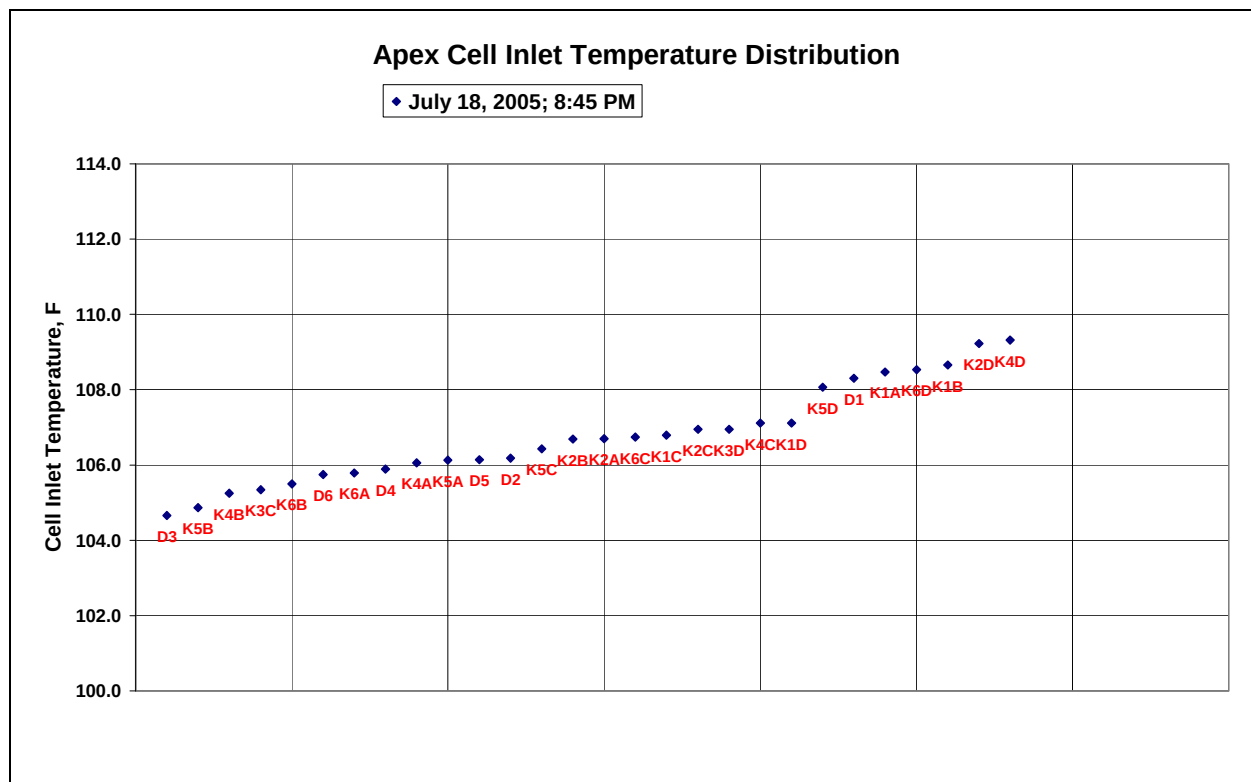


Figure 6-69. Cell inlet temperature distribution, July 18, 2005, 8:50 p.m.

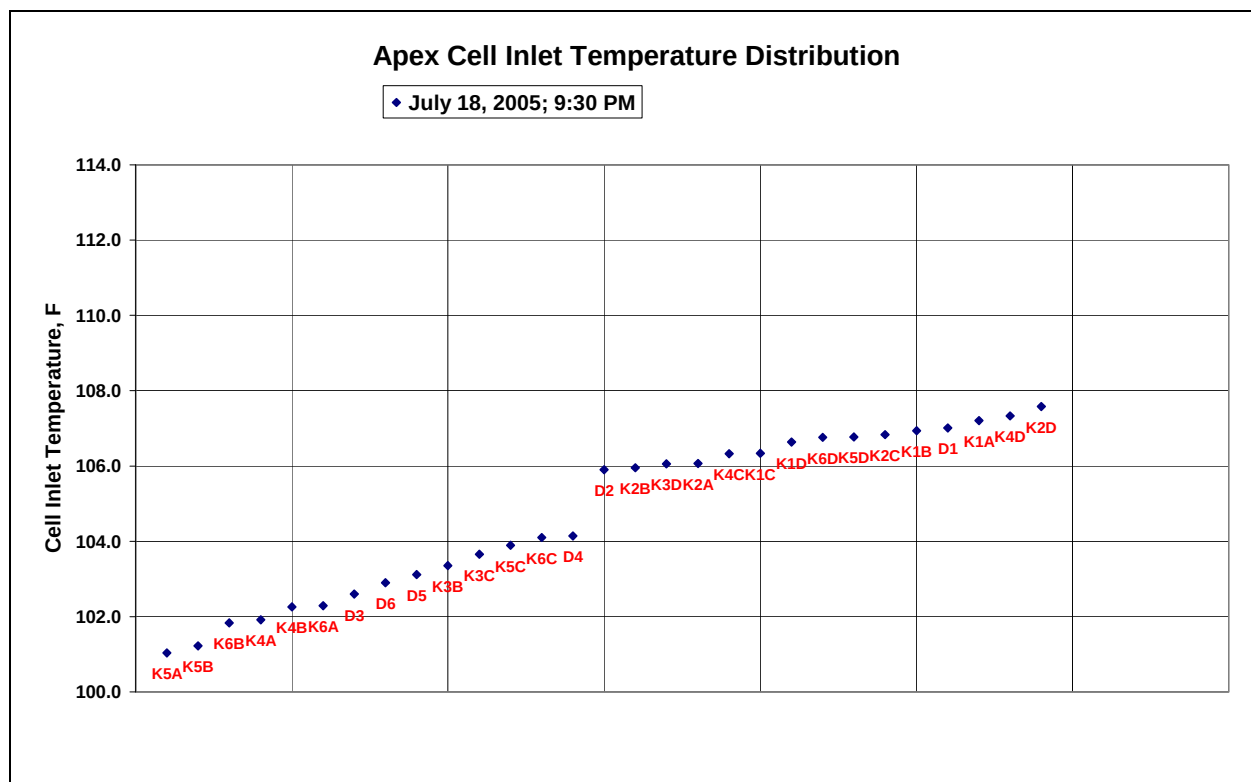


Figure 6-70. Cell inlet temperature distribution, July 18, 2005, 9:30 p.m.

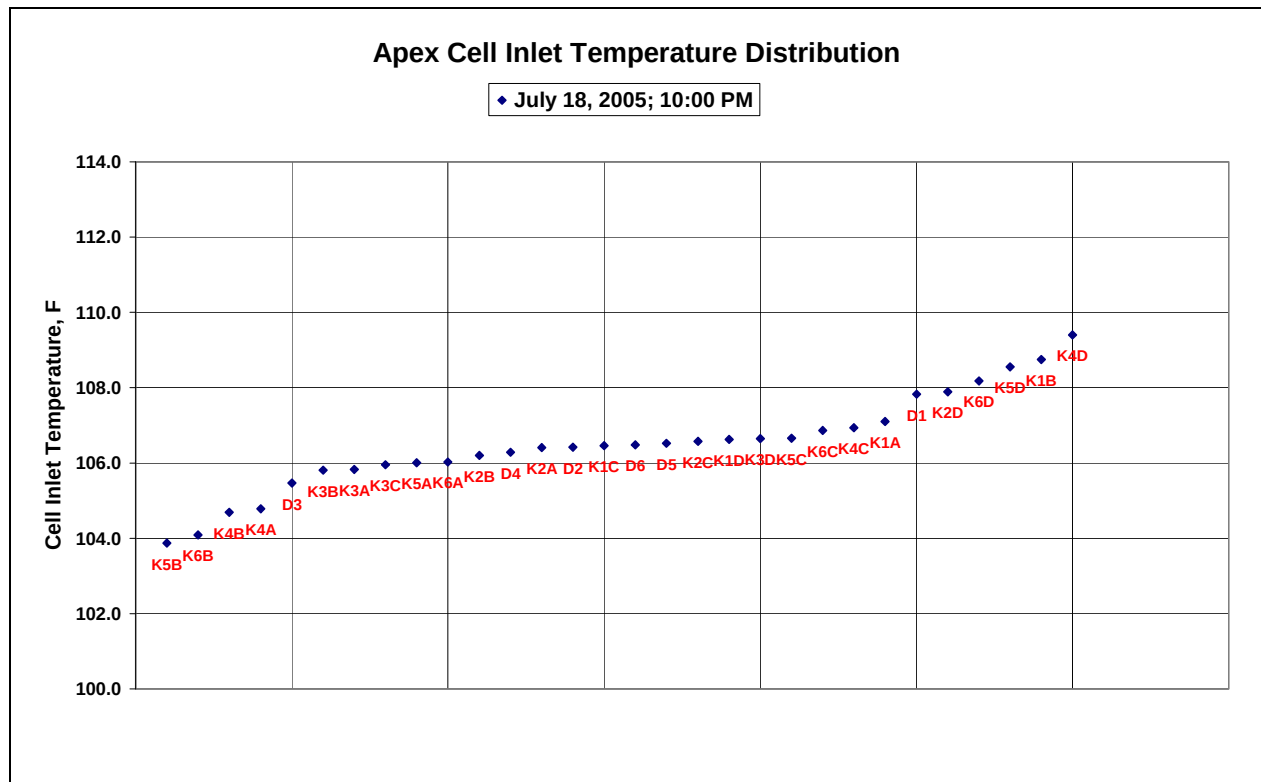


Figure 6-71. Cell inlet temperature distribution, July 18, 2005, 10:00 p.m.

6.5 Recirculation Tests—Front Range

The test period at Front Range ran from July 27, 2005, at 5:48 p.m. until August 4, 2005, at 6:43 a.m. Figure 6-4 displays the average recirculation for all times during the test period when all 40 fans were running at full speed. Figure 6-72 shows the wind speed and direction throughout the test period.

Prevailing winds are from the south and southeast and range from ~5 to ~15 mph. Higher winds frequently come from the northwest and north and are sometimes accompanied by squalls or storms characteristic of mountain weather patterns, often in the late afternoon or early evening.

Plant instrumentation records the ambient temperature at one location on the administration building and another under the ACC. Figure 6-73 shows excellent agreement between the two plant readings. Figure 6-74 compares the plant measurements under the ACC with the minimum cell inlet temperature measured with project test instrumentation.

During most of the test period, the agreement with the plant readings is excellent. Exceptions occur in the late afternoon on several days when the minimum cell inlet temperature continues to rise after the plant readings have peaked. There are also a few instances (around midnight on July 31 and again on August 4) when all the temperature readings exhibit rapid, unsteady changes. Some of these periods (perhaps all but this is not confirmed) are coincident with rain. It is likely that the test probes, mounted inside the cells, are protected from wetting by the rain

and record higher temperatures. In any case, the minimum cell inlet temperature will be used as the recirculation baseline for the Front Range tests.

Figure 6-75 plots all the data as average recirculation vs. wind speed. Although Front Range exhibits higher levels of recirculation compared to some of the other plants, the general relationship with wind speed is similar. The highest wind speeds (> 25 mph) exhibit low average recirculation, while the highest recirculation is associated with wind speeds below 10 mph.

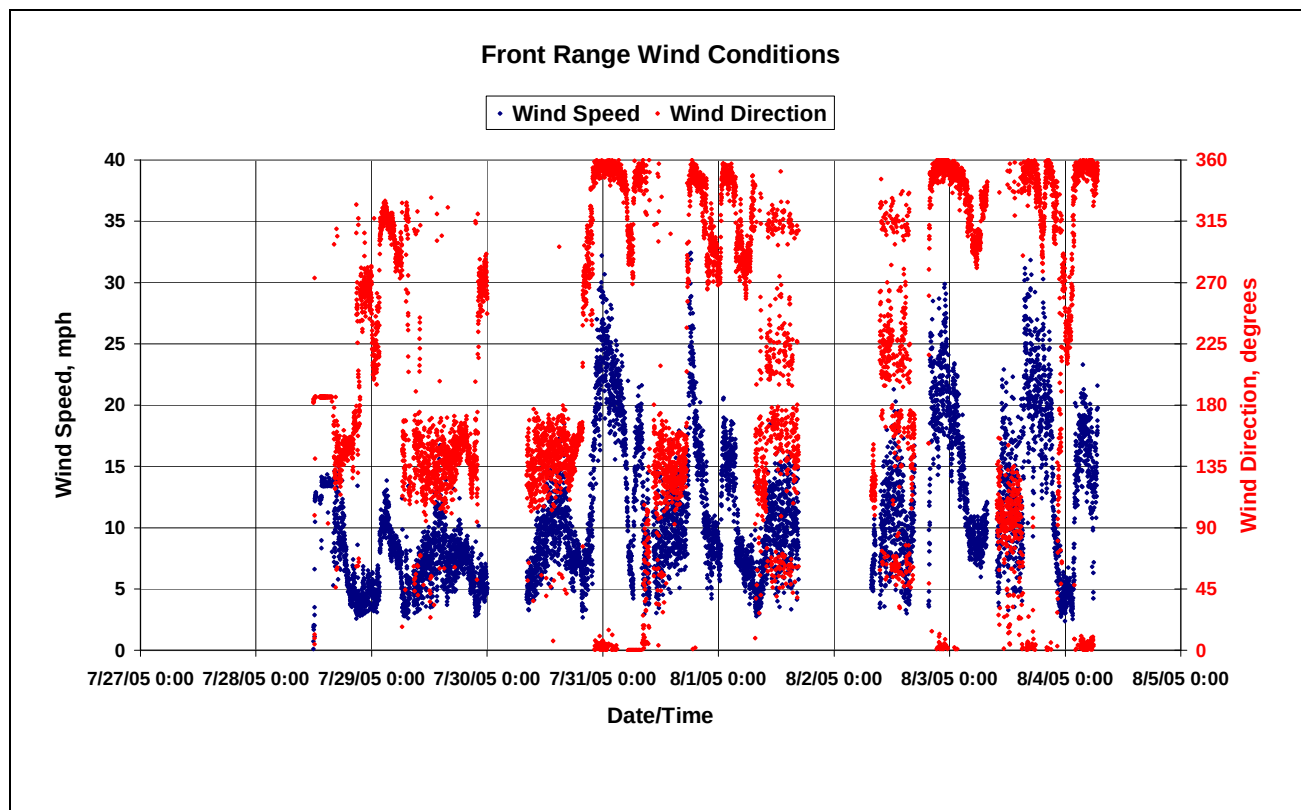


Figure 6-72. Front Range wind conditions during test period.

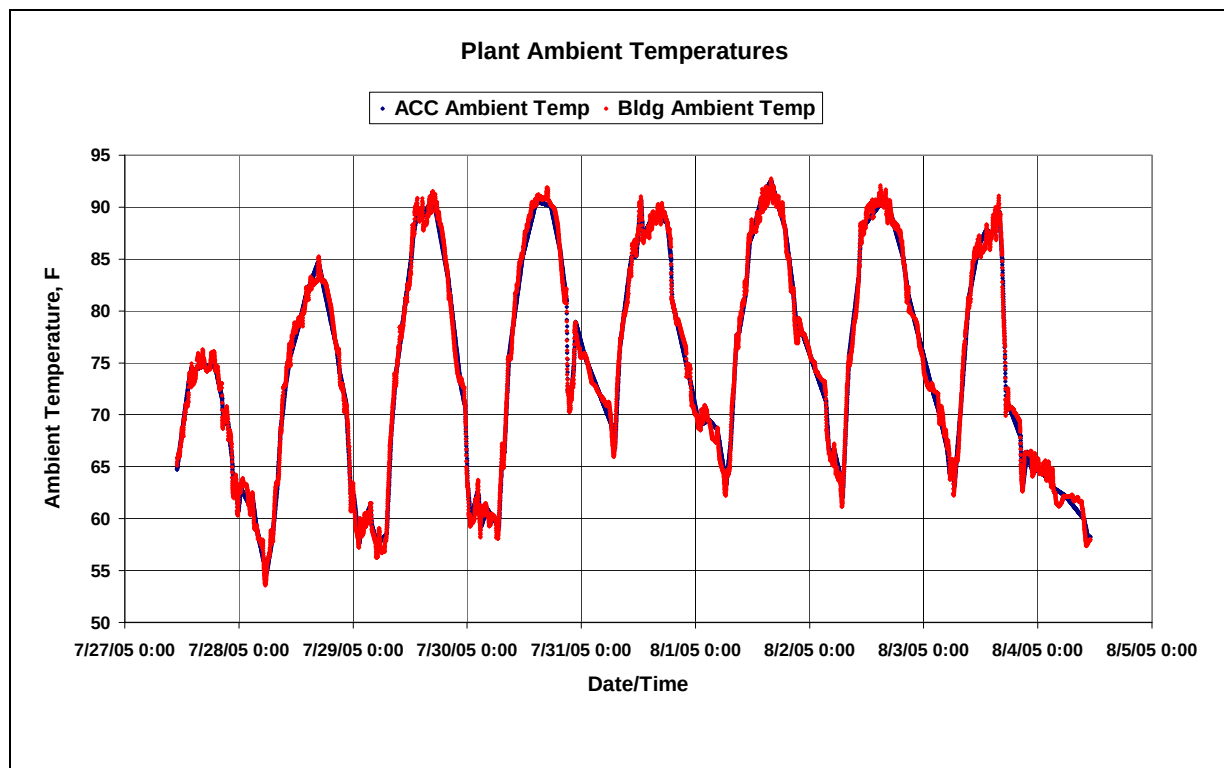


Figure 6-73. Front Range ambient temperature comparisons.

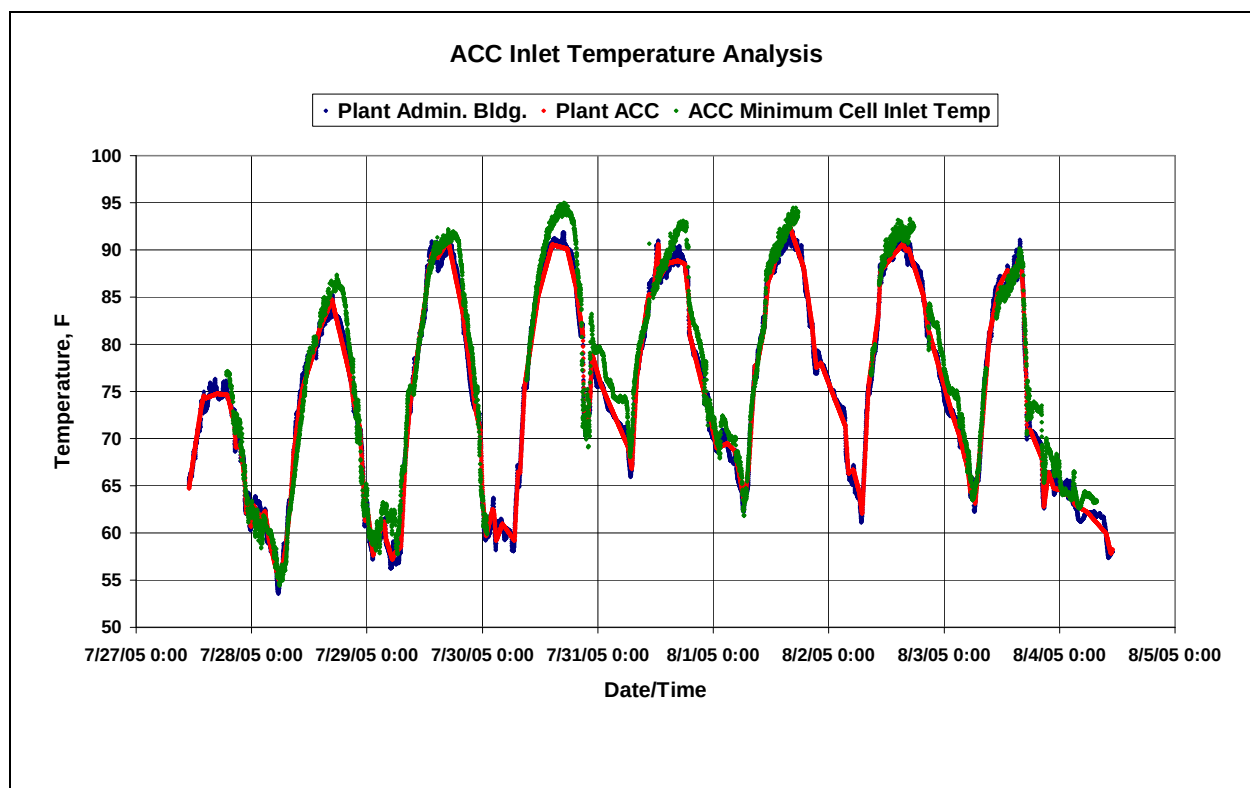


Figure 6-74. ACC inlet temperature analysis.

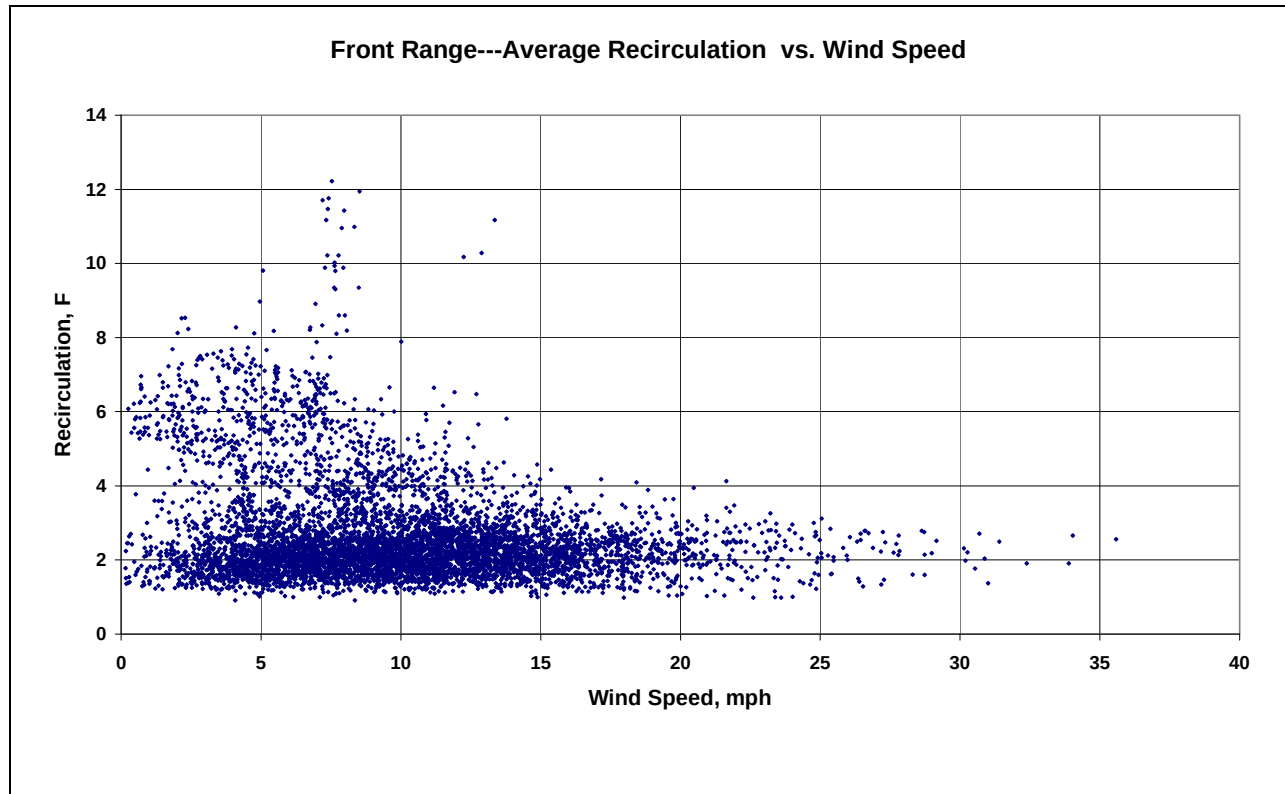


Figure 6-75. Front Range-average recirculation vs. wind speed.

Given the results presented in Figure 6-4, three periods will be examined in more detail:

- Late afternoon and evening of August 28—moderate to high recirculation with low winds.
- Late afternoon of July 30—high recirculation with low wind.
- Midnight of July 30/early morning July 31—high wind with low recirculation.

6.5.1 July 30, 2005; 5:00 p.m. to Midnight

Figure 6-76a and 6-76b display the average recirculation and the wind speed and direction for the same time period. The variation in recirculation is dramatic. Before approximately 6:15 p.m. and after approximately 10:30 p.m., the recirculation is low and reasonably constant between 2.0° and 2.5°F. Beginning at 6:30 p.m. the recirculation increases steadily to about 5.0° to 6.0°F over a one-hour period and then very rapidly to 12°F over the next 20 minutes between 7:30 and 7:50 p.m. It declines rapidly to ~6.0°F between 7:50 and 8:10 p.m., varies erratically between 4.0° and 8.0°F until about 9:20 p.m., drops rapidly to 1.0 ° to 2.0°F by 9:30, and finally stabilizes at 2.0°F from 10:45 p.m. until midnight.

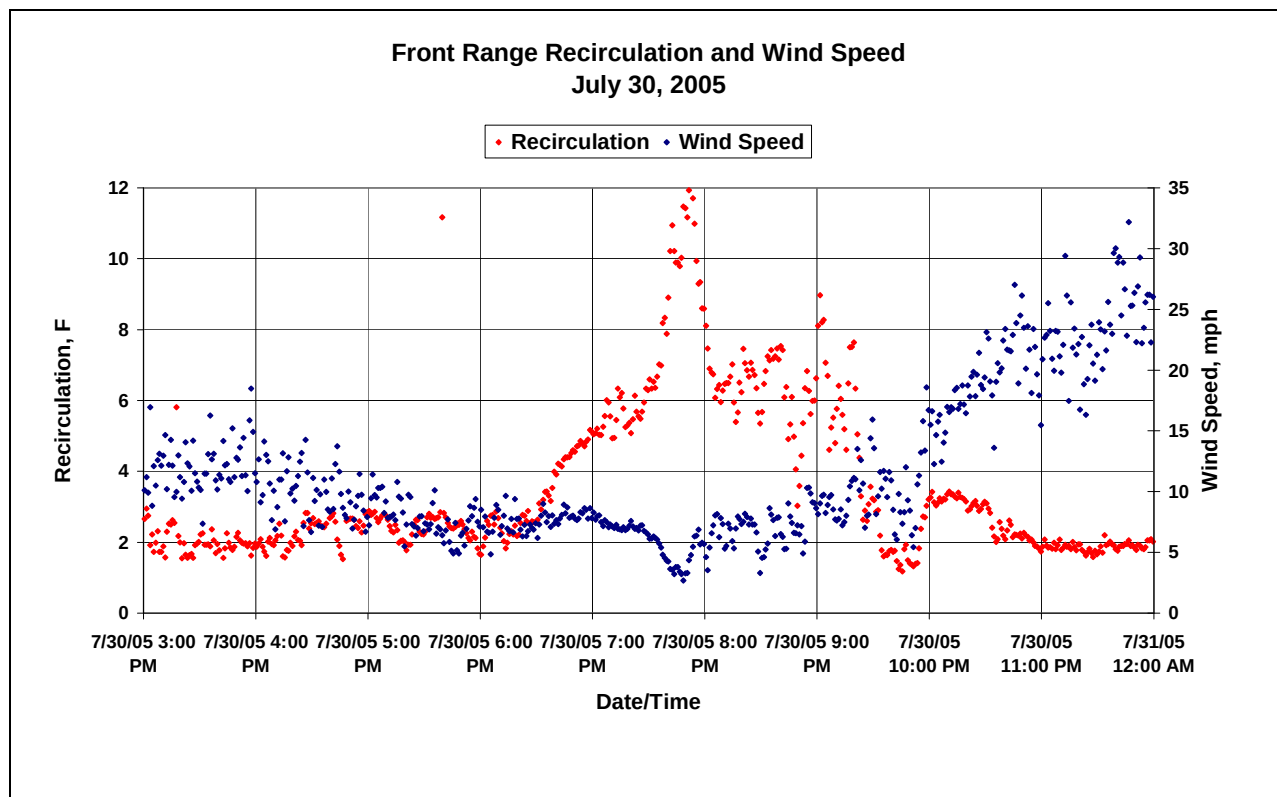


Figure 6-76a. Recirculation and wind speed, July 30, 2005, 5:00 p.m. to midnight.

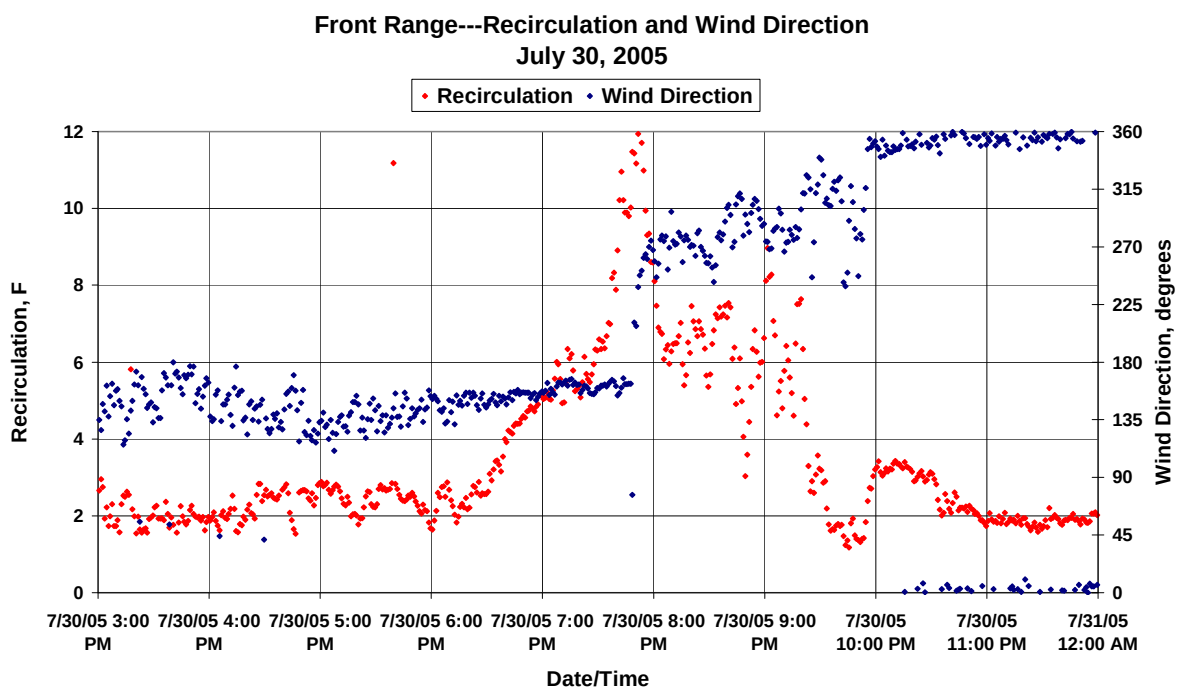


Figure 6-76b. Recirculation and wind direction, July 30, 2005; 5:00 p.m. to midnight.

It is noteworthy that:

- Wind *speed* effects, to the extent that there are any, are subtle and difficult to discern.
 - o The initial increase in recirculation at 6:30 p.m. is not related to any change in wind speed with the possible exception of a slight decrease in variability followed by a decrease until about 7:50 p.m.
 - o The sharp peak in recirculation (from ~7:30 p.m. ending at ~8:15 p.m.) is accompanied by a downward “peak” in wind velocity
 - o For the rest of the day the wind speed increases, reaching 25.0 to 30.0 mph at midnight, while the recirculation decreases, finally stabilizing at a low level.
- Wind *direction*, while exhibiting a major shift from southeasterly to northerly, appears to affect recirculation only during periods of rapid change or high variability.

Some insight into the reasons for the variation in recirculation may be obtained from Figure 6-77, which displays the minimum, average, and maximum cell inlet temperatures for the period.

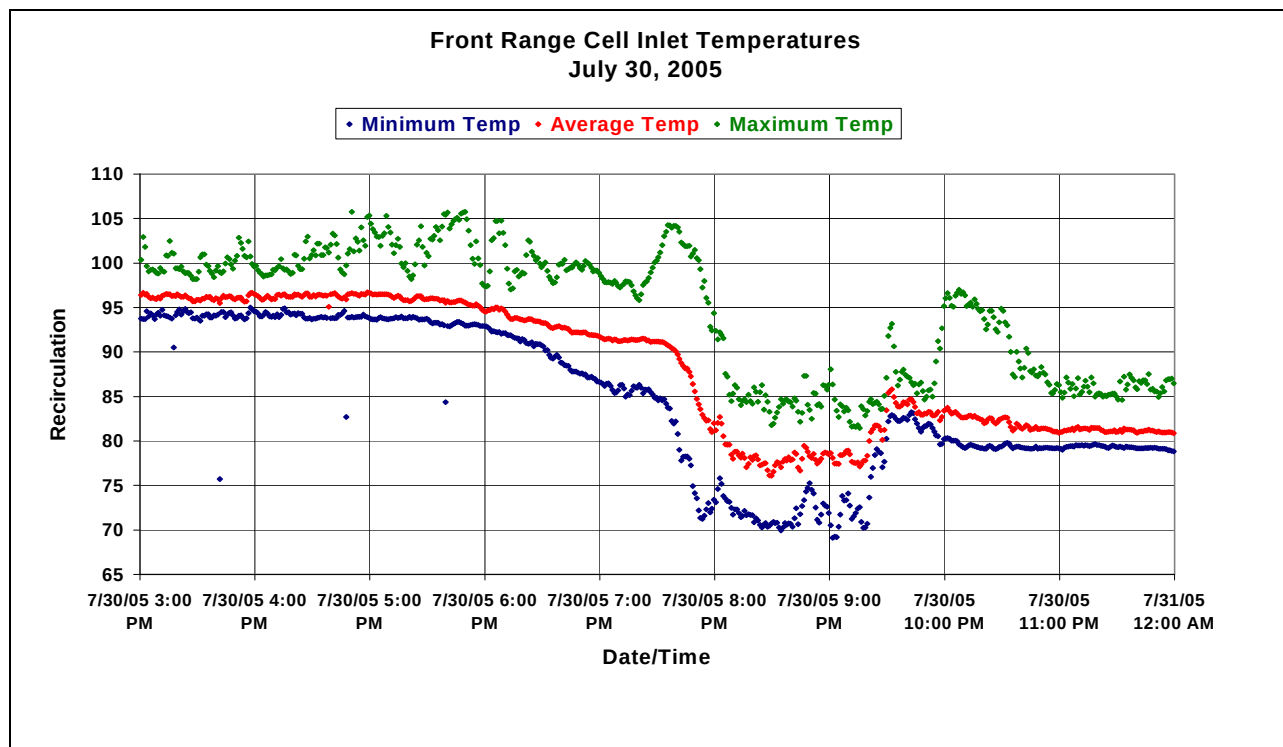


Figure 6-77. Front Range cell inlet temperatures, July 30, 2005.

During the period of initial increase in the average recirculation (from 6:30 to 7:30 p.m.), the minimum cell inlet temperature begins a slow decrease as the ambient temperature begins to

fall with the advent of a cold front coming into the area. During the same period, the average temperature declines as well but more slowly, as existing recirculation into the higher-temperature cells supports the average temperature.

At about 7:30 p.m., two things happen. The ambient and the minimum cell inlet temperature decline suddenly as the front arrives. At the same time, the wind direction shifts from southeasterly to westerly. As discussed at other sites, the previously established hot air plume may be forced back into the ACC with the observed increase in the maximum cell inlet temperature which maintains the average temperature and indicates a higher recirculation.

As the new wind direction stabilizes from the west, the influx of warm air into the ACC stops with the observed sudden decrease in the maximum cell inlet temperature and the associated drop in the average inlet temperature and the recirculation. This is followed by a period of erratically variable wind conditions and recirculation that gradually decreases as a quasi-equilibrium plume condition is reestablished.

Additional insight is gained from Figures 6-78 through 6-85 which display the distributions of all cell inlet temperatures at intervals though the late afternoon and evening of July 30, 2005. All of these plots are plotted on the same scale with the temperatures on the ordinates running from 70° to 100°F. The intent is to provide a clearer comparison of the shapes of the temperature distributions at the expense of some loss of precision in the display of the individual temperatures.

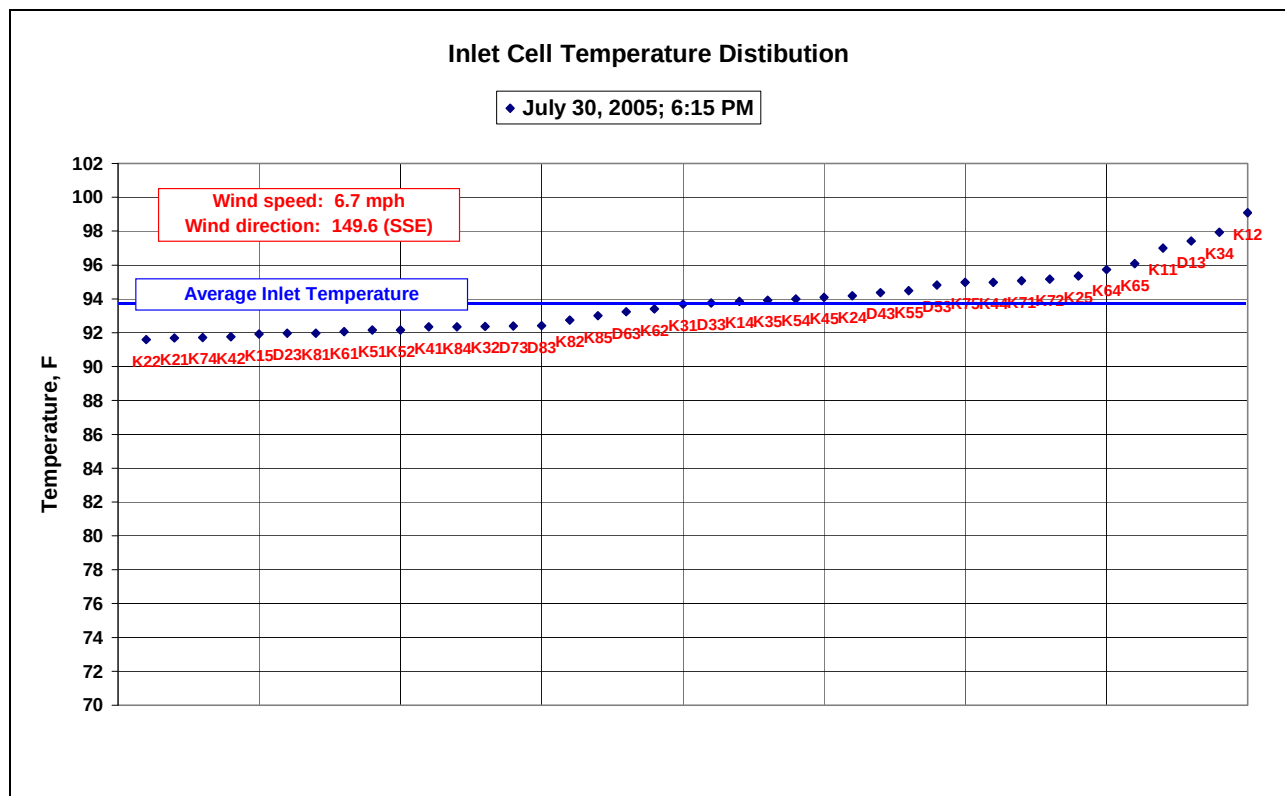


Figure 6-78. Front Range cell inlet temperatures, July 30, 2005, 6:15 p.m.

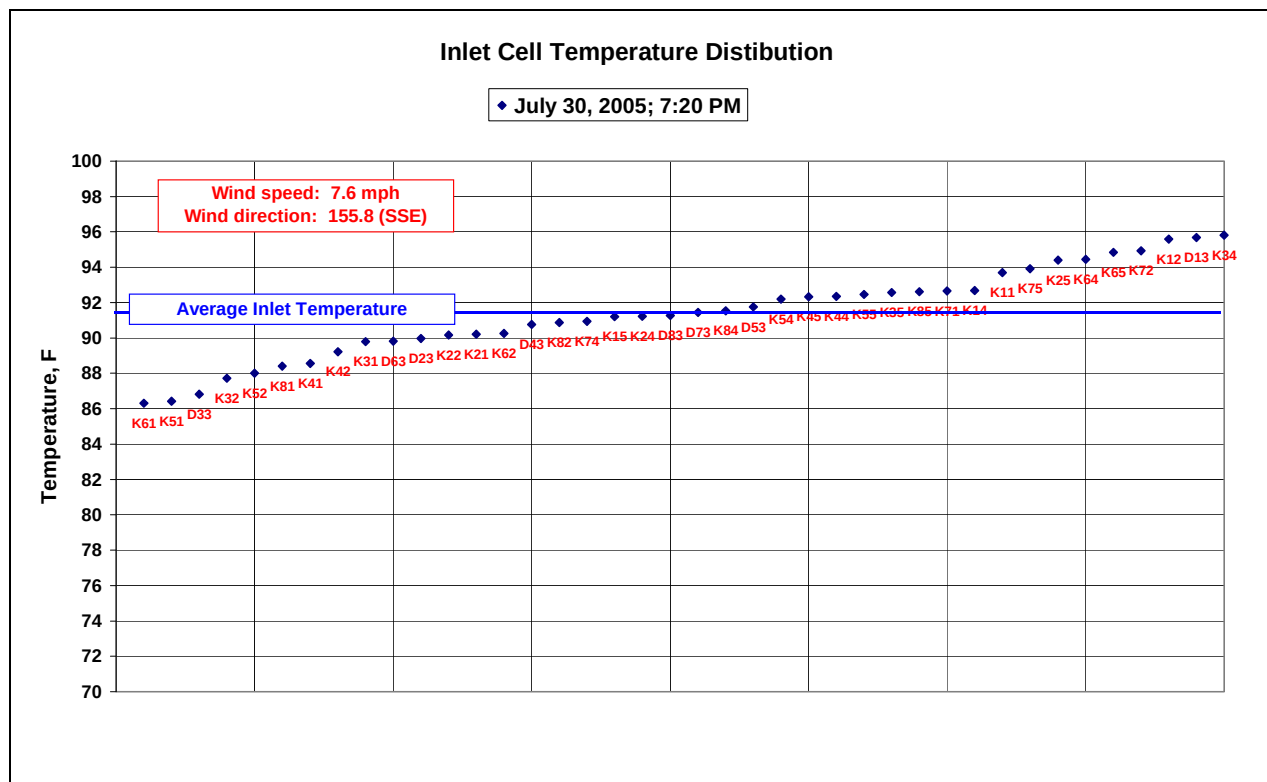


Figure 6-79. Front Range cell inlet temperatures, July 30, 2005, 7:20 p.m.

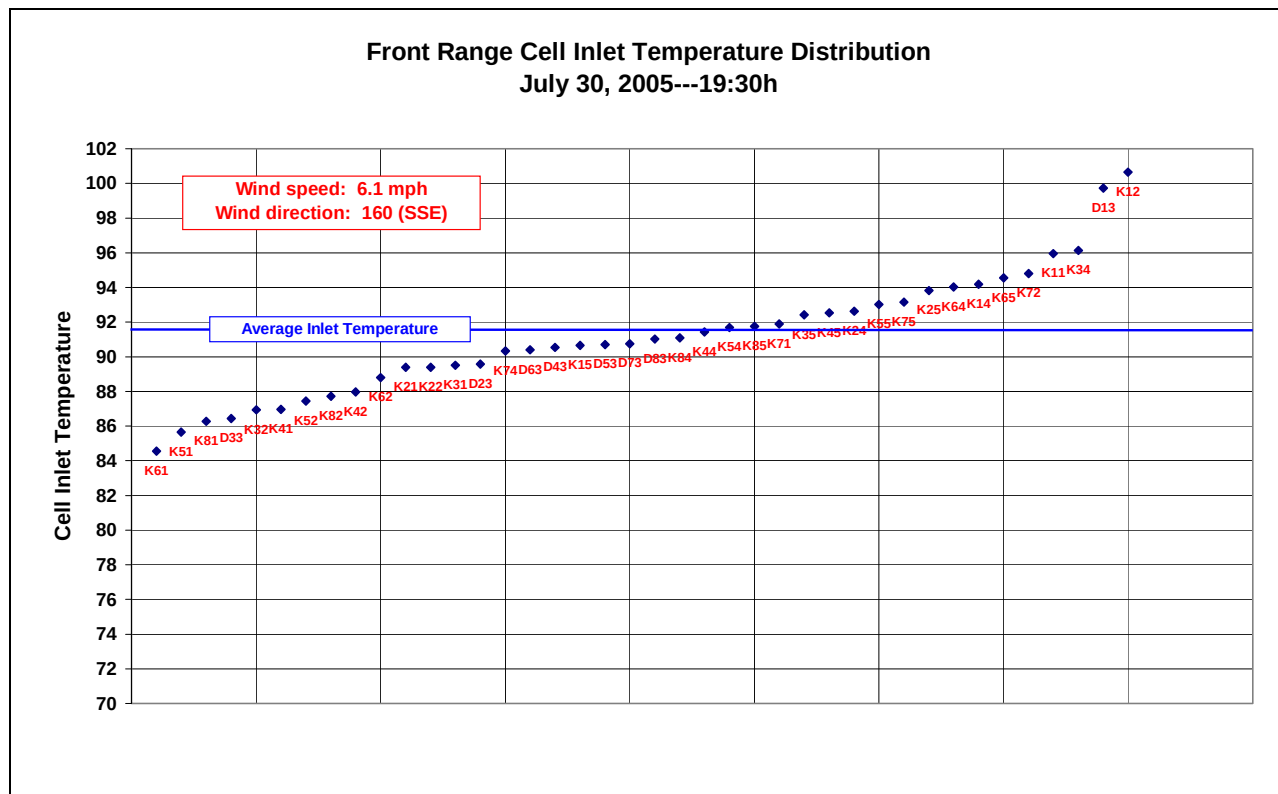


Figure 6-80. Front Range cell inlet temperatures, July 30, 2005, 7:30 p.m.

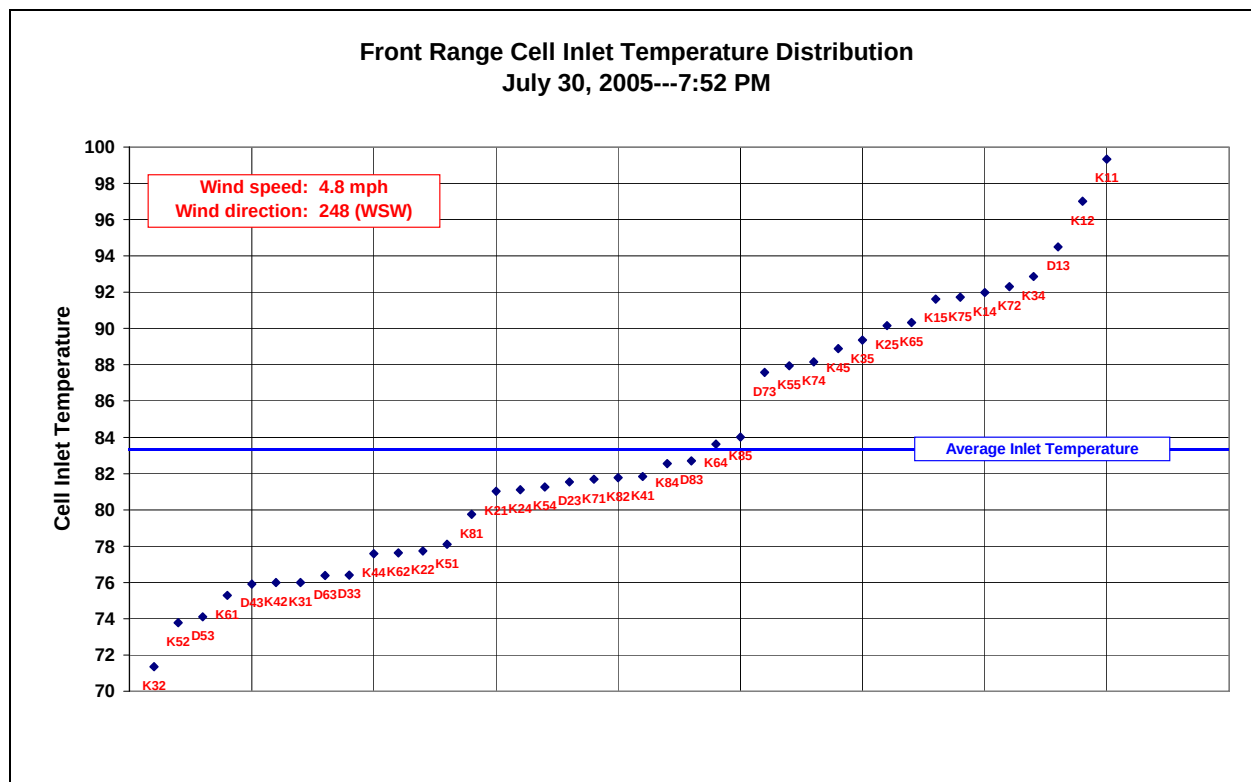


Figure 6-81. Front Range cell inlet temperatures, July 30, 2005, 7:50 p.m.

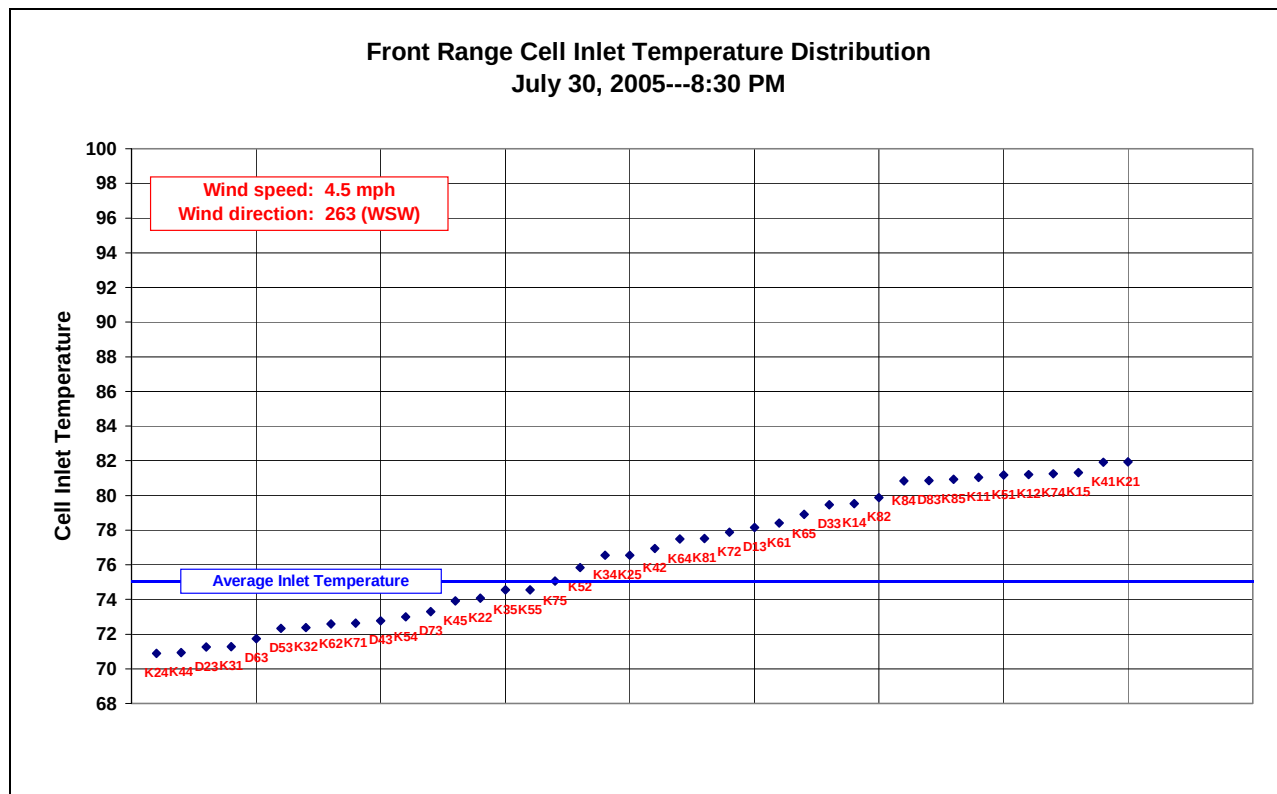


Figure 6-82 Front Range cell inlet temperatures, July 30, 2005, 8:30 p.m.

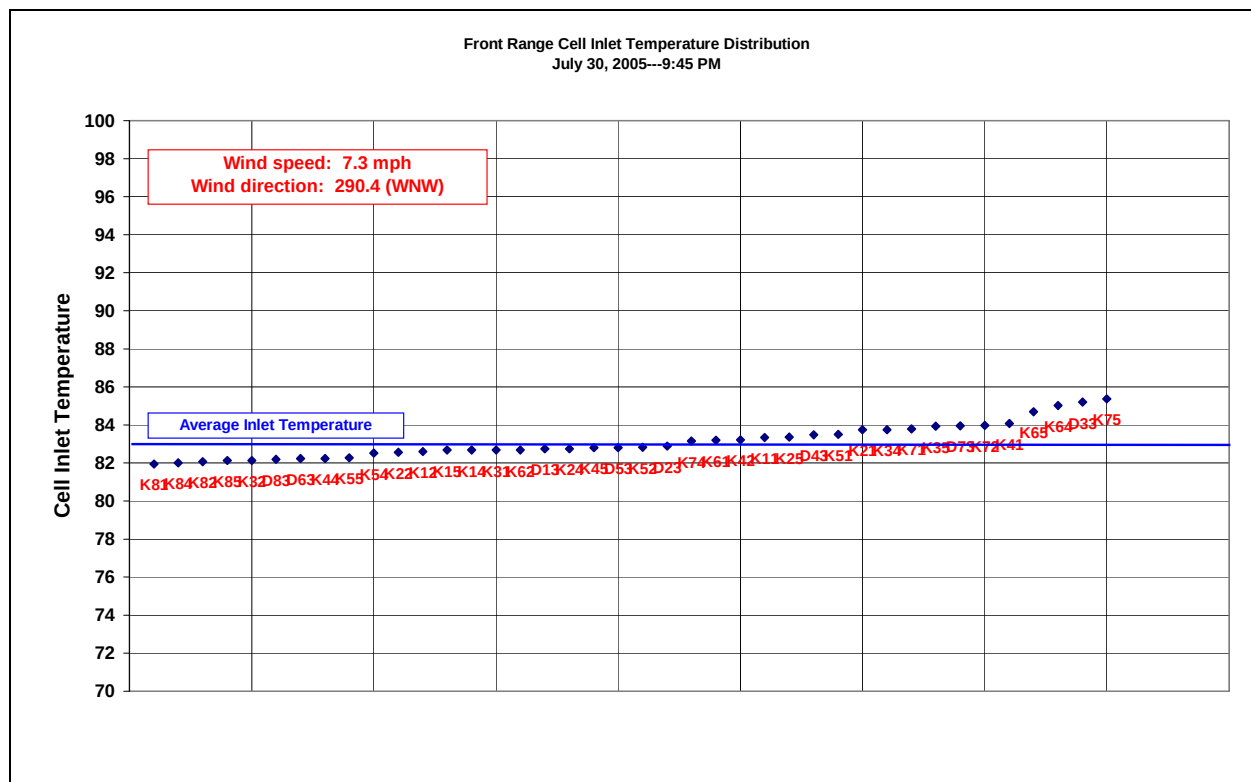


Figure 6-83. Front Range cell inlet temperatures, July 30, 2005, 9:45 p.m.

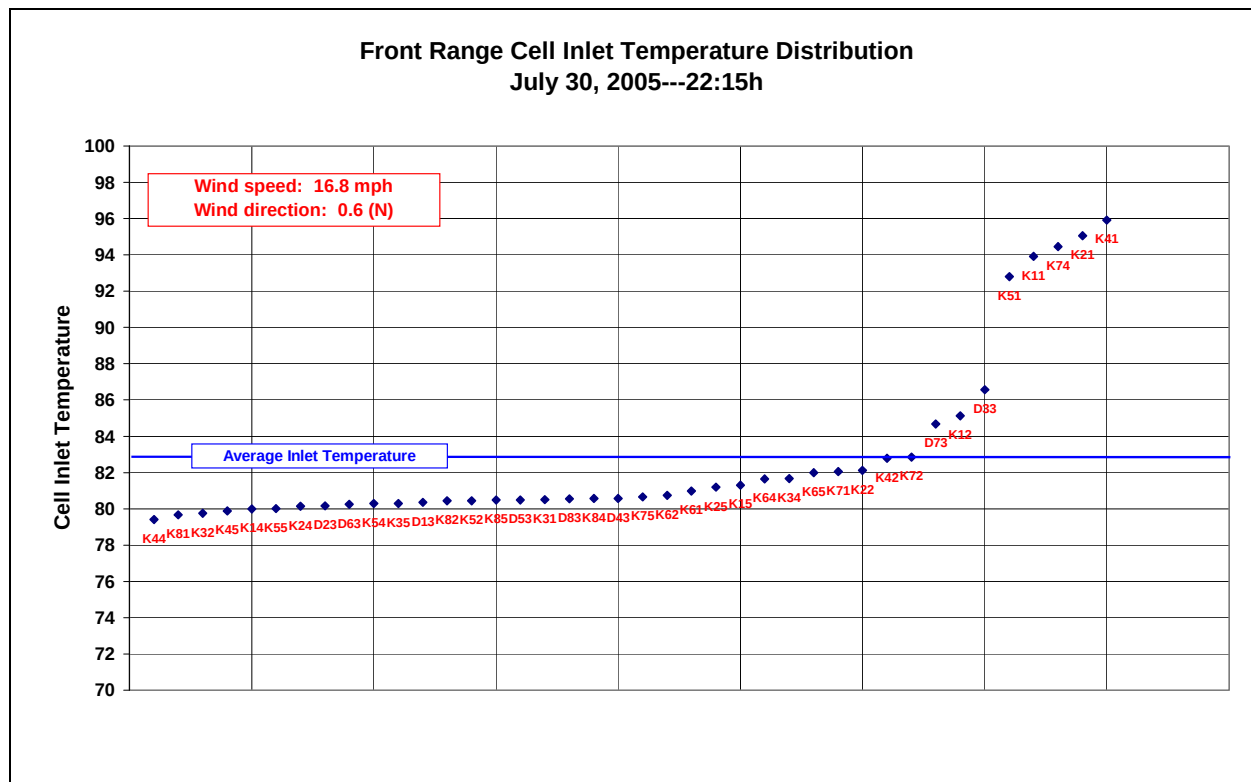


Figure 6-84. Front Range cell inlet temperatures, July 30, 2005, 10:15 p.m.

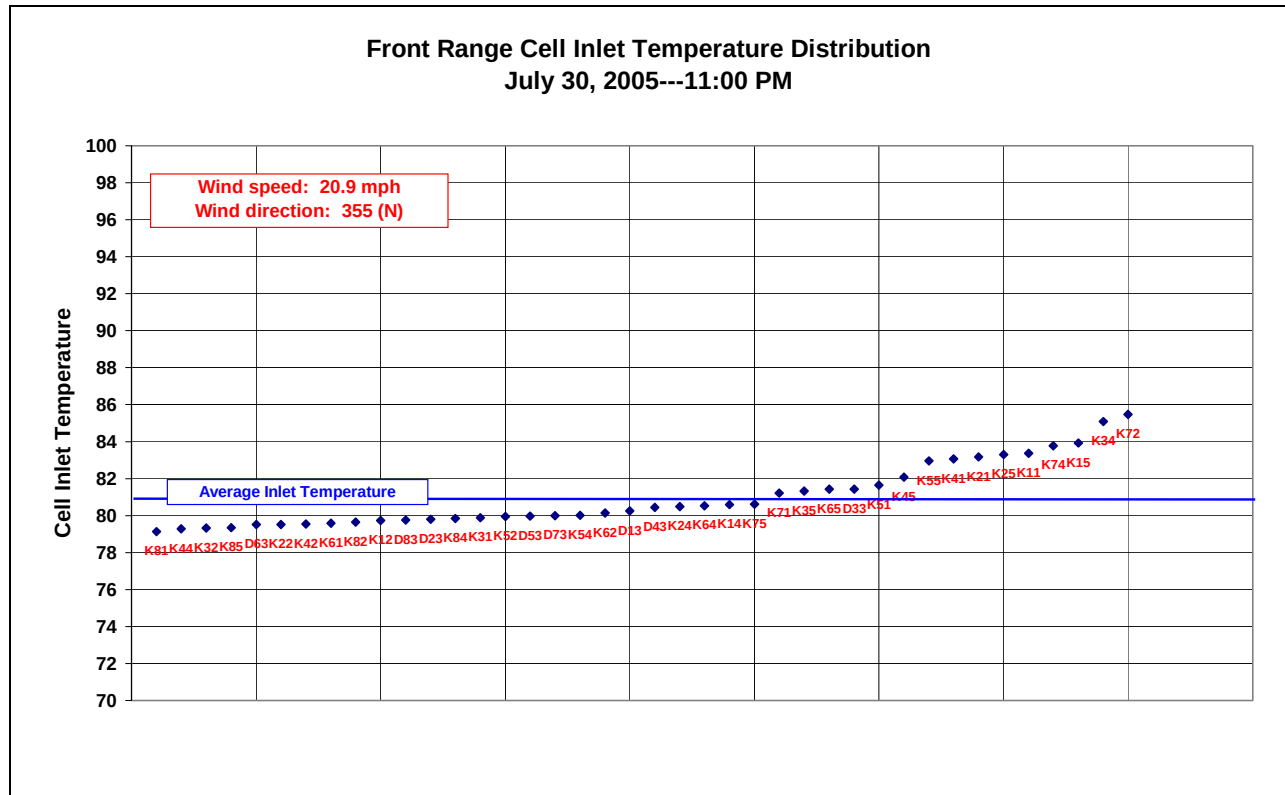


Figure 6-85. Front Range cell inlet temperatures, July 30, 2005, 11:00 p.m.

Conditions progress as follows:

- 6:15 p.m. (Figure 6-78): This time is at the beginning of increasing verge recirculation after a long period of relatively constant, low recirculation. The temperature distribution is essentially uniform across the entire ACC. Modest recirculation is indicated for Cells K12, D13, and K11, which are all in the north street and would be expected to experience the most recirculation from winds from the south quadrant. The modest elevation in inlet temperature in Cell K34 is not understood.
- 7:20 p.m. (Figure 6-79): This time is at the end of approximately one hour of steady increase in the average recirculation. During that time, both the wind speed and direction have remained essentially constant, but the ambient temperature has fallen by about 6°F. The shape of the inlet temperature distribution has changed. While it is still uniform over much of the ACC, there is more of a “drop-off” in the lower-temperature cells and less of a “recirculation rise” region in the high-temperature cells. The seven highest-temperature cells are the same seven as were the warmest at 6:15 p.m. The location of the coolest cells has changed, with a concentration of the lowest inlet temperatures near the center of the eastern rows (Rows 1 and 2). That these cells were the first to respond to the falling ambient temperature is not understood but may be related to the fact that the eastern rows are upwind to winds from the SSE and that the

cells at the south end of these rows (Streets 6 and 7) are partially blocked by the condensate return equipment under the ACC.

- 7:30 p.m. (Figure 6-80): At this time, the rapid increase in recirculation is just beginning. The wind speed and direction are still unchanged, but the decline in ambient temperature has temporarily stabilized. With the exception of the two hottest cells (K12 and D13), the general distribution of inlet cell temperatures is essentially unchanged from the earlier pattern at 7:20 p.m. (Figure 6-79). The abrupt increase in the inlet temperatures of Cells K12 and D13 signals the beginning of a recirculating plume pattern with the greatest effect on cells in the north street. As yet it has little effect on the average recirculation.
- 7:50 p.m. (Figure 6-81): From 7:30 p.m. to 7:50 p.m., significant changes occur in ambient temperature, wind direction, and inlet cell temperature distribution pattern. With the arrival of the storm front, the ambient temperature drops by about 15°F and the wind shifts from SSE to WSW. The inlet temperature distribution shows a substantial change across the ACC and significant recirculation in the four or five hottest cells. The hottest cells remain at the north end (K11, K12, and D13). Cell K34 remains high. The coldest cells shift from the southeast edge (Row 1) to a cluster of interior cells in Rows 2 and 3 from Streets 3 to 5. The rapid, significant increase in average recirculation is caused by a precipitous drop in the minimum and lower cell temperatures, while the hottest 15 cells are relatively unaffected, due perhaps to a persistent plume pattern established by the earlier wind patterns.
- 8:30 p.m. (Figure 6-82): Following the reestablishment of a relatively steady wind pattern (low speed from the WSW), the cell inlet temperature distribution pattern returns to one similar to the early evening (6:15 p.m.) but at a significantly lower ambient temperature.
- 9:45 p.m. (Figure 6-83): From 8:30 p.m. to 9:45 p.m. the squall subsides, the ambient temperature increases rapidly from 70°F to about 82°F, the wind shifts slightly to the north (WNW), and the wind speed increases somewhat erratically. The cell inlet temperature distribution remains uniform with all cells within +/-1.5°F.
- 10:15 p.m. (Figure 6-84): By 10:15 p.m. the wind has shifted further to northerly and the wind speed increased to over 16 mph. Some recirculation is established in five cells. The affected cells have shifted from the north street to the east row (Cells K41, K21, K11, and K51), with an anomalously high inlet temperature at Cell K74. The four cells are not sufficient to affect the average recirculation, which increases only slightly and remains at or below 3.5°F.
- 11:00 p.m. (Figure 6-85): By 11:00 p.m., steady winds from the north—although at increased speeds of over 20 mph—show a stabilized pattern similar to the early evening with only modest recirculation to the hottest cells and an average recirculation of under 2 °F.

6.6 Recirculation Tests—Black Hills

The test period at the Wygen plant ran from August 9, 2005, at 10:16 a.m. until August 18, 2005, at 9:38 a.m. Figure 6-86 displays the average recirculation for the entire test period. All of the fans were operating continuously the entire time, although they may not have been always at full speed. The level of recirculation is generally low between 0.5° and 1.5°F for all conditions, with rare excursions above 3°F.

Figure 6-87 shows the wind speed and direction for the corresponding period. The periods of greatest recirculation (just before and after midnight on the 15th and 16th and later in the days of the 9th and 10th) do not coincide with the times of highest wind velocity or with any particular wind direction, although they seem to be associated primarily with winds from the northwest.

Figure 6-88 plots wind speed vs. wind direction and shows a tri-modal wind pattern where the strongest winds during that week came from the south (~ 180°), the northwest (~290° to ~350°), or the southeast (~110° to ~135°).

Plots of recirculation vs. wind speed (Figure 6-89) and wind direction (Figure 6-90) indicate that maximum recirculation occurs at low to moderate wind speeds (~2 to 8 mph) from the northwest (~210° to ~295°).

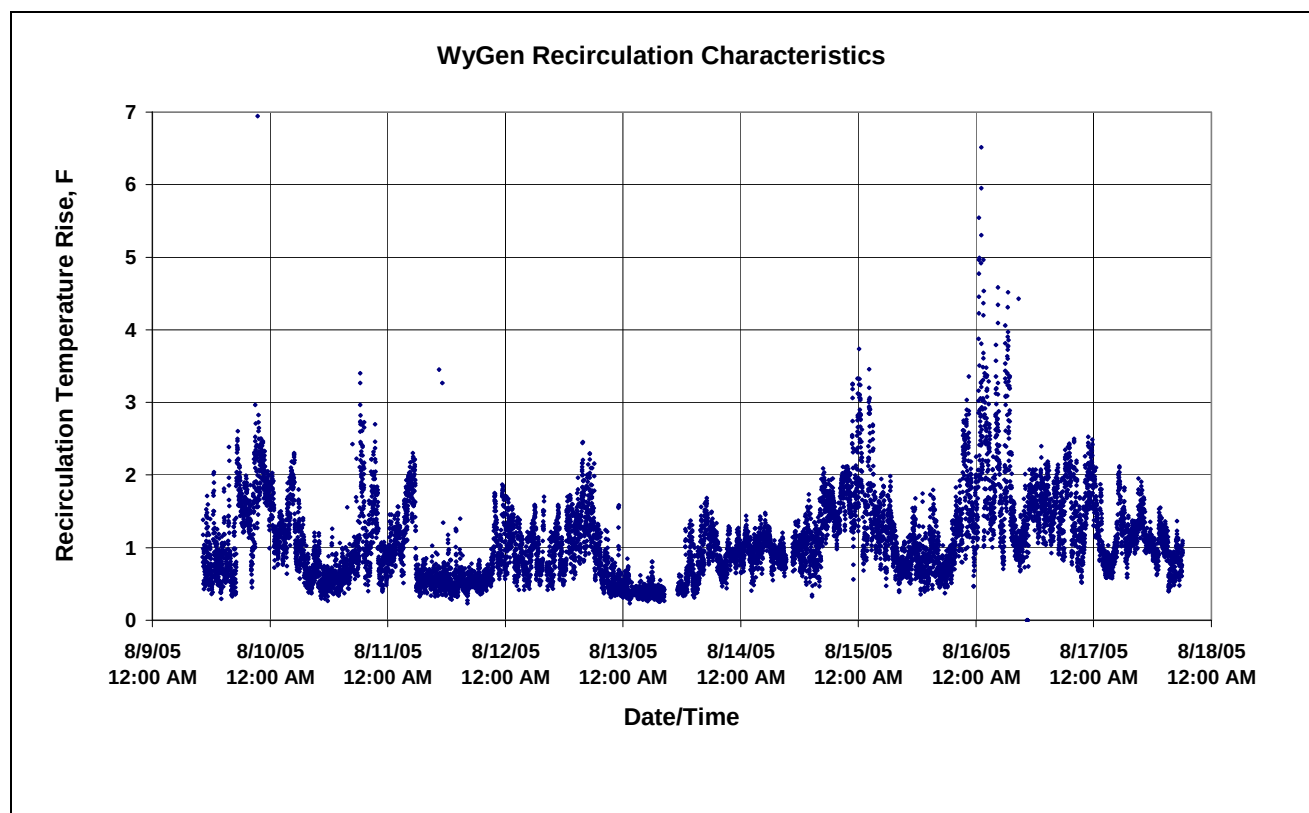


Figure 6-86. Wygen recirculation—full test period.

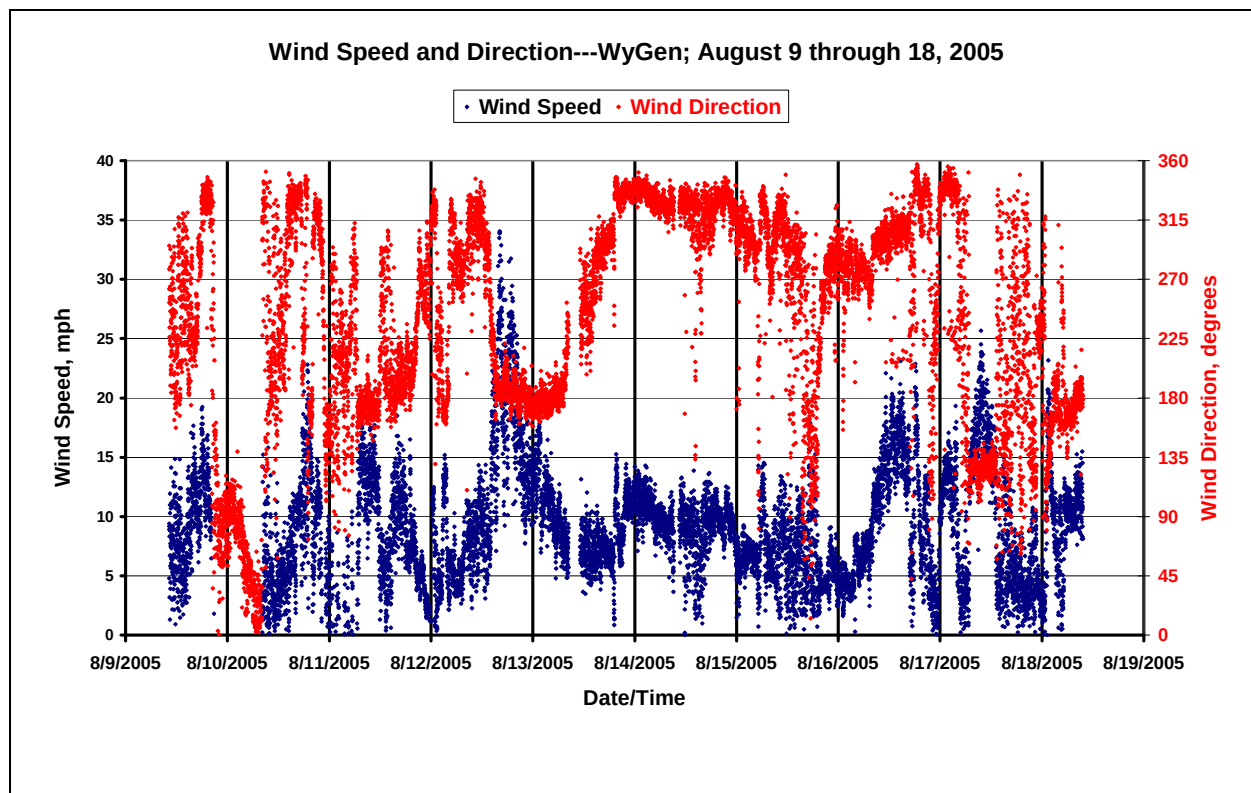


Figure 6-87. Black Hills wind conditions—full test period.

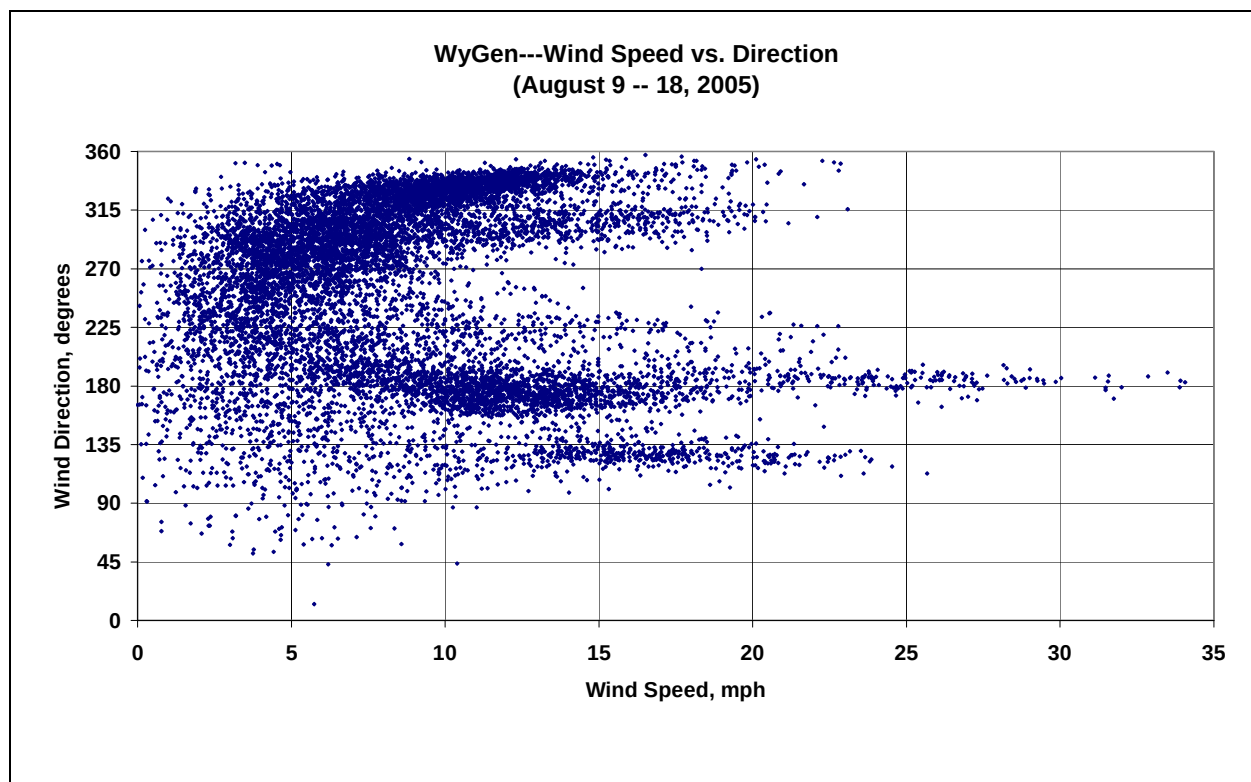


Figure 6-88. Black Hills wind speed vs. direction—full test period.

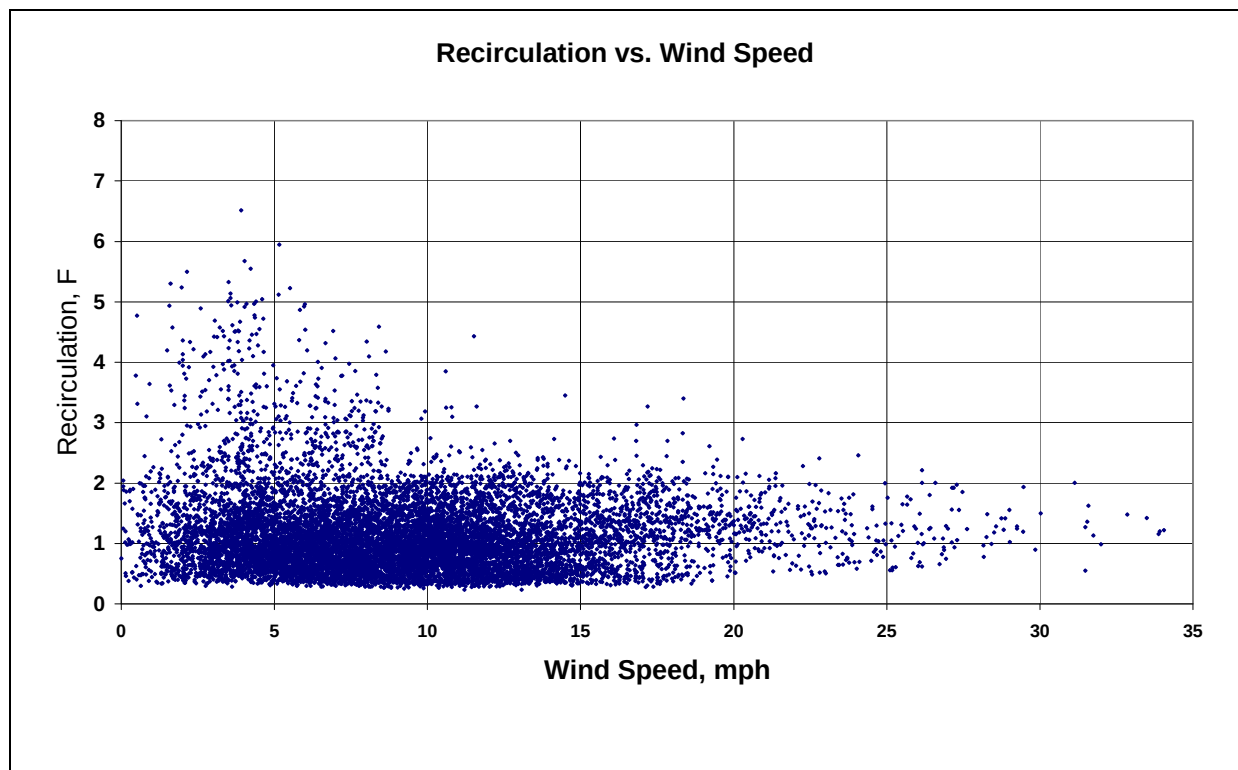


Figure 6-89. Wygen recirculation vs. wind speed.

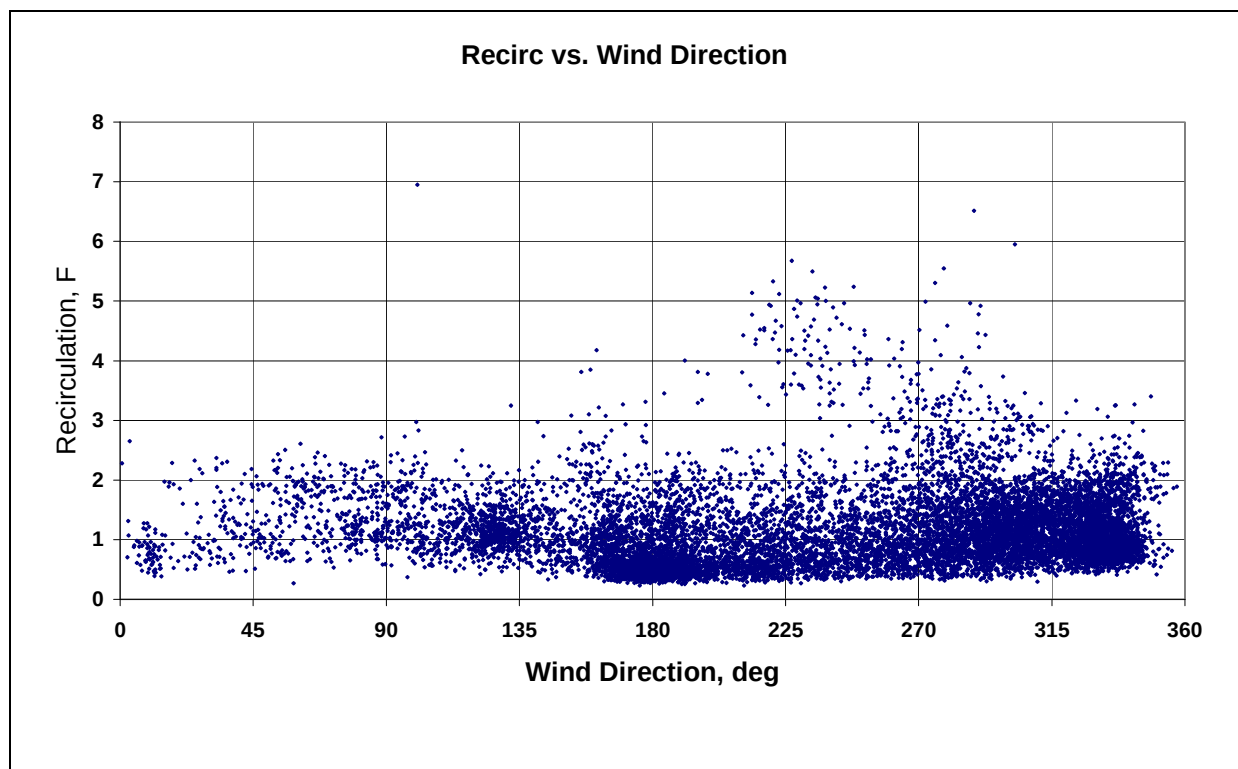


Figure 6-90. Wygen recirculation vs. wind direction.

This result is unexpected and may be investigated further by a detailed look at the period of greatest recirculation from 6:00 p.m. on August 15 to 10:00 a.m. on August 16.

6.6.1 August 15, 6:00 p.m. to August 16, 10:00 a.m.

Figures 6-91 and 6-92 show the average recirculation and the concurrent wind speed (6-91) and wind direction (6-92). There are three times during this period where significant variations in recirculation occur; after 9:00 p.m. on August 15, between 11:00 p.m. on August 15 and 3:00 a.m. on August 16, and just after 6:00 a.m. on August 16. From about 7:30 a.m. until noon on August 16, the wind speed steadily increases from below 5 mph to nearly 20 mph. At none of these times is there any consistent, discernible effect of wind speed on recirculation. A similar conclusion can be drawn from Figure 6-92 for the influence of wind direction.

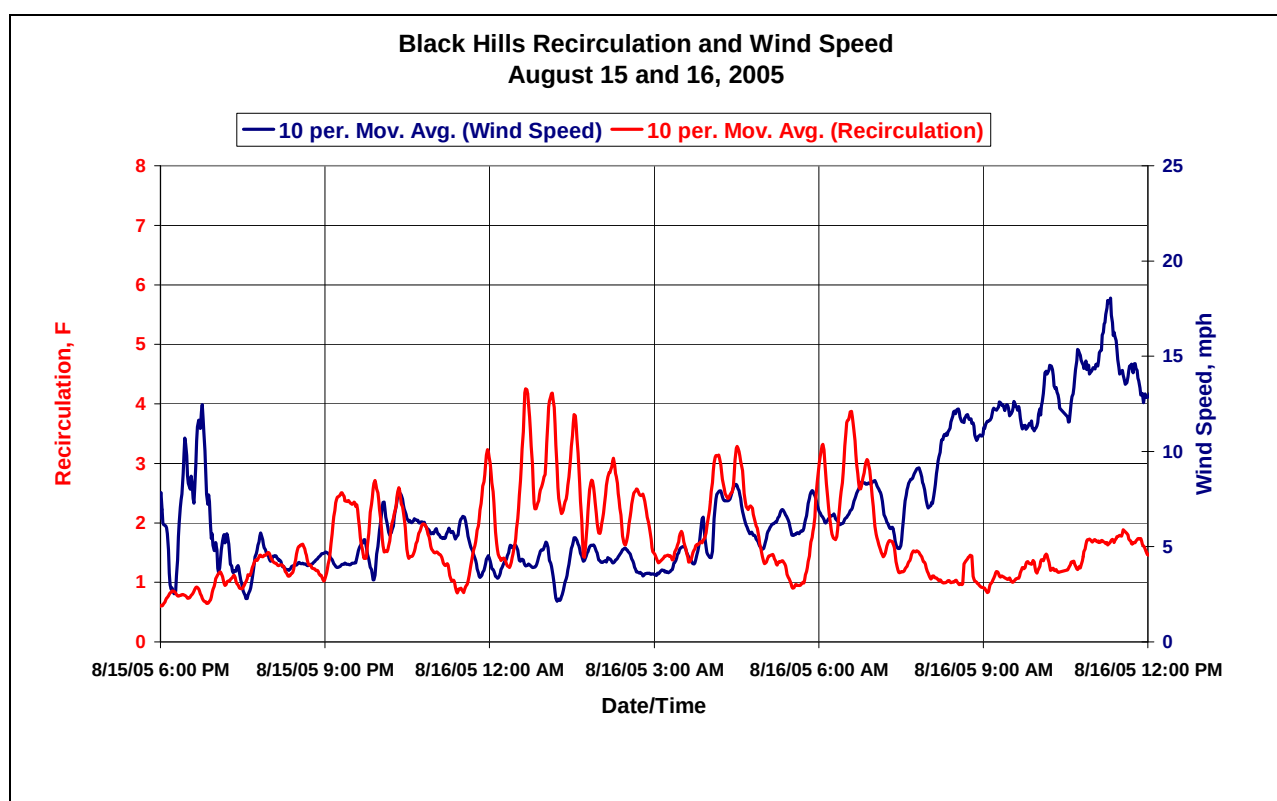


Figure 6-91. Black Hills recirculation and wind speed, August 15 and 16.

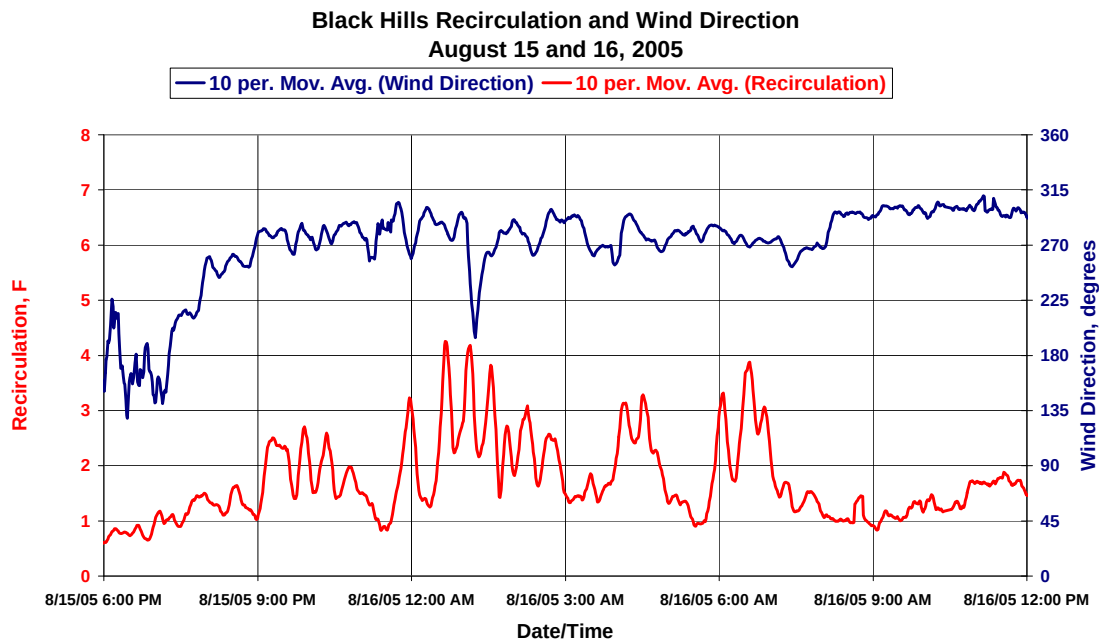


Figure 6-92. Wygen recirculation and wind direction, August 15 and 16.

6.6.2 August 12, 12:00 a.m. to August 13, 12:00 a.m.

Similar observations are displayed in Figures 6-93 and 6-94 for August 12.

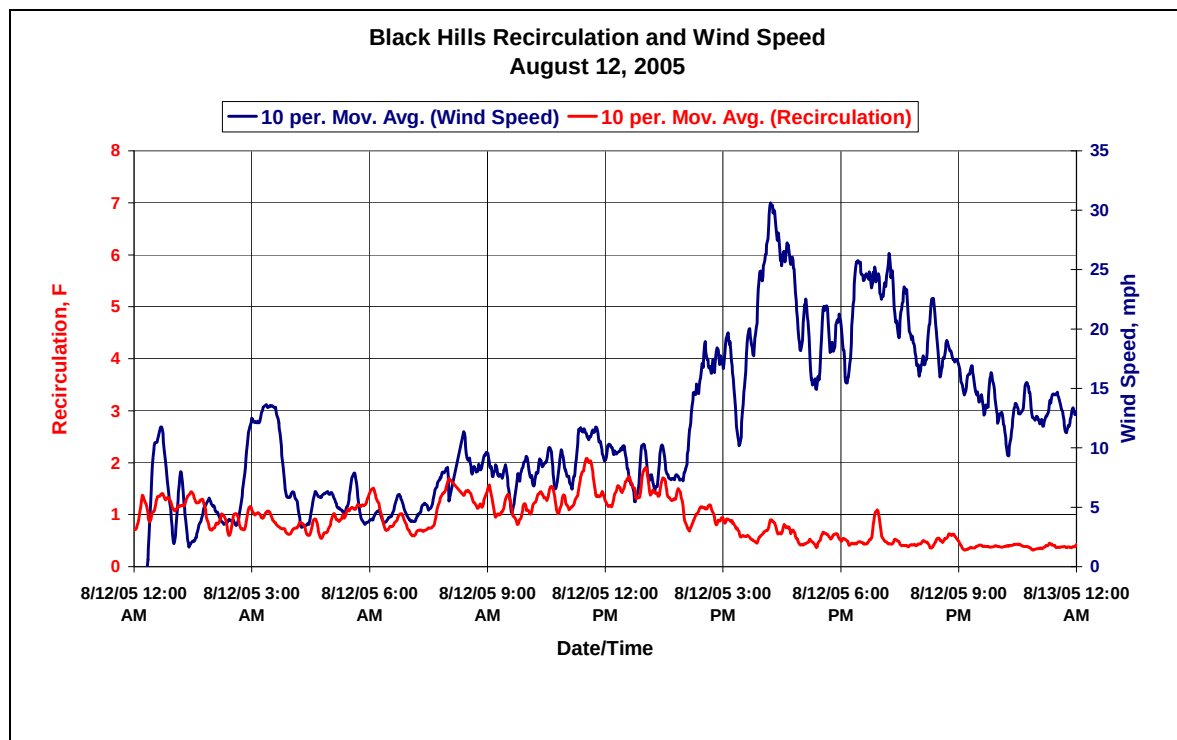


Figure 6-93. Wygen recirculation and wind speed, August 12.

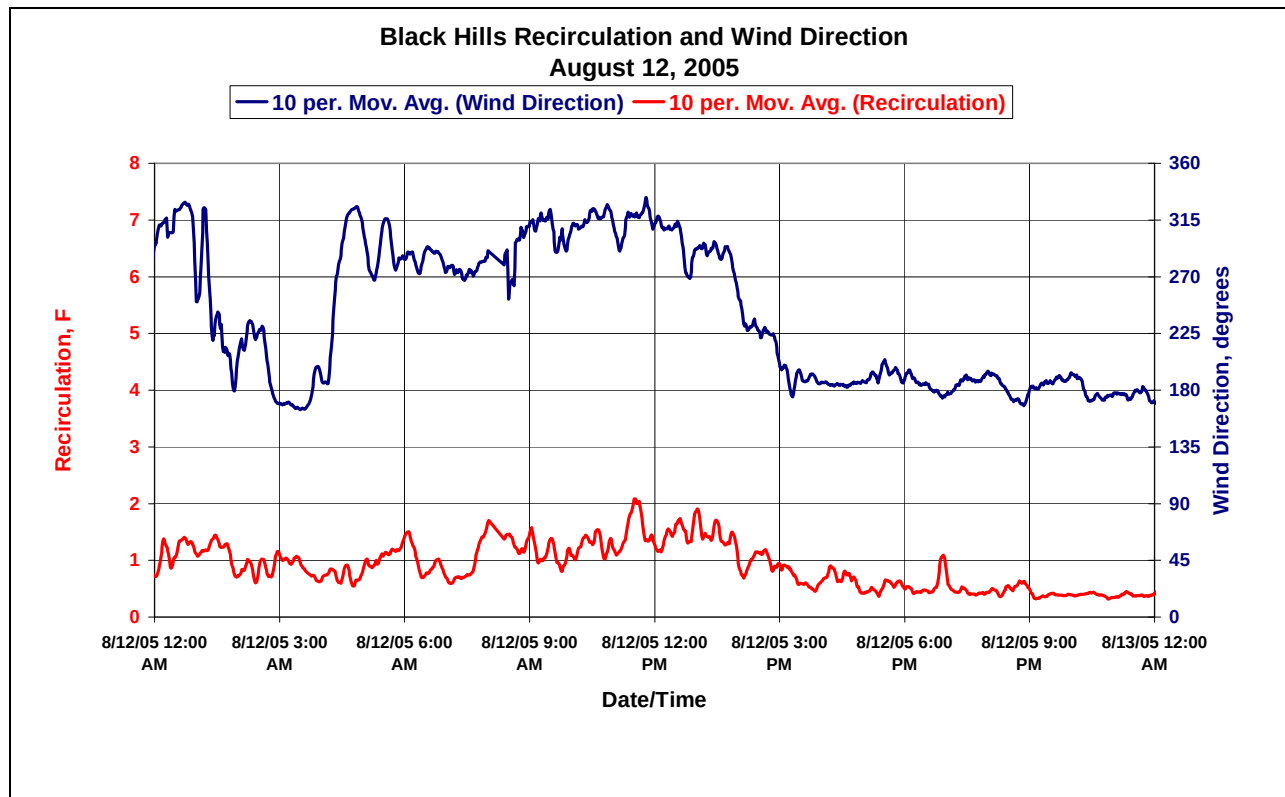


Figure 6-94. Wygen recirculation and wind direction, August 12.

A more fine-structured examination of the period from 7:30 a.m. to 1:30 p.m. on August 12 in Figure 6-95 shows some slight correlation between variations in recirculation (all between 0.75° and 2.0°F) and wind speed (all between 5.0 and 10.0 mph). Figure 6-96 shows no discernible effect of wind direction.

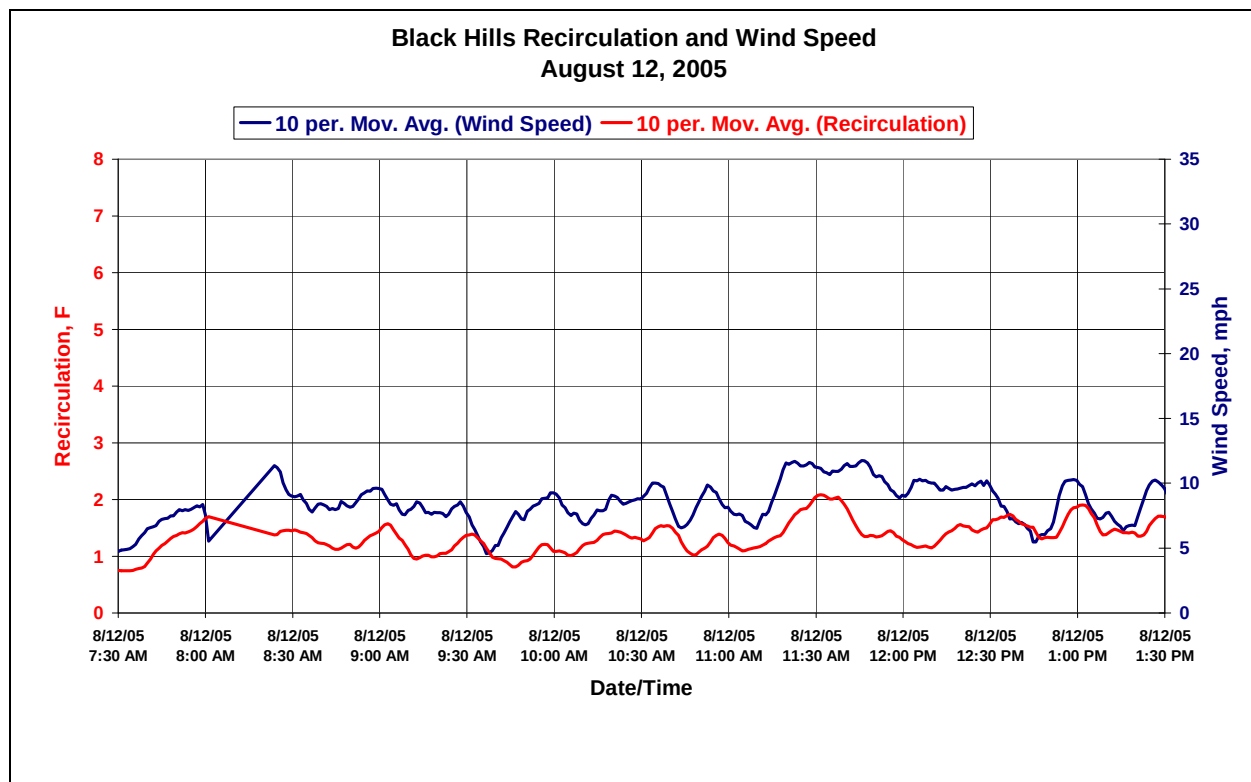


Figure 6-95. Wygen recirculation and wind speed, August 12, 7:30 a.m. to 1:30 p.m.

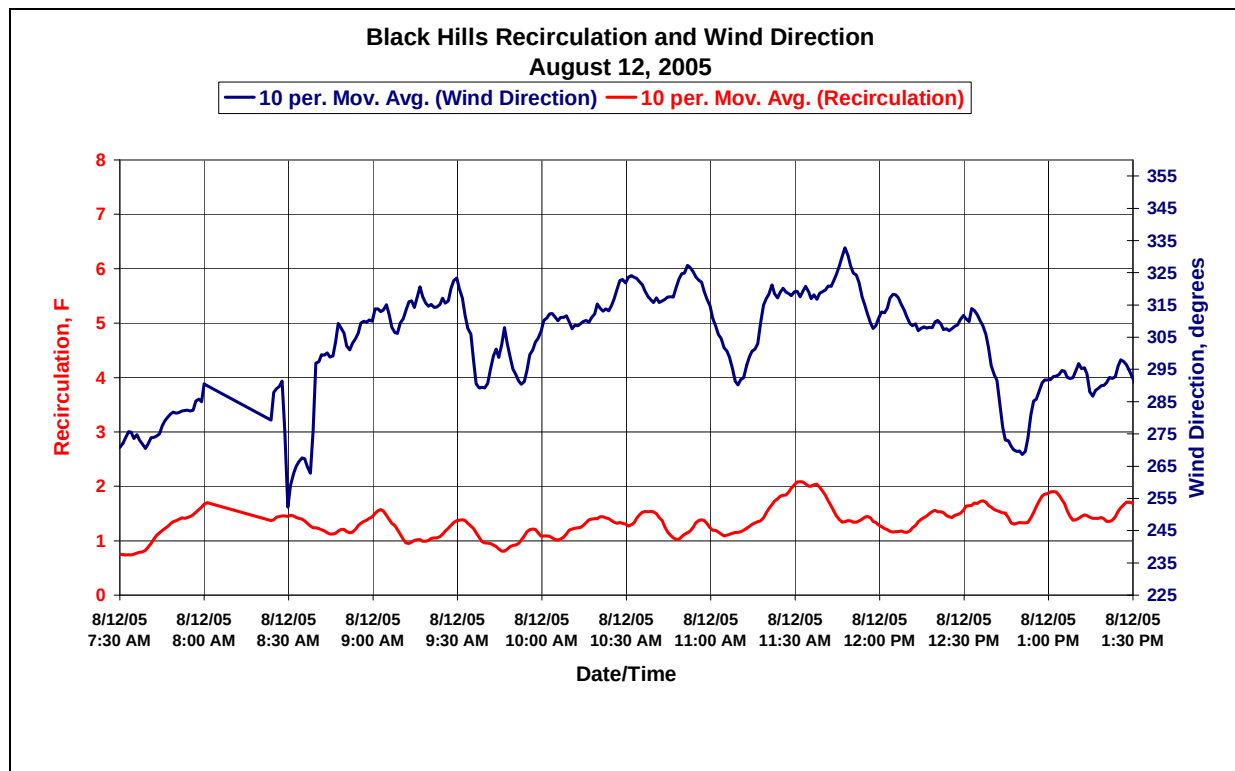


Figure 6-96. Wygen recirculation and wind direction, August 12, 7:30 a.m. to 1:30 p.m.

Looking more closely at the period from midnight to 3:00 a.m. August 16, rapid, significant variations are observed in the recirculation with little change in wind conditions. Figure 6-97 displays the minimum, average, and maximum cell inlet temperatures, and Figure 6-98 shows the average recirculation and wind speed for this three-hour period.

The temperature variation pattern shows three events in which the minimum temperature declines rapidly and then suddenly reverses and increases. The three “turnaround” points in minimum temperature correspond to the three peaks in recirculation. The maximum temperature also exhibits some declines and abrupt reversal and recovery. Five points of interest are shown on Figure 6-98 at 12:25, 12:37, 12:57, 1:03, and 1:13 a.m. The points at 12:25 and 12:37 a.m. correspond to the beginning and peak of the first recirculation excursion. Points 12:57 and 1:03 are the same for the second excursion, and 1:13 a.m. corresponds to the point at which the maximum temperature goes to a temporary minimum. Figure 6-99 shows the cell inlet temperature distribution at 12:25 and 12:37 a.m. The location on the ACC of the various cells referred to in the following discussion can be found in Figure 4-10 in Section 4.4.5. The increase in the calculated recirculation in this instance is due to the significant drop in the minimum inlet temperatures in the southeast (downwind) cells (K24 and K25) as opposed to an abrupt increase in maximum-temperature cells as might result from hot-plume recirculation. Most cell temperatures are relatively unchanged during this event.

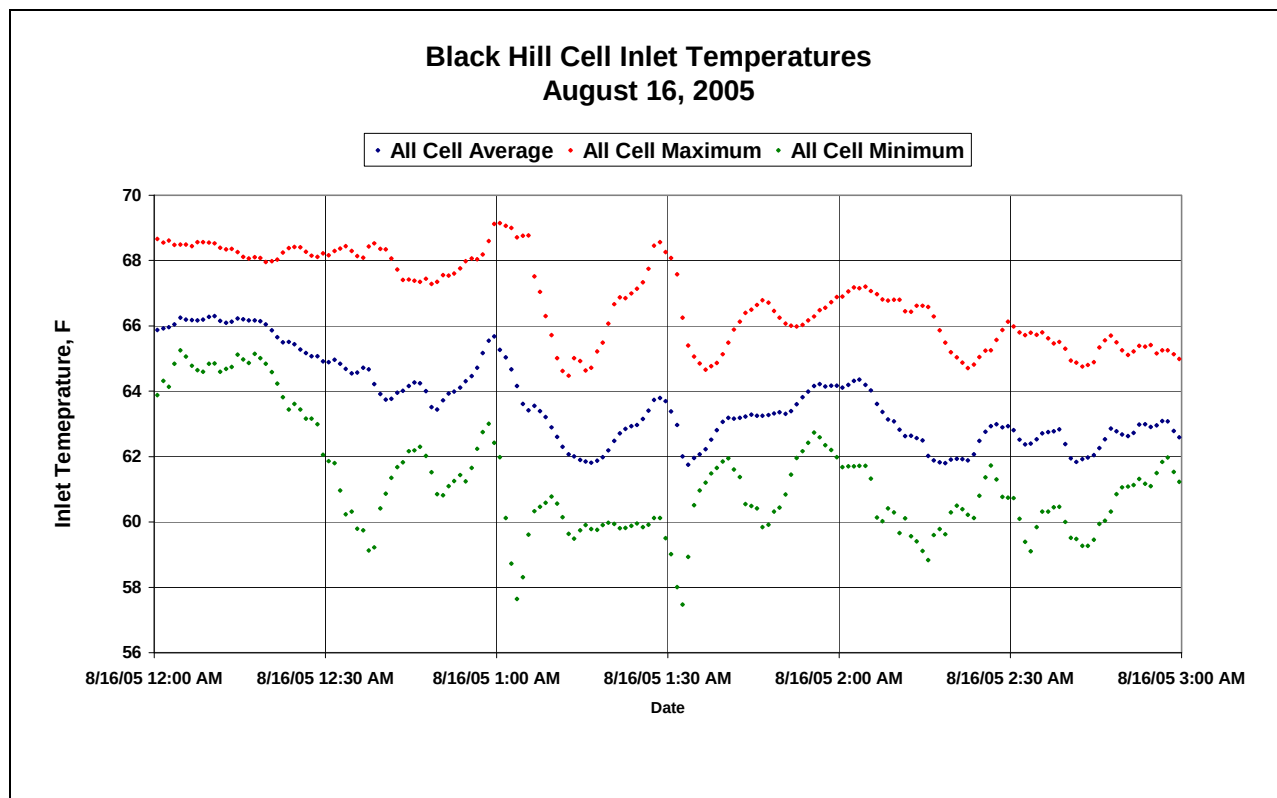


Figure 6-97. Cell temperature range and wind speed, August 16, 12:00 a.m. to 3:00 a.m.

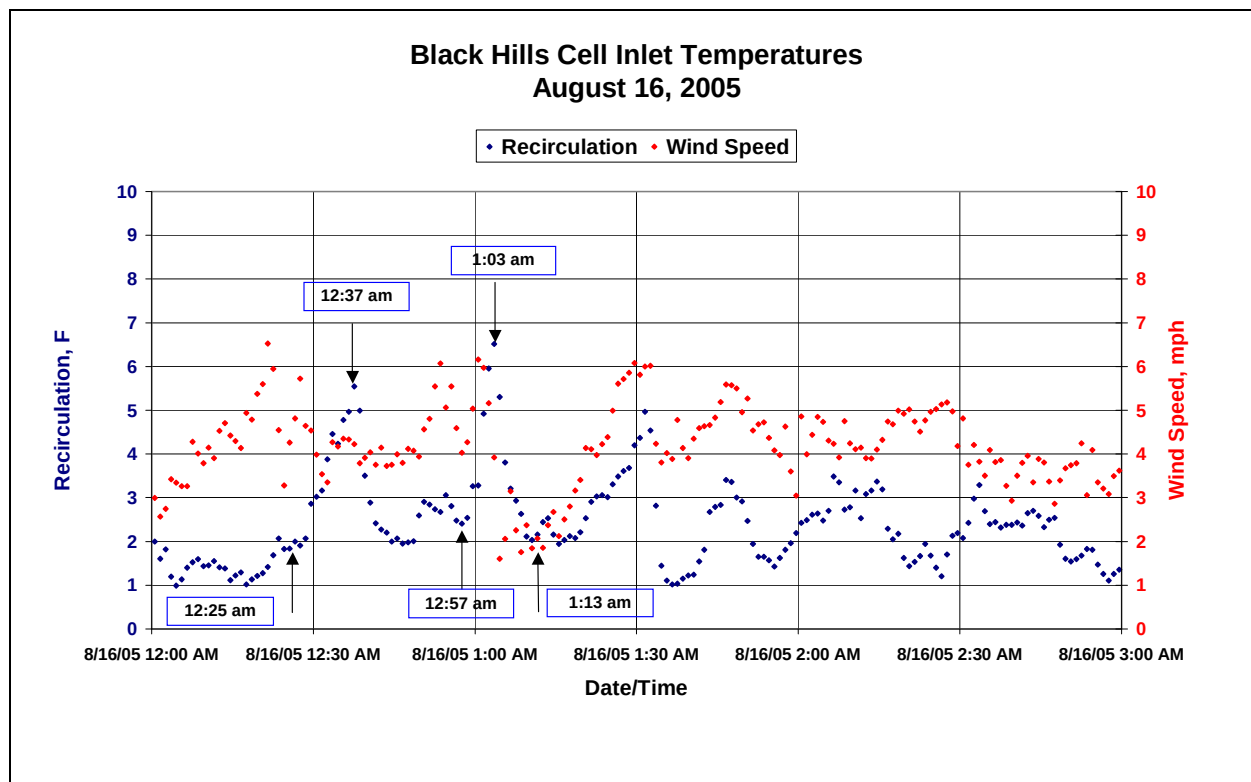


Figure 6-98. Recirculation and wind speed, August 16, 12:00 a.m. to 3:00 a.m.

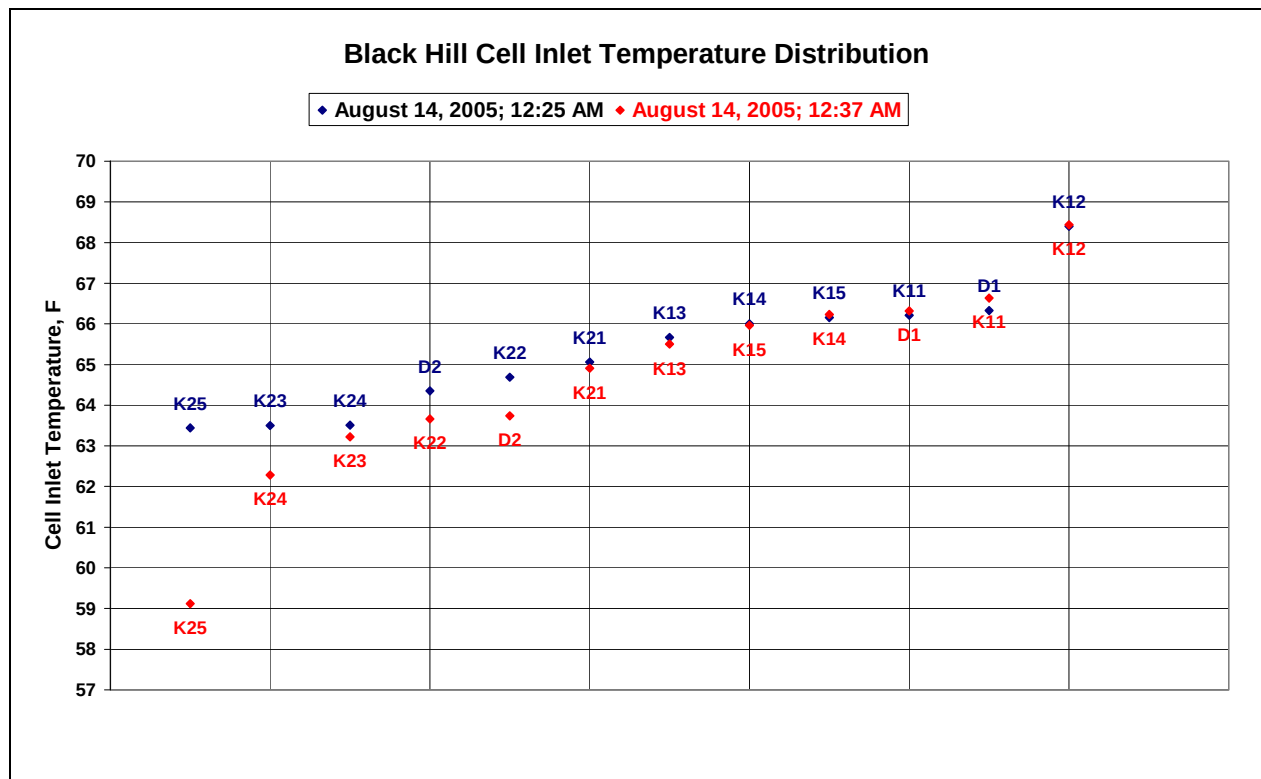


Figure 6-99. Inlet cell temperature distribution, August 16; 12:00 a.m. to 3:00 a.m.

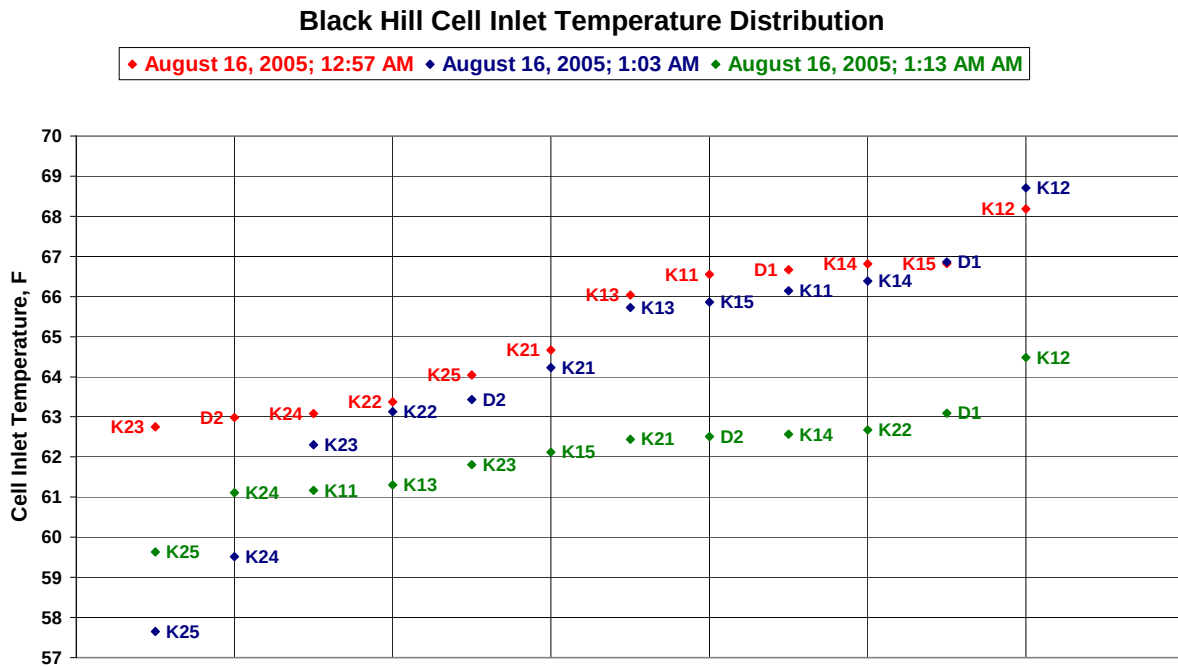


Figure 6-100. Inlet cell temperature distribution, August 16, 12:00 a.m. to 3:00 a.m.

Figure 6-100 displays similar data for 12:57 to 1:13 a.m. The two earlier times exhibit essentially the same behavior discussed above for the earlier event. In addition, the distribution a few minutes later is shown for the time when the maximum cell inlet temperature has dropped significantly. Several observations are noteworthy:

- The average temperature at 1:13 a.m. is significantly lower than for the earlier times, as is the maximum temperature as shown in Figure 6-100.
- At both earlier times, the temperatures were segregated by street with all the cells from Street #1, the western side, having higher cell inlet temperatures than any of the cells in Street #2 on the eastern side. At 1:13 a.m., the cell inlet temperature for the two sides overlapped, although the two warmest cells were still in the northwest corner (Cells K11 and K12) while the two coldest cells were in the southeast corner (Cell K24 and K25).

There is insufficient detailed information to explain all aspects of the observed behavior. It is reasonably consistent that winds from slightly north of westerly result in the upwind cells being warmer than the downwind cells. This may be the result of wind from that direction (WNW) coming around the corner from the side of the plant boiler/turbine building, which would make this inlet air warmer than the far-field ambient air. The eastern cells, because of the solid windwall under the ACC, will draw air primarily from the eastern side. When, at 1:13 a.m., the wind has shifted to south of westerly, the incoming air to the west of the ACC may be cooler ambient air. In no case does there appear to be any significant amount of real recirculation of the hot discharge plume into the inlet air stream.

6.6.3 Temperature Variability Within Cells

Some indication of the effect of wind on recirculation can be discerned from measuring variations in inlet temperature at different locations within an individual cell. The four corner cells (K11, K15, K21, and K25) were instrumented with four inlet air temperature probes arranged as shown in Figure 4-10.

Figures 6-101 through 6-104 display the variations of inlet temperature with time in those four corner cells over a 24-hour period (9:00 a.m. on August 9 to 9 p.m. on August 10). Figure 6-105 shows the wind conditions for the same period.

The readings in all four cells show generally excellent agreement, with the maximum difference among the four probes nearly always between 0.5° and 1.5°F. Maximum variations in all four cells occur around 3:00 to 6:00 p.m. on August 9 and around 6:00 p.m. on August 10. Both times coincide with rapid shifts in wind direction and rapid drops in ambient temperature due to the arrival of storm fronts.

The cells at the south end of the ACC (Cells K15 and K25) are furthest from the shelter of the plant and the most exposed. The variability in these cells is consistently higher even during periods of relatively steady winds. The cell in northeast corner (Cell K21) is particularly sensitive to winds from the east, with the temperature probe in the east part of the cell running as much as 7° to 8°F higher than the other three probes. These observations are not understood at this time.

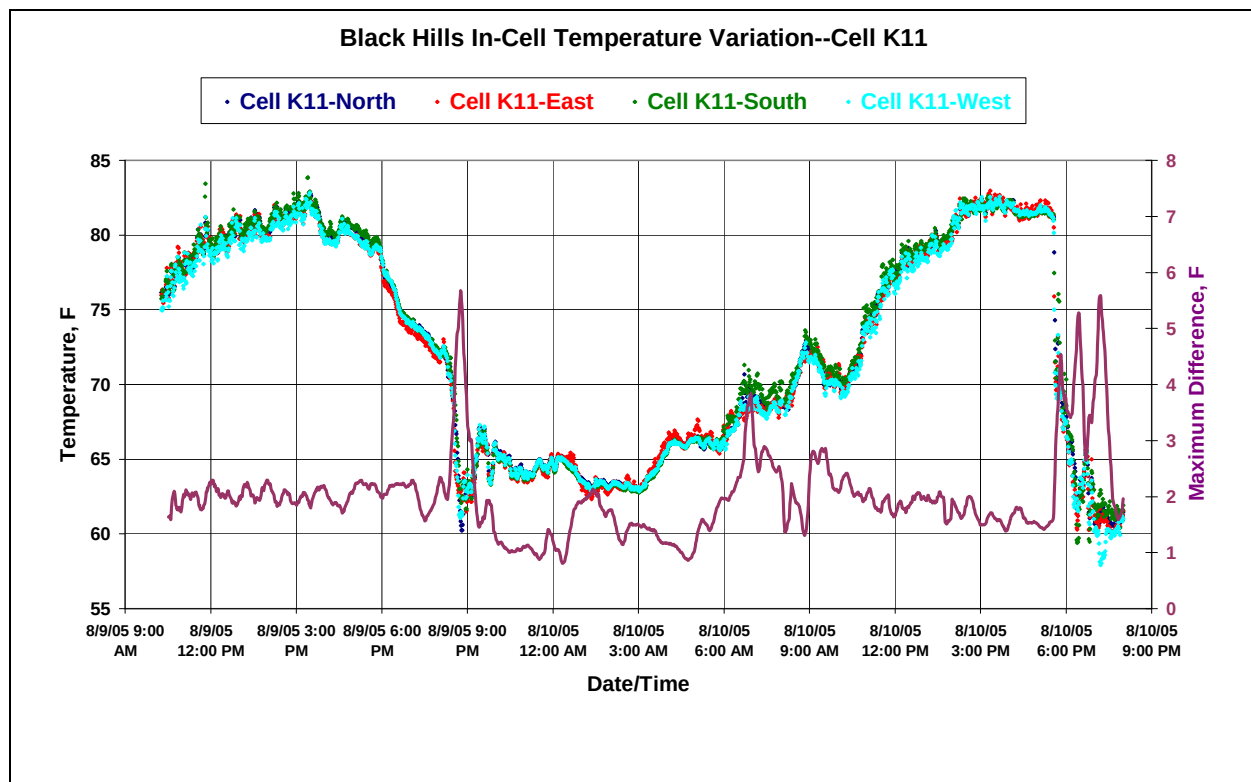


Figure 6-101. Inlet air temperature variations inside Cell K11.

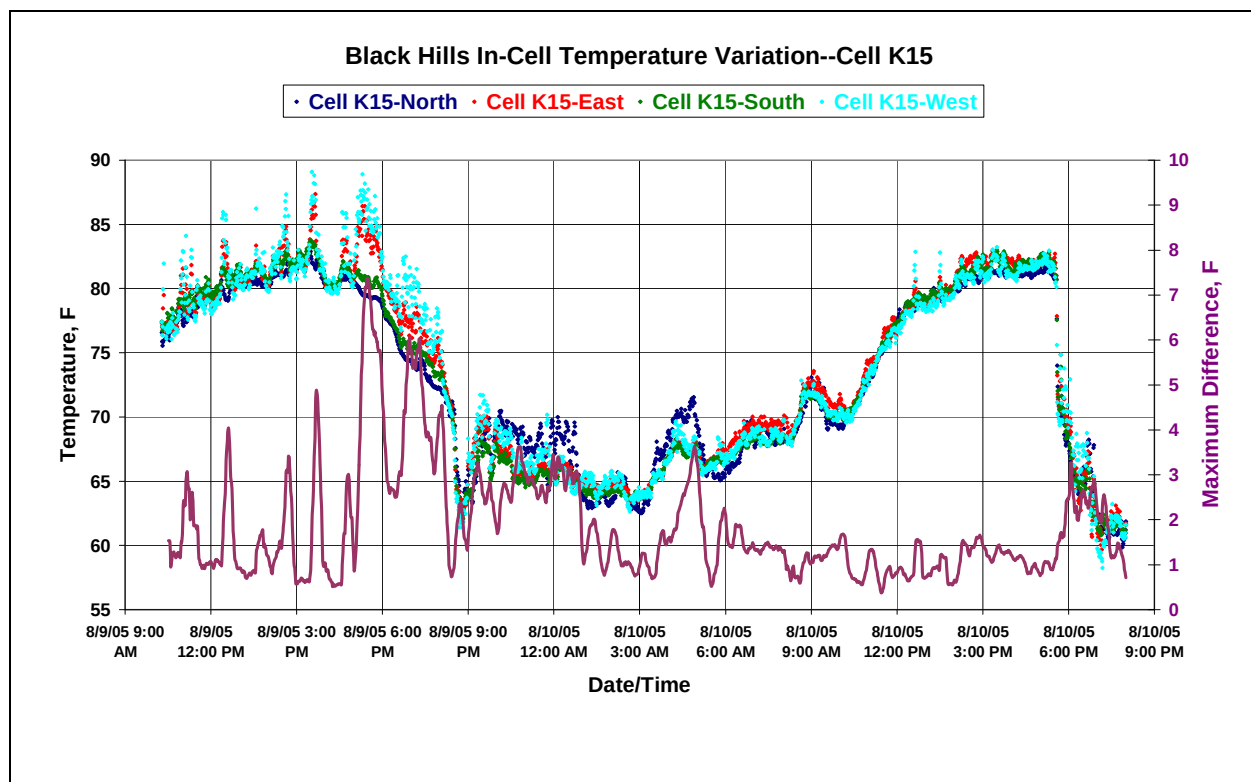


Figure 6-102. Inlet air temperature variations inside Cell K15.

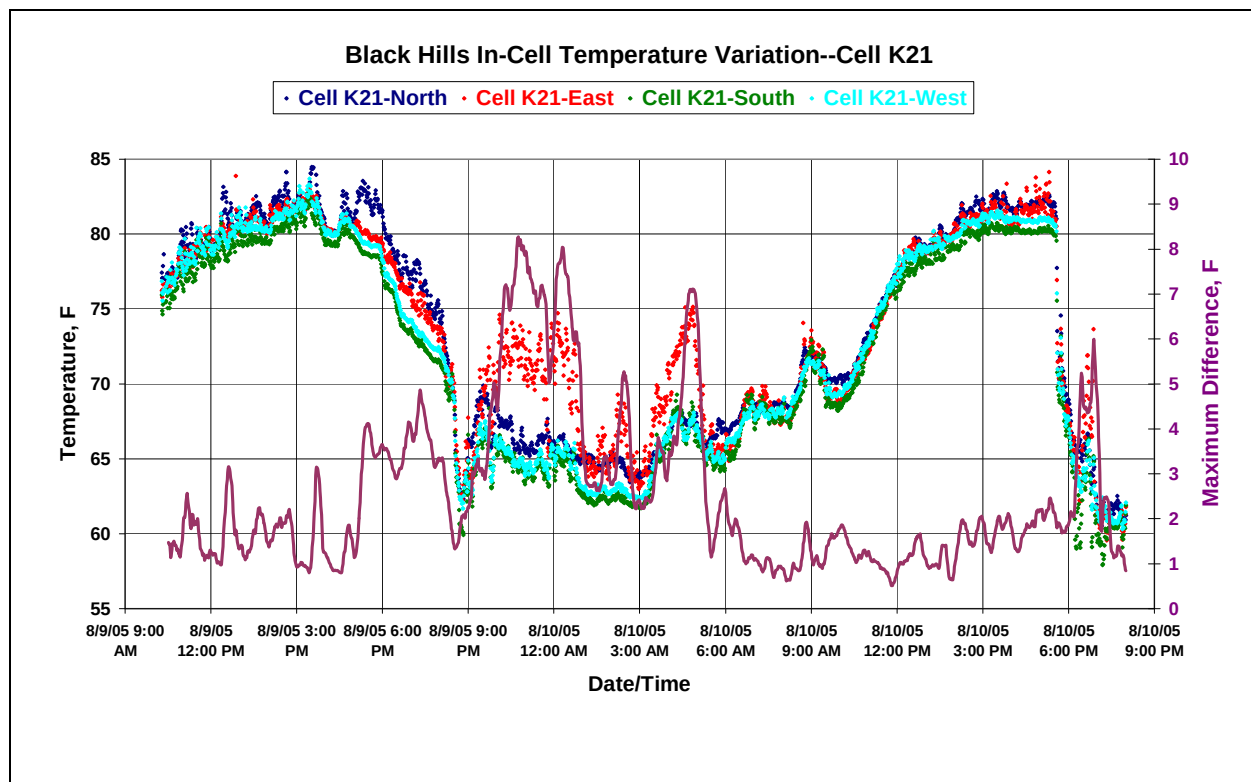


Figure 6-103. Inlet air temperature variations inside Cell K21.

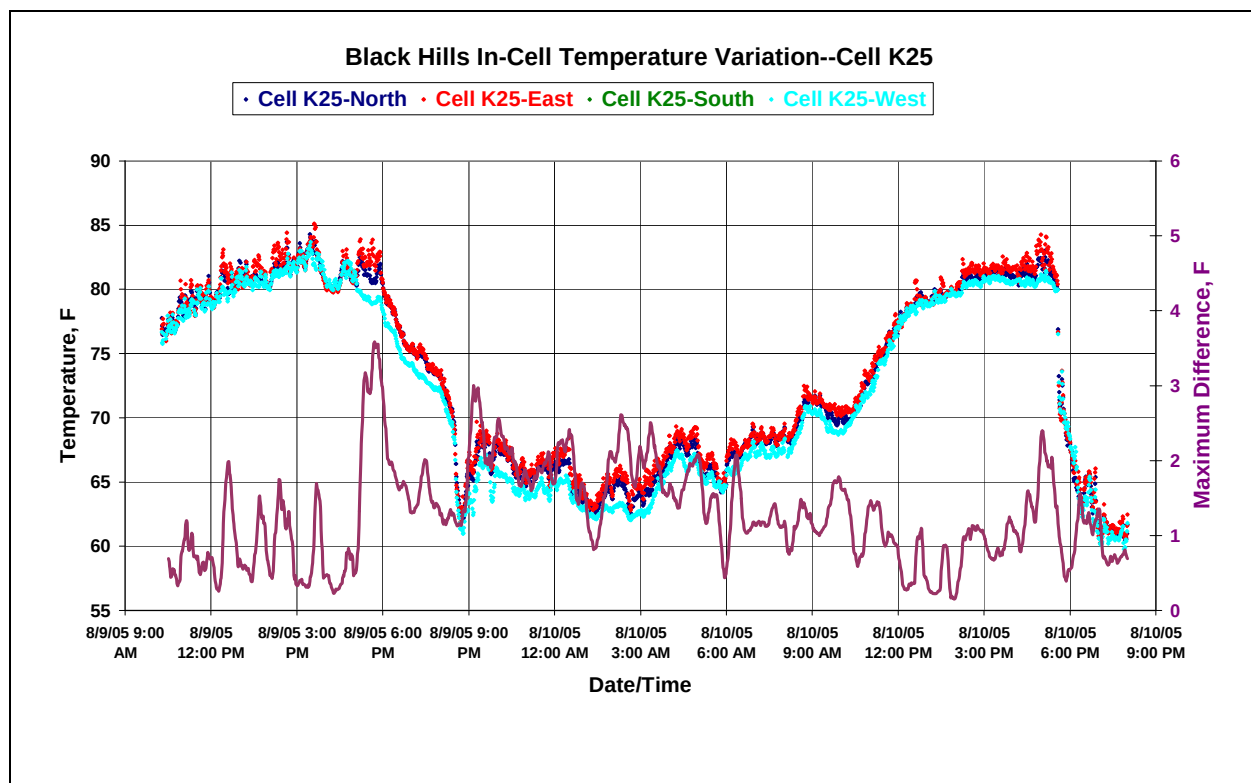


Figure 6-104. Inlet air temperature variations inside Cell K25.

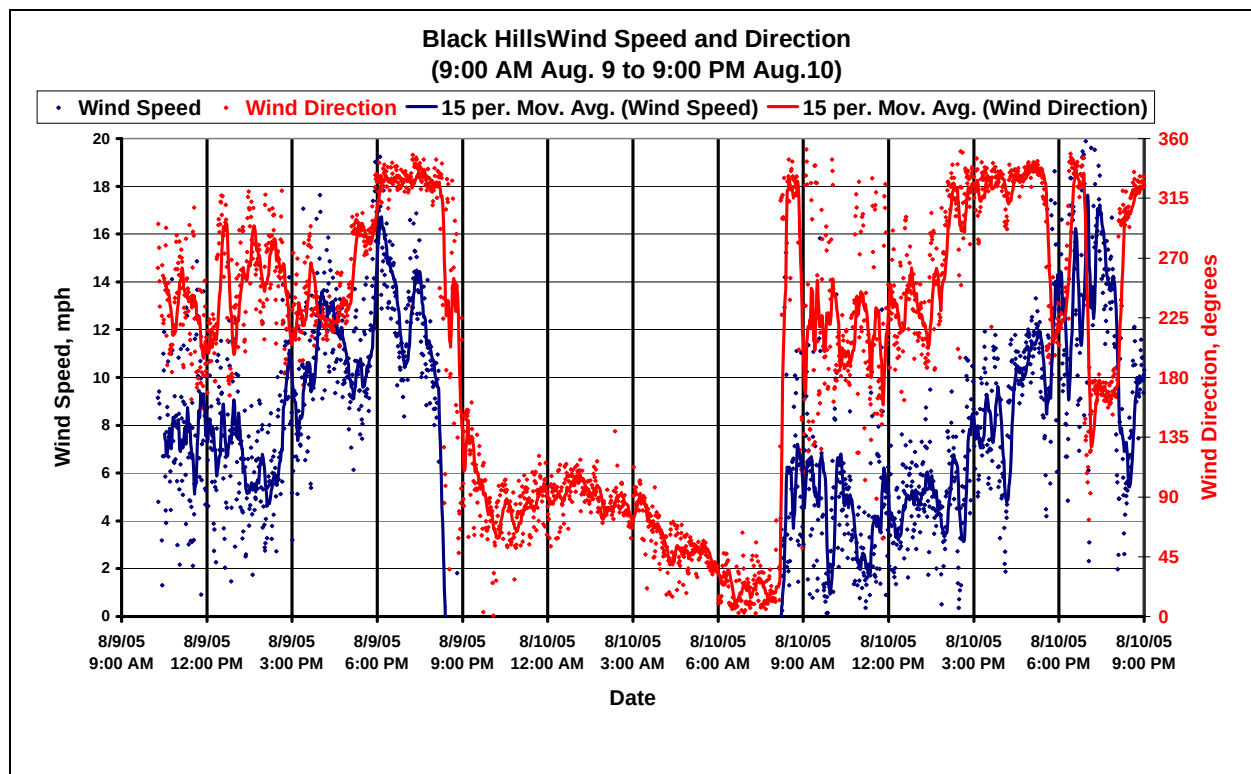


Figure 6-105. Black Hills wind conditions—August 9 and 10, 2005.

CHAPTER 7:

Analysis of Results: Fan Performance Degradation

7.1 Analysis of Results: Fan Performance Degradation

This chapter discusses the effect of wind on fan performance. The underlying hypothesis is that crosswinds at the fan inlets degrade the performance of the fans, leading to lower airflows than the ACC was designed for. The following sections present evidence for this hypothesis and discuss the relationship of fan performance to wind speed and direction at each of the five test sites.

7.2 Bighorn Cell Inlet Velocity

Cells 3J, 3I, and 3F were instrumented with propeller anemometers suspended below the fans to monitor airflow into the cells as a function of wind conditions.

Figure 7-1 shows the average velocity in each of the three cells for the entire test period when all fans were at full speed. The design flow for the fans is 1,250 actual cubic feet per minute (acfm) which would correspond to an inlet velocity of approximately 1,320 ft/min as shown on the figure. The average data show reasonable agreement with this value for most of the time.

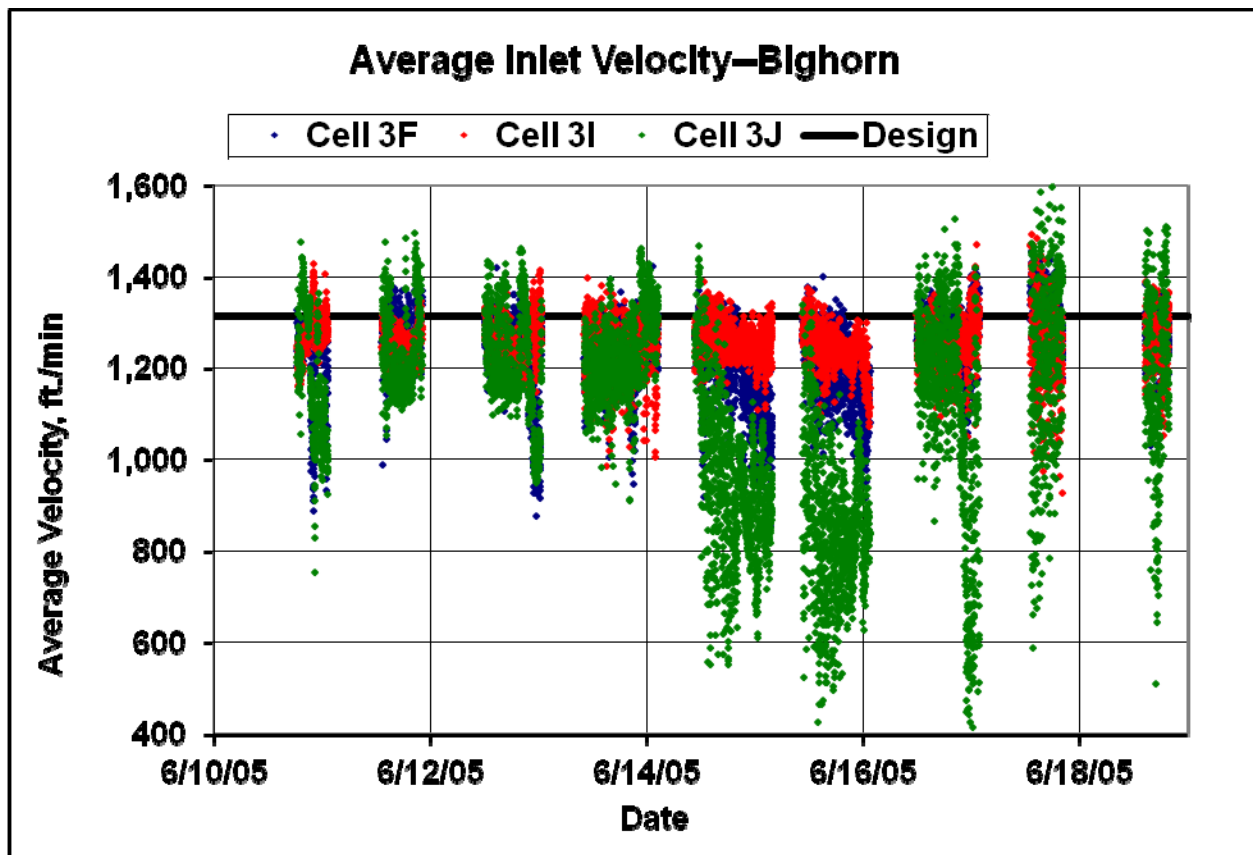


Figure 7-1. Average cell inlet velocity of Cells 3F, 3I, and 3J: entire test period, all fans full speed.

The periods of significantly decreased airflow into Cell 3J on June 14 and 15, around midnight on June 17 and, to a lesser extent, on June 17 and 18, correspond with the wind coming from the southeast. Winds from the southwest to northwest have less effect. Figure 73a, Bighorn Wind Conditions, is reproduced here as Figure 7-2 for convenient reference.

Figure 7-3 illustrates this in a plot of average inlet velocity into Cell 3J against wind direction. For winds from the southwest (220°) around to the east (90°), the velocity is relatively constant at a value close to the design value. For winds from the east (90°) to the southwest (220°) through the southeastern quadrant, there is considerable degradation in fan performance and significant variability. The variability in that region is related to wind speed as shown in Figure 7-4.

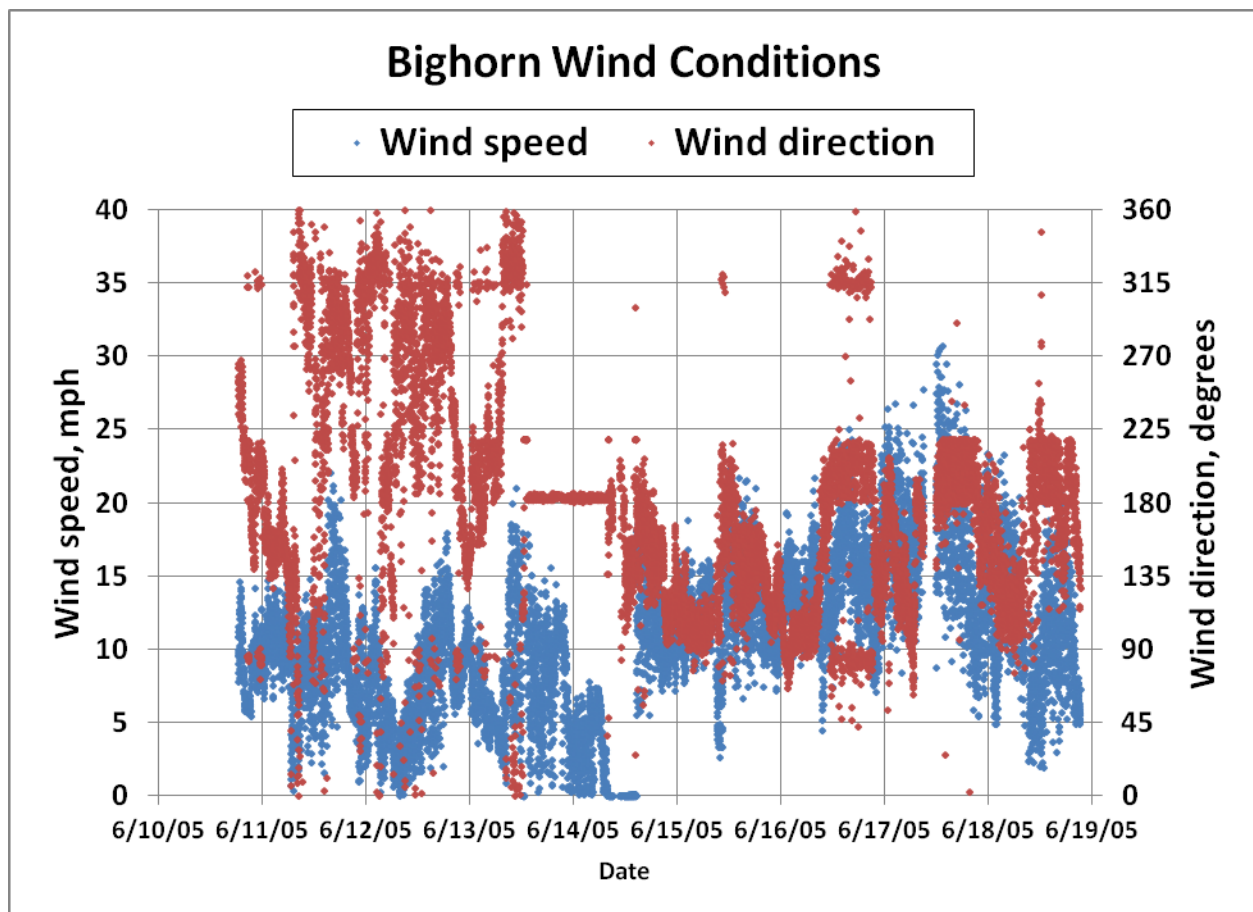


Figure 7-2. Bighorn wind conditions (from Figure 73a).

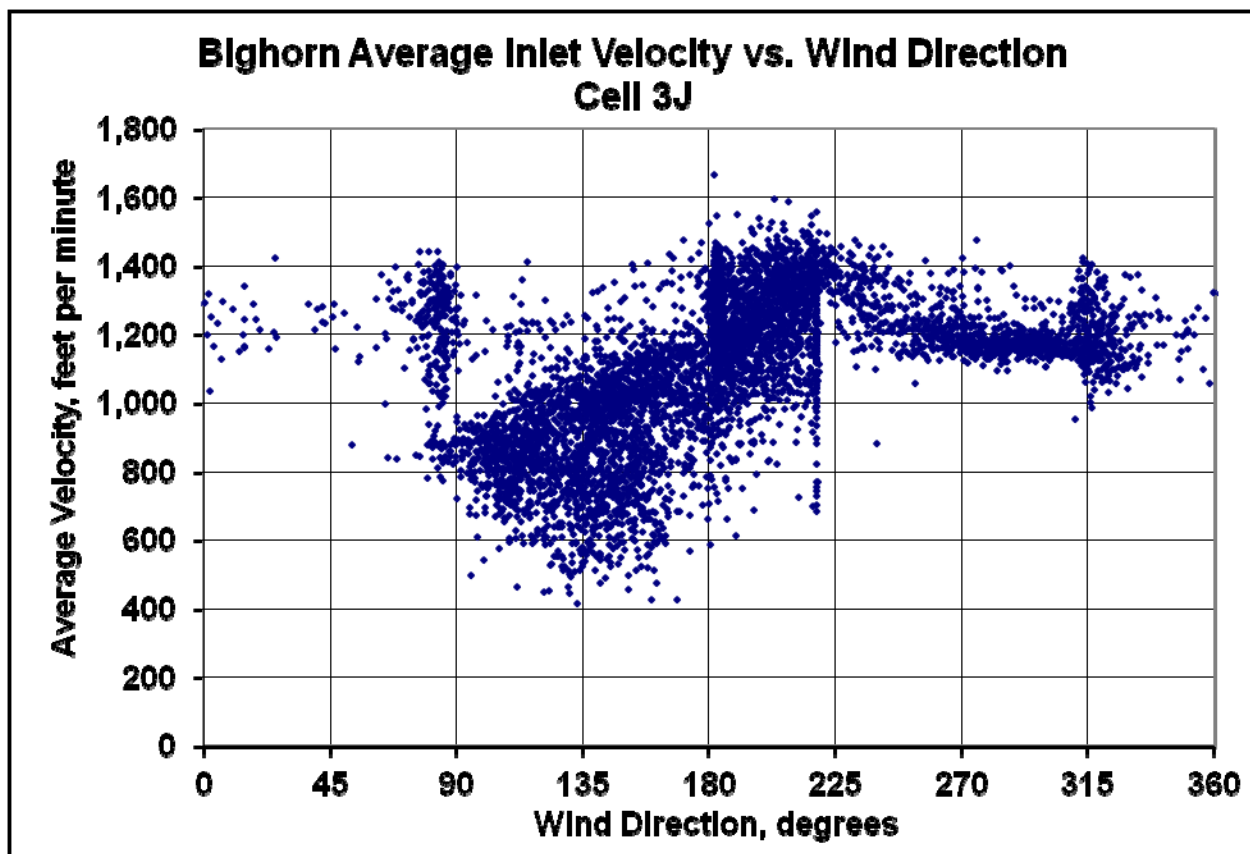


Figure 7-3. Cell 3J inlet velocity vs. wind direction—entire test period.

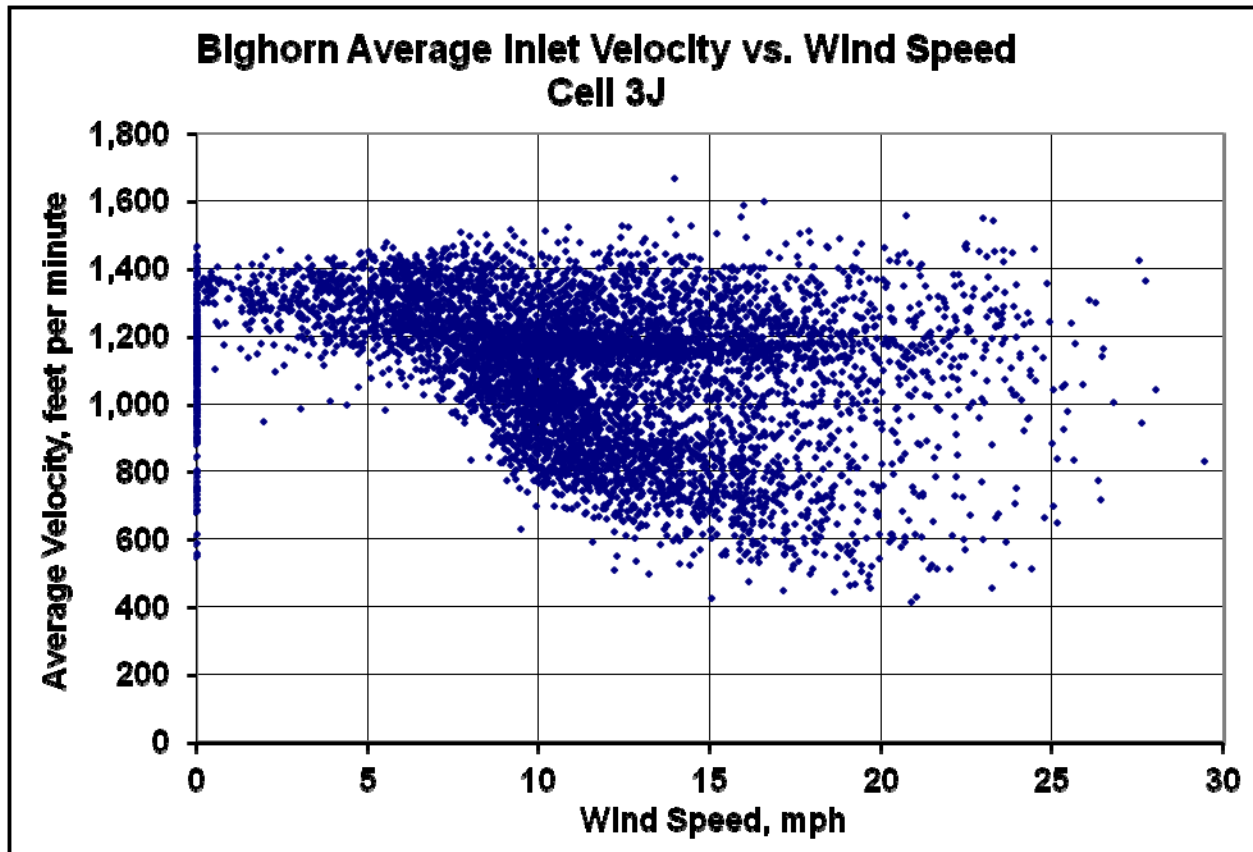


Figure 7-4. Cell 3J inlet velocity vs. wind speed—entire test period.

The demarcation lines between regions of little wind effect and significant wind effect are surprisingly sharp. In the case of the transition at $\sim 220^\circ$, this is believed to be related to the presence of the windscreen. Winds from more westerly directions are intercepted by the screen directly to the east of Cell 3J, creating a slightly higher static inlet pressure and augmenting fan performance. Winds from the south flow unimpeded past the fan inlet or create a separated region at the bottom of the windwall, which in turn creates an unfavorable inlet velocity distribution and degrades fan performance. Winds from the southeast may create a separated turbulent area at the end of the windscreen under Cell 3J with similar effect, but this is not well understood at this time.

The transition at 90° where winds from the north of east appear to have less of an effect on fan performance is likely due to the fact that all winds in the northeast quadrant (0° to 90°) are at very low speed, with the exception of a few between $\sim 70^\circ$ and 90° as shown in Figure 73b. The lesser effect from northeasterly winds may also be related to interference or blockage of the incoming winds by plant structures (HRSGs) located to the east of the ACC, but this is speculative at this time.

A more detailed look is taken at wind effects in Cells 3F, 3I, and 3J during three time periods selected to illustrate the differing effects of wind from different directions.

June 15, 2005, from noon to midnight

June 18, 2005, from 2:30 p.m. to 7:30 p.m.

June 12, 2005, from noon to midnight

7.2.1 June 15, Noon to Midnight

Figures 7-5 and 7-6 show the average inlet air velocities for Cells 3F, 3I, and 3J plotted concurrently with wind speed (Figure 7-5) and direction (Figure 7-6). For the entire period, the wind speed is moderate, mostly between 7 and 17 mph, from the south to southeast with brief excursions to near easterly in the mid-afternoon and late evening. There is very little effect on Cells 3F and 3I at any time. Cell 3I exhibits minor decreases in flow when the wind is more directly from the east. This is consistent with Cell 3F's location at the north end of the north cluster (see Figure 37) adjacent to the opening between the north and south clusters. This opening creates a passage running east-west through which easterly winds may be funneled. Cell 2I is an interior cell and sheltered from most wind effects.

Cell 3J, at the south end of the south cluster, is the most exposed to winds from the east to west through the southern half of the wind rose. This results in degraded performance of fan 3J for essentially the entire period. Prior to about 8:45 p.m., the wind speed and direction are both erratically variable, reflected in the high variability in airflow to Cell 3J. After 8:45, both wind speed and direction become much steadier, as does the inlet airflow. Between 10:00 p.m. and midnight, average Cell 3J inlet velocity increases from ~800 feet per minute (fpm) to 1,100 fpm and then decreases again. There is no obvious reason for this increased inlet velocity, although it may be related to the very modest decrease and increase of the wind speed over the same time period.

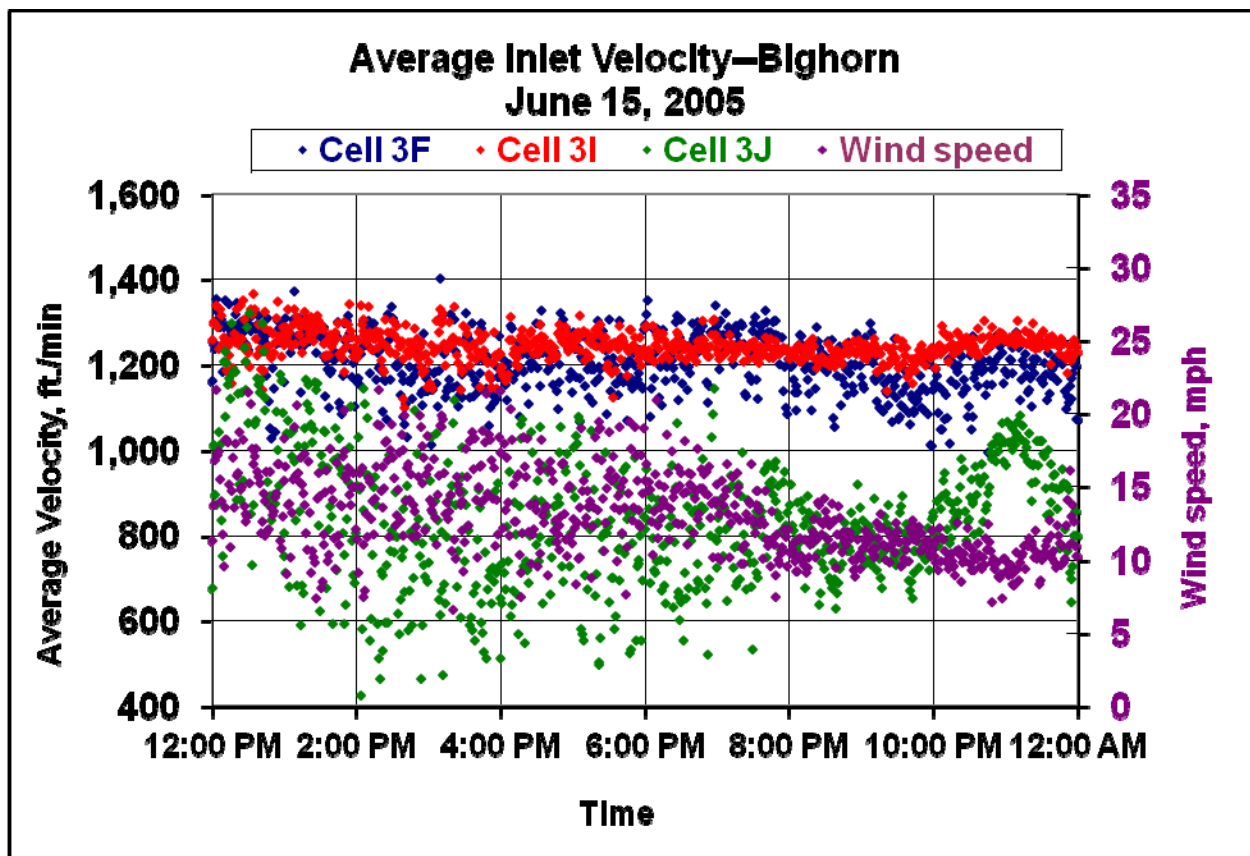


Figure 7-5. Bighorn inlet velocity vs. wind speed: Cells 3F, 3I and 3J and Wind Speed, June 15, 2005.

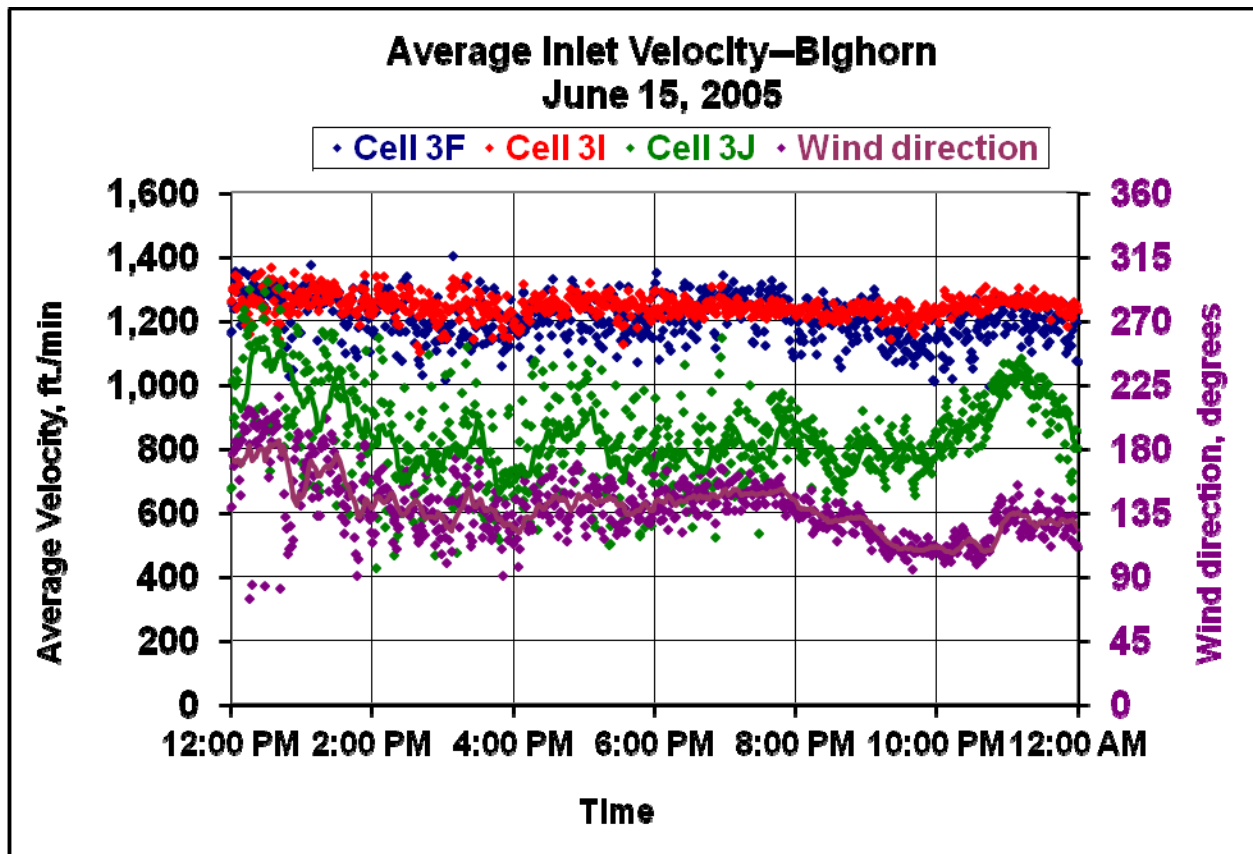


Figure 7-6. Blghorn inlet velocity vs. wind speed: Cells 3F, 3I and 3J and Wind Direction, June 15, 2005.

7.2.2 June 18, 2:30 p.m. to 7:30 p.m.

Figures 7-7 and 7-8 present similar plots for June 18 between 2:30 p.m. and 7:30 p.m. The wind speed is modest from 7 to 17 mph with direction, varying slowly between southeast and southwest.

Cells 3F and 3I are unaffected for the entire period. Cell 3J varies in response to both wind speed and direction. The greatest effect is seen between 5:00 p.m. and 6:00 p.m. when the average velocity decreases from 1,200 fpm by 6:00 p.m. This variation coincides with a wind direction shift from 180° to 140° and back again. During that same period there is an increase in wind speed from 10 to 15 mph and back again to 10 mph synchronously with the variation in wind direction.

Smaller but similar variations are seen between 3:30 p.m. and 4:30 p.m. and again between 7:00 p.m. and 7:45 p.m. when winds from the southwest slightly augment fan performance in Cell 3J.

7.2.3 June 12, Noon to Midnight

Figures 7-9 and 7-10 display similar data for June 12 between noon and midnight. For most of this period, the wind is from the west to northwest and all three fans perform at or near their design flow with even some indication of slightly augmented flow.

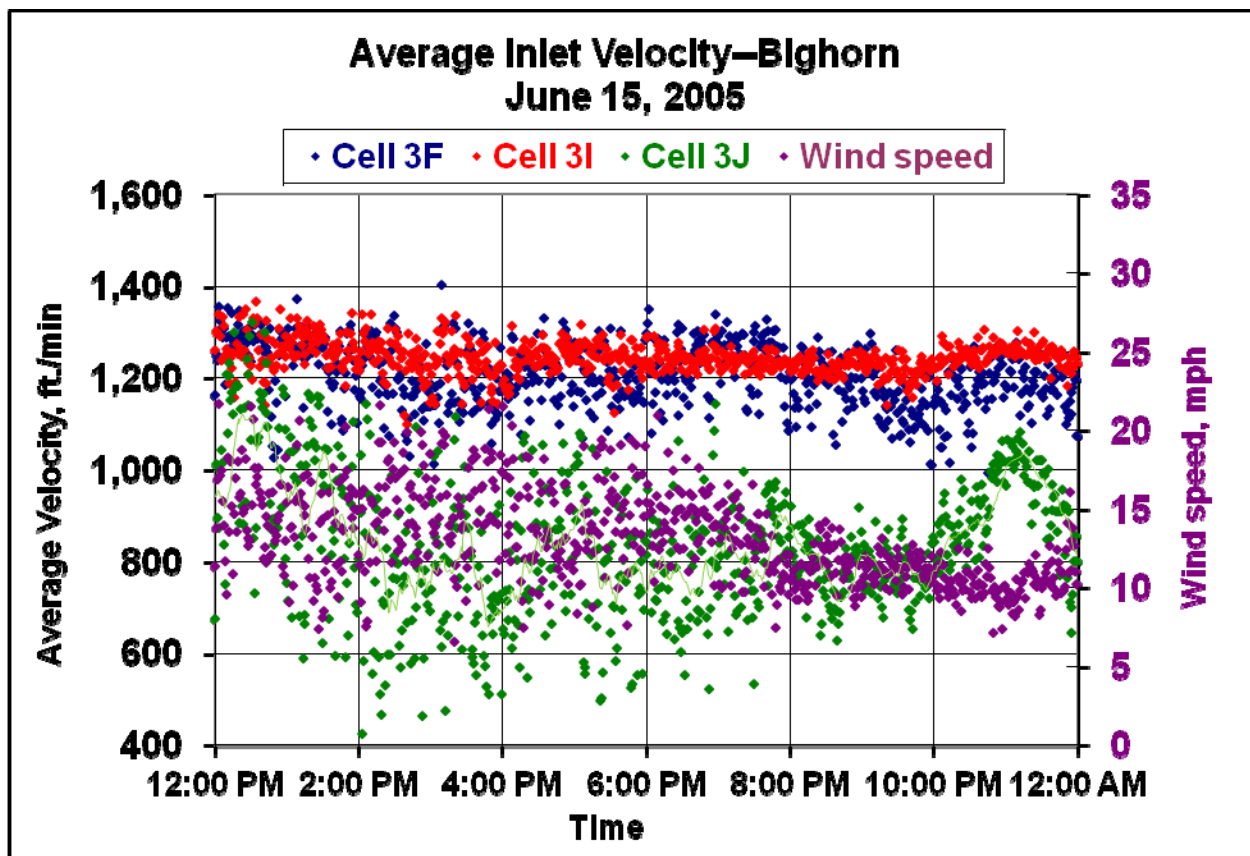


Figure 7-7. Bighorn inlet velocity vs. wind speed: Cells 3F, 3I and 3J and Wind Speed, June 18, 2005.

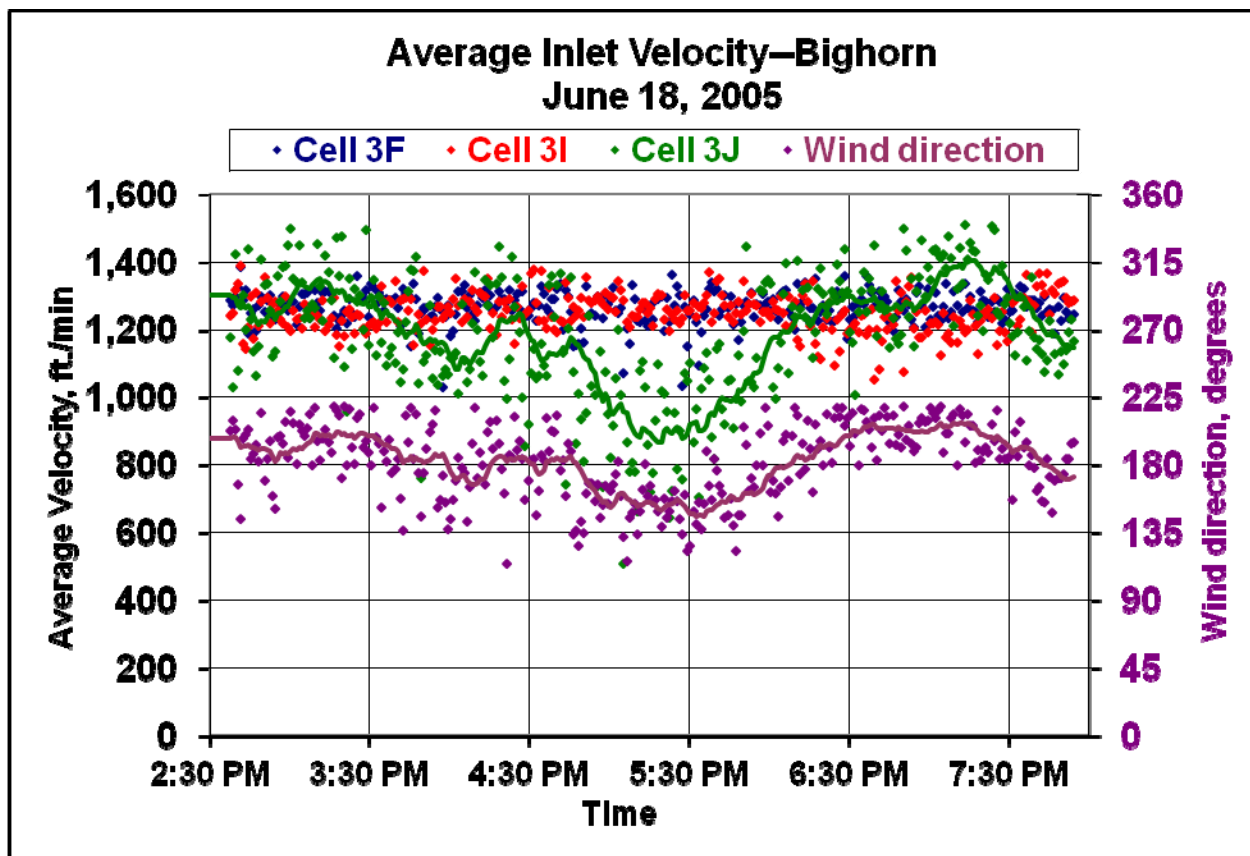


Figure 7-8. Bighorn inlet velocity vs. wind speed: Cells 3F, 3I and 3J and Wind Direction, June 18, 2005.

Just before 8:00 p.m., the wind decreases and begins to shift to the south and, by 11:00 p.m., is from the southeast. At approximately 9:45 p.m., as the wind passes through the northwesterly direction (225°), the performance of Fan 3J begins to fall off and by midnight, with wind steady from the southeast, Fan 3J is operating at a reduced flow of about 25% below its design flow (~1,000 fpm vs. ~1,300 fpm).

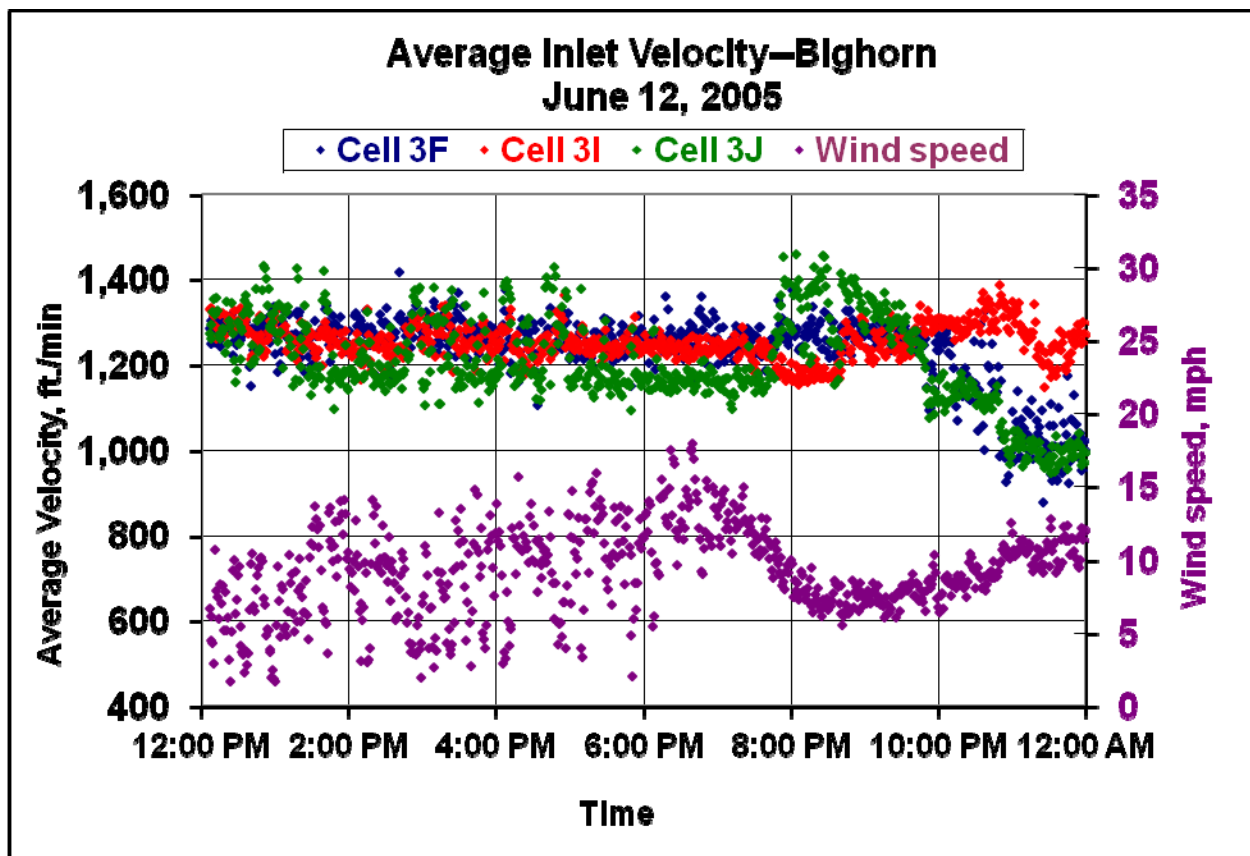


Figure 7-9. Bighorn inlet velocity vs. wind speed: Cells 3F, 3I and 3J and Wind Speed, June 12, 2005.

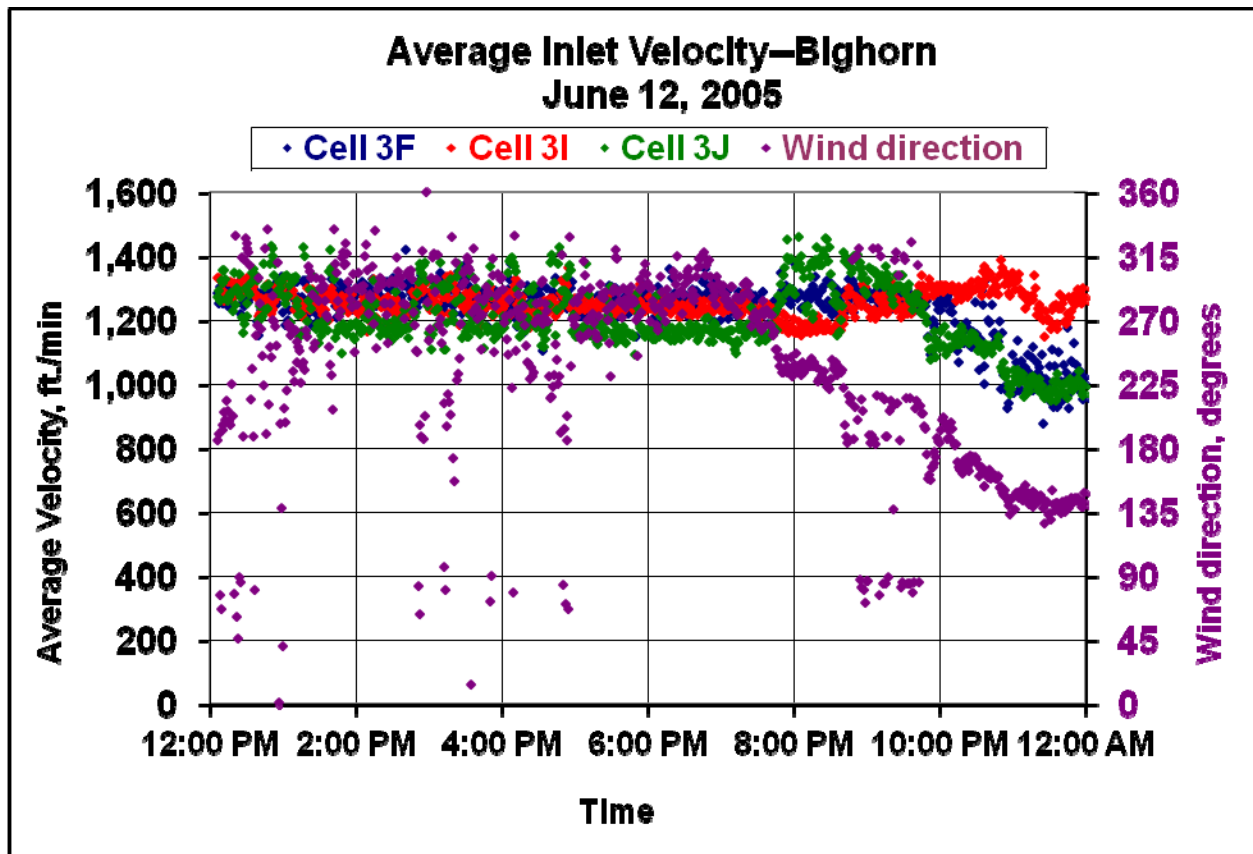


Figure 7-10. Bighorn inlet velocity vs. wind speed: Cells 3F, 3I and 3J and Wind Direction, June 18, 2005.

7.2.4 Bighorn Static Pressure

Measurements of static pressure rise, expressed in units of inH₂O (inches of water) across the fans in six cells (2A, 2J, 3A, 3F, 3I, and 3J) were made over the entire test period. The results are shown in Figure 7-11. Periods of lower static pressure rise on June 14 and 15, around midnight on June 16 and, to a lesser extent, on June 17 and 18 and coincide with periods of reduced airflow shown in Figure 7-1. Periods of high static pressure rise in Cell 3A at the north end of the north cluster and just to the west of the windscreen appear to coincide with periods of wind from the southwest at higher-than-usual speeds. Other periods of southwesterly winds on June 14 and 15 are at lower speeds and do not exhibit this effect. However, the behavior is not fully understood.

Figures 7-12 and 7-13 show the static pressure rise in Cell 3J vs. wind speed (Figure 7-12) and direction (Figure 7-13). The significant effects exhibit the same pattern as the effects of wind on inlet velocity illustrated in Figures 7-3 and 7-4.

Figures 7-14 and 7-15 display the static pressure rise in Cells 2A, 2J, 3A, 3I, and 3J on June 15 from 1:00 p.m. to midnight along with wind speed and direction. Several items are noteworthy. Cells 2A and 3A, both at the north end of the north cluster, show the least variability and the

highest static pressure rise, suggesting the least influence of wind. This is as would be expected with winds from the south to southwest.

Cell 3J shows an intermediate effect with more performance degradation evidenced at higher wind speeds (before 8:00 p.m.) and more southerly winds.

Cells 2I and 2J—at the south end of the south cluster like Cell 3J but on the east side of the windscreen—are the most exposed to southeasterly winds. Wind shifts to easterly result in slightly higher static pressure rises. This may be explained by the impinging of the easterly winds against the east side of the windscreen with some augmentation in performance of the nearby fans.

Figure 7-16 displays fan static pressure and wind direction for June 18 between 2:30 and 8:00 p.m. The singularly high static pressure rise of fan 3A during this time was noted earlier and is not well understood. The other fans show little effect of wind with the exception of a modest decrease in 2J, 3I, and 3J coincident with the wind shift further to the east and a slightly augmented performance of fan 3J with wind from the west of south.

Figure 7-17 displays similar data for June 12 from 1:00 p.m. to midnight. During this process, the winds are primarily from the west to northwest and augment the performance of the Street 3 fans. In particular, Fan 3F, which borders the gap between the clusters, shows a significantly higher static pressure rise for most of the day. This is not fully understood. Later in the day, as the wind shifts back to the south and southeast, the fans' behavior becomes more uniform as was the case for such winds on June 18 (see Figure 7-16). The notable exception is Cell 3A, which does not exhibit the same higher static pressure rise as it did on June 18. This may be due to the difference in wind speeds, but this is speculative at this time.

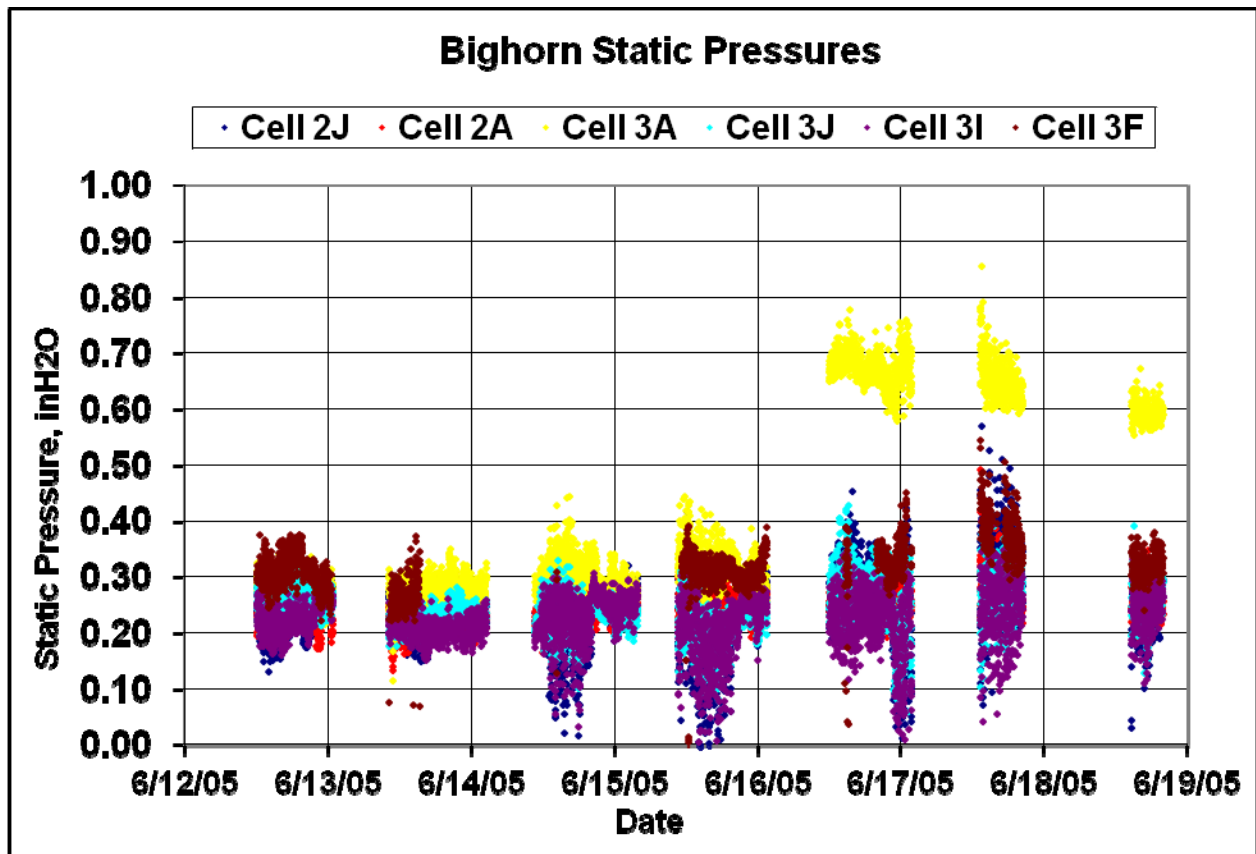


Figure 7-11. Static pressure rise variation at Bighorn---All fans on.

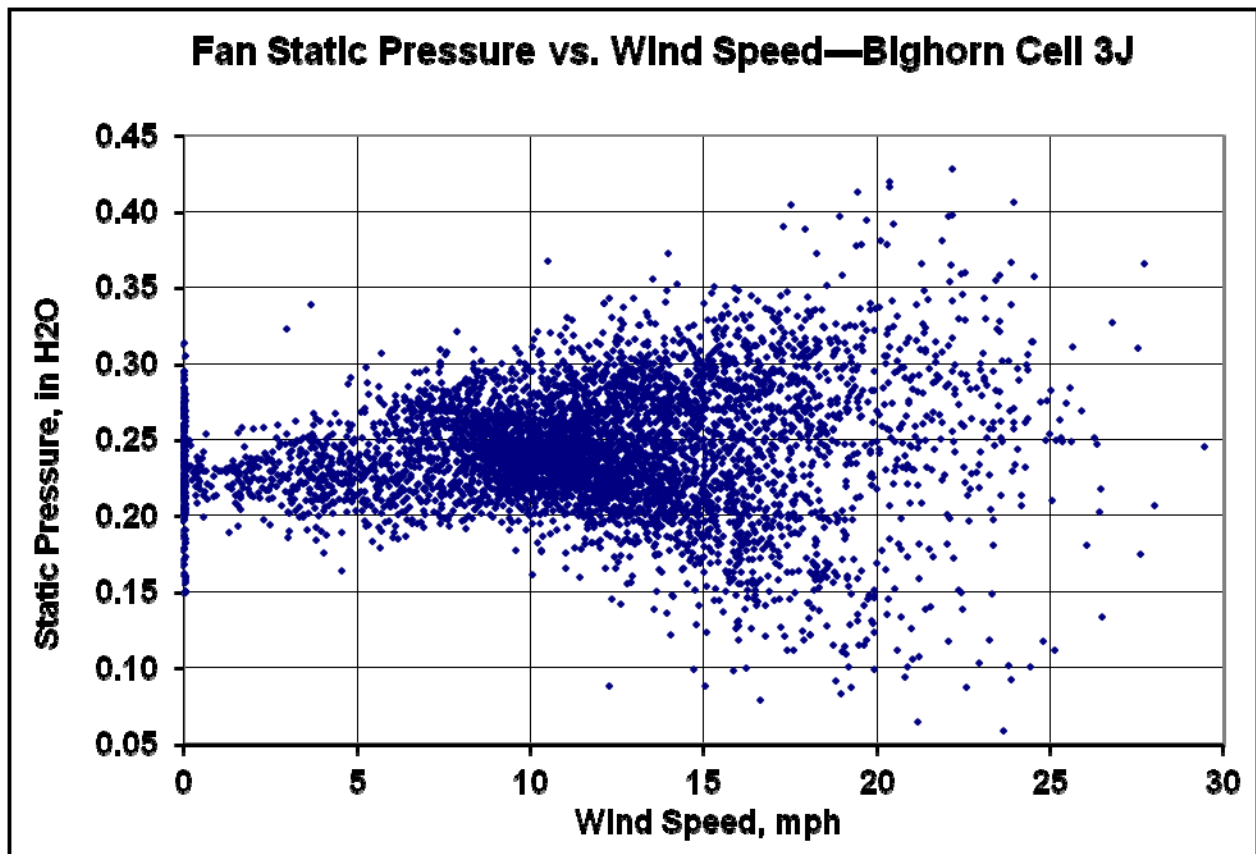


Figure 7-12. Static pressure rise vs. wind speed in Cell 3J---entire test period.

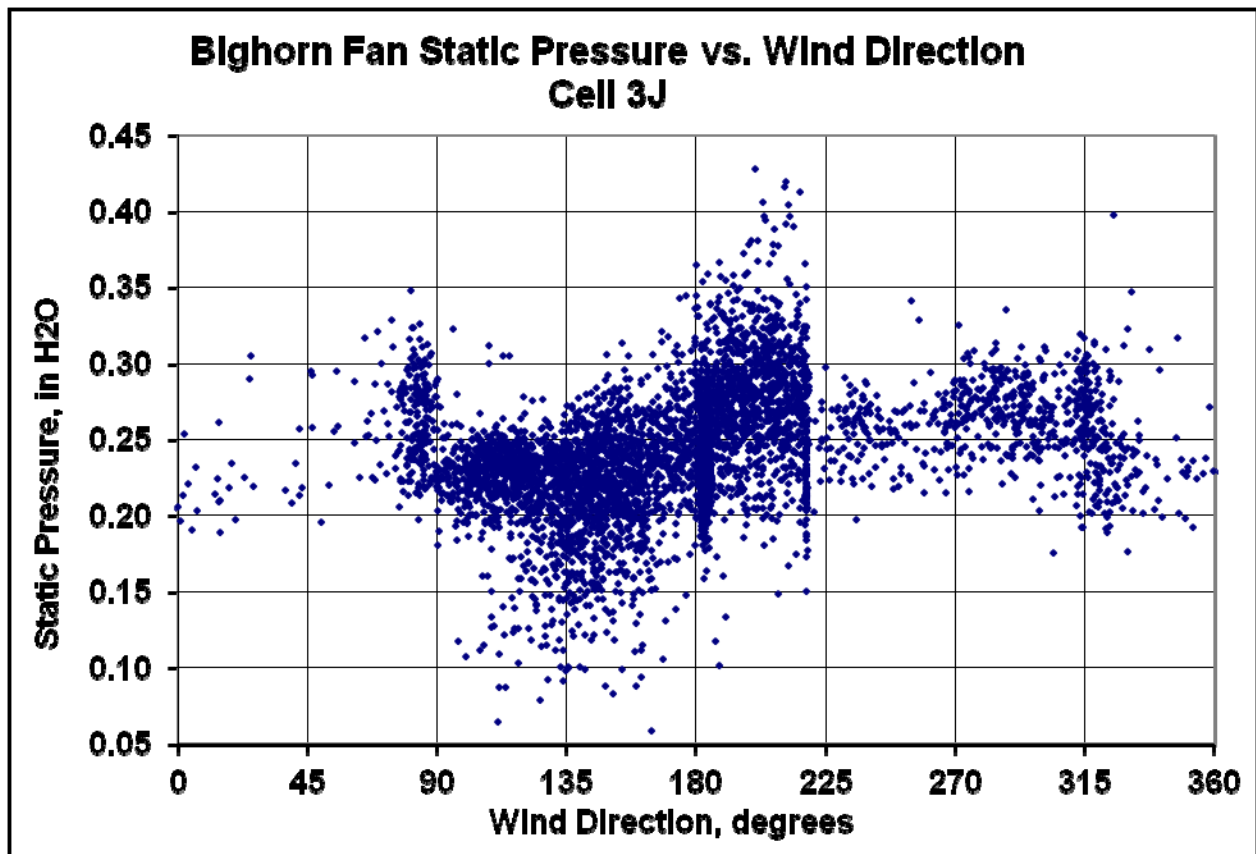


Figure 7-13. Static pressure rise vs. wind direction in Cell 3J---entire test period.

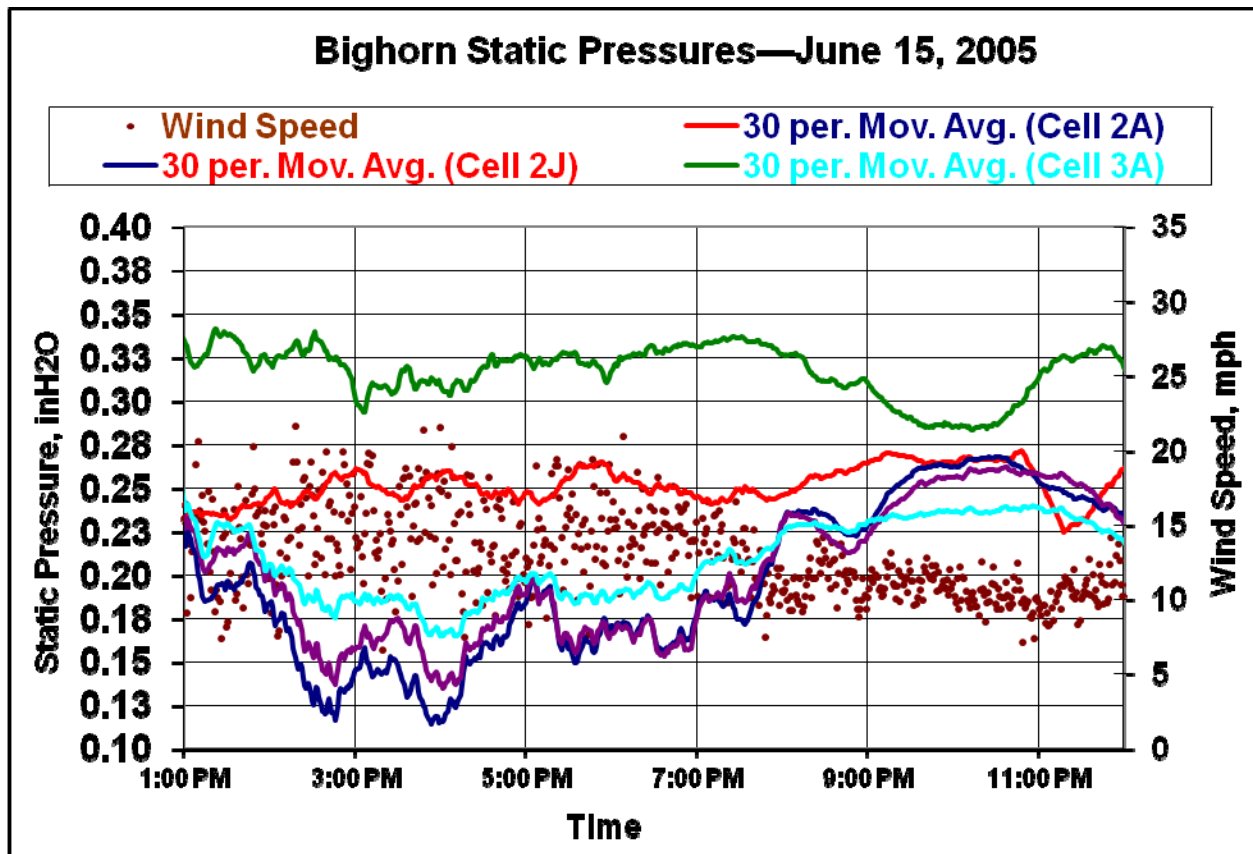


Figure 7-14. Static pressure rise in selected cells with wind speed---June 15, 2005.

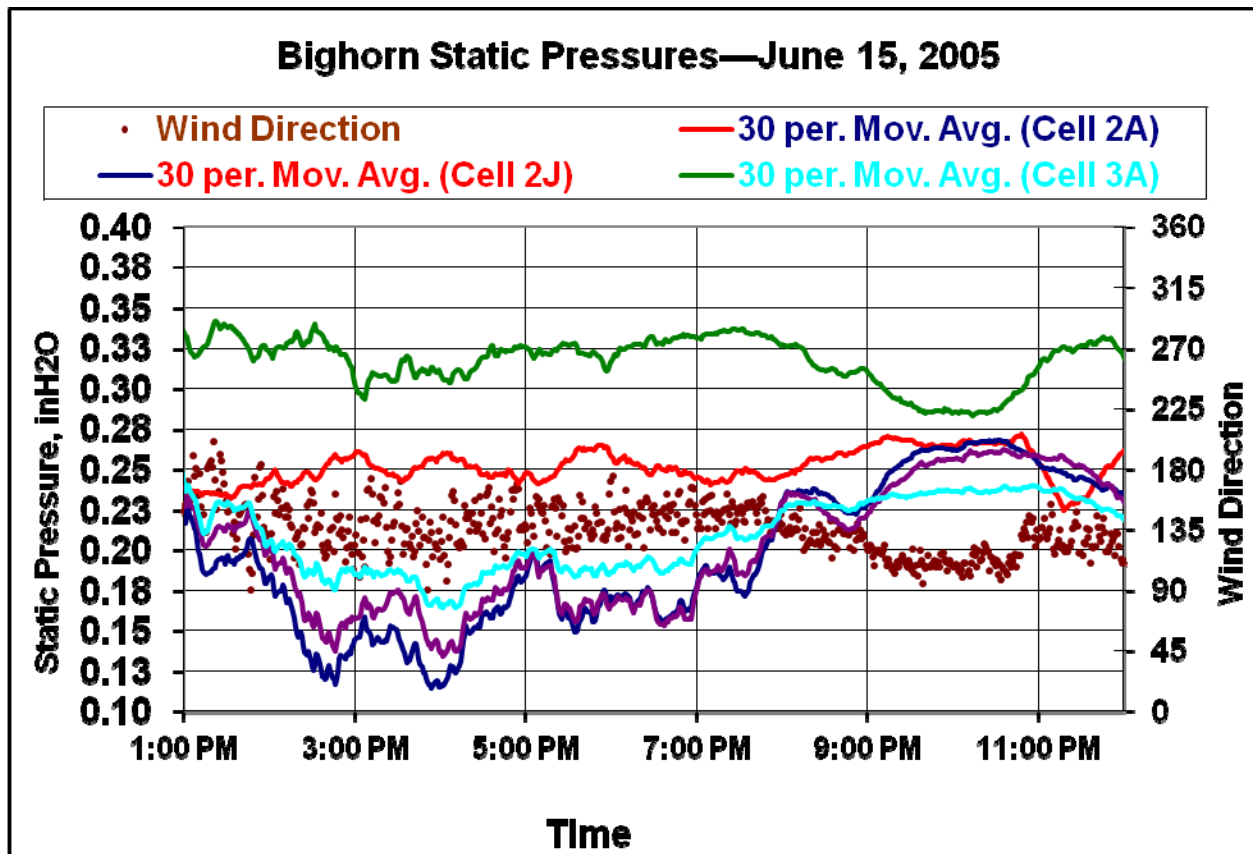


Figure 7-15. Static pressure rise in selected cells with wind direction---June 15, 2005.

Bighorn Static Pressures---June 18, 2005

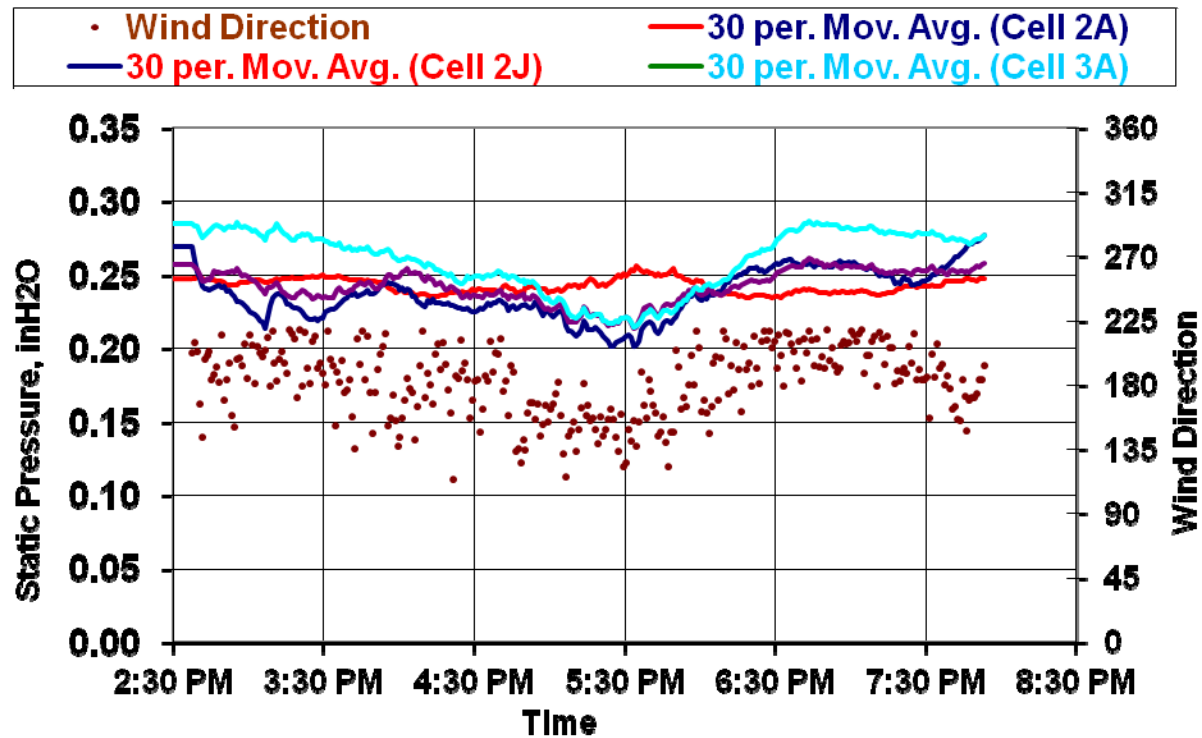


Figure 7-16. Static pressure rise in selected cells with wind direction---June 18, 2005.

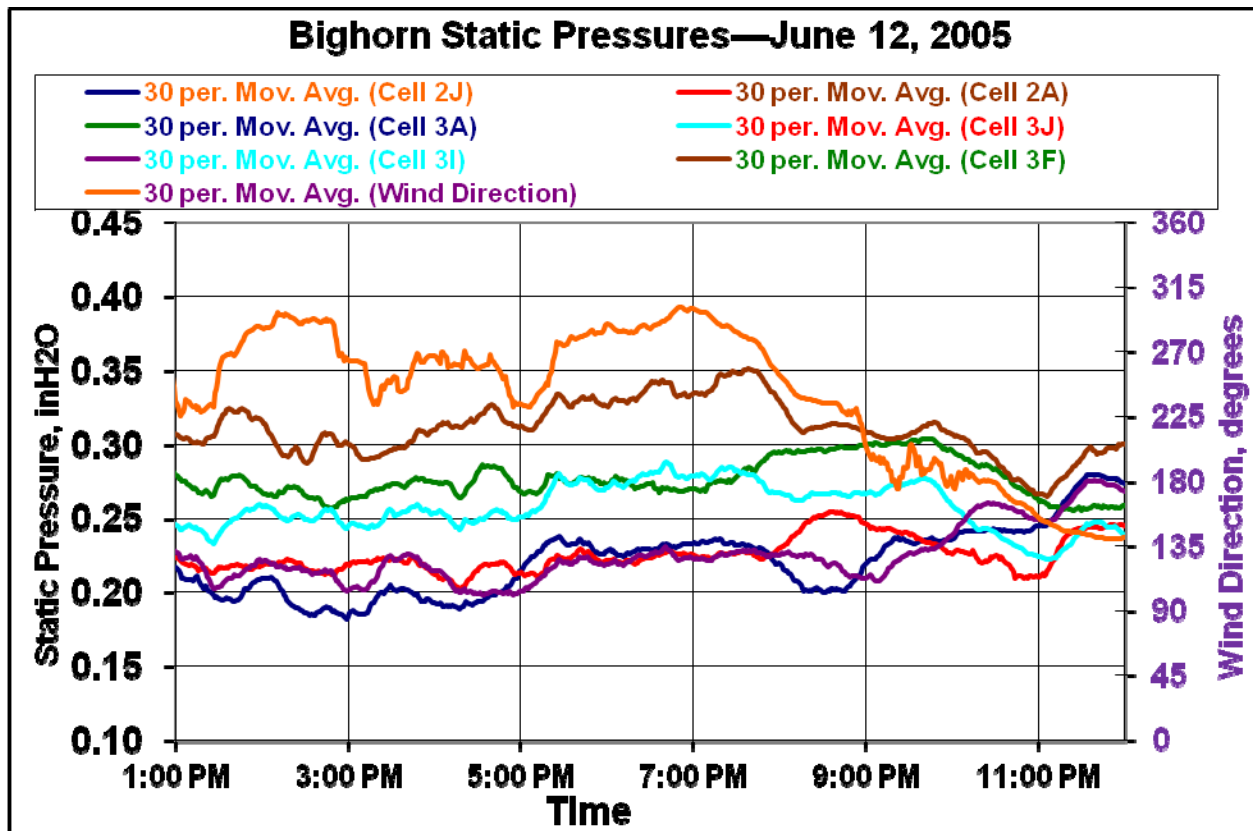


Figure 7-17. Static pressure rise in selected cells with wind direction---June 12, 2005.

7.2.5 Bighorn Motor Currents

Motor current measurements were made on one leg of each of six fan motors (2A, 2J, 3A, 3H, 3I, and 3J) for the entire test period. These data are displayed in Figure 7-18.

There are significant cell-to-cell differences with the currents ranging from 135 to 180 amperes for much of the test period. While each motor exhibits a wide range of currents as shown for Cell 3J in Figures 7-19 and 7-20, there is also a consistent difference from motor to motor with no apparent relationship to wind conditions. For example, Cell 2J seems to consistently have the lowest motor current throughout the test period. Figures 7-21 and 7-22 show the average motor currents along with wind direction on June 15 and June 18, the same time periods that were discussed earlier in the sections on inlet velocity and static pressure rise. No consistent or explainable behavior could be discerned in the motor current data, and motor current measurements were discontinued for the subsequent sites.

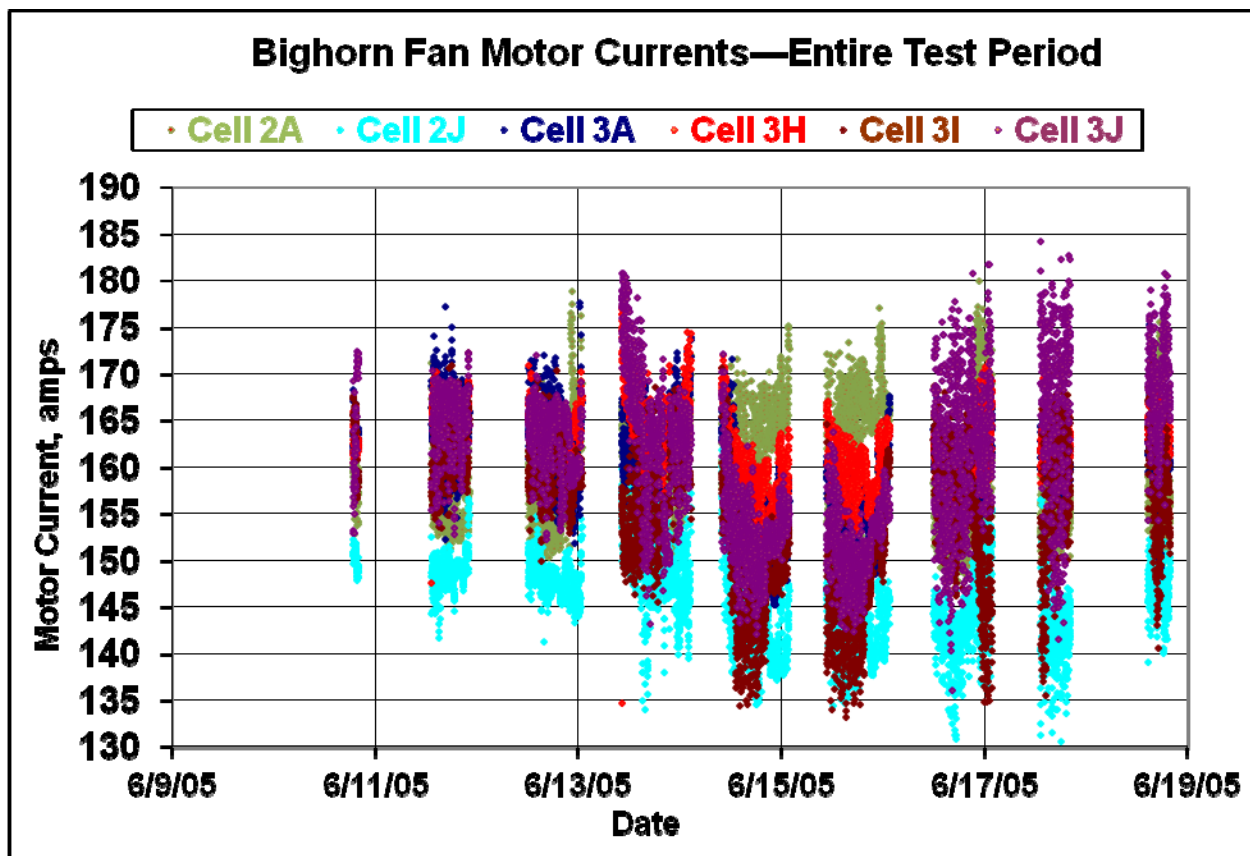


Figure 7-18. Bighorn motor current variation over entire test period---all fans on.

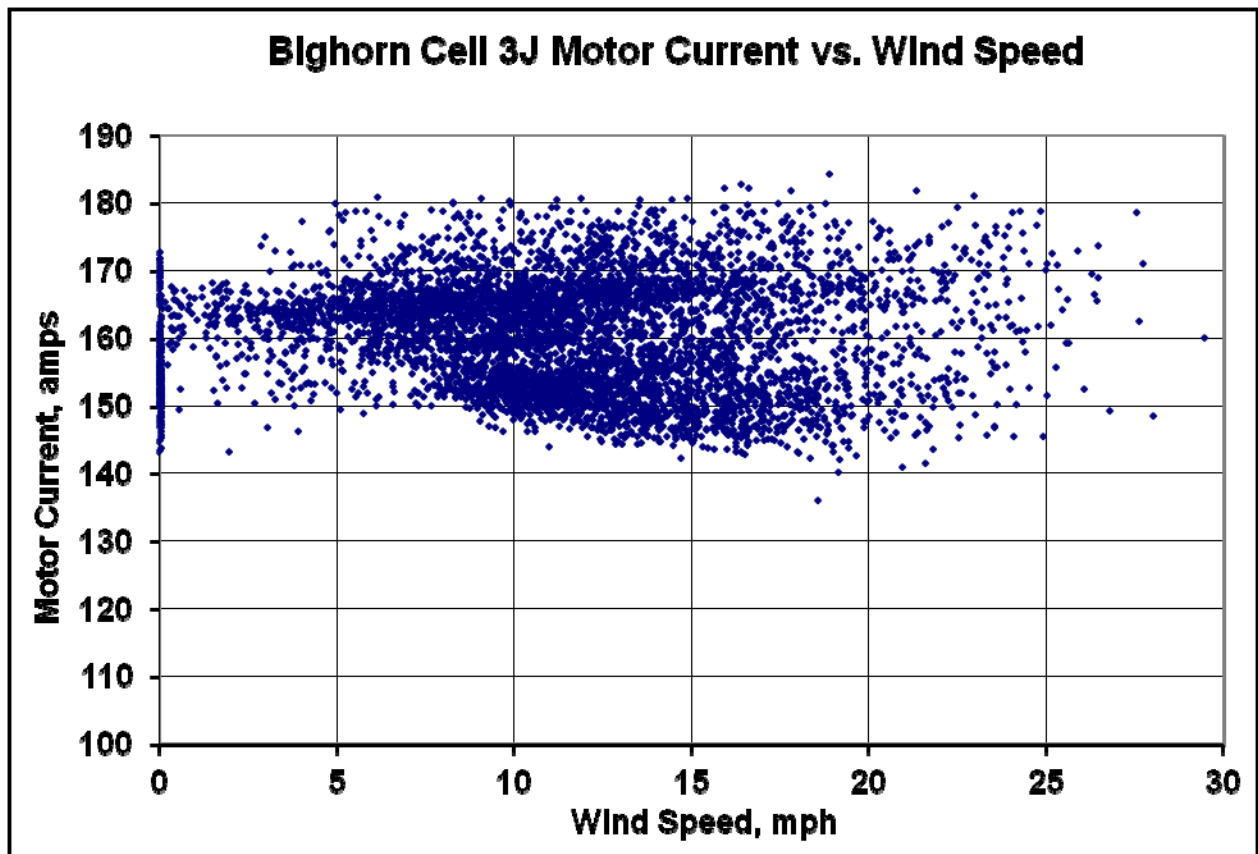


Figure 7-19. Motor current variation in Cell 3J vs. wind speed---entire test period.

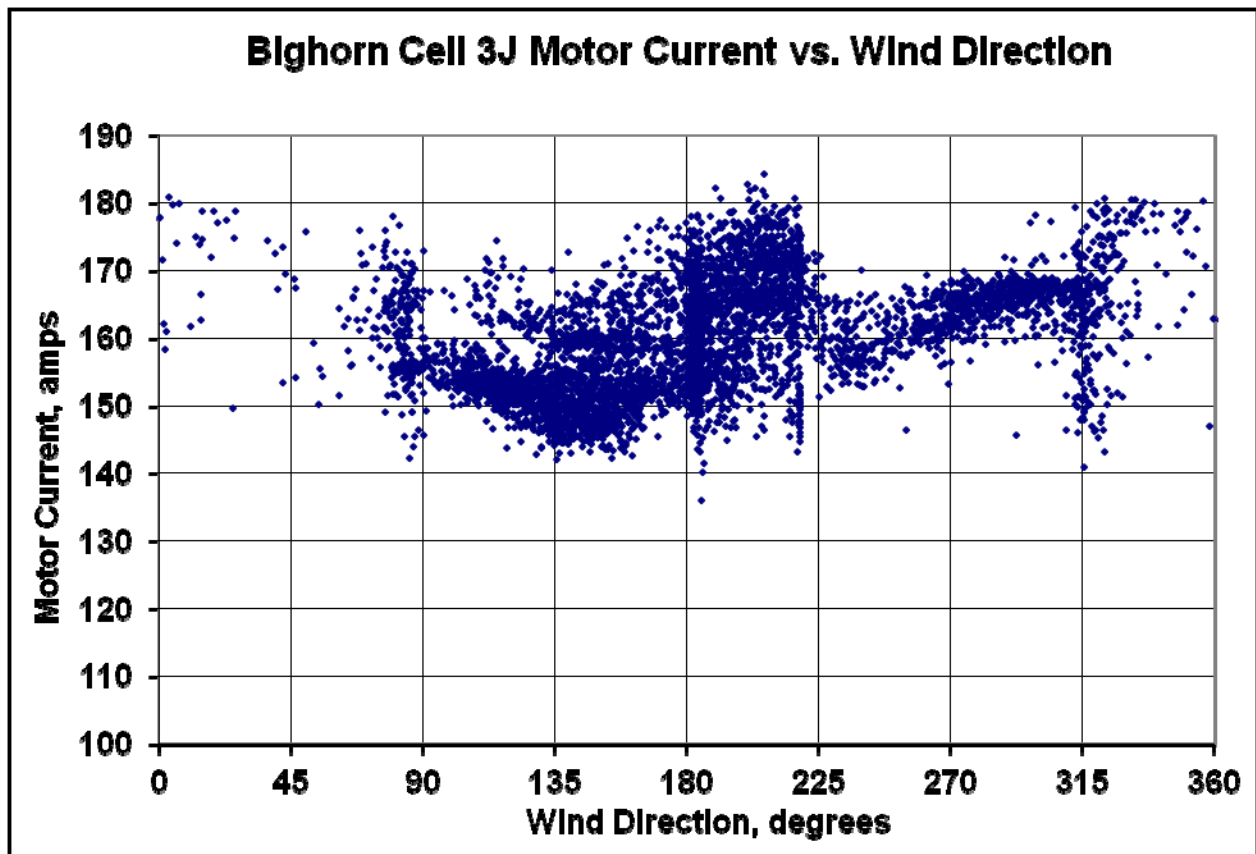


Figure 7-20. Motor current variation in Cell 3J vs. wind direction---entire test period.

Bighorn Fan Motor Currents—July 15, 2005

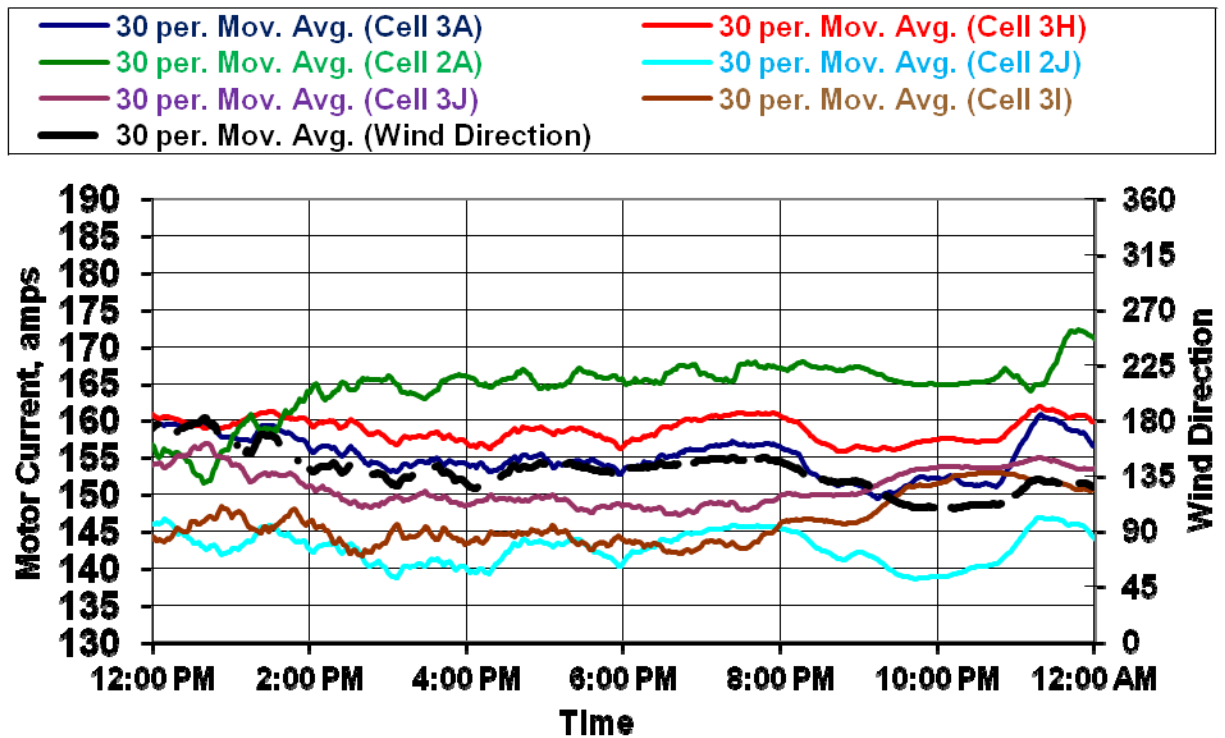


Figure 7-21. Motor current in selected cells---June 15, 2005.

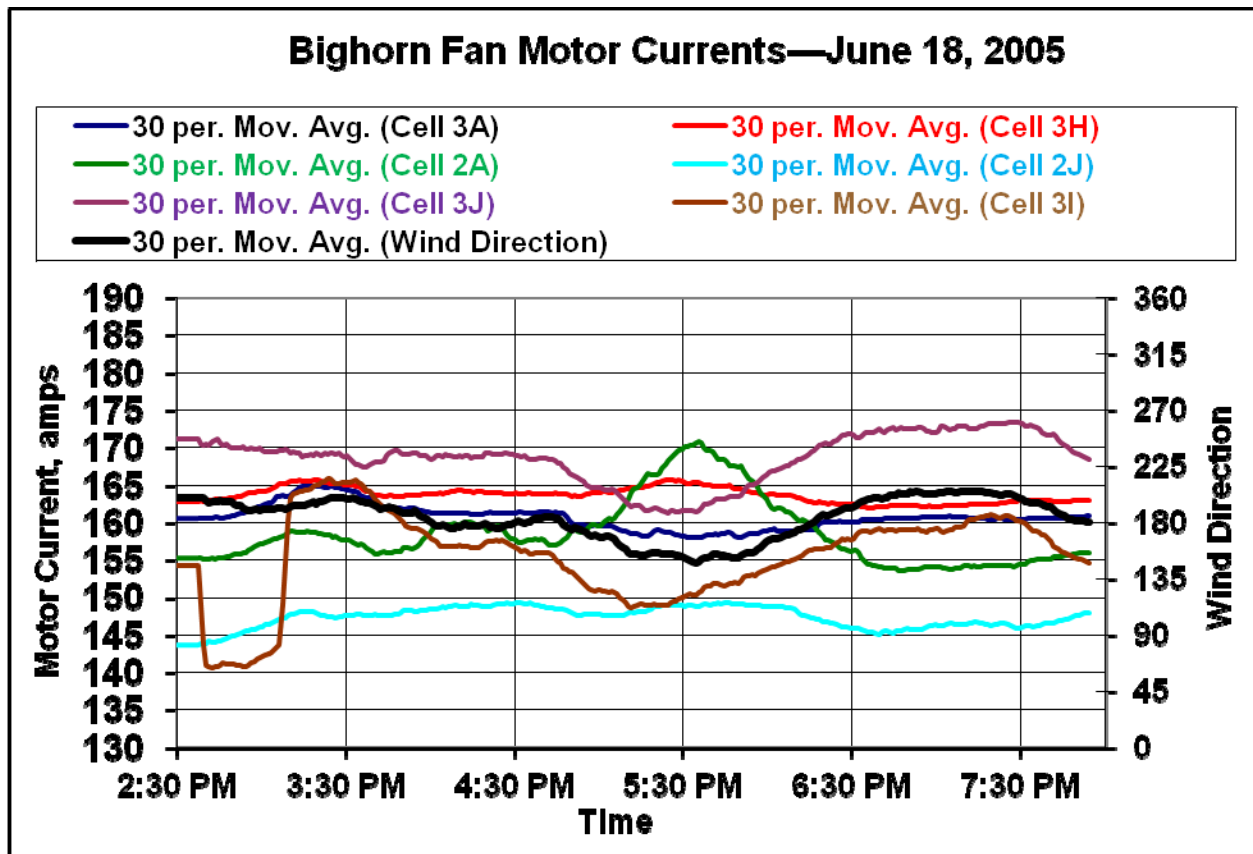


Figure 7-22. Motor current in selected cells---June 18, 2005.

7.3 El Dorado Fan Performance

Fan performance degradation at El Dorado was tested with measurements of cell inlet velocity (Cells 19, 22 and 24), and fan static pressure rise (Cells 1, 15, 19, 22 and 24). Fan motor current measurements were found to be of little value at Bighorn, and these measurements were discontinued at El Dorado and for all future sites. The locations of the cells on which inlet velocity and static pressure measurements were made and their position relative to the windscreens are shown on Figure 42. The location of the El Dorado windscreens is also displayed in Figure 3-11.

7.3.1 Cell inlet velocity

Cells 1, 19, 22, and 24 were instrumented with propeller anemometers suspended under the fans to monitor cell inlet velocity. The location of these cells on the ACC is shown in Figure 4-7.

Figure 7-23 plots the results of these measurements during the entire test period for those hours when all 30 fans were operating at full speed. The design flow for the fans is 1,332,000 acfm which corresponds to a velocity at the shroud inlet plane where the measurements are made of approximately 1060 fpm as shown on the plot. The average measured values are generally lower than the design value most of the time.

The periods of reduced cell inlet velocities correspond well with periods where the wind is blowing strongly from the south, as seen by comparison to the plot of wind speed and direction for the entire test period in Figure 89 (reproduced here as Figure 7-24 for convenient reference). Figure 7-25 shows wind speed vs. wind direction, confirming that the strongest winds are primarily southerly.

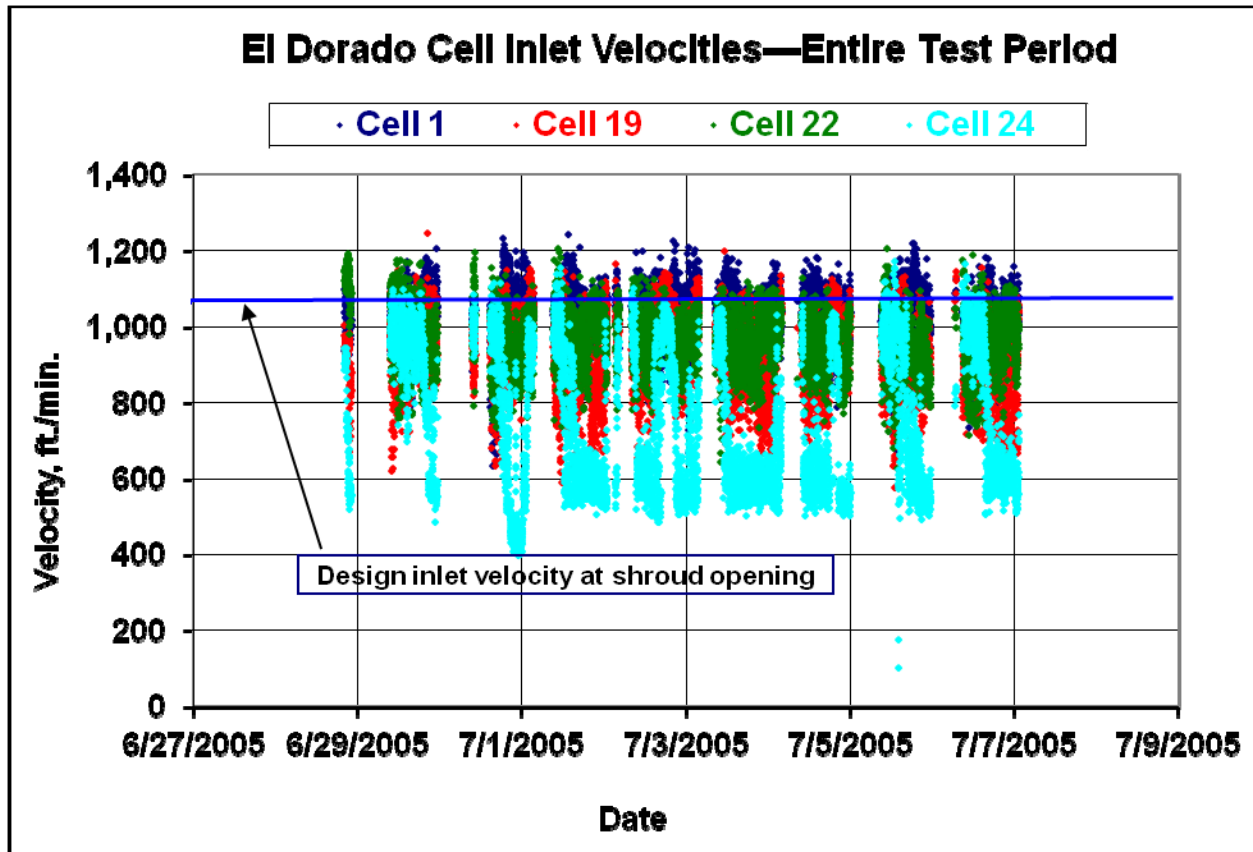


Figure 7-23. Average cell inlet velocity in selected cells---all fans on---entire test period.

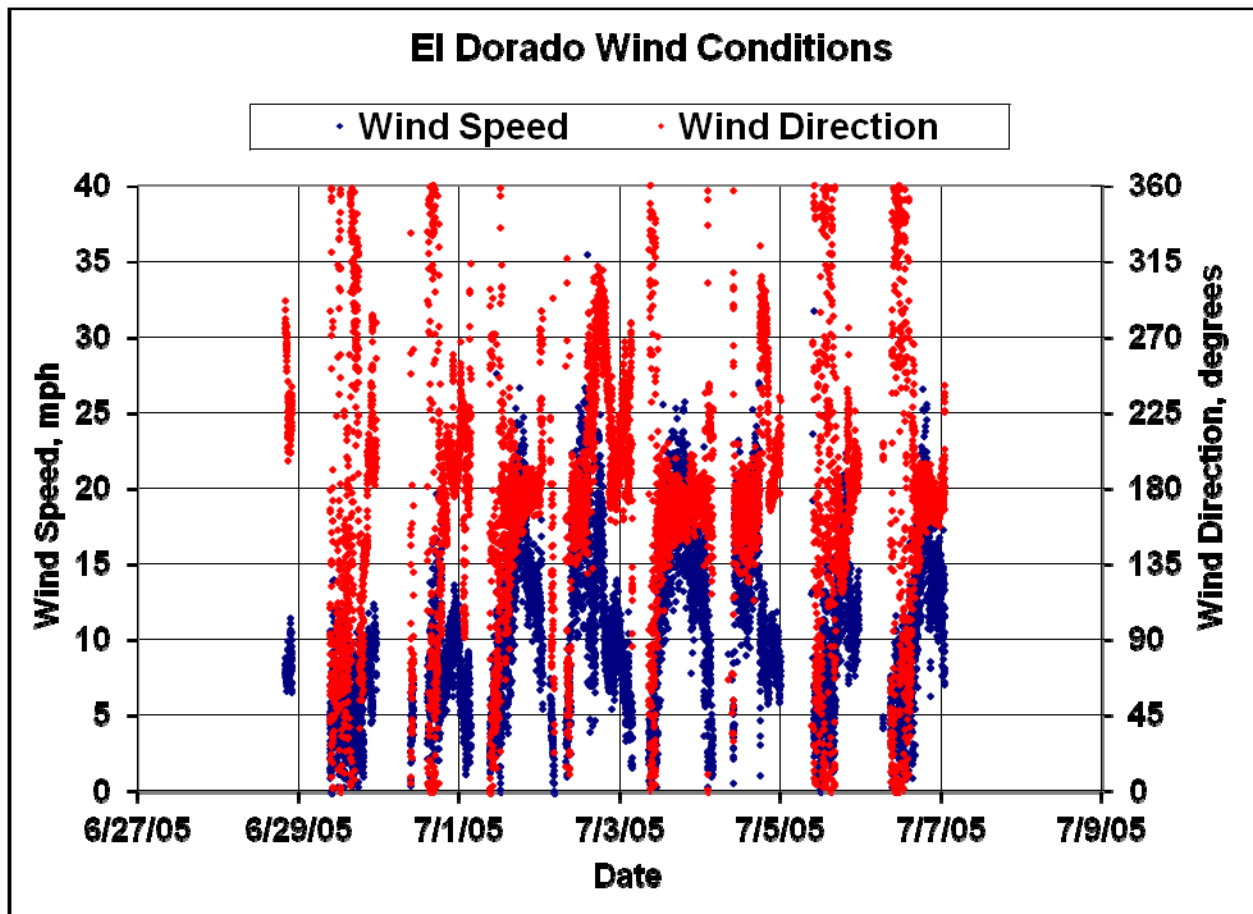


Figure 7-24. El Dorado wind speed and direction during test period (from Figure 6-29).

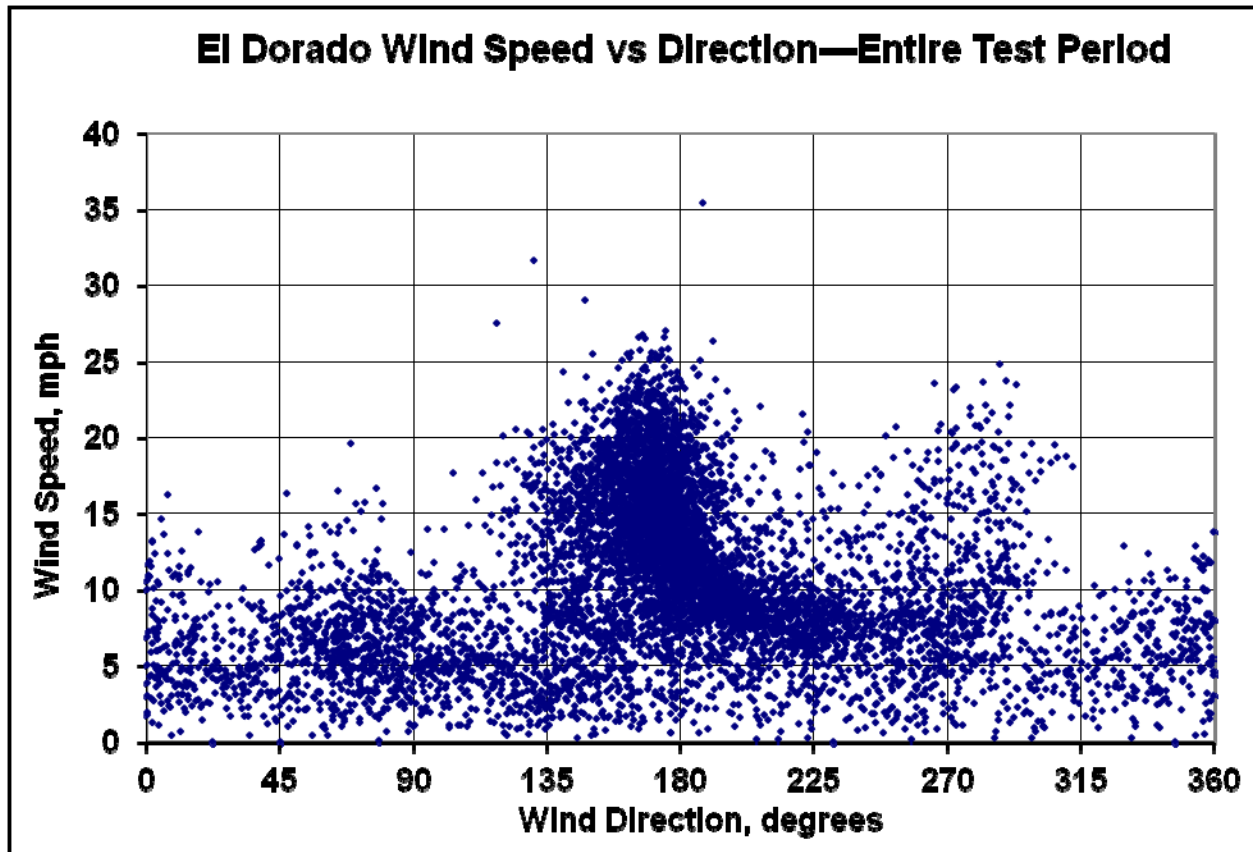


Figure 7-25. Wind speed vs. wind direction at El Dorado---entire test period.

Cell 24, located at the south end of Street 4, just west of the N-S windscreen, is the most exposed of the four cells to the prevailing southerly winds. Figure 7-26 shows the significant drop-off in airflow in the presence of southerly winds. The effect is nearly symmetrical around the southerly direction. It might be argued that the effect persists somewhat longer as the wind shifts to the east and that the effect ends more abruptly for easterly wind shifts at about $\sim 135^\circ$ than for westerly wind shifts. This may be the result of the windscreen which may allow more disturbances for winds slightly to the east of south but then shield the fan from winds from more easterly directions. Cell 22, shown in Figure 7-27, is located in the southwest corner of the intersection of the N-S and E-W windscreens and shows less effect of wind with little dependence on wind direction.

Cell 1 (Figure 7-28) is located at the northeast corner of the ACC and is most exposed to northerly and easterly winds. Some reduction in average inlet velocity is seen during northerly winds from 340° to 30° . However, as seen in Figure 7-29, the northerly winds are quite mild—typically less than 5 mph and only occasionally as high as 10 mph so their effect on fan performance is significantly less than the effect of the stronger southerly winds on Cell 24.

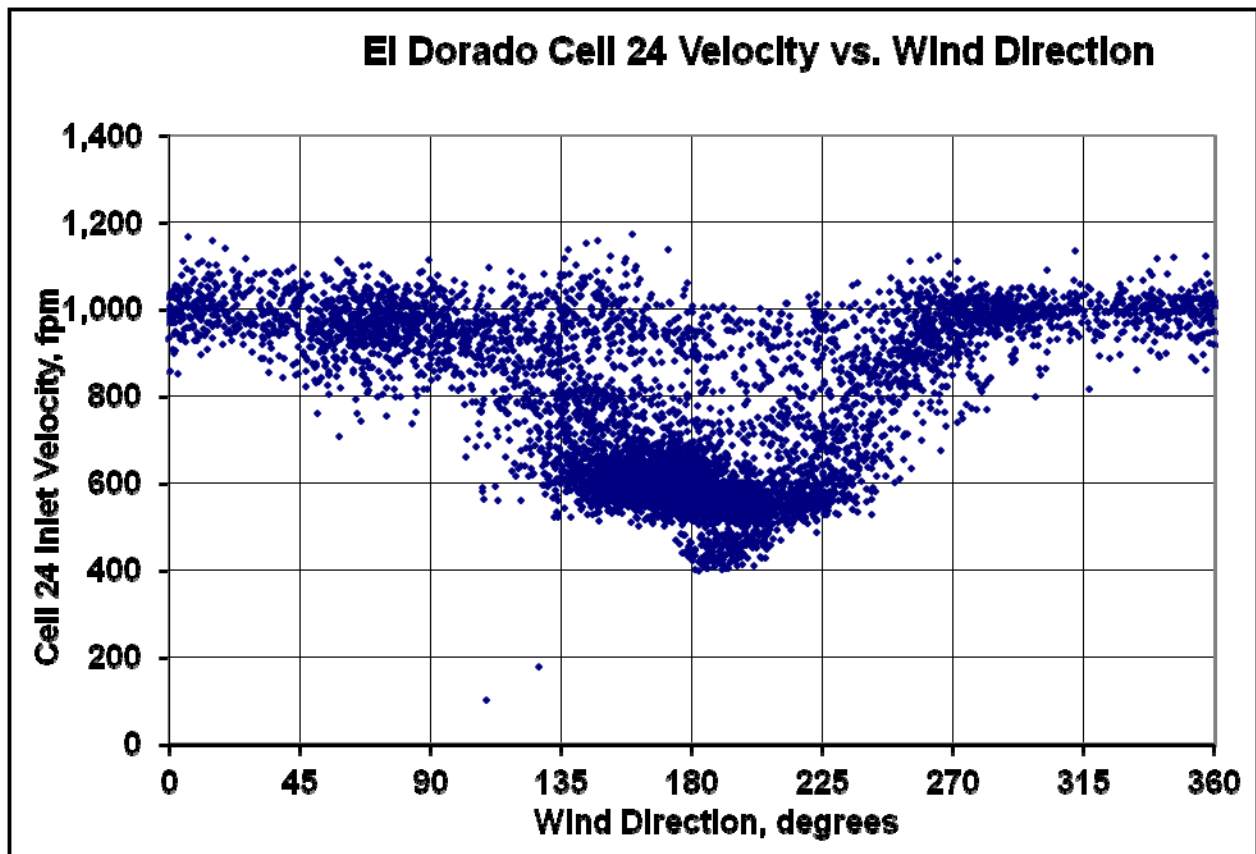


Figure 7-26. Cell 24 inlet velocity vs. wind direction---entire test period.

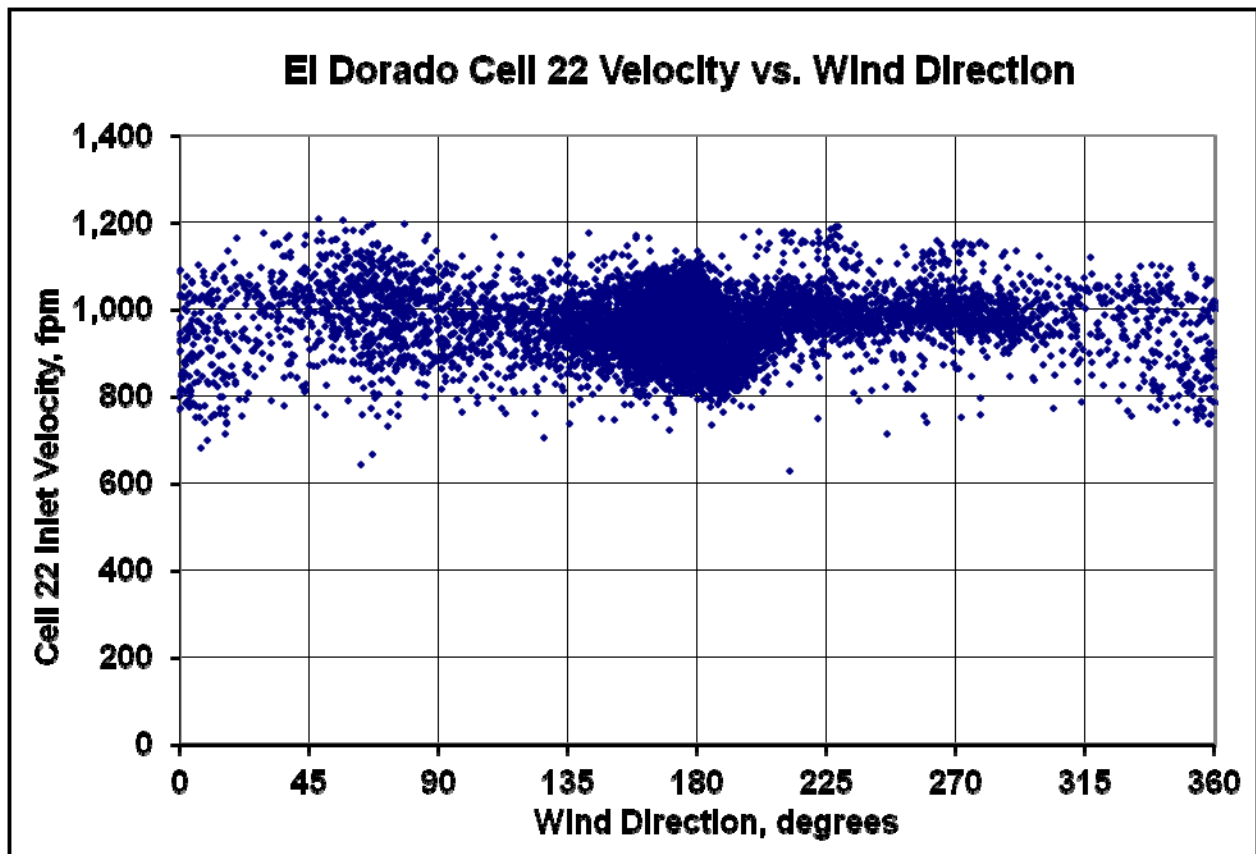


Figure 7-27. Cell 22 inlet velocity vs. wind direction---entire test period.

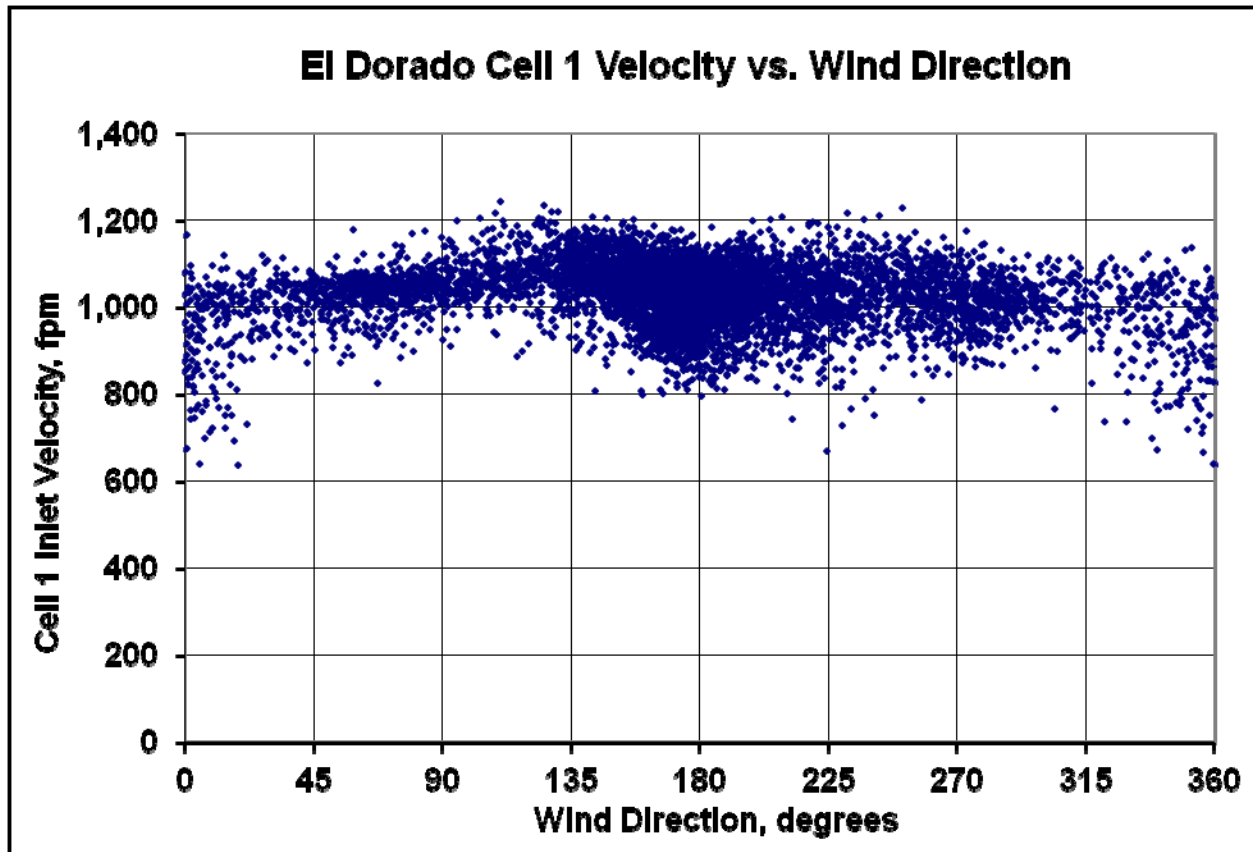


Figure 7-28. Cell 1 inlet velocity vs. wind direction---entire test period.

Figures 7-29 and 7-30 plot the inlet velocities for the four cells along with concurrent wind direction for July 2 and 3 from 8:30 a.m. to midnight.

On July 2, the effect is restricted almost entirely to Cell 24. When the wind is primarily from the south (9:30 a.m. to 2:30 p.m. and again from 9:00 p.m. until midnight), the average inlet velocity in Cell 24 is reduced by 50 %. At a few times during the day, the wind was from the east (8:30 a.m. to 9:00 a.m.) and from the northwest (4:30 p.m. to about 8:00 p.m.). Under those conditions, the airflow in Cell 24 returned essentially to the design level, consistent with the other fans.

The absence of any discernible effect on the performance of Cell 1 during the periods of easterly and northwesterly winds is likely attributable to the low wind velocities during those periods as shown in Figure 7-30.

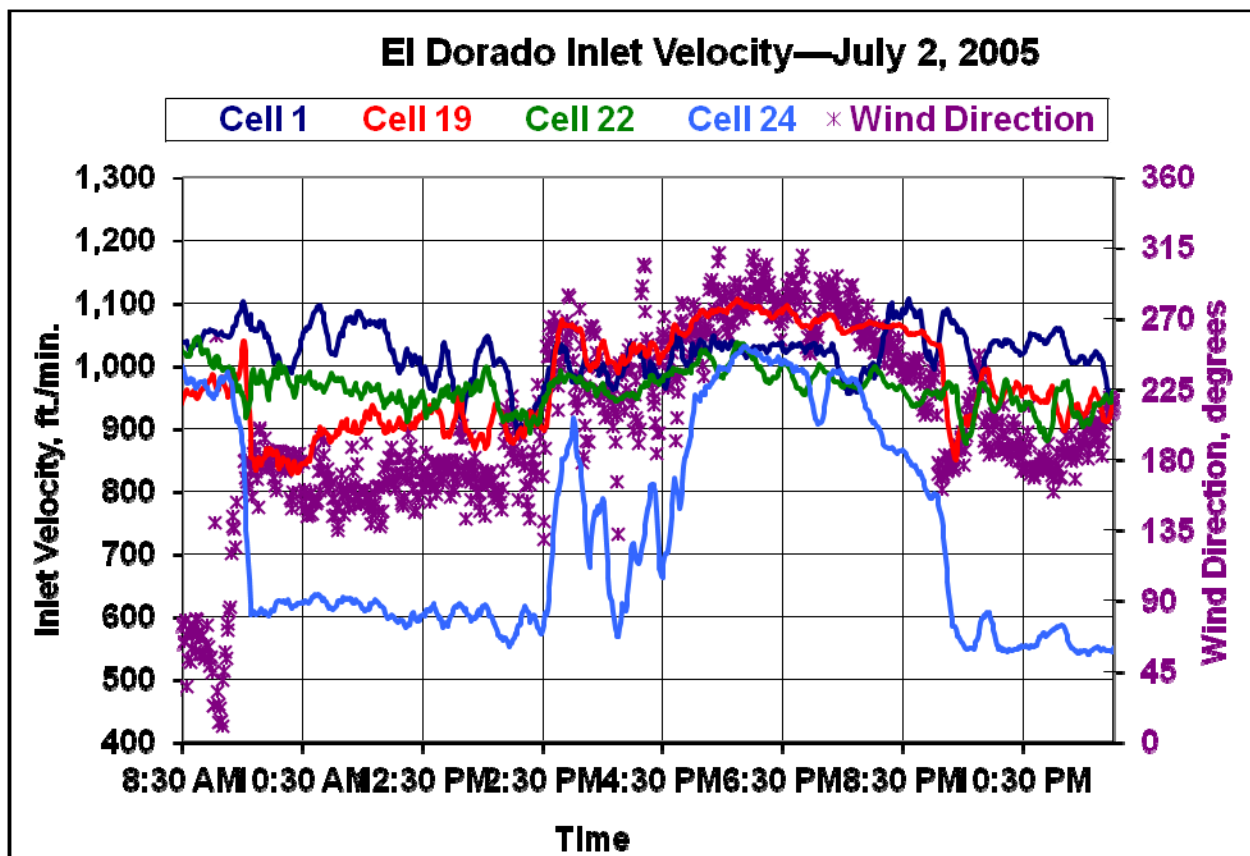


Figure 7-29. Inlet velocity in selected cells with wind direction---7/2/2005; 8:30 am to midnight

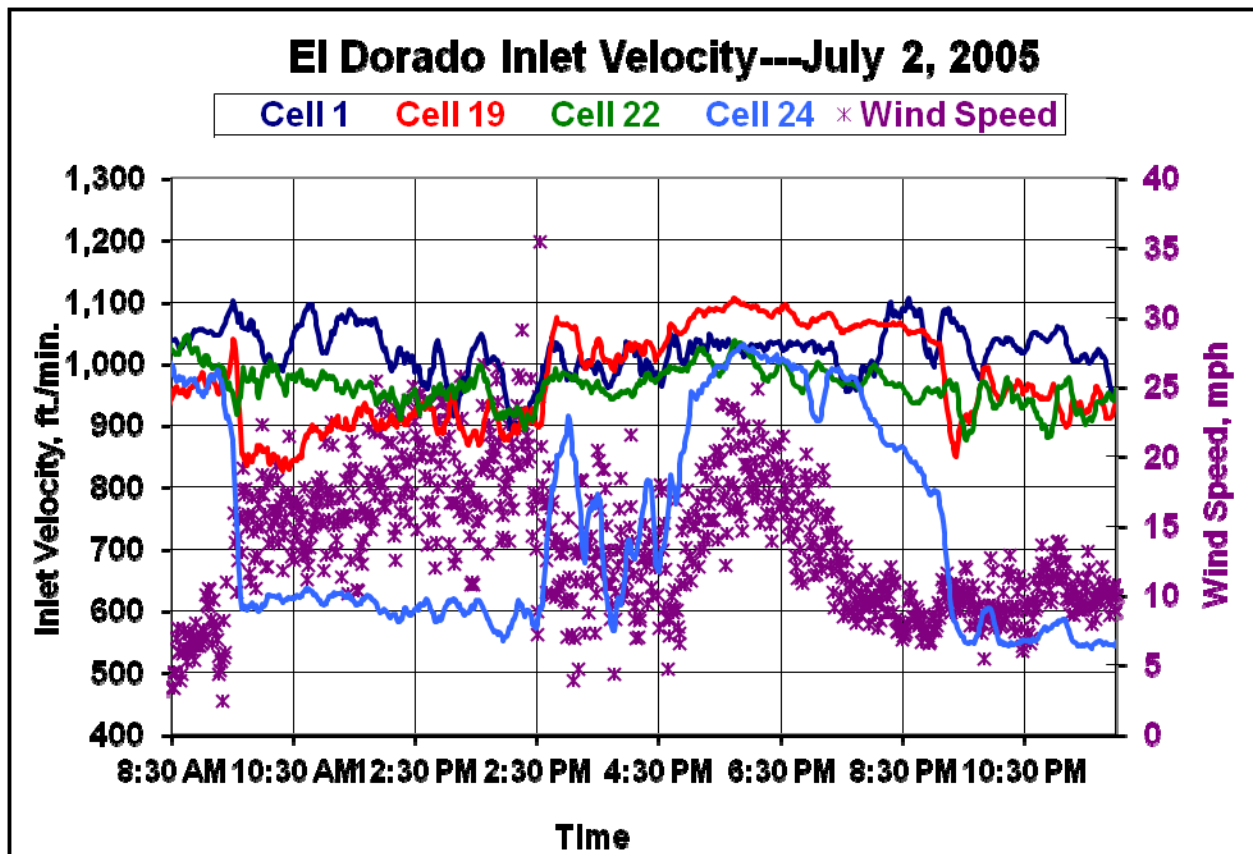


Figure 7-30. Inlet velocity in selected cells with wind speed---7/2/2005, 8:30 a.m. to midnight.

Figure 7-31 plots similar data for July 3. The results are the same with slight differences. Again, the wind for most of the day is directly from the south with a significant degrading effect on the performance of Cell 24 except early in the day when the winds were from the east to northwest. However, on July 3 there is a slightly noticeable effect on the performance of Cells 1 and 19. Cell 1 shows an abrupt increase in flow at about 11:00 a.m. when the wind shifts to a steady southerly direction. Similarly, around 8:30 p.m. the performance of Cell 19 drops abruptly. This appears to coincide with a very light shift in wind direction to the west of south and a decrease in wind speed. The reason for this altered behavior is not understood.

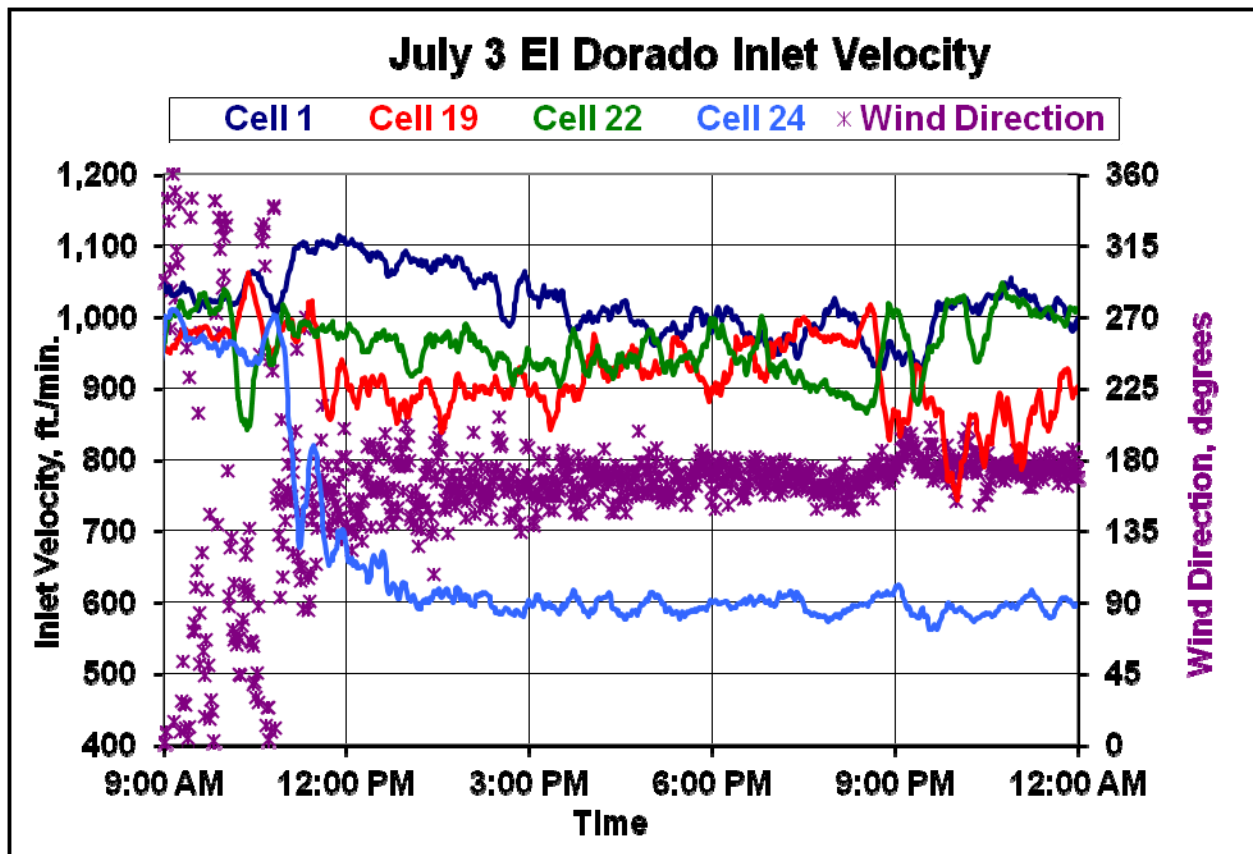


Figure 7-31. Inlet velocity in selected cells with wind direction---July 3, 2005; 9:00 a.m. to midnight.

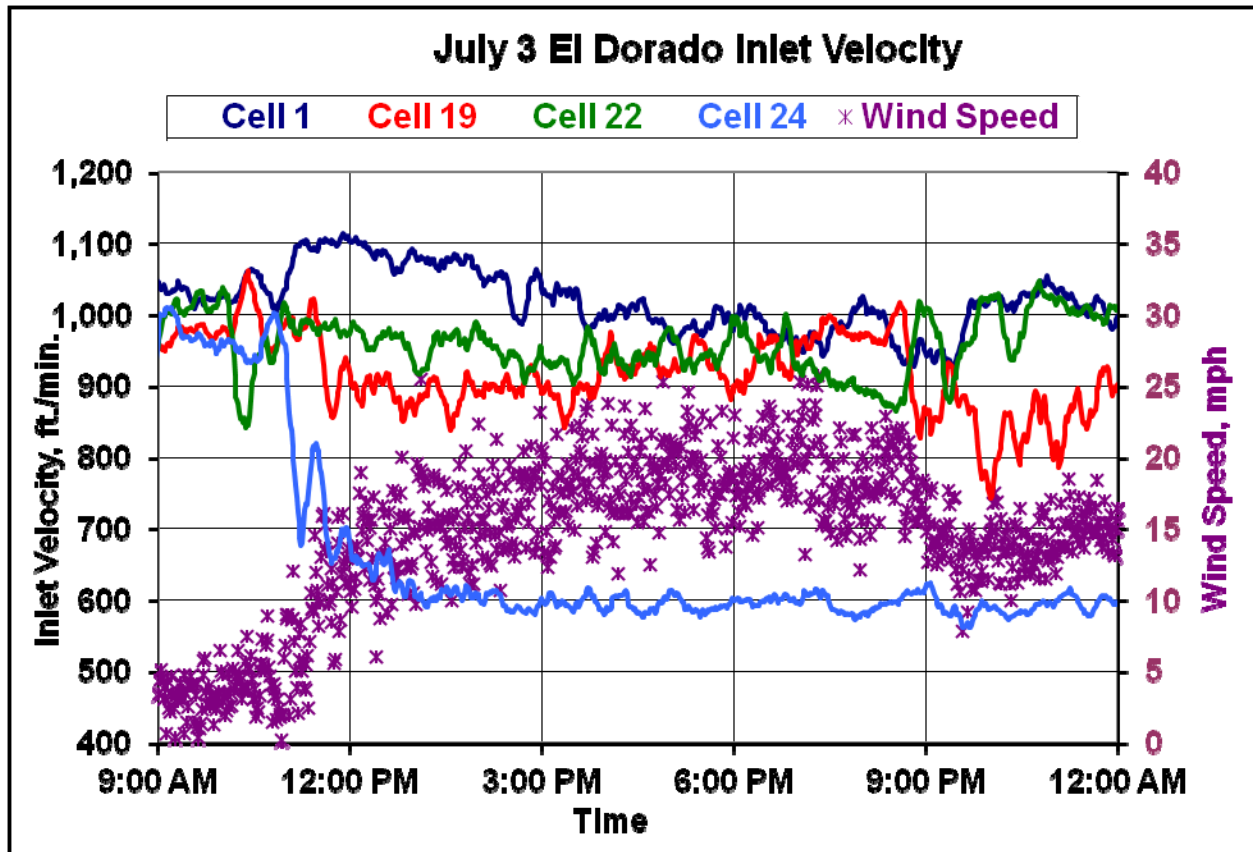


Figure 7-32. Inlet velocity in selected cells with wind speed---July 3, 2005; 9:00 a.m. to midnight.

7.3.2 El Dorado Static Pressure

Static pressure rise measurements were made in five cells, Cells 1, 15, 19, 22, and 24. These are the same cells for which inlet velocity measurements were made, plus Cell 15. The location of these cells is shown in Figure 42.

Data for the entire test are shown in Figure 7-33. For most of the time, the static pressure rise in Cell 24 is the highest during those periods when the flow into Cell 24 is the least, as would be expected.

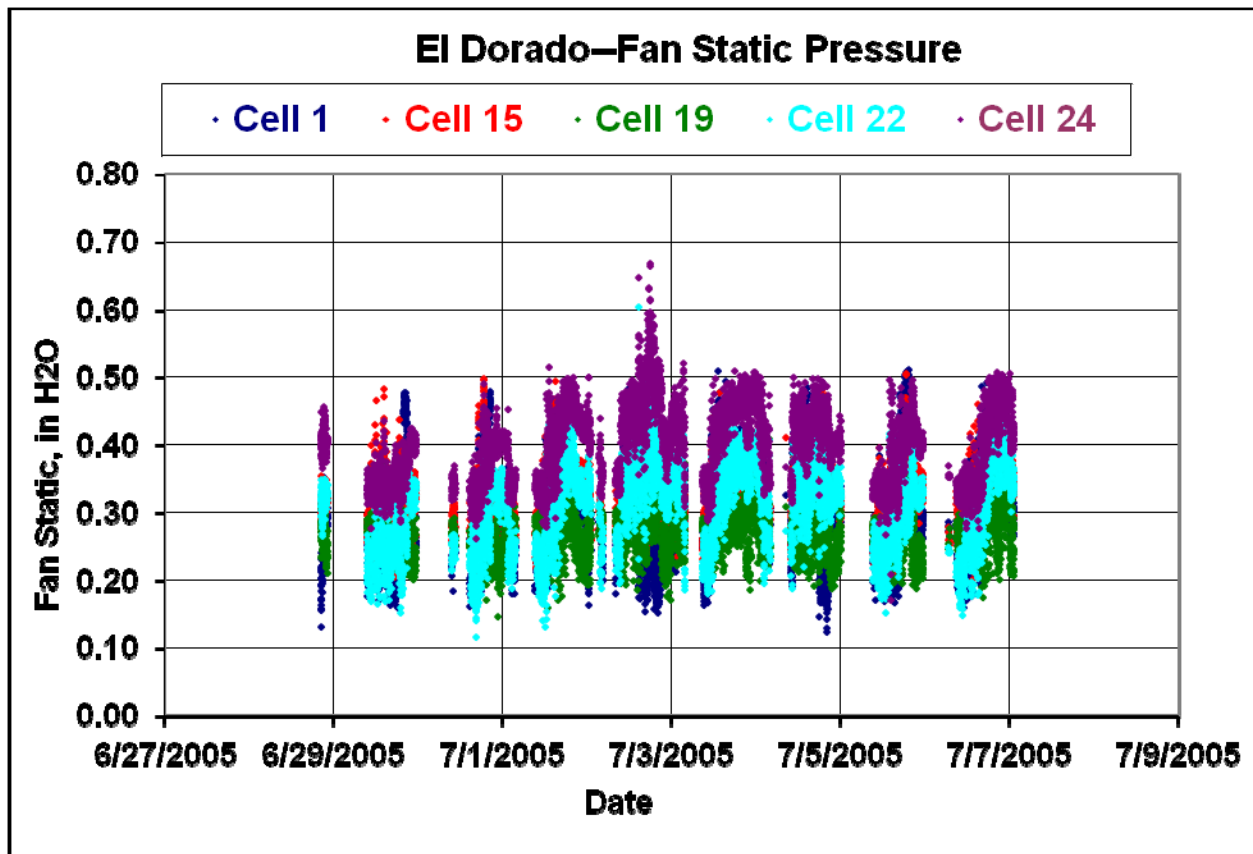


Figure 7-33. Static pressure rise variation at El Dorado---All fans on —entire test period.

Figures 7-34 and 7-35 display the static pressure rise variations concurrently with wind direction for July 2 and July 3, corresponding to the similar plots of inlet velocity shown in Figures 7-29 through 7-31. On July 2, Cell 24 consistently has the highest static pressure rise. The variations in Cells 1, 15, and 19 are not understood since they exhibit significant changes in the static pressure rise at times where there are no recorded variations in airflow. It may be that wind patterns under the ACC cause small variations in the reference probes used in determining the static pressure rises in the several cells. However, at this time there is no way to verify this hypothesis.

The behavior of July 3 is more consistent, with Cell 24 having the highest static pressure rise and the lowest airflow while Cell 1 has the lowest static pressure rise and the highest air flow. The readings of all the cells are quite steady and no unexplained excursions are noted.

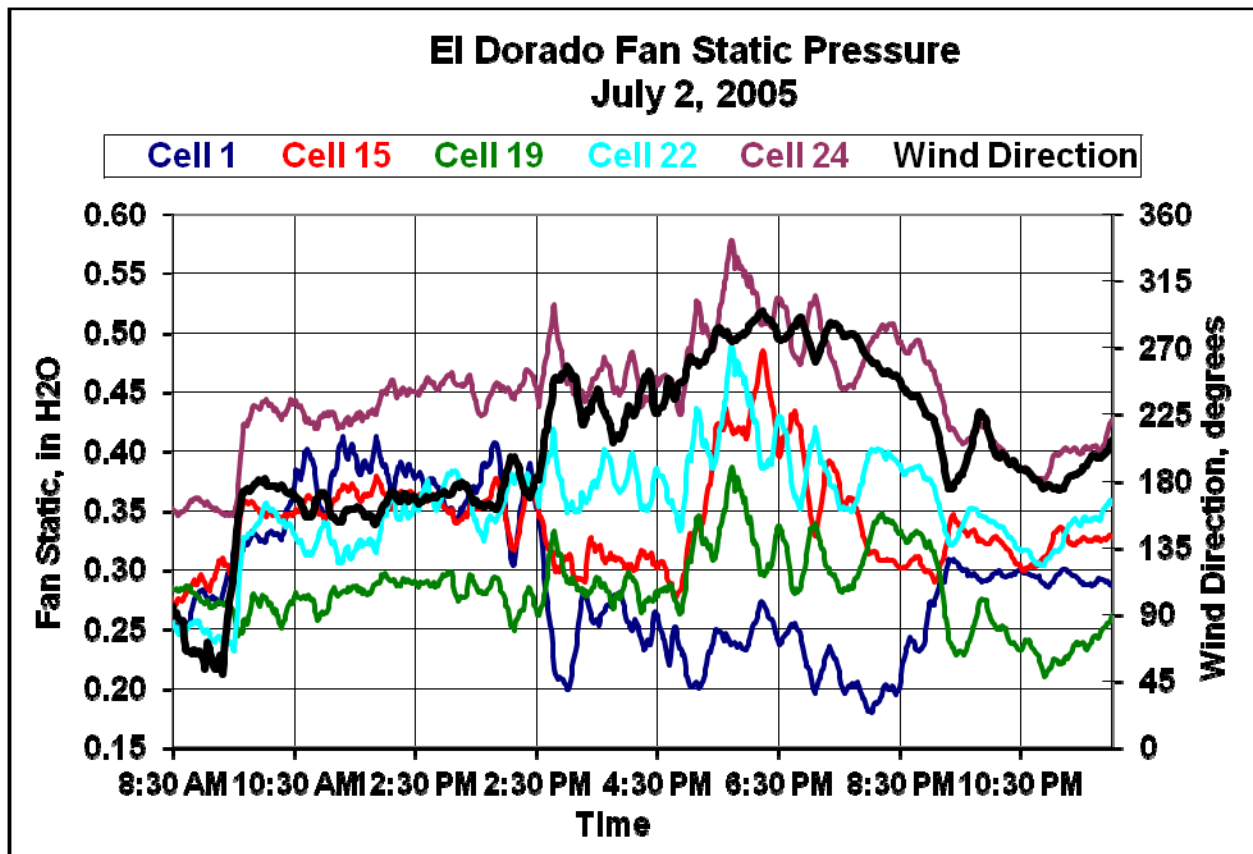


Figure 7-34. Static pressure rise in selected cells with wind direction---July 2, 2005; 8:30 a.m. to midnight.

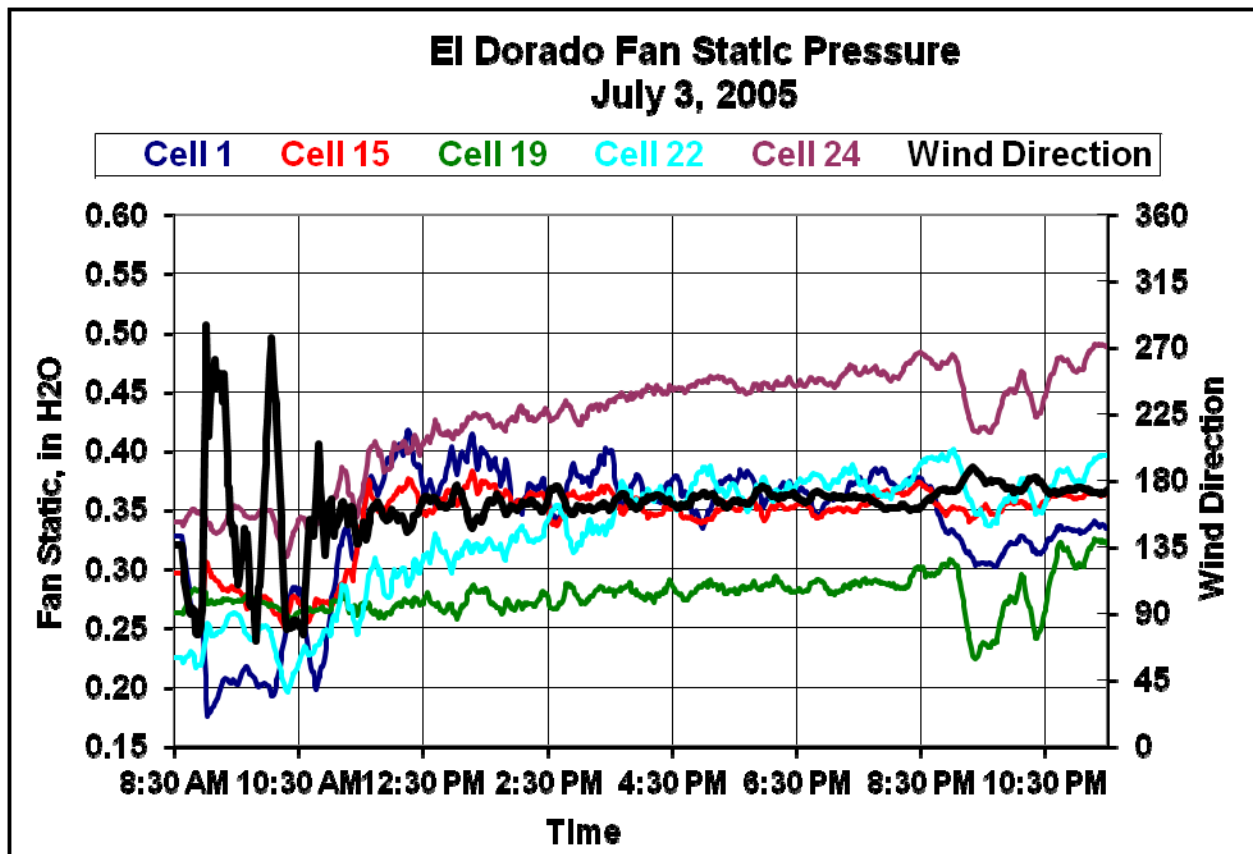


Figure 7-35. Static pressure rise in selected cells with wind direction---July 3, 2005; 8:30 a.m. to midnight.

7.4 Apex Fan Performance

Fan performance degradation at Apex was tested with measurements of cell inlet velocity (Cells K1D, K5D and D4). Fan static pressure measurements were found to add little additional insight beyond that gained from the inlet velocity measurements at Bighorn and El Dorado and were discontinued at Apex and for all future sites. The locations of the cells on which the inlet velocity measurements were made are shown on Figure 4-8. There are no windscreens at Apex.

7.4.1 Inlet velocity

Cells 1D, 4 and 5D were instrumented with propeller anemometers. At Apex and at future sites, four anemometers in each of the instrumented cells were suspended above the fans by hanging them from the edge of the fan support bridge inside the cells. They were positioned in two pairs on opposite sides of the fan bridge and on both sides of the motor hub, approximately half way from the blade root to the tip. While it is recognized that the swirling of the air flow above the fans may have some effect on the absolute value of the measured velocity, the value at very low ambient wind speeds is close to the fan design value. It is further assumed that the comparative values at differing wind speeds will remain valid.

Figure 7-36 plots the results of these measurements during the entire test period for those hours when all 30 fans were operating at full speed. The design flow for the fans is 1,610,000 acfm which corresponds to a velocity in the plane of the fan of approximately 1,775 fpm as shown on the plot. The average measured values are generally lower than the design value most of the time.

The periods of reduced cell inlet velocities correspond well with periods where the wind is blowing strongly from the south, as seen by comparison to the plot of wind speed and direction for the entire test period in Figure 109 (reproduced here as Figure 7-37 for convenience of reference).

Figure 7-38 shows the relationship between wind speed and direction during the test period. While at the lower wind speeds, the wind direction is reasonably uniformly distributed across all directions, the higher wind speeds come primarily from the southeast (135 to 180 degrees) and the west-southwest (225 to 270 degrees). Figures 7-39 and 7-40 show the effect of wind speed (Figure 7-39) and wind direction (Figure 7-40) on the average air inlet velocity in Cell 1D. There is a clear fall-off in air flow with increasing wind speed from nearly all directions between 10 and 20 mph. For speeds below 10 mph, there are many instances where the flow remains high. Inspection of Figure 7-40 suggests that this occurs for winds which are generally northerly (45 to 90 degrees and 315 to 360 degrees). The ACC is well shielded on the north side by the plant buildings.

The general shape of the airflow vs. wind direction mirrors the pattern of wind speed vs. wind direction seen in Figure 7-38. This indicates that the combined effect of wind speed and direction results from the fact that the higher wind speeds are nearly always from the southwest and southeast.

The corresponding results for Cells 4 and 5D are similar but less distinct with variations in wind speed and direction. The performance measurements of Cell 4 are clouded by the fact that it frequently is run at half-speed, but no detailed fan status information is available from the plant. Cell 5D is somewhat more protected from wind effects since, unlike Cell 1D, it is not a corner cell.

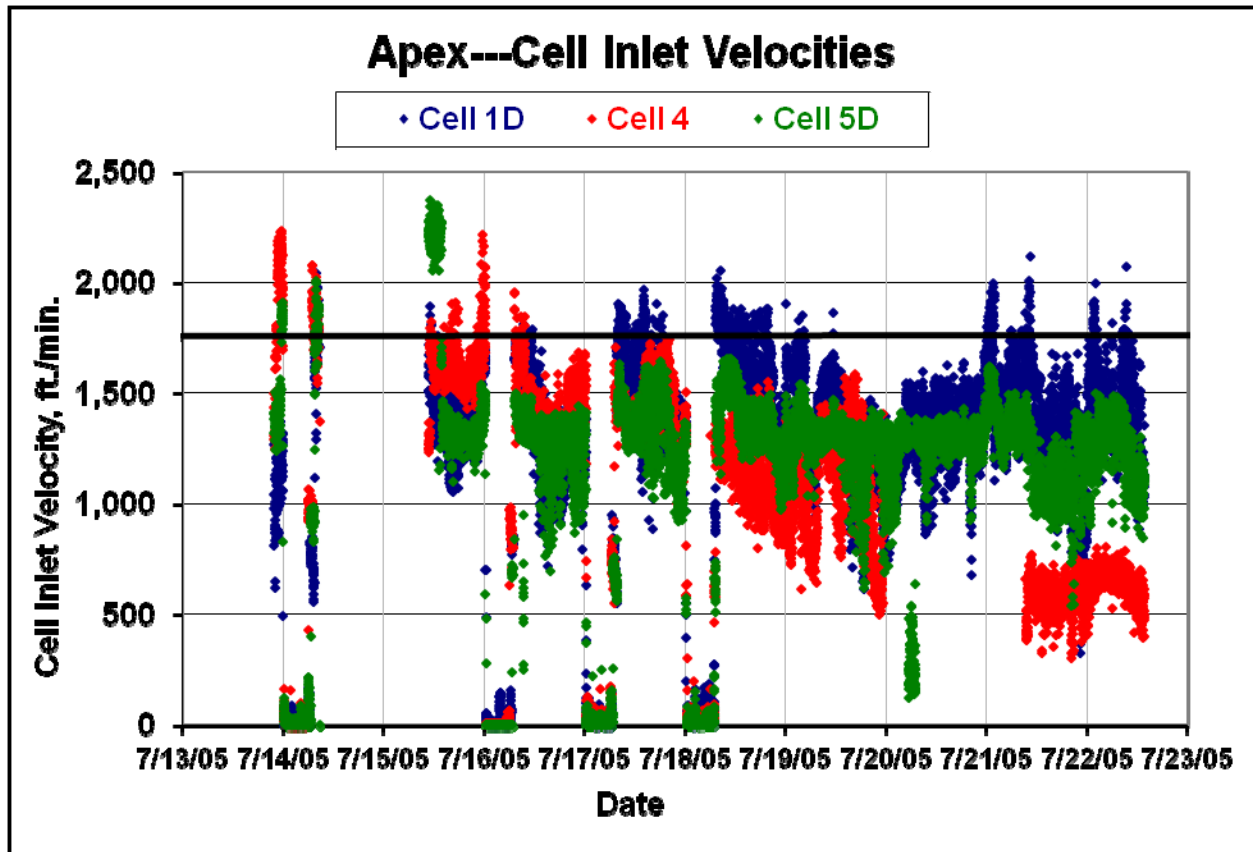


Figure 7-36. Apex Cell Inlet Velocities; Cells 1D, 4 and 5D---Entire Test Period.

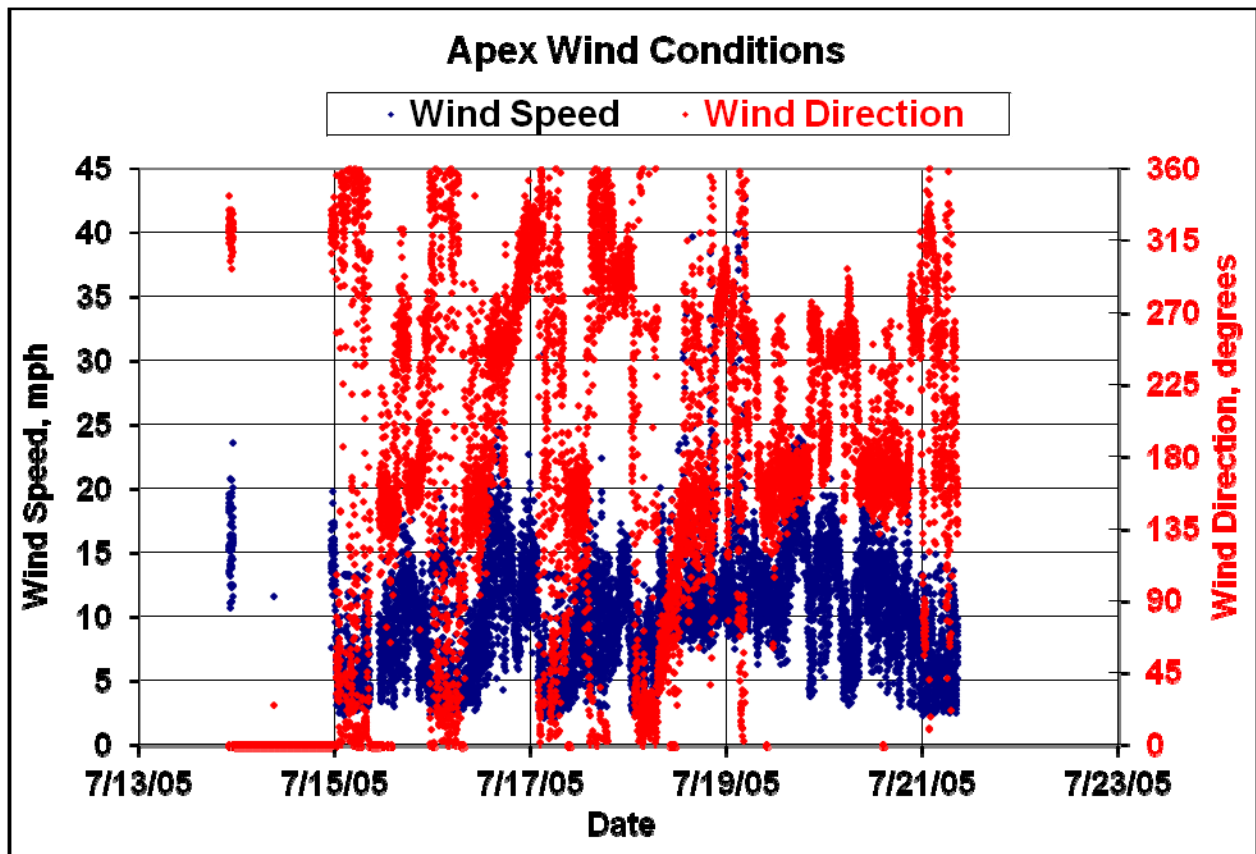


Figure 7-37. Apex Wind Conditions—Entire Test Period.

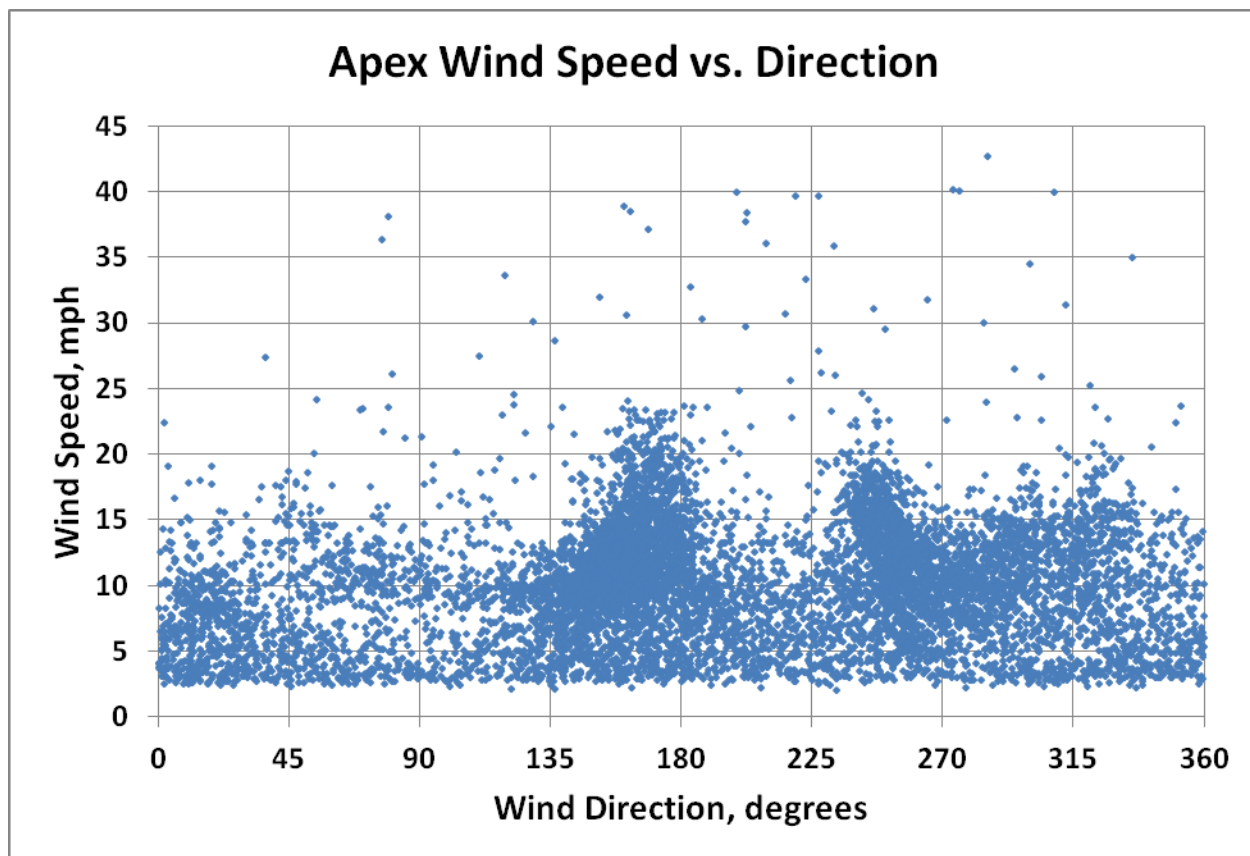


Figure 7-38. Apex Wind Speed vs. Direction—Entire Test Period.

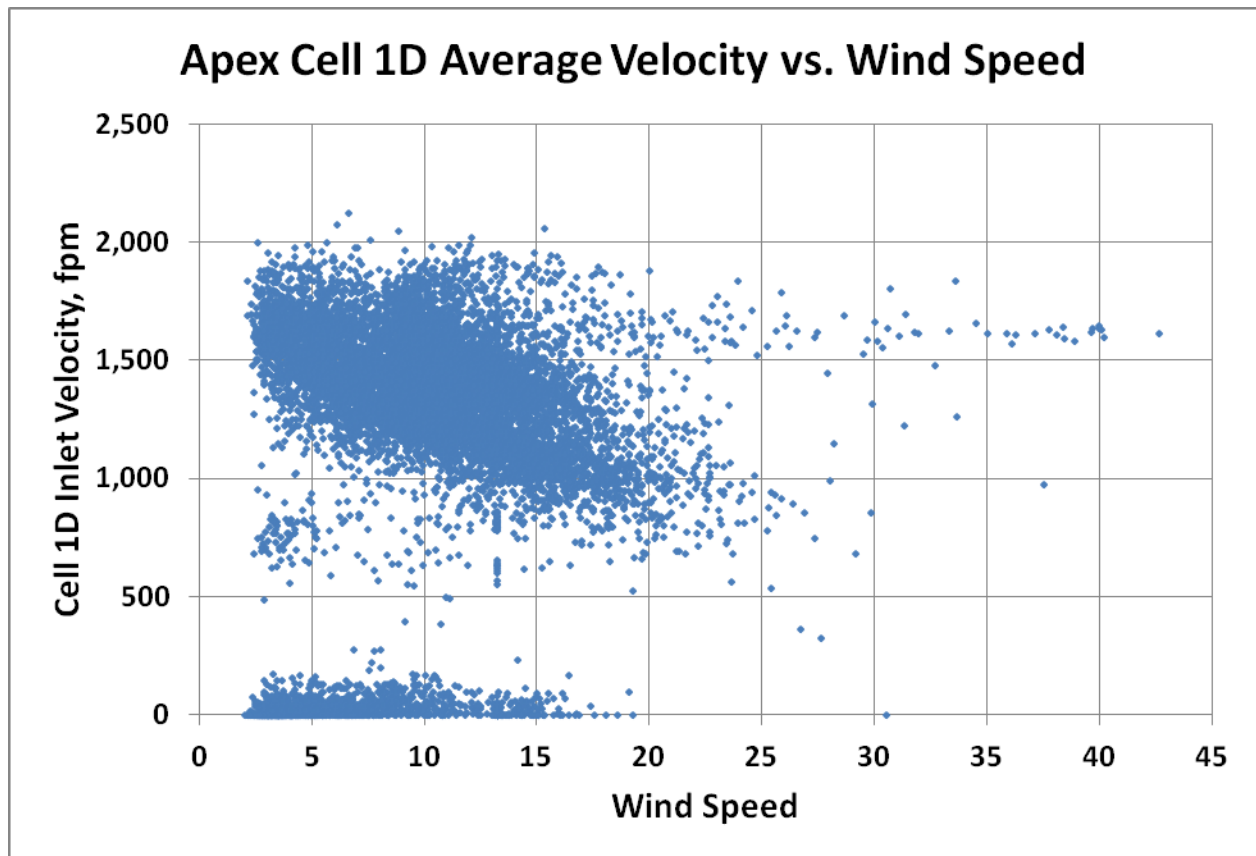


Figure 7-39. Apex Cell 1D Inlet Velocity vs. Wind Speed.

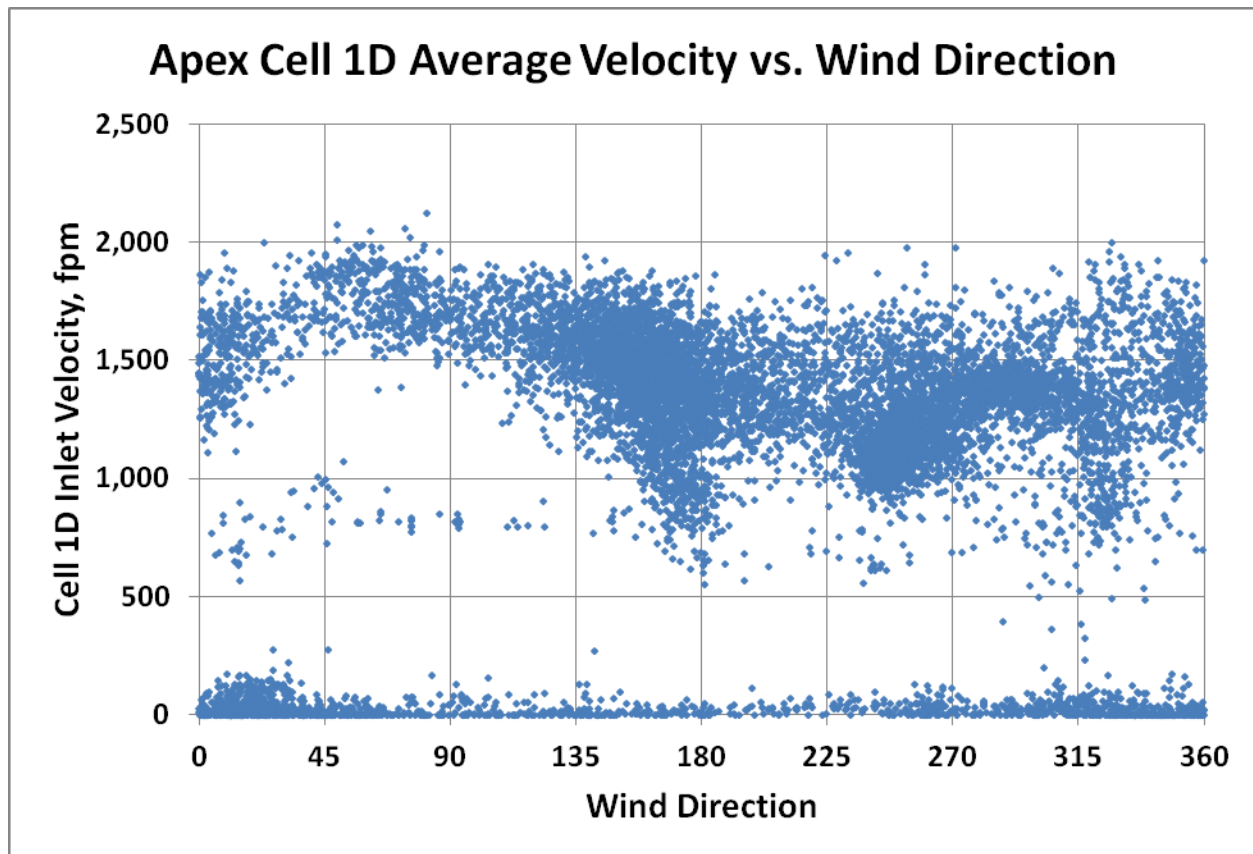


Figure 7-40. Apex Cell 1D Inlet Velocity vs. Wind Direction.

Figures 7-41 through 7-46 illustrate these wind effects in more detail for each cell during plant operating hours on July 16, 18 and 19. It is noted that on July 16 and 18 (Figures 7-41 through 209) Cells 4 and 5D are being brought up to speed in the earliest morning hours and the reduced inlet velocities are unrelated to wind effects.

On July 16, there is a continuous decrease in air inlet velocity in all cells as the wind speed increases during the morning. The decrease is greatest in Cell 1D from about 1 pm until 8 pm during which time the wind is from the south to southwest. A sudden decrease in wind speed around 8 pm is accompanied by a uniform increase in air flow to all cells. From 9 pm until midnight the wind picks up again but this time from the northwest. In this case, the greatest effect is on Cell 5D which is on the west side of the ACC. For winds from that direction, Cells 1D and 4 are more shielded by the plant buildings than is Cell 5D.

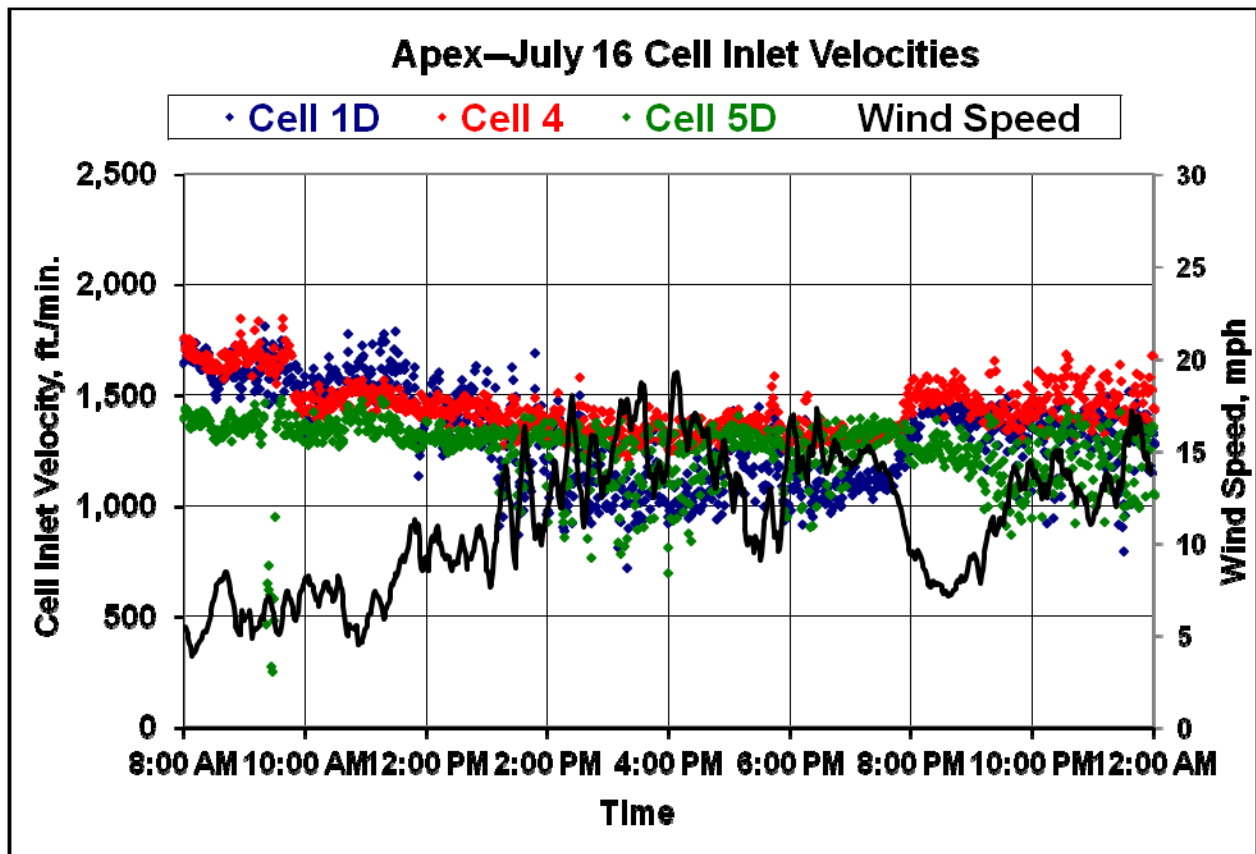


Figure 7-41. Apex Inlet Velocities; Cells 1D, 4 and 5D and Wind Speed—July 16, 2005.

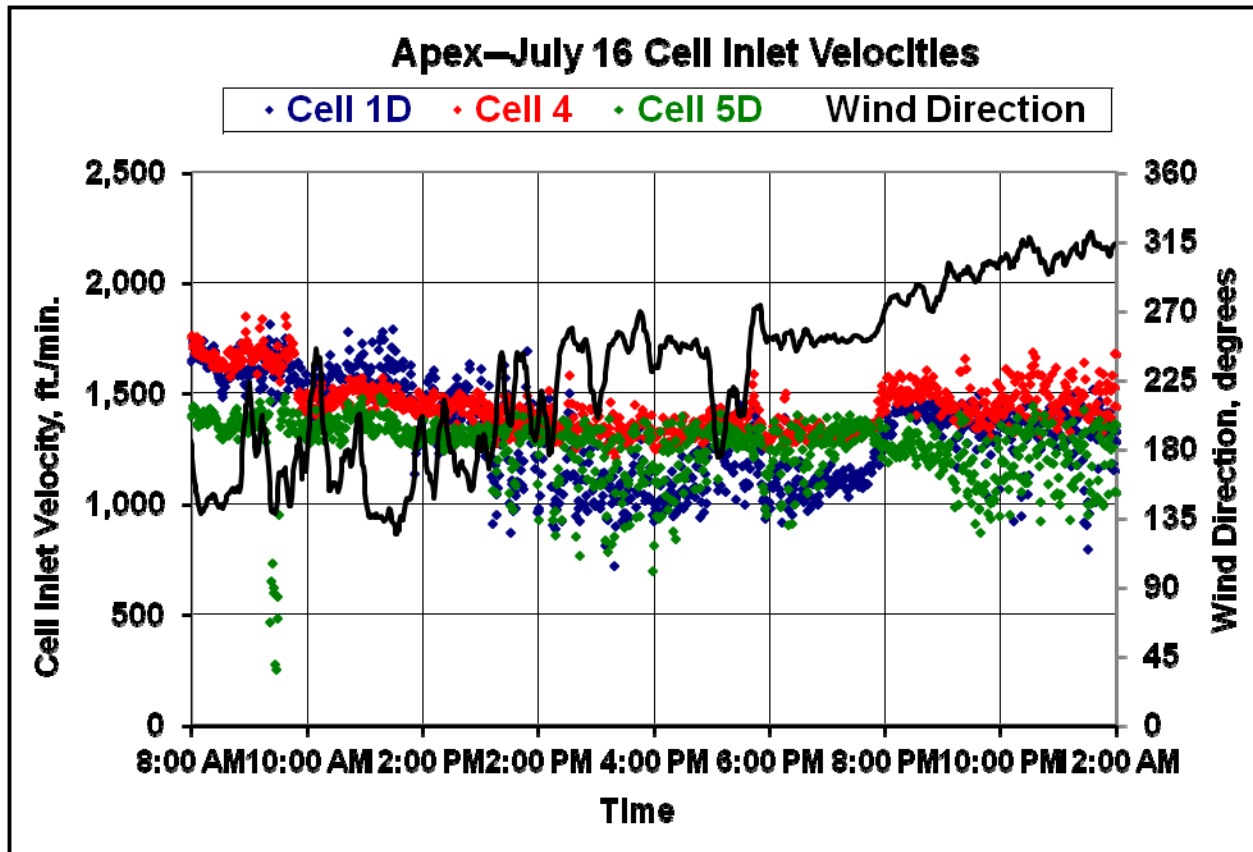


Figure 7-42. Apex Inlet Velocities; Cells 1D, 4 and 5D and Wind Direction—July 16, 2005.

On July 18, the winds are uniformly low to moderate for the entire day with the exception of a brief spike at about 3:30 pm and a longer lasting increase between 8 and 9 pm. In both instances there is a brief reduction in air flow to the cell. The wind direction is highly variable during much of the afternoon and again between 8 and 9 pm but the fluctuations are too rapid for any short term effect on air flow to be discerned. The reduction in flow to Cells 4 and 5D after 10:30 pm until midnight is believed to be due to the fans' being reduced to half speed, although this cannot be documented.

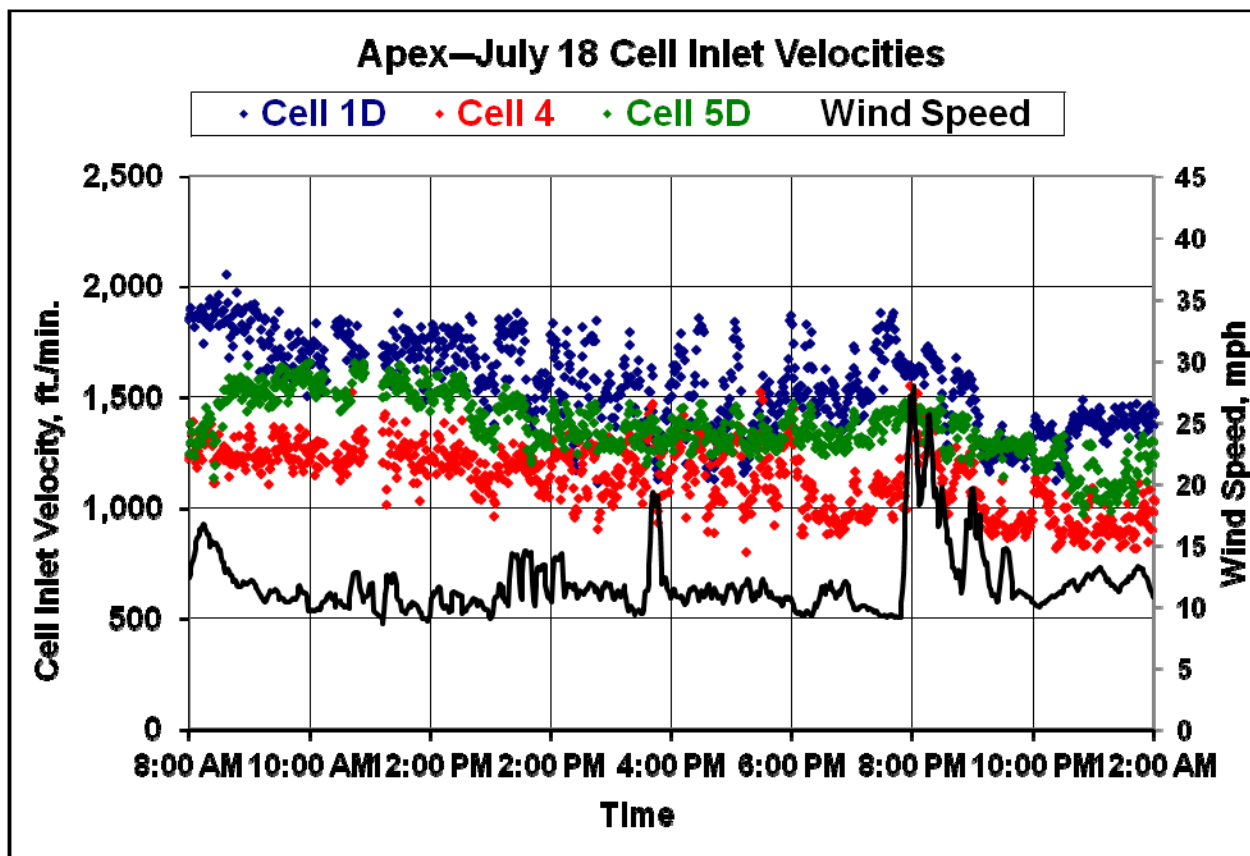


Figure 7-43. Apex Inlet Velocities; Cells 1D, 4 and 5D and Wind Speed—July 18, 2005.

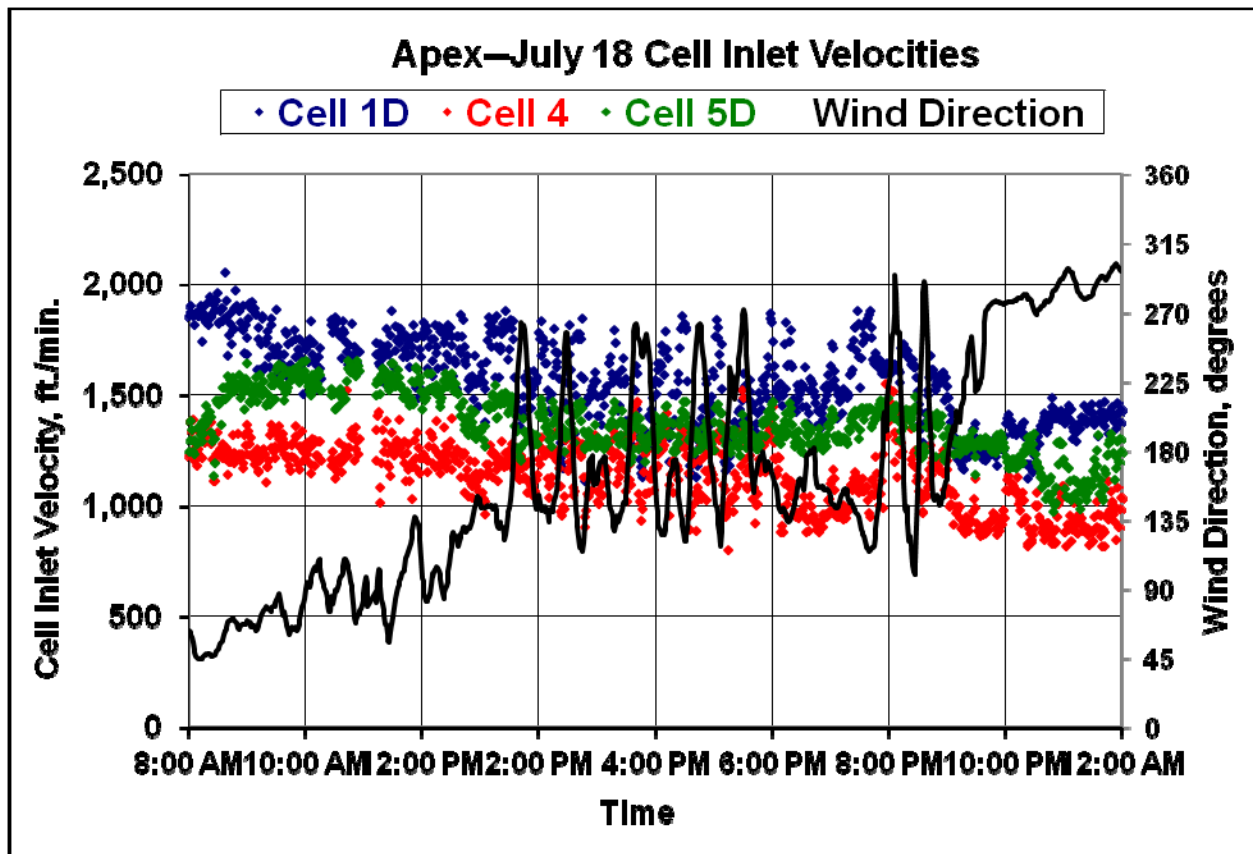


Figure 7-44. Apex Inlet Velocities; Cells 1D, 4 and 5D and Wind Direction—July 16, 2005.

The behavior on July 19 is similar to that on July 16. The variation with wind speed looks to be less, but the variation is between 10 and 20, whereas, on the 16th, there were periods of very low wind (3 to 7 mph) with correspondingly higher air flows so the change appeared more dramatic. The abrupt change in wind direction at about 8:30 pm is accompanied by a countervailing decrease in wind speed which results in little effect on Cells 1D and 5D. the decrease in the air flow to Cell 4 between 8 pm and 11 pm followed by an increase between 11 pm and midnight appears related to the changes in wind direction but the reason why Cell 4 is more affected than Cell 5D is not understood.

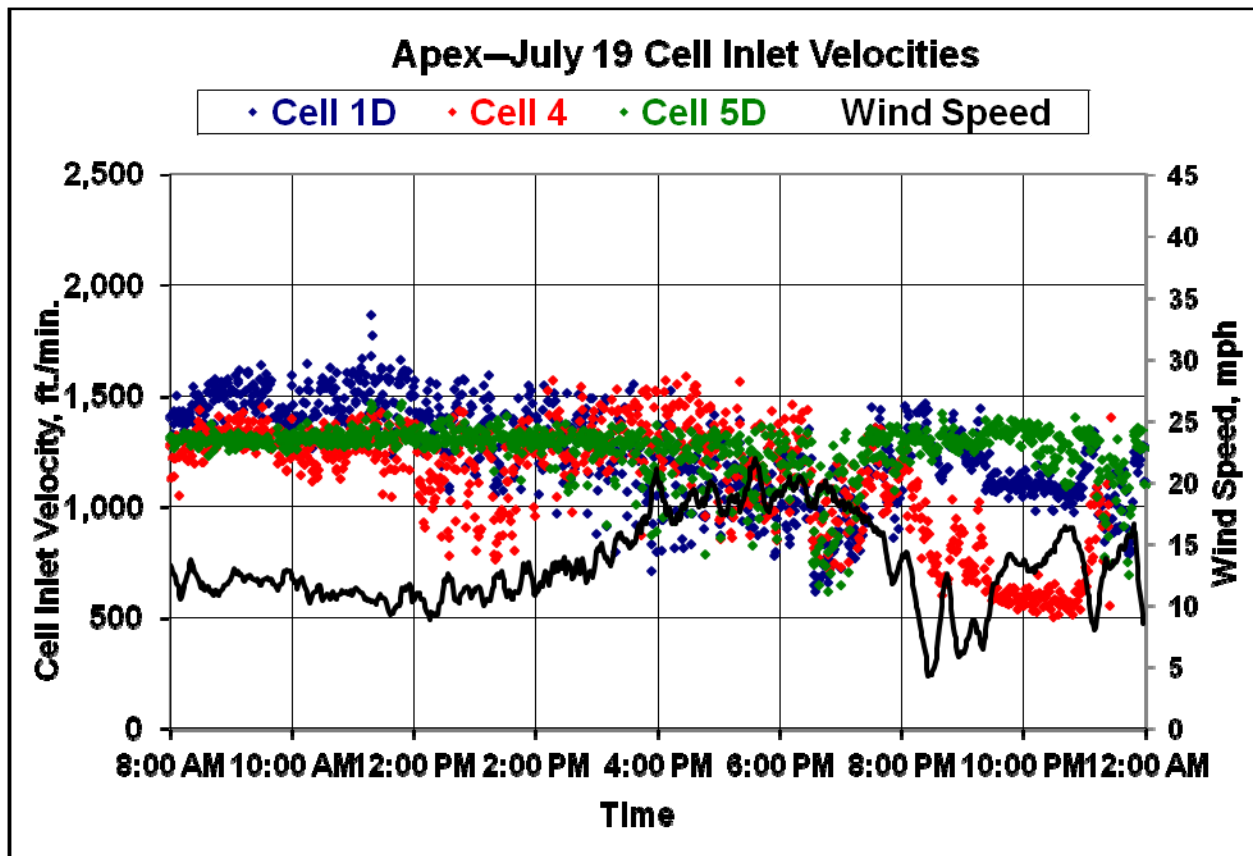


Figure 7-45. Apex Inlet Velocities; Cells 1D, 4 and 5D and Wind Speed—July 19, 2005.

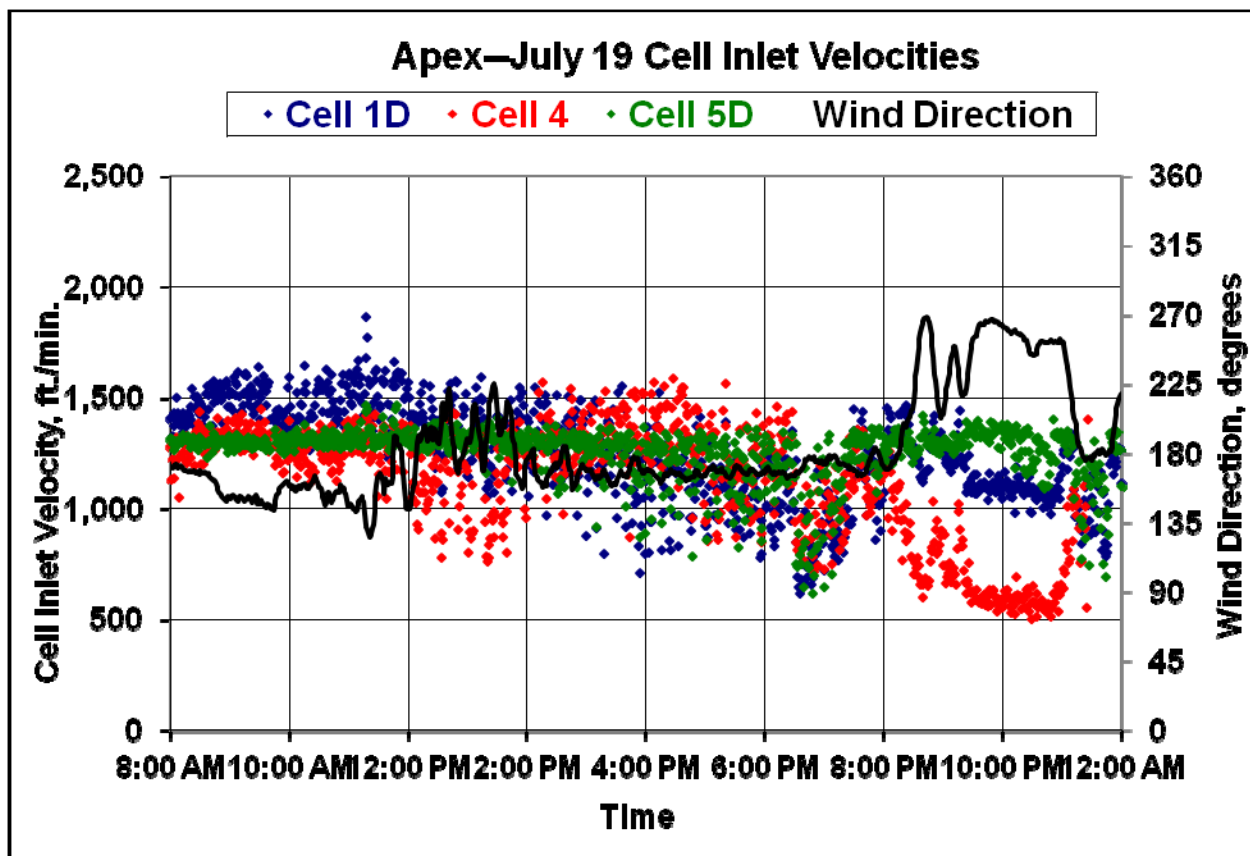


Figure 7-46. Apex Inlet Velocities; Cells 1D, 4 and 5D and Wind Speed—July 19, 2005.

7.5 Front Range Fan Performance

Fan performance degradation at Front Range was tested with measurements of cell inlet velocity (Cells K11, D43 and K83). The locations of the cells on which the inlet velocity measurements were made are shown on Figure 4-9. There are no windscreens at Front Range.

7.5.1 Inlet velocity

Cells K11, D43 and K83 were instrumented with propeller anemometers. At Front Range four anemometers in each of the instrumented cells were suspended above the fans by hanging them from the edge of the fan support bridge inside the cells. As at Apex, they were positioned in two pairs on opposite sides of the fan bridge and on both sides of the motor hub, approximately half way from the blade root to the tip.

Figure 7-47 plots the results of these measurements during the entire test period for those hours when all 30 fans were operating at full speed. The design flow for the fans is 1,291,881 acfm which corresponds to a velocity in the plane of the fan of approximately 1,425 fpm as shown on the plot. The average measured values are generally lower than the design value most of the time.

The periods of reduced cell inlet velocities correspond well with periods where the wind is blowing from the easterly quadrant, as seen by comparison to the plot of wind speed and direction for the entire test period in Figure 132 (reproduced here as Figure 7-48 for convenience of reference).

Figure 7-49 shows the relationship between wind speed and direction during the test period. At the higher wind speeds, the wind direction is reasonably uniformly distributed across all directions. The lower wind speeds come primarily from the east (90 to 150 degrees) and the northwest (280 to 340 degrees). Figures 7-50 and 7-51 show the effect of wind speed (Figure 7-50) and wind direction (Figure 7-51) on the average air inlet velocity in Cell K11.

The Cell K11 inlet velocities fall into three clusters. The lowest speed cluster with inlet velocities between 600 and 800 fpm are assumed to be at half-speed operation, although detailed fan status information was not available. The highest cluster shows a clear fall-off in airflow at wind speeds above 15 mph. The majority of the points in the "> 15 mph" range are for winds from the northwest. Cell K11, located on the northeast corner of the ACC, would be expected to be directly affected by such winds. The intermediate cluster, showing reduced air flow at wind speeds between 5 and 15 mph, is for winds primarily from the southeast which impinge directly on Cell K11 and have an apparent effect at even lower wind speeds.

The same three cluster grouping is evident in the inlet velocity vs. wind direction plot (Figure 7-51). It is reasonable that winds from the southeast might have more effect on Cell K11 than those from the northwest because of the presence of the stairwell structure located on the north face of the cell. This structure provides some shielding of the fan for winds from the north to northwest directions.

The general shape of the airflow vs. wind direction mirrors the pattern of wind speed vs. wind direction seen in Figure 7-49. This indicates that the combined effect of wind speed and direction results from the fact that the higher wind speeds are nearly always from the southwest and southeast.

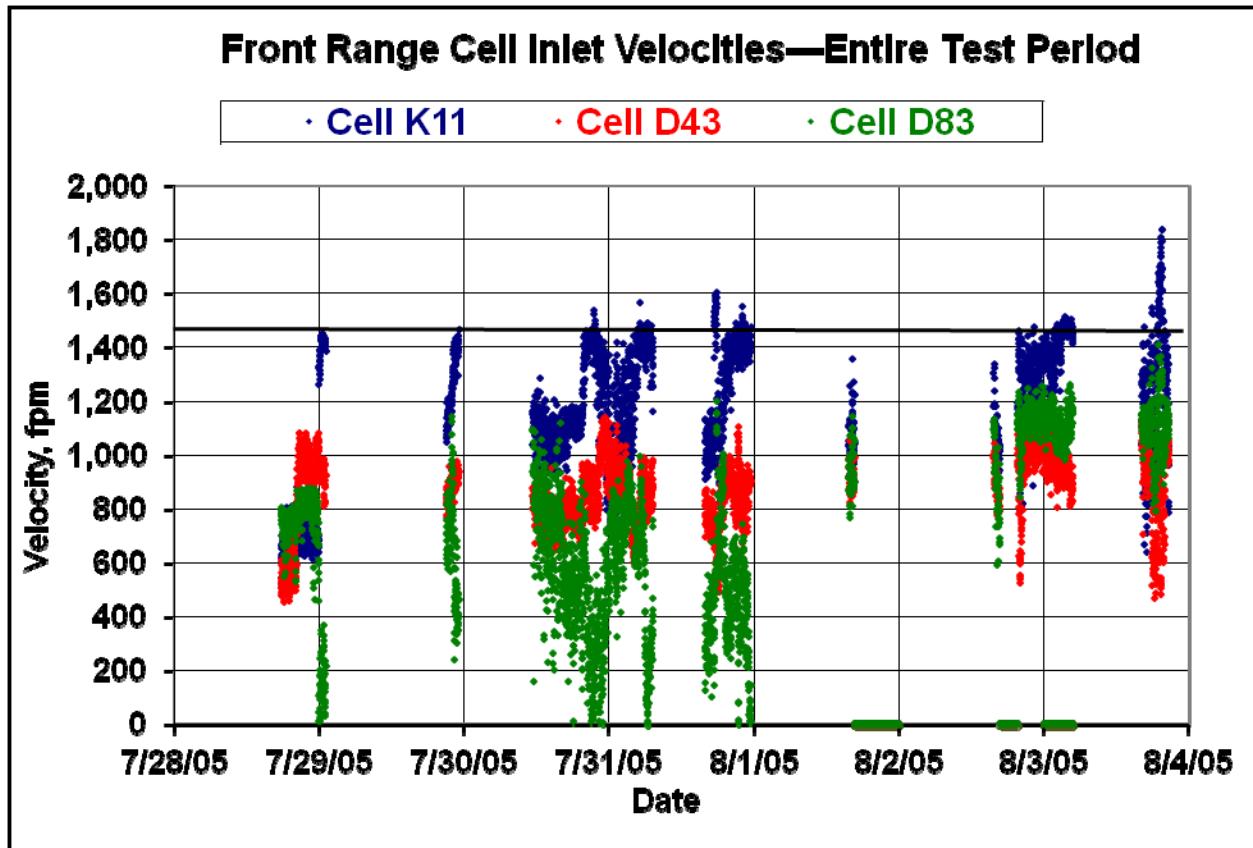


Figure 7-47. Front Range Inlet Velocities; Cells K11, D43 and D83—Entire Test Period.

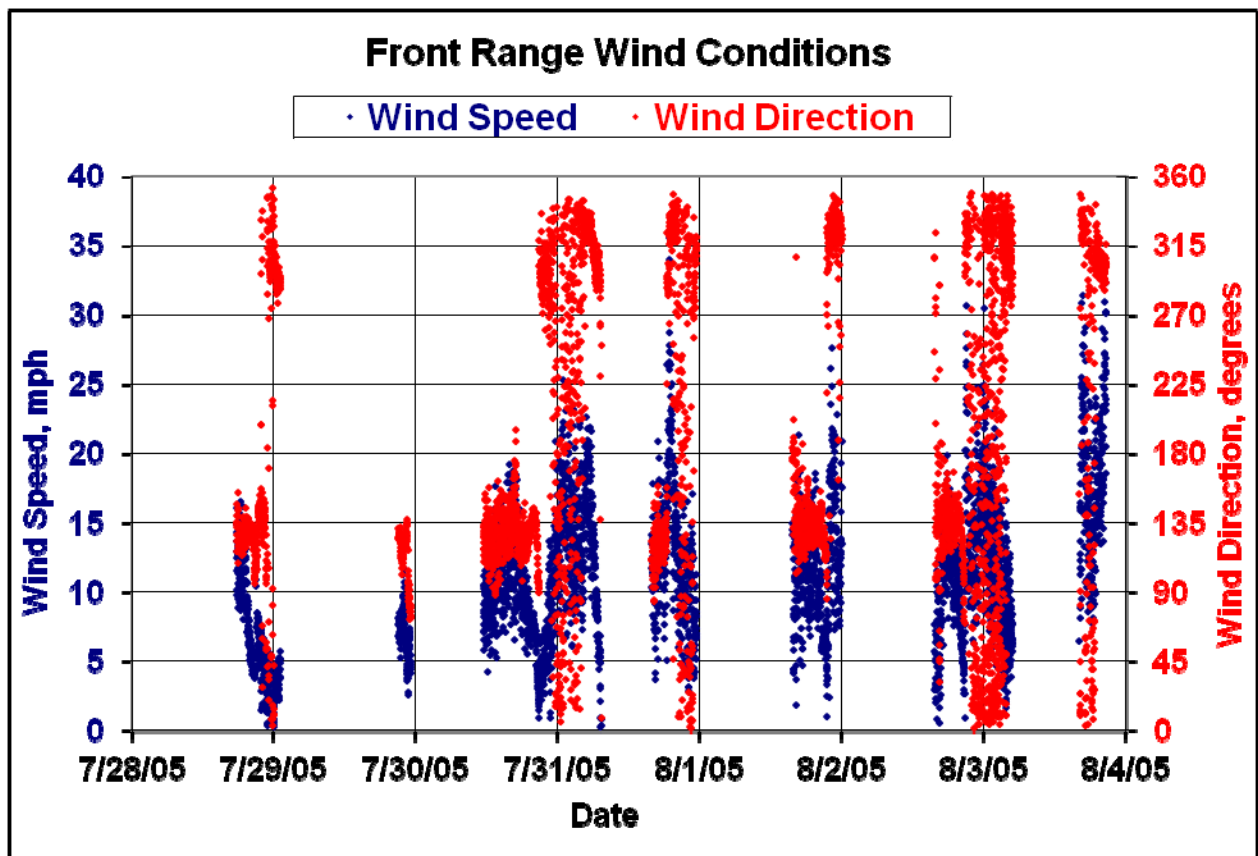


Figure 7-48. Front Range Wind Conditions—Entire Test Period.

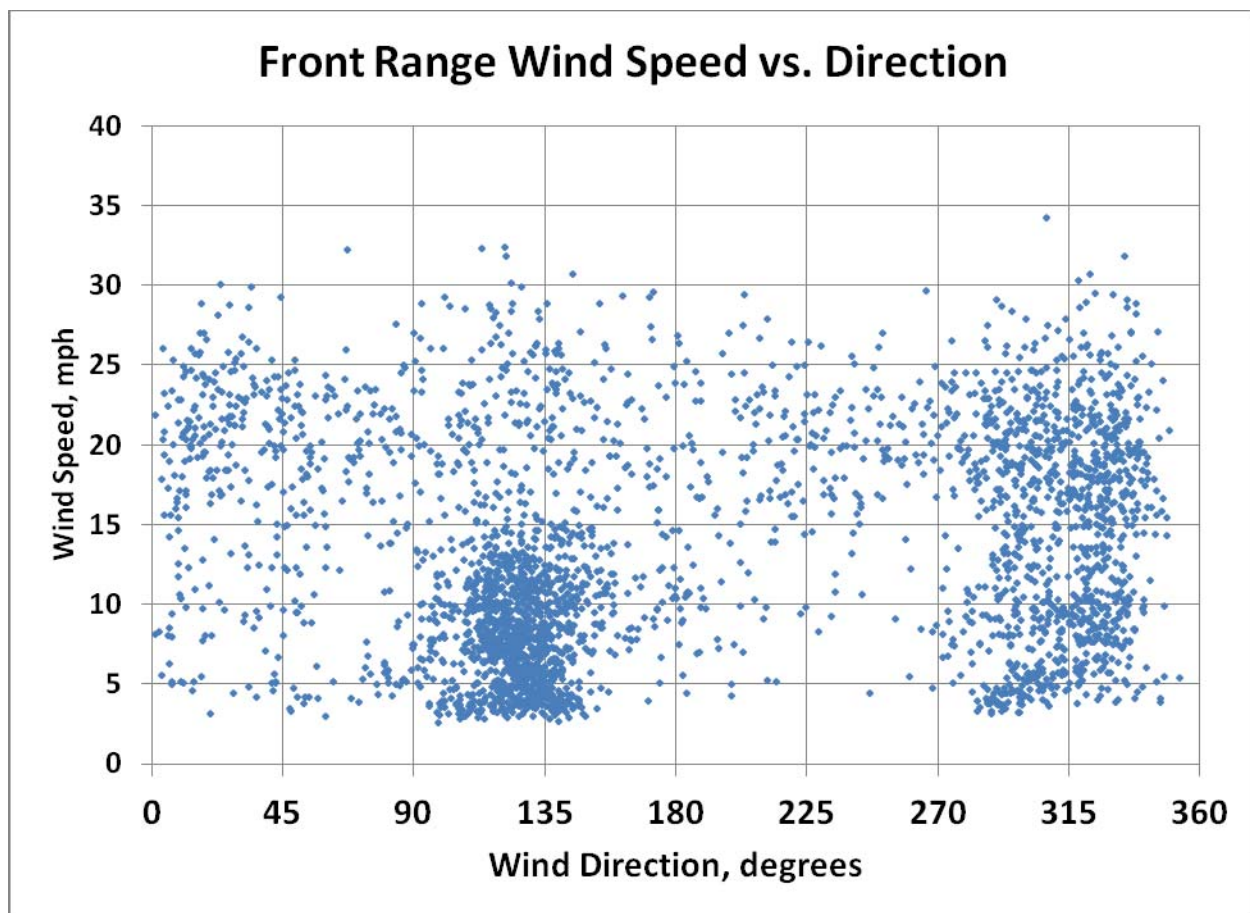


Figure 7-49. Front Range Wind Speed vs. Wind Direction.

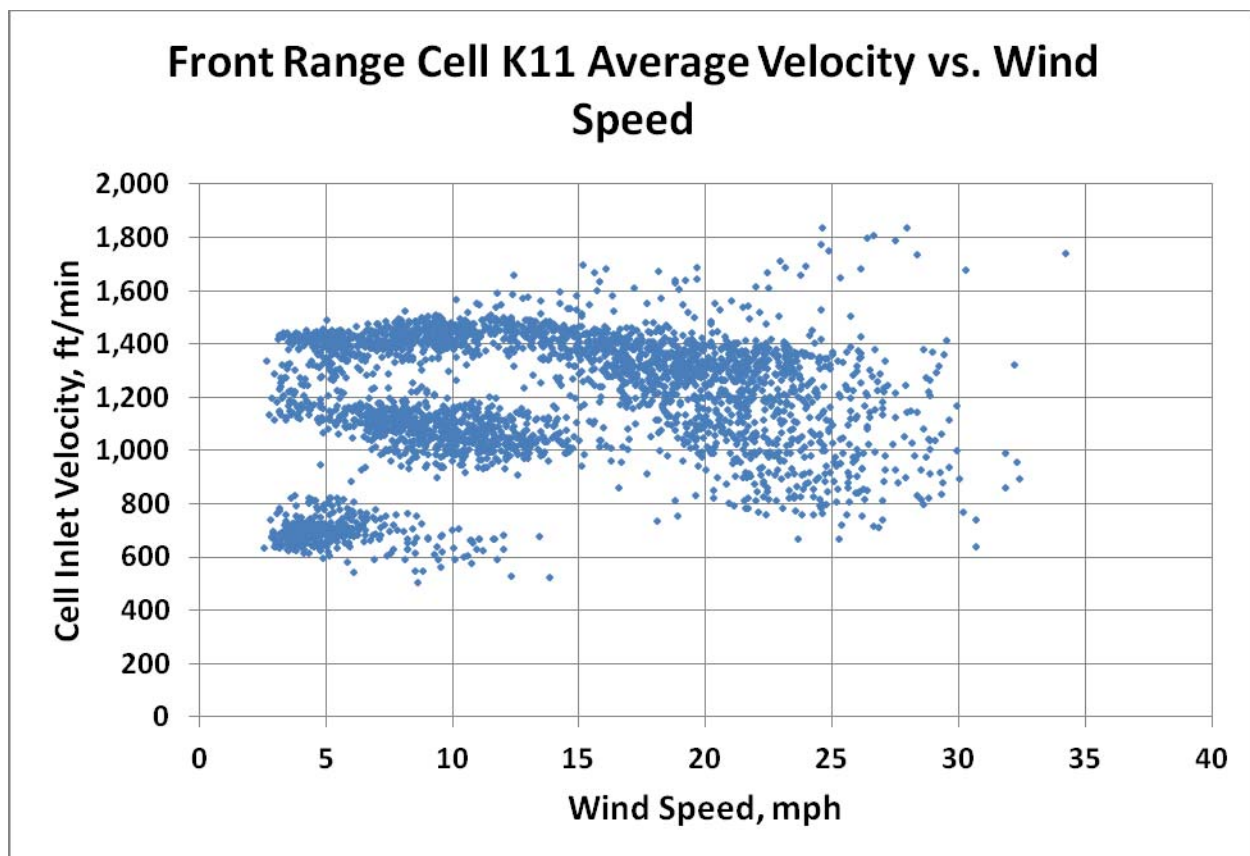


Figure 7-50. Front Range Cell K11 Inlet Velocity vs. Wind Speed.

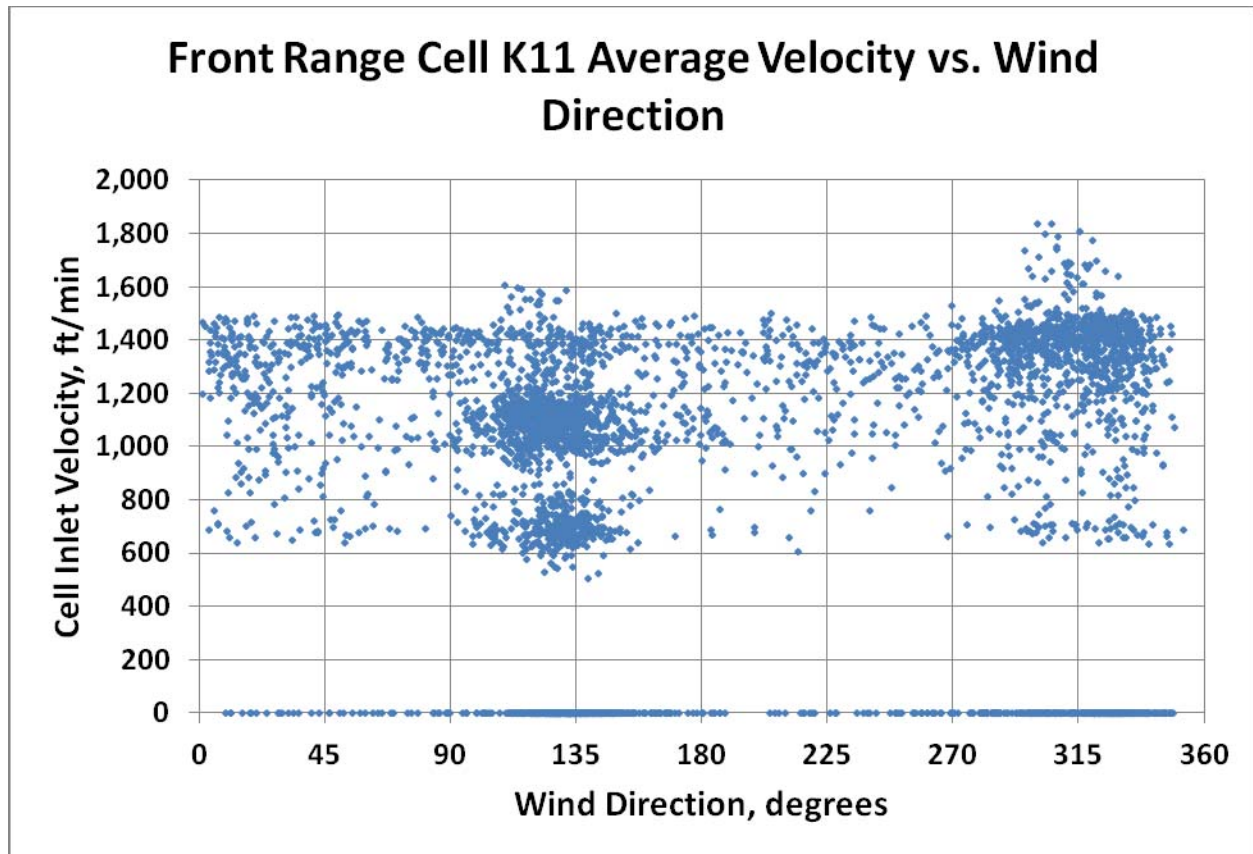


Figure 7-51. Front Range Cell K11 Inlet Velocity vs. Wind Direction.

The corresponding results for Cell D43 are shown in Figures 7-52 and 7-53. The design airflow in the D-cells is lower than it is for the K-cells, with an expected average inlet velocity measured in the plane of the fan of about 1,215 fpm. Even so there is a consistent reduction in the inlet velocity below what would be expected from the design value. It may be that the blade pitch has been changed from the design point, but no information was available from the plant. However, the important observation is that the fan performance is almost completely insensitive to wind speed or direction. The location of Cell D43 in the center of the ACC suggests that the wind effects are completely dissipated by the time the air reaches the inner cells.

Cell D83, also with a design inlet velocity of 1,215 fpm, is an edge cell in the center of the south street. It is exposed to winds from the southern half of the wind rose and well shielded from winds from the northern half. This is illustrated in Figures 7-54 and 7-55 which shows an array of points at the highest inlet air velocity for all wind speeds with a cluster at reduced speeds corresponding to winds from the south and south east.

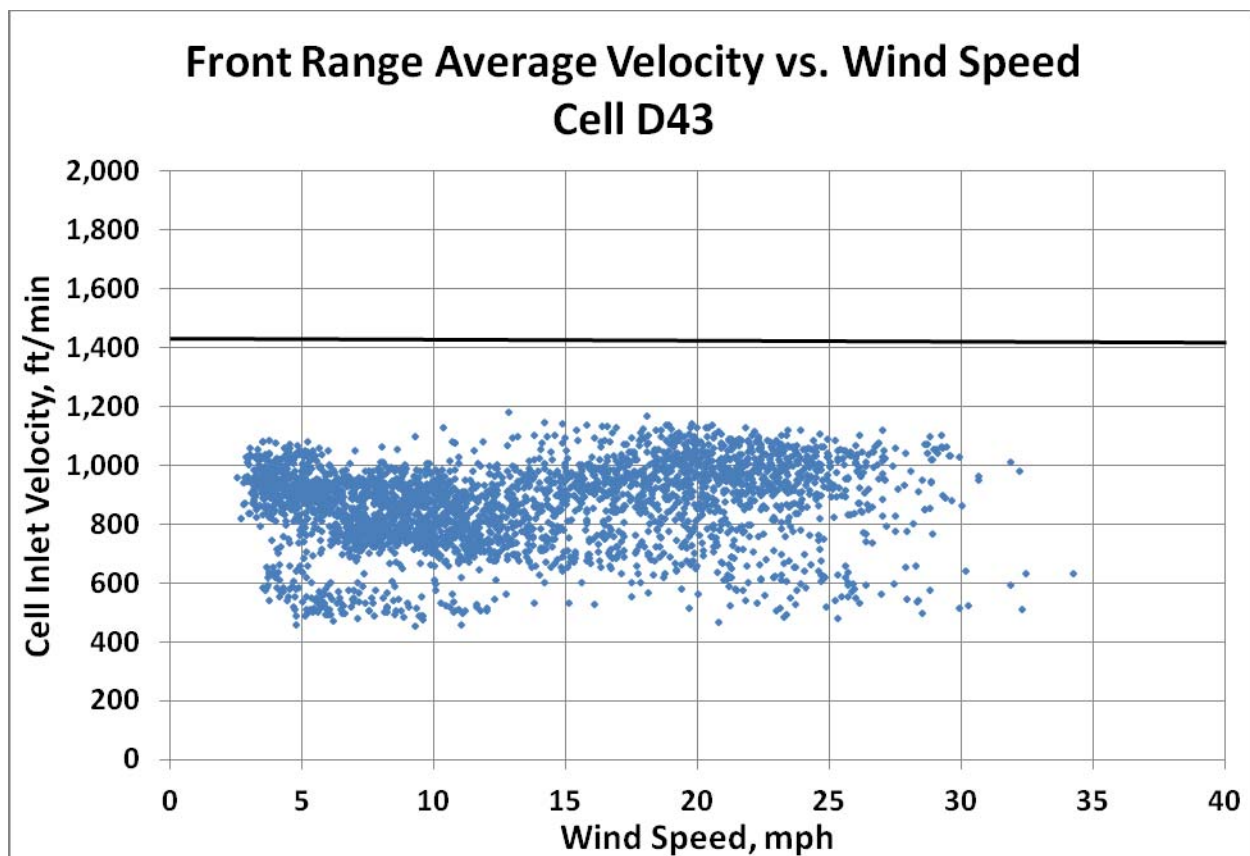


Figure 7-52. Front Range Cell D43 Inlet Velocity vs. Wind Speed.

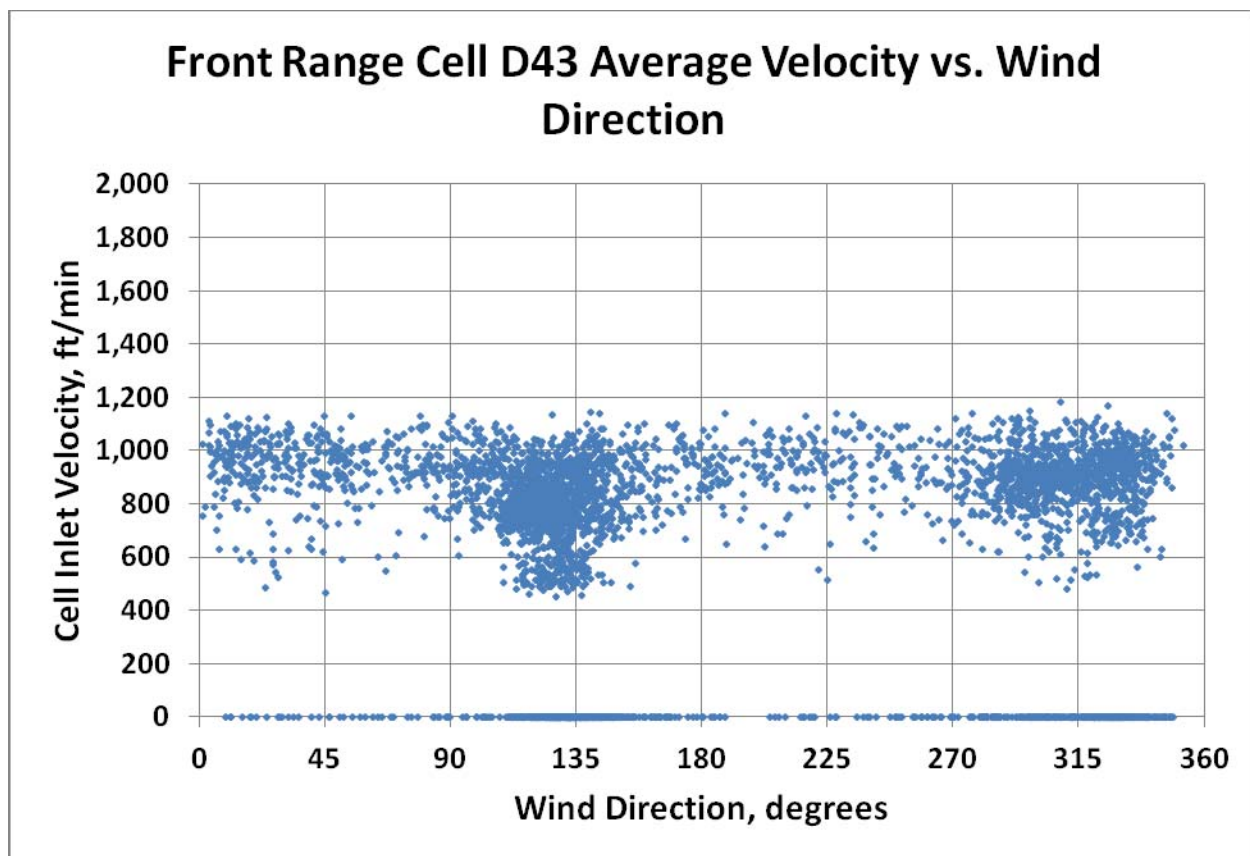


Figure 7-53. Front Range Cell D43 Inlet Velocity vs. Wind Direction.

Front Range Average Velocity vs. Wind Speed Cell D83

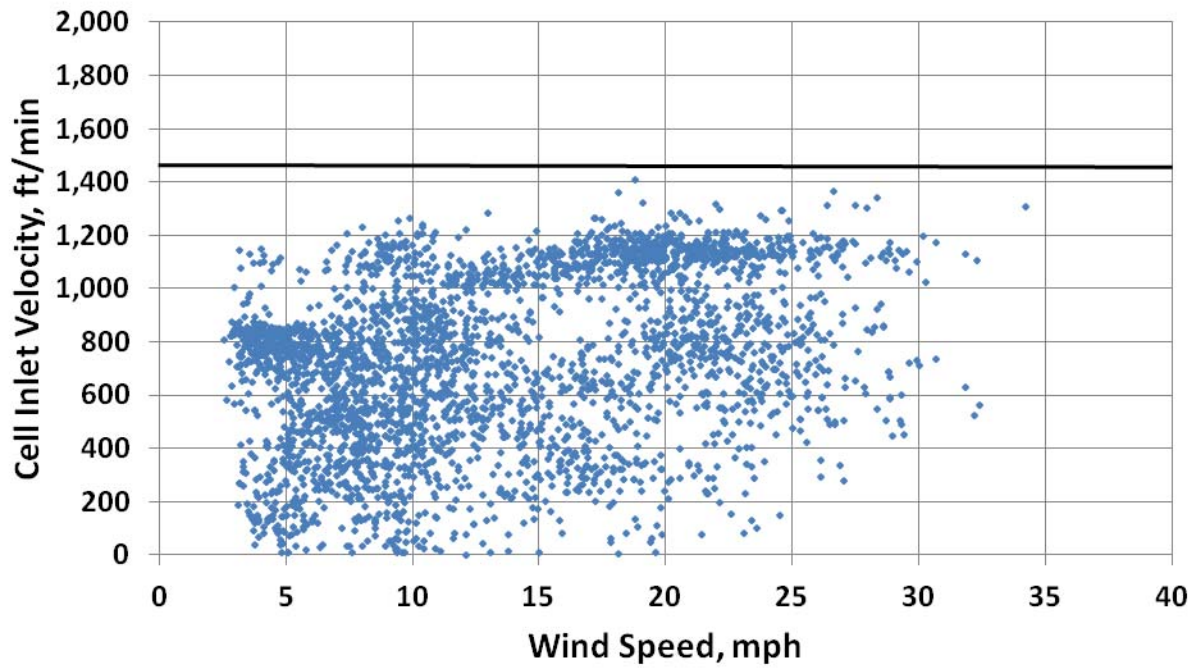


Figure 7-54. Front Range Cell D83 Inlet Velocity vs. Wind Speed.

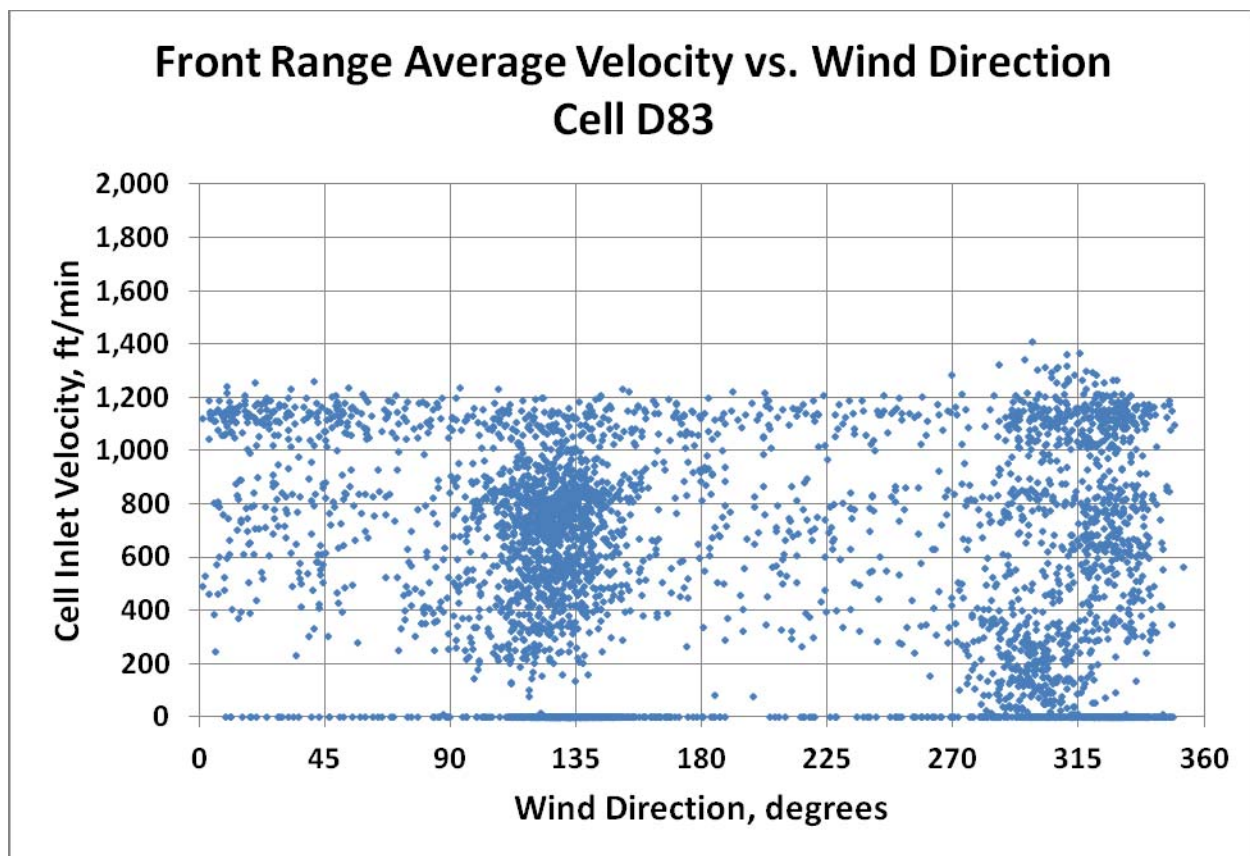


Figure 7-55. Front Range Cell D83 Inlet Velocity vs. Wind Direction.

Figures 7-56 through 7-59 illustrate these wind effects in more detail for each instrumented cell during late afternoon and evening hours on July 30 and 31.

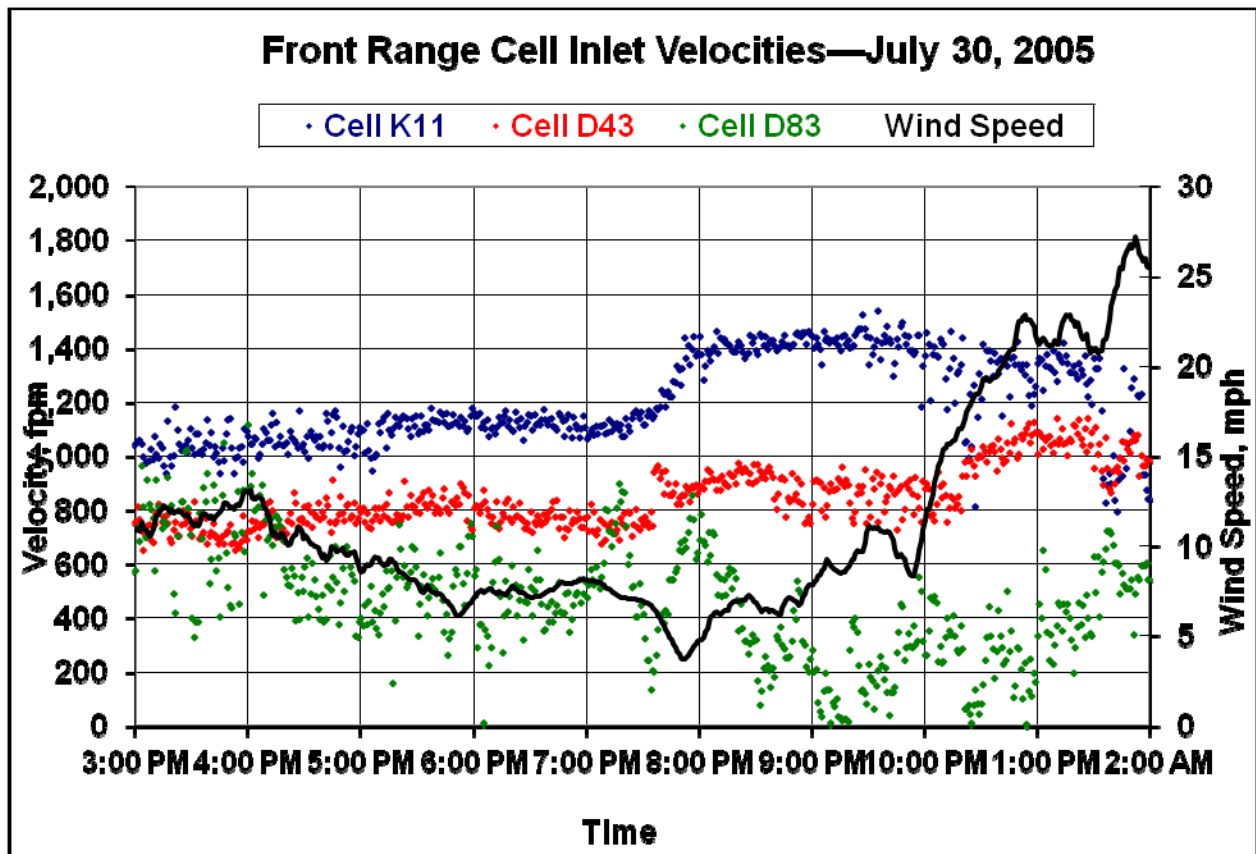


Figure 7-56. Front Range Inlet Velocities; Cells K11, D43 and D83 and Wind Speed—July 30, 2005.

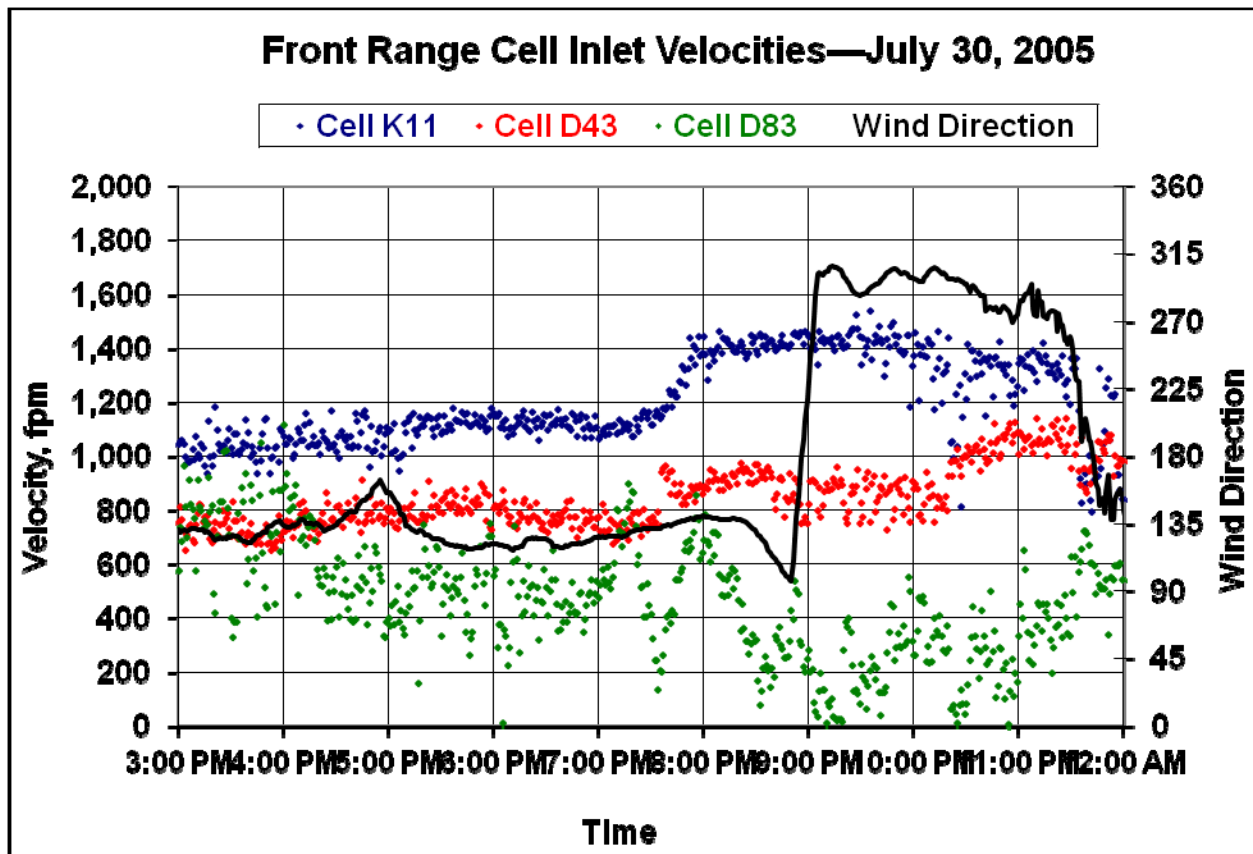


Figure 7-57. Front Range Inlet Velocities; Cells K11, D43 and D83 and Wind Direction—July 30, 2005.

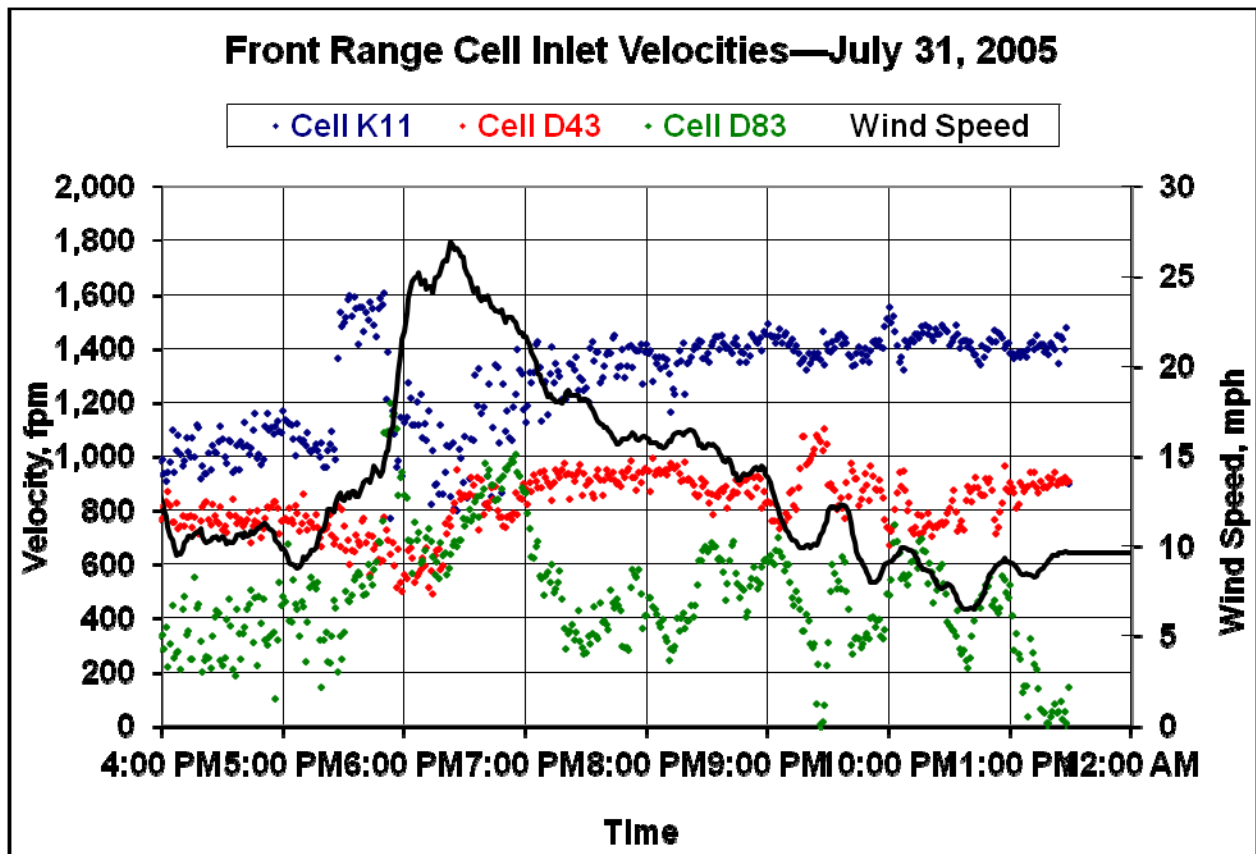


Figure 7-58. Front Range Inlet Velocities; Cells K11, D43 and D83 and Wind Speed—July 31, 2005.

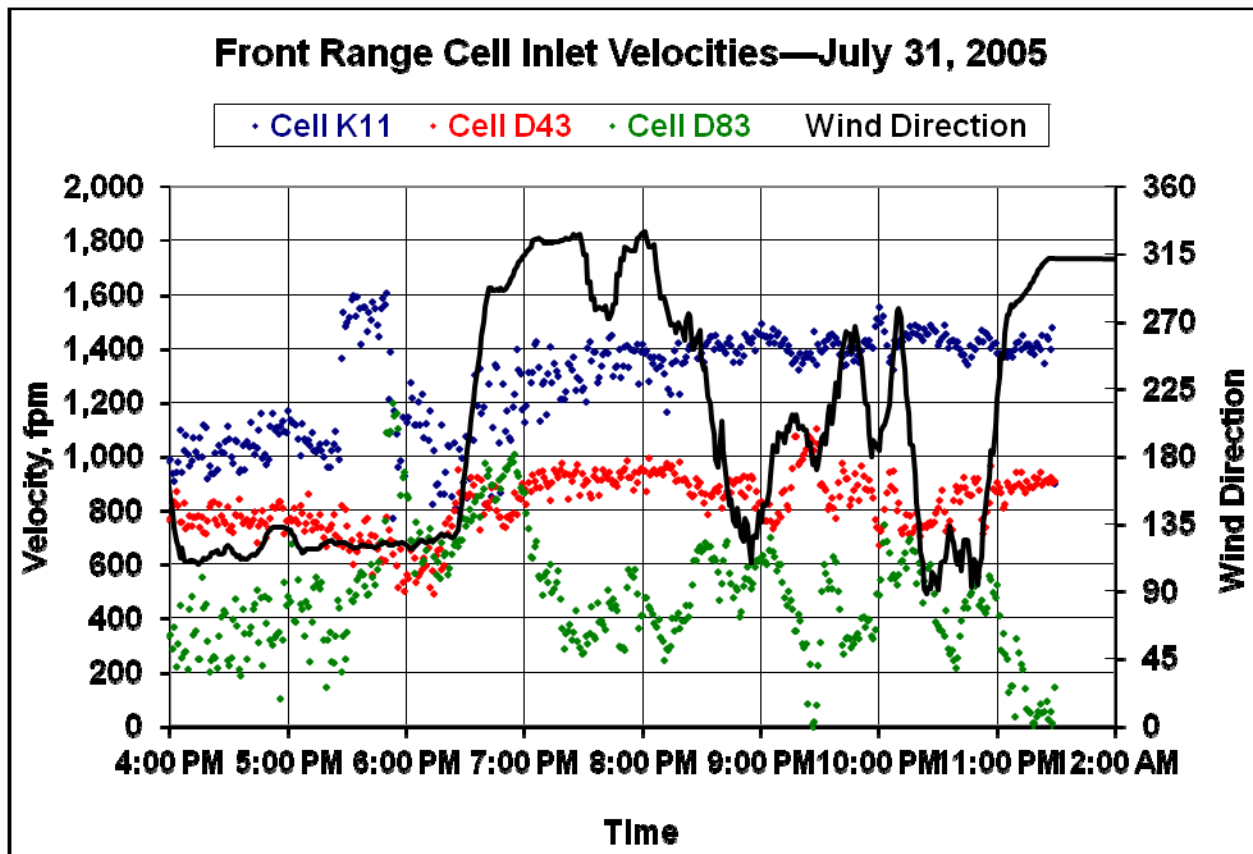


Figure 7-59. Front Range Inlet Velocities; Cells K11, D43 and D83 and Wind Direction—July 31, 2005.

The behavior of Cell D83 during the late evening hours (after about 8 pm) on July 30 is particularly erratic and does not correlate well with either wind speed or direction. It may be that the fan was reduced to half-speed or even shut off but this could not be confirmed. The interior cell, D43, exhibits little sensitivity to wind conditions but runs consistently at a relatively low flow but exhibits a slight increase after 10 pm with increasing wind speeds from the west. This is not understood. Cell K11 shows an increase in air flow at about 7:30 pm with a fall-off in wind speed to below 5 mph. It remains at this higher flow as the wind speed increases to over 10 mph with a concurrent shift in direction from southeast to northwest. This shift in wind direction and the shielding of the fan by the stair structure discussed above may be a compensating factor that makes the cell less sensitive to winds from the northwest than to winds from the southeast. The shift in direction back to southeasterly after 11 pm is accompanied by a decrease in Cell K11 airflow.

The behavior on July 31 is similar but the greater variation in both wind speed and direction leads to more fluctuations in the airflow to all cells. Cell D43 remains reactively insensitive to wind. Cell K11 again exhibits little sensitivity to winds below 15 mph with somewhat more sensitivity to those from the southeast. The brief excursion of a higher inlet velocity between 5:30 and 6 pm is not understood.

Cell D83 is most affected to winds from the southeast and shielded from northerly winds as indicated by the increase in airflow at about 6:30 pm when the wind shifts to then northwest. The fall-off after about 6:45 pm is presumed to be associated with half-speed operation although this could not be confirmed.

7.6 WyGen Fan Performance

Fan performance degradation at WyGen was tested with measurements of cell inlet velocity (Cells K11, K15 and K25). The locations of the cells on which the measurements were made and their location relative to the windwall under the ACC are shown schematically on Figure 4-10 and in the photograph in Figure 31.

7.6.1 Inlet velocity

Cells K11, K15 and K25 were instrumented with propeller anemometers. Four anemometers in each of the instrumented cells were suspended above the fans by hanging them from the edge of the fan support bridge inside the cells. As at Apex and Front Range, they were positioned in two pairs on opposite sides of the fan bridge and on both sides of the motor hub, approximately half way from the blade root to the tip.

Figure 7-60 plots the results of these measurements during the entire test period for those hours when all 12 fans were operating at full speed. The design flow for the fans is 1.050,000 acfm which corresponds to a velocity in the plane of the fan of approximately 1300 fpm as shown on the plot.

At Wygen, the measured average values are lower than the design value most of the time. It is also noteworthy that while the average inlet velocities of the three individual fans fluctuate significantly, the average of the three together is reasonably constant around the design value with the exception of a brief excursion to a lower flow on the night of August 12 through the morning of August 13. As will be discussed in more detail in a later section, this is believed to be the result of the effect of the windwall which runs between the two streets. (See Figures 3-25 and 4-10). Winds perpendicular to the axis of the ACC and to the windwall (i.e., winds from the east or the west) may enhance the performance of the fans on the upwind side of the wall while degrading the performance of those on the downwind side.

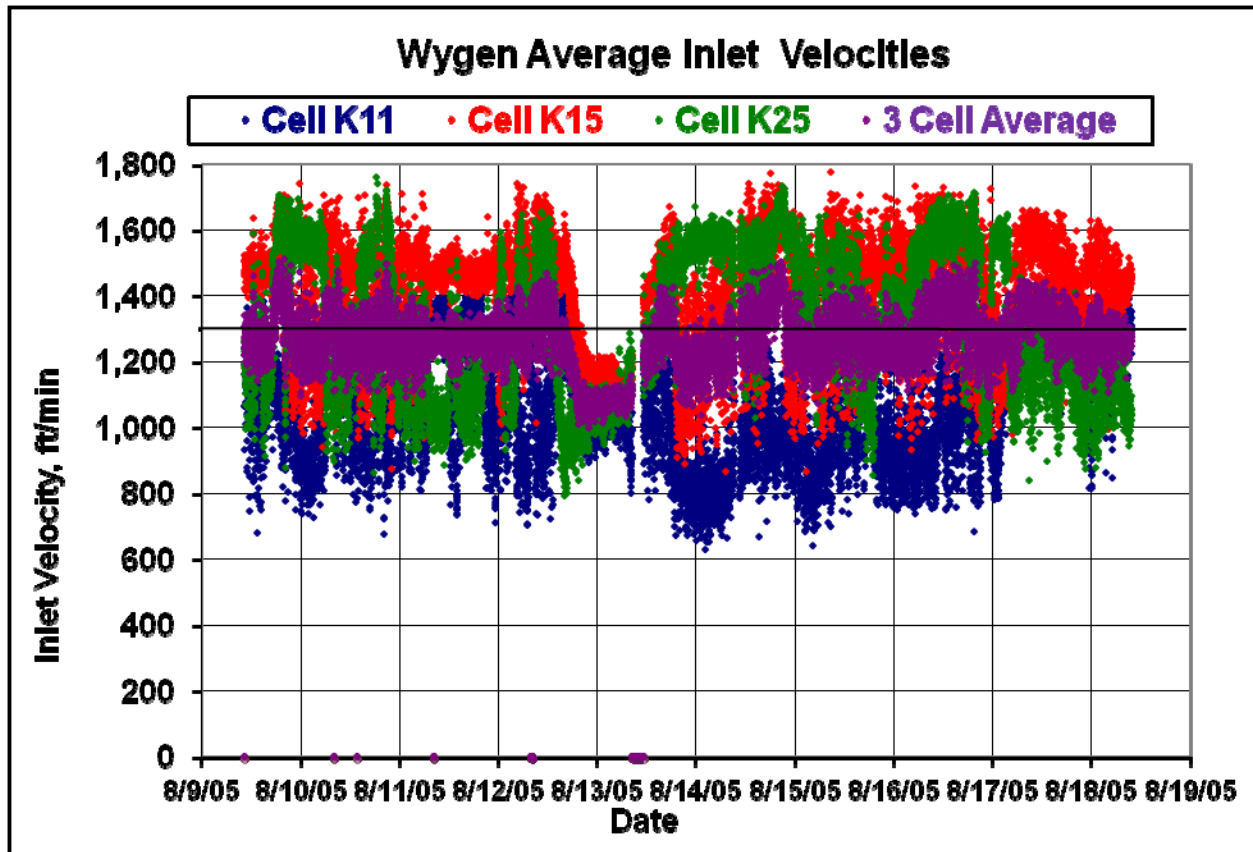


Figure 7-60. Wygen Inlet Velocities; Cells K11, K15 and K25—Entire Test Period.

Figure 7-61 (reproduced from Figure 6-87 for convenience of reference) shows the wind speed and direction for the entire test period. The distribution of wind direction is nearly bi-modal oscillating between southerly and northwesterly with a brief period from late night on August 9 until early morning on August 10 when the winds east and northeast. The wind speeds were typically moderate, less than 15 mph, with excursions to higher speeds on the night of August 12 and midday on August 16 and 17. These times correspond generally to time of reduced flow in Cells K11 and K25. The single period of wind from the east/northeast corresponds to a period of significant flow reduction in Cell K15.

Note that the time period from the night of August 12 through the morning of August 13, during which the average airflow of the three cells decreased markedly (See Figure 7-60 above), corresponds to the only period of strong winds from the south (parallel to the ACC axis and the windwall.)

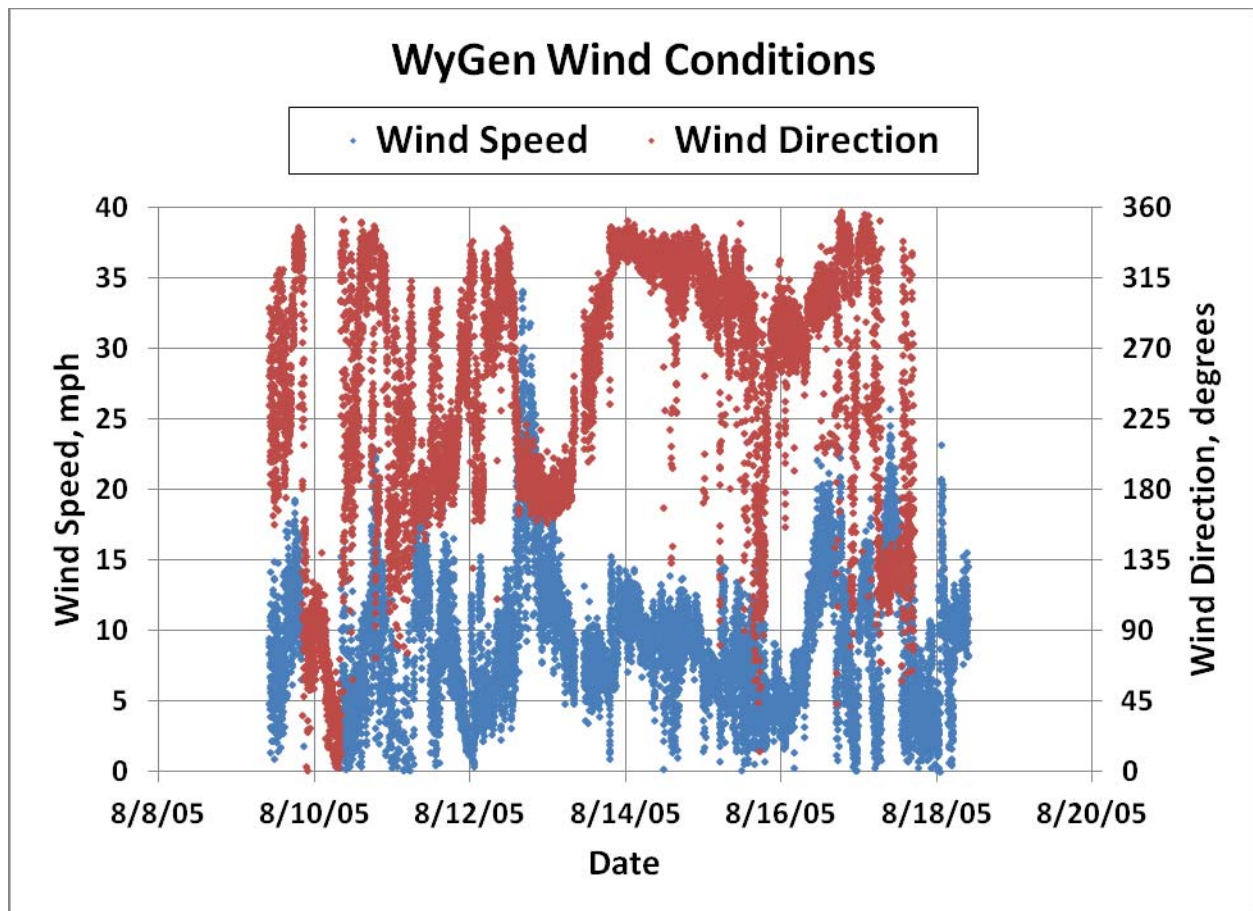


Figure 7-61. Wygen Wind Conditions—Entire Test Period.

Figure 7-62 shows the relationship between wind speed and direction. The highest wind speeds (> 25 mph) are from the south. Moderately high winds (20 to 25 mph) come from either the east-southeast or the north-northwest. Winds below 15 mph are reasonably uniformly distributed from all directions with the exception of the northeast quadrant.

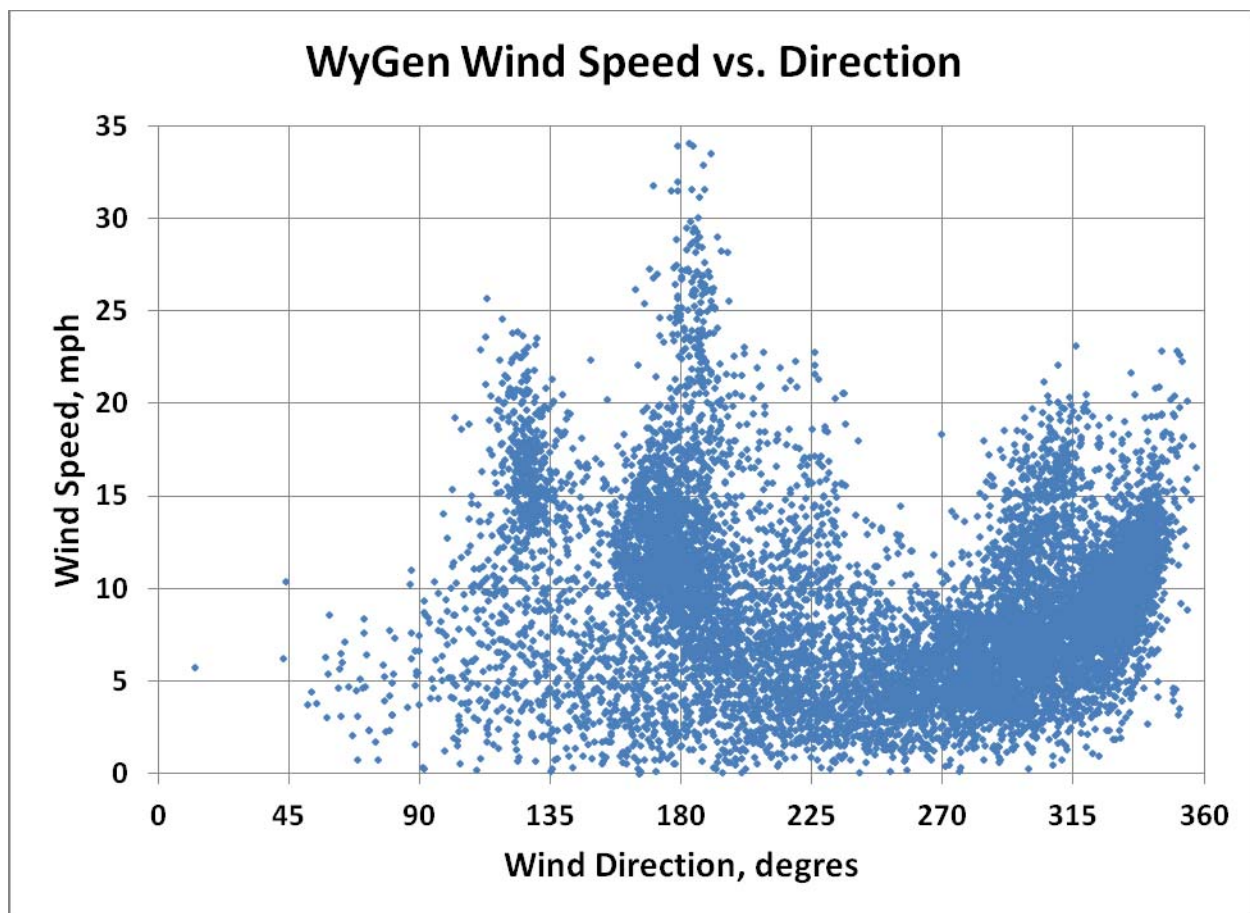


Figure 7-62. Wygen Wind Speed vs. Direction.

Figures 7-63 through 7-68 display the average inlet velocities for each of the three monitored cells vs. wind speed (Figures 7-63, 7-65 and 7-67) and wind direction (Figures 7-64, 7-66 and 7-68).

Cell K11 is relatively insensitive to wind speed except at the highest levels. At all speeds below about 20 mph the airflow can be either at the design level or down to 50% of design. All wind speed occurrences above 20 mph are from the south and Cell K11 is shielded by the plant building from southerly winds.

Cell K11 shows a general decline in performance as the wind shifts from south to northwest at the higher wind speeds. As the wind shifts, the shielding of the plant building is no longer a factor and the wind impinges directly on the cell. The cluster of low flow points at 180 degrees and the scattering of points in the range of northeast to easterly are not understood. These occasions were in the late night hours and may be related to a reduction in fan speed with lower ambient temperature or load demand, but this could not be confirmed.

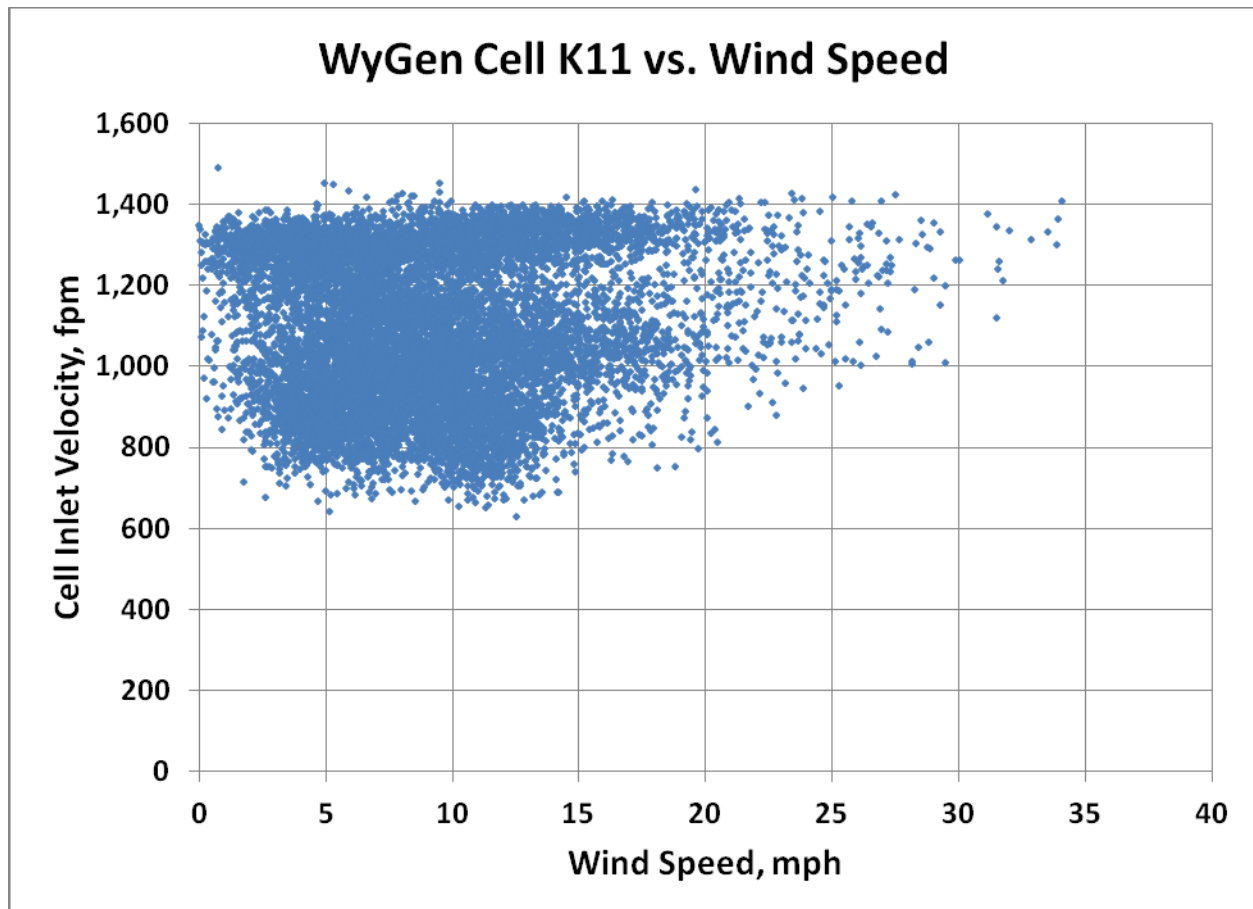


Figure 7-63. Wygen Inlet Velocity vs. Wind Speed; Cell K11—Entire Test Period.

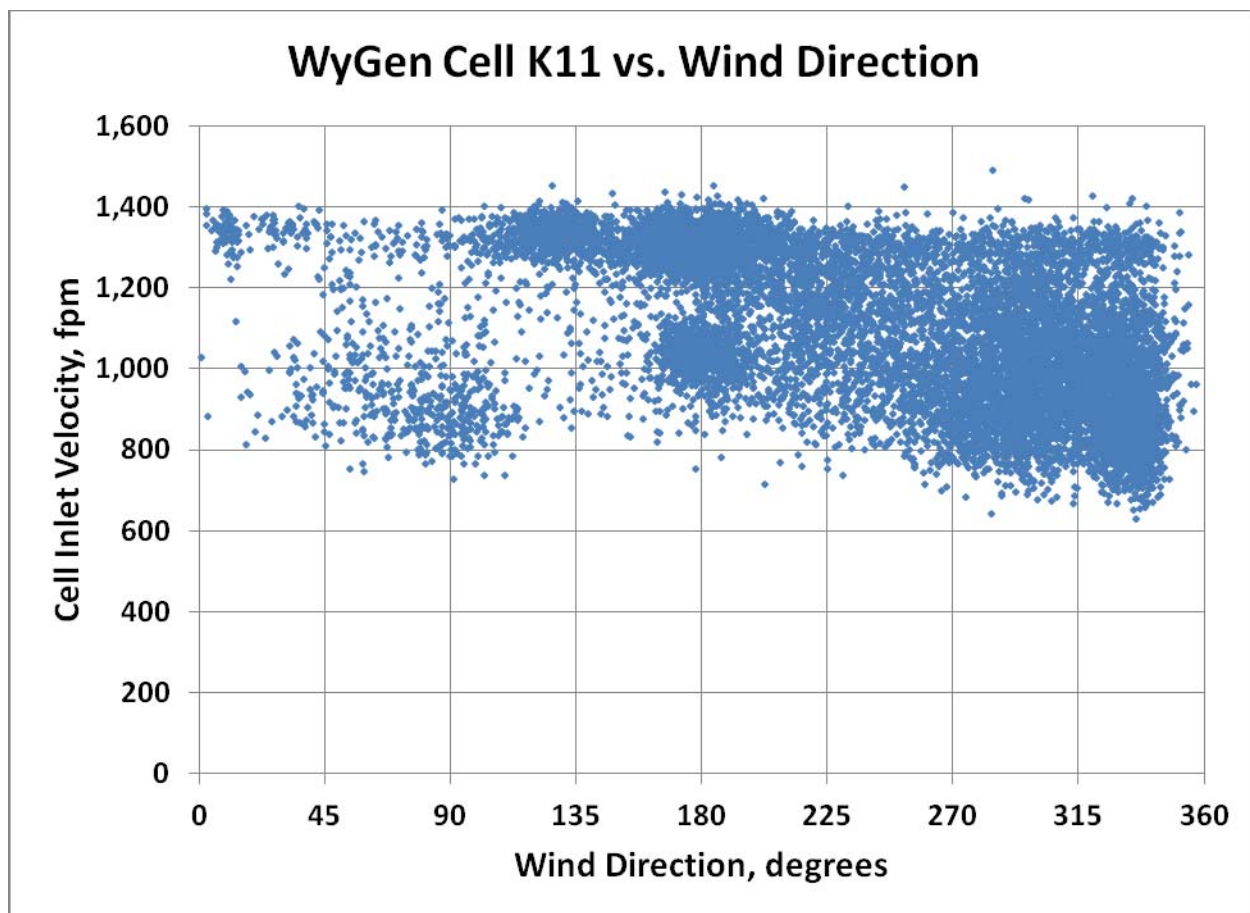


Figure 7-64. Wygen Inlet Velocity vs. Wind Direction; Cell K11—Entire Test Period.

Cell K15 shows a similar insensitivity to wind speed. The data points appear to group into two clusters with the majority of the points at or above design flow and a smaller group at a lower flow. Figure 7-66 indicates a lack of sensitivity to wind direction until the winds shift to the north-northwest at which time the higher wind speeds degrade the fan performance. The clusters of lower flow points for southerly or easterly winds are similar to those discussed above for Cell K11 and are also not understood. Again, a reduction in fan speed, while unconfirmed, may be responsible.

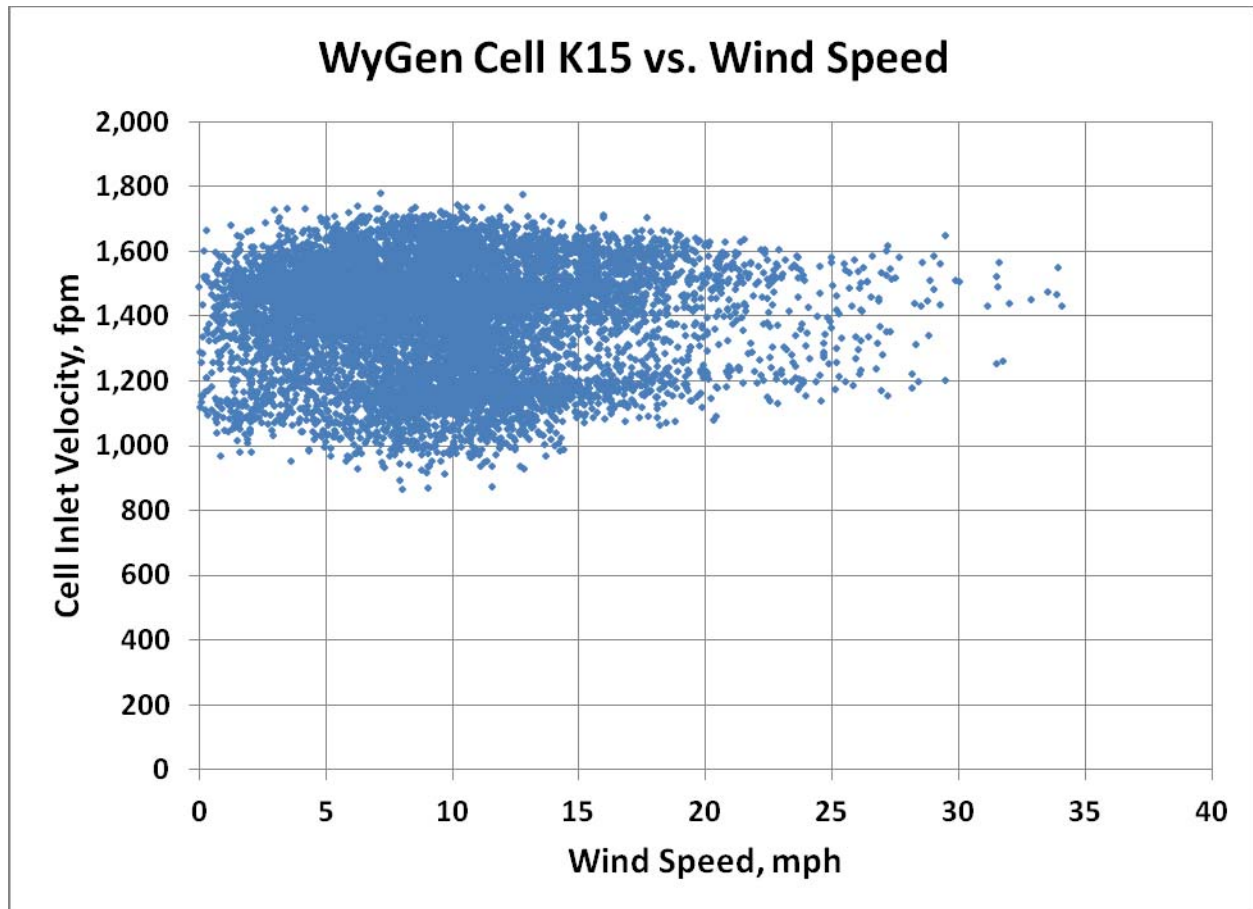


Figure 7-65. Wygen Inlet Velocity vs. Wind Speed; Cell K15—Entire Test Period.

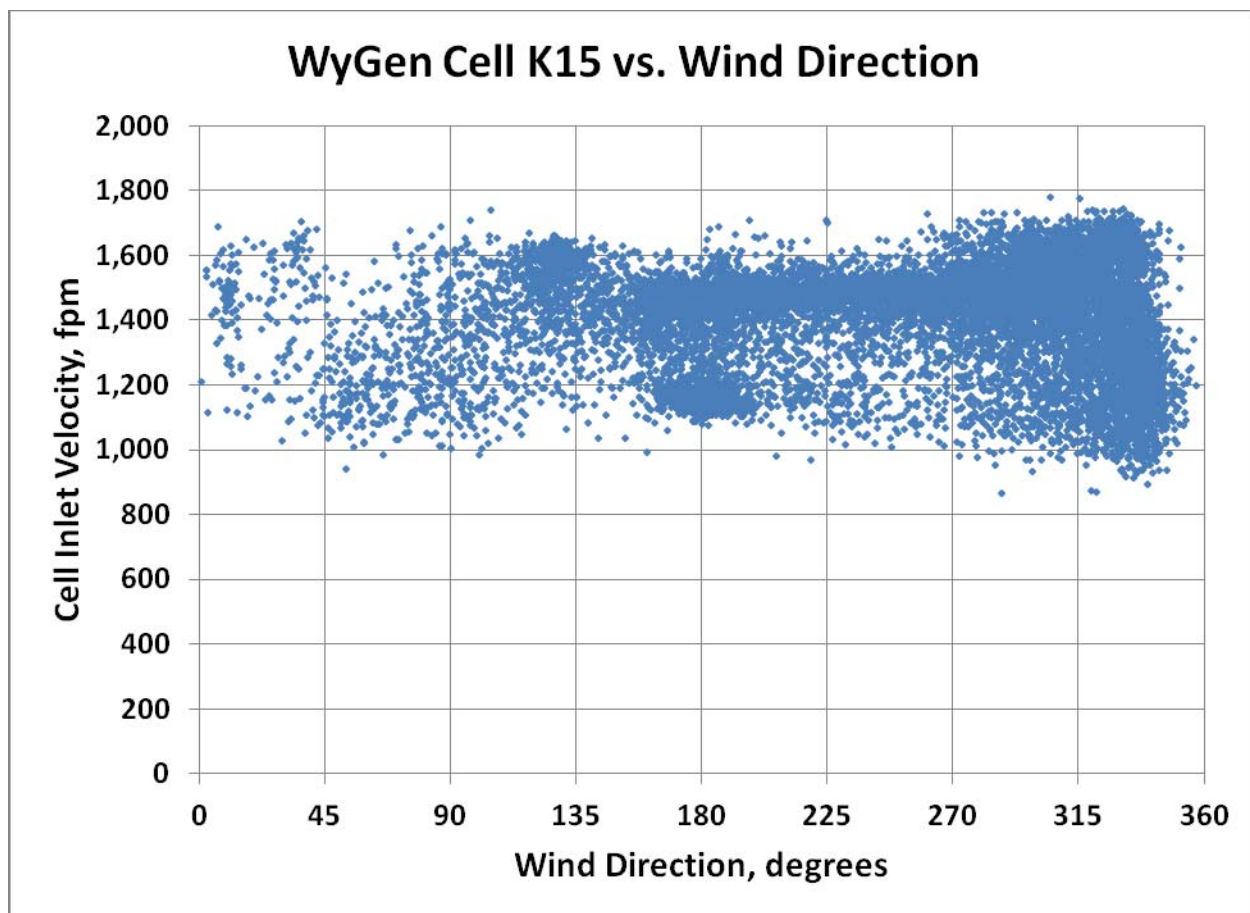


Figure 7-66. Wygen Inlet Velocity vs. Wind Direction; Cell K15—Entire Test Period.

Cell K25 on the northeasterly end of the ACC shows a definite bifurcation with wind speed up to the highest wind speeds. At the north end of the unit there is little benefit of shielding from the plant building and the performance is degraded at higher speed winds from the south.

Figure 7-73 shows a continuous increase in performance as the wind shifts from the south to the northwest. This may be due to the presence of the wind wall which blocks the winds from the southwest and west and puts the K25 fan in a reduced pressure wake region. As the wind turns more from the north, the effect is reduced.

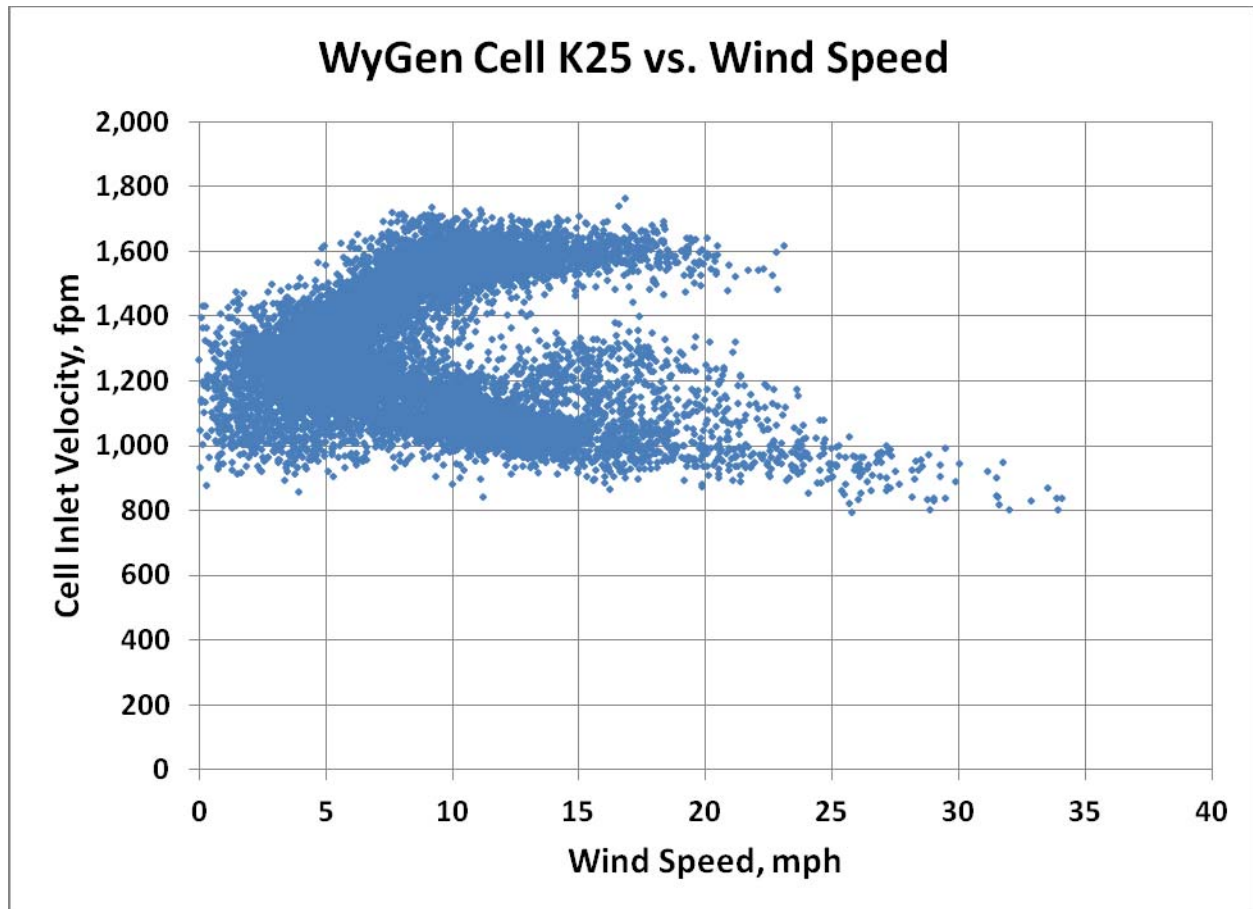


Figure 7-67. Wygen Inlet Velocity vs. Wind Speed; Cell K25—Entire Test Period.

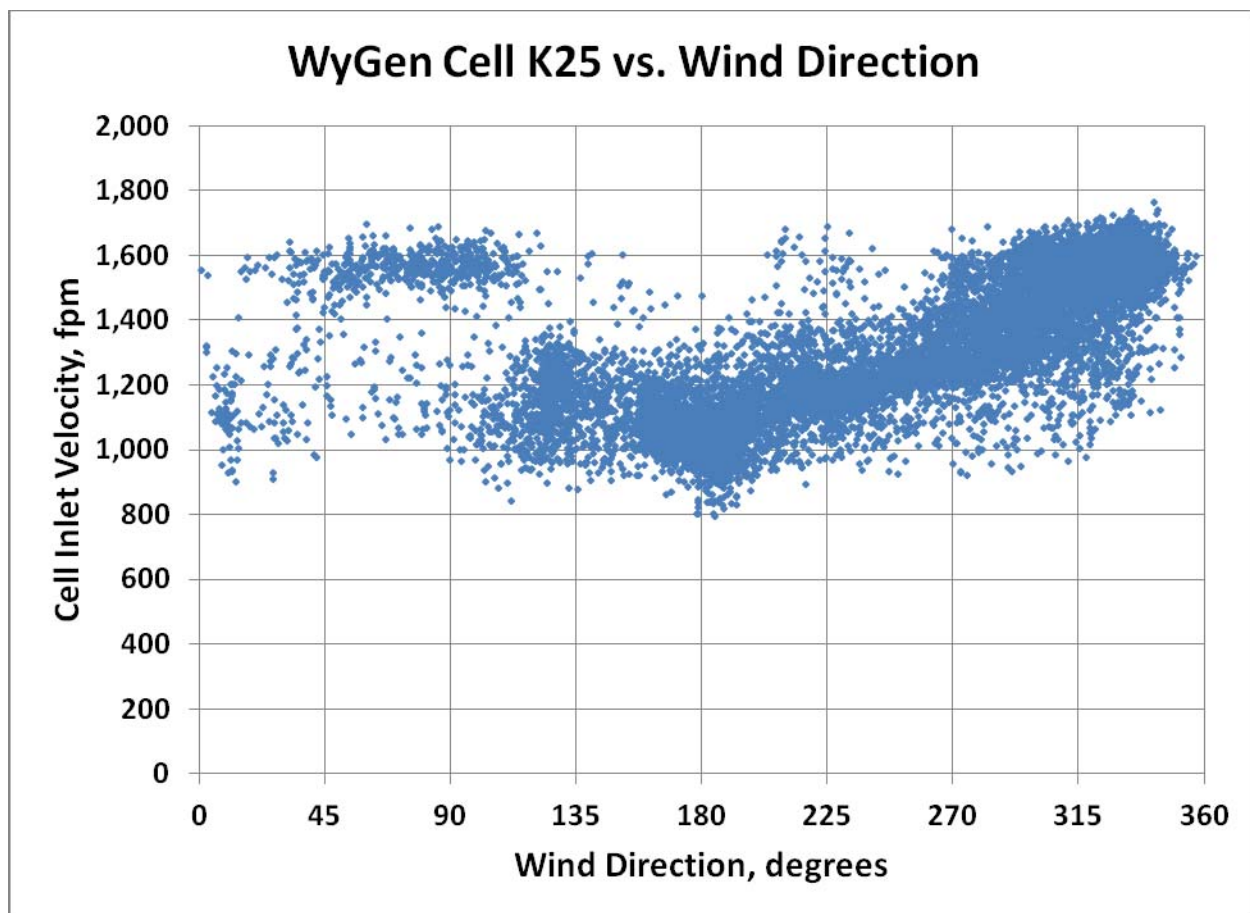


Figure 7-68. Wygen Inlet Velocity vs. Wind Direction; Cell K25—Entire Test Period.

These observations are discussed in more detail in Figures 7-69 through 7-74 displaying the average performance of each of the cells for the days of August 11, August 12 and August 17 with the wind speeds and directions shown on the charts.

On August 11, with low to moderate winds all day, Cells K15 and K25 are relatively unaffected at all speed and all directions. Cell K11 shows reduced behavior for brief periods between midnight and dawn, from noon to 3 pm and after 9 pm. Figure 7-70 indicates that these periods correspond to winds which are highly variable in direction switching from southerly to northwesterly. Cell K11 is shielded by the building from southerly winds but not from westerly to northwesterly winds so it experience intermittently varying wind exposure during those periods which is reflected in the erratic performance.

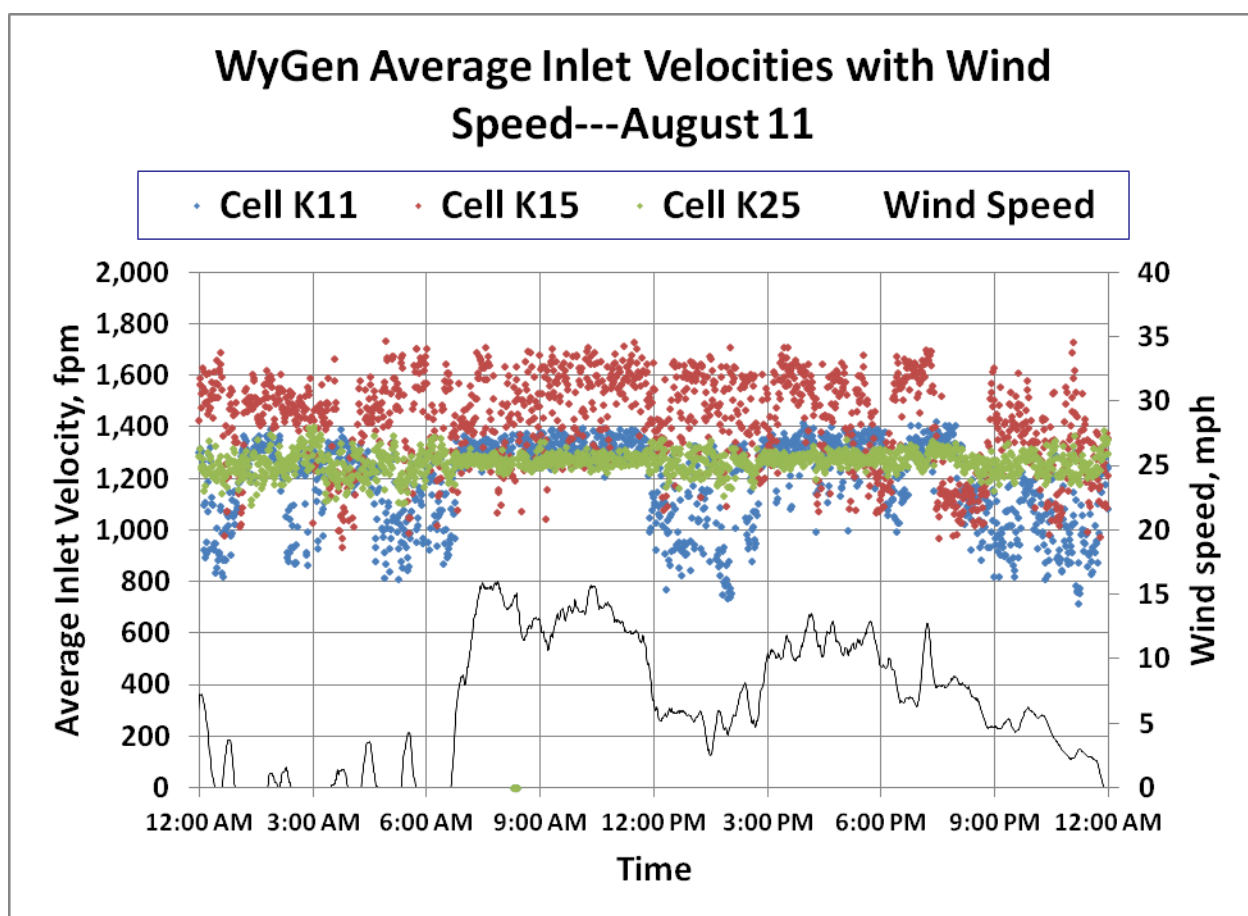


Figure 7-69. Wygen Inlet Velocity; Cells K11, K15 and K25 and Wind Speed—August 11, 2005.

WyGen Average Inlet Velocities with Wind Direction--August 11

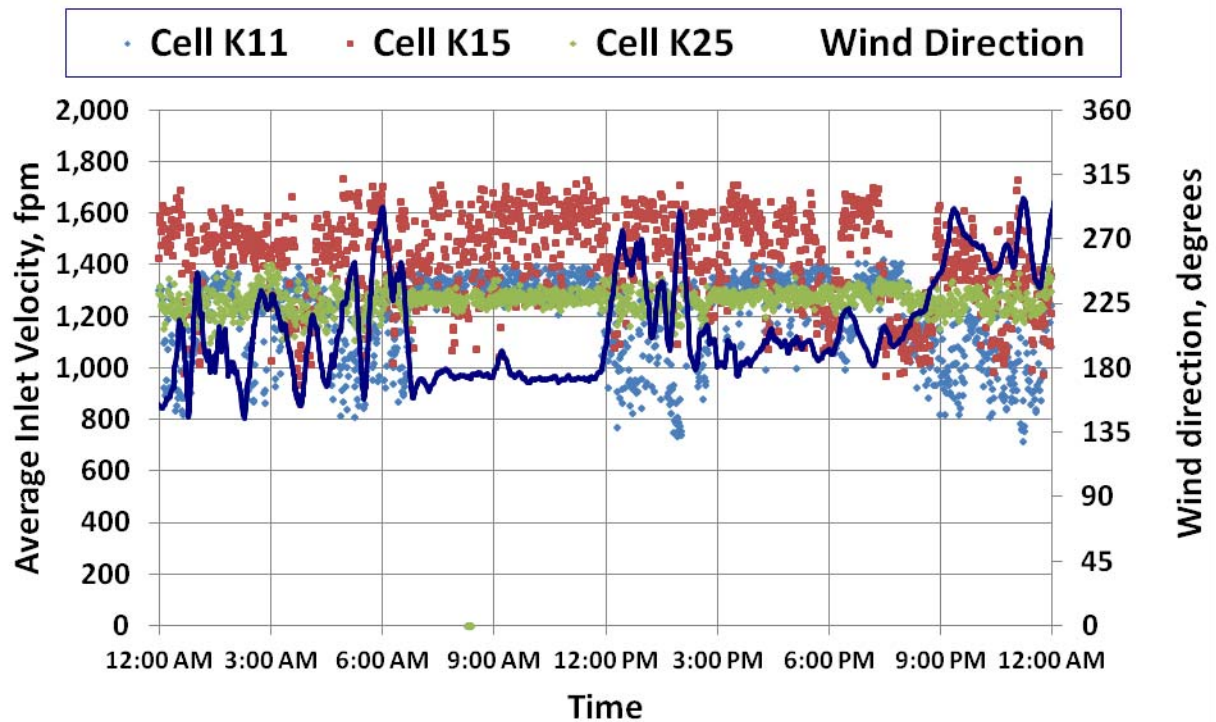


Figure 7-70. Wygen Inlet Velocity; Cells K11, K15 and K25 and Wind Direction—August 11, 2005.

August 12 shows similar behavior but with instances of much higher wind speeds. Again, Cell K11 is most affected with wind from the west-to northwest. At the higher speeds above 20 mph, the shielding effect of the building is insufficient to prevent some performance degradation on Cell K11. Cell K15 is free from wake effects of the building and appears to be slightly aided by the windwall which decelerates the wind and leads to a slight increase in static pressure on the upwind side. The opposite effect is true for Cell K25 which is in a lower static pressure wake region until the wind get well to the northwest.

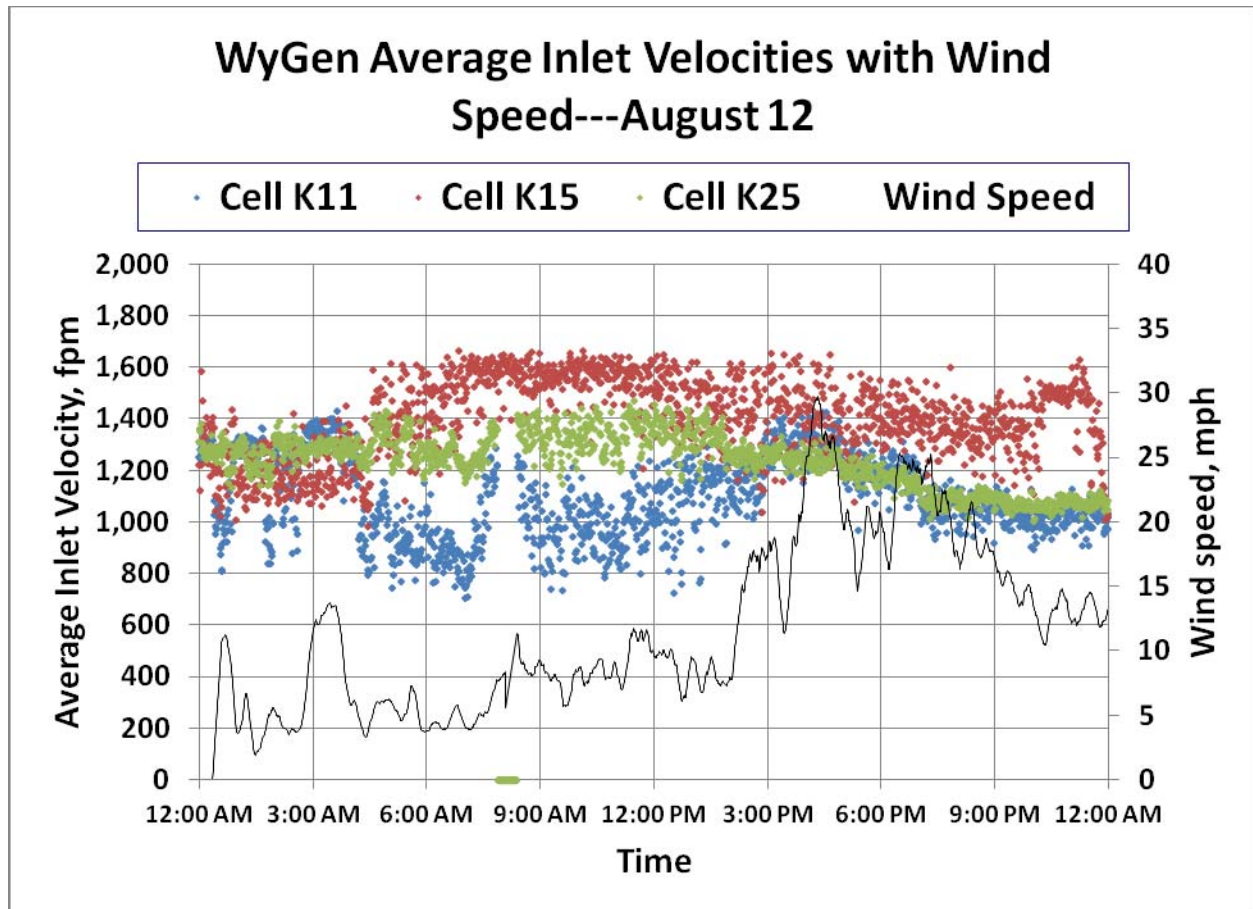


Figure 7-71. Wygen Inlet Velocity; Cells K11, K15 and K25 and Wind Speed—August 12, 2005.

WyGen Average Inlet Velocities with Wind Direction--August 12

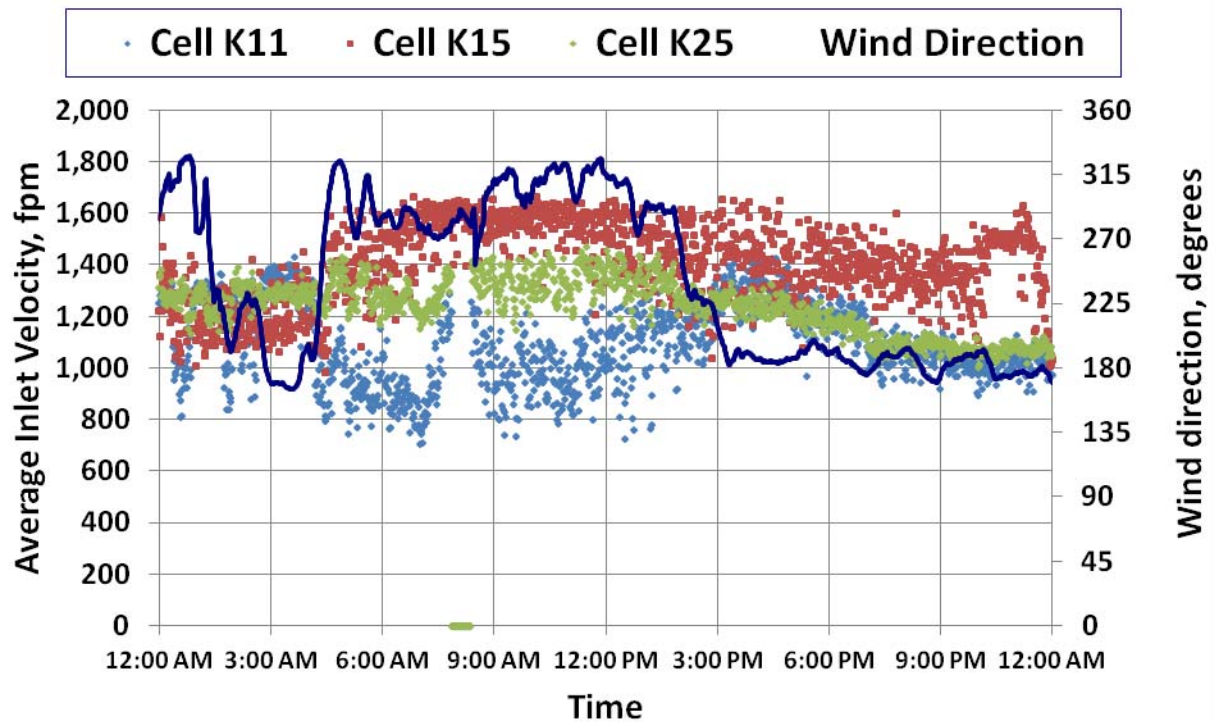


Figure 7-72. Wygen Inlet Velocity; Cells K11, K15 and K25 and Wind Direction—August 12, 2005.

On August 17, winds shift to the southeast with different effects. It appears that Cell K15 benefits from the protection of the wind screen, while Cell K25 does not as would be expected. Cell K11 exhibits a degraded performance which is not fully understood. It may be that the combined effect of interference from the building and the protection of the windwall lead to a reduced static pressure under Cell K11.

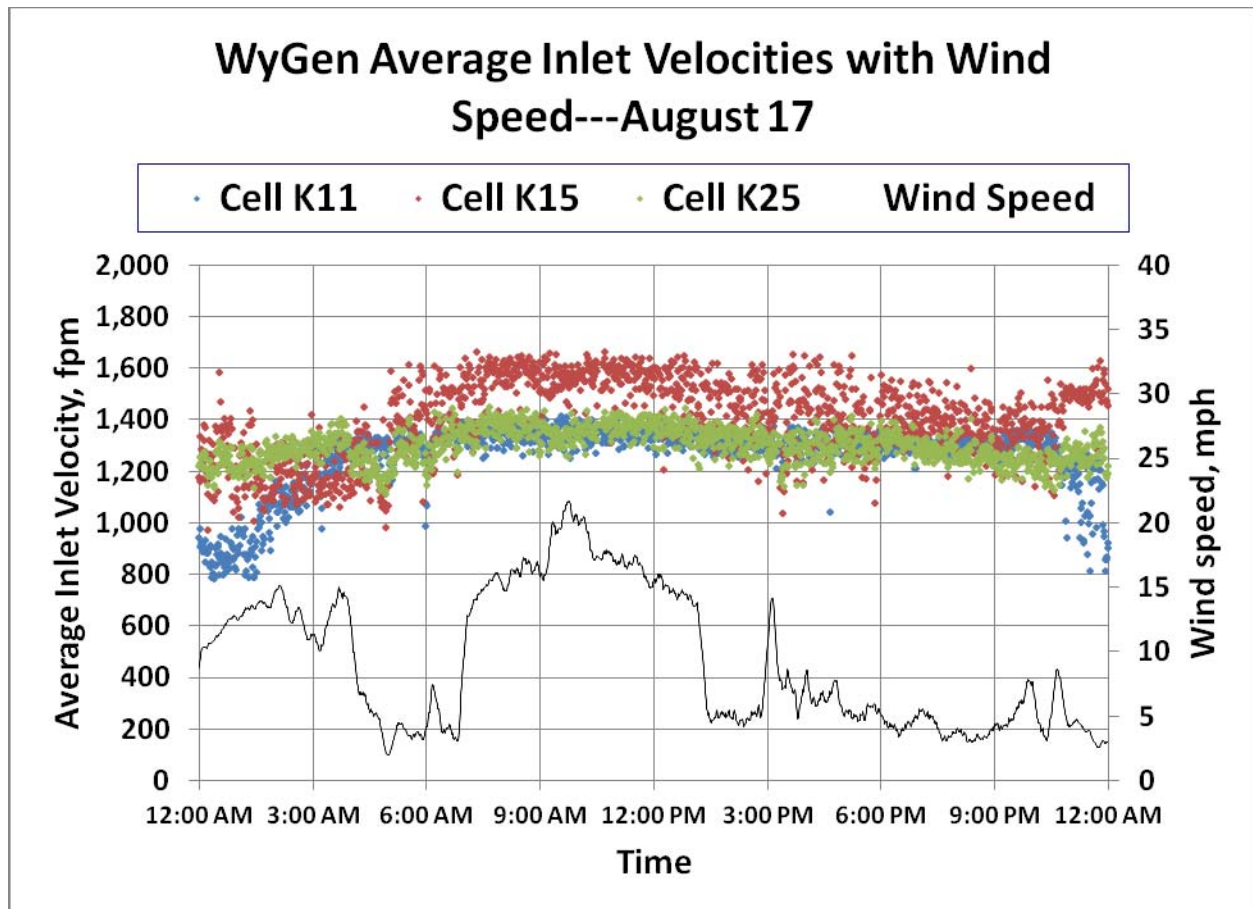


Figure 7-73. Wygen Inlet Velocity; Cells K11, K15 and K25 and Wind Speed—August 17, 2005.

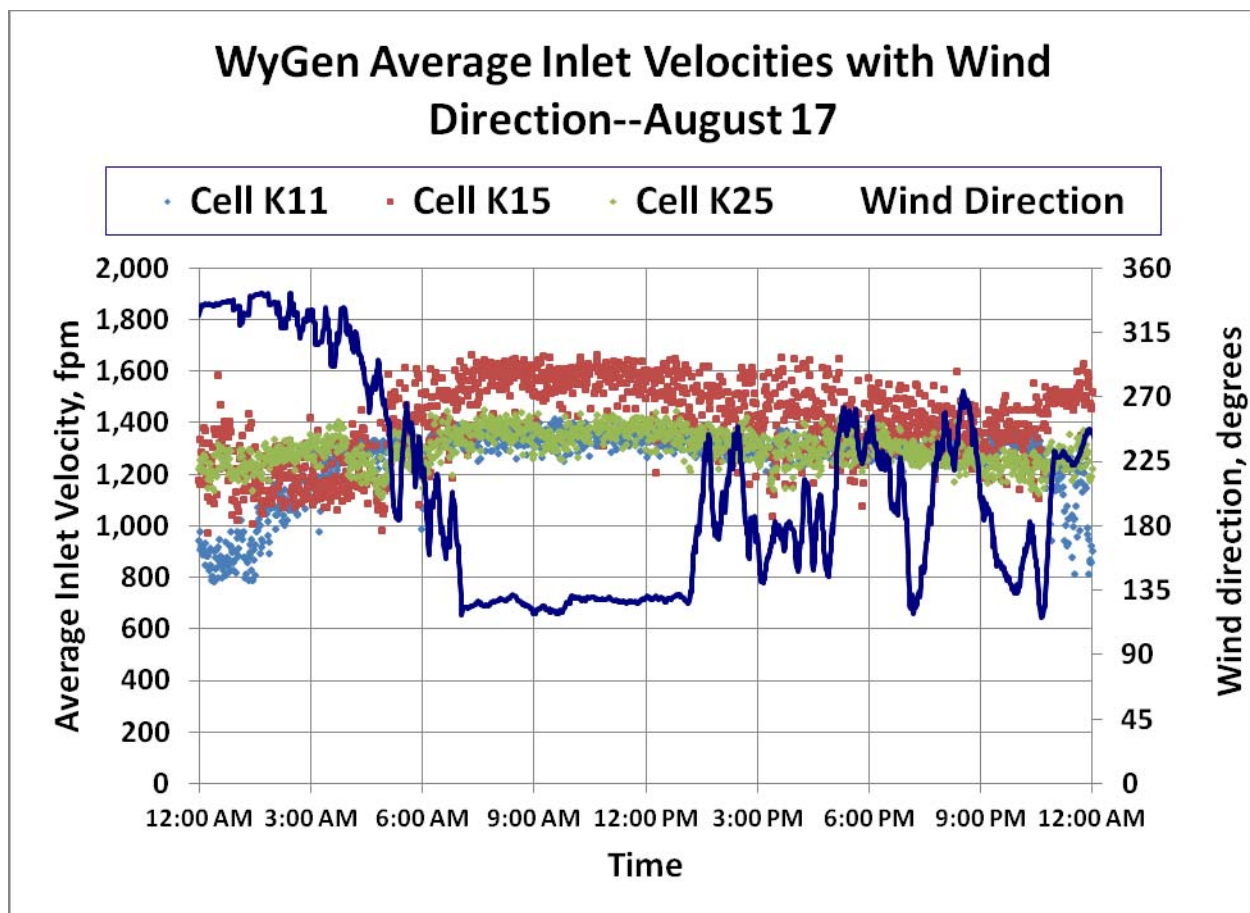


Figure 7-74. Wygen Inlet Velocity; Cells K11, K15 and K25 and Wind Direction—August 17, 2005.

The effect of varying winds on fan performance at WyGen is more complex than at other sites because of the proximity of the pant building, the presence of the solid cruciform wind wall under the ACC and the fact that all the cells are edge or corner cells with heightened sensitivity to wind. Figures 7-75 and 7-76 display the average of the three cells vs. wind speed (Figure 7-75) and wind direction (Figure 7-76). The average fan performance is essentially insensitive to wind speed or direction suggesting that the combination of shielding of the wind from the building and the protective effect of the windwall balances out the effects of wind on the two sides of the ACC and maintains a consistent performance for the range of winds encountered at the site.

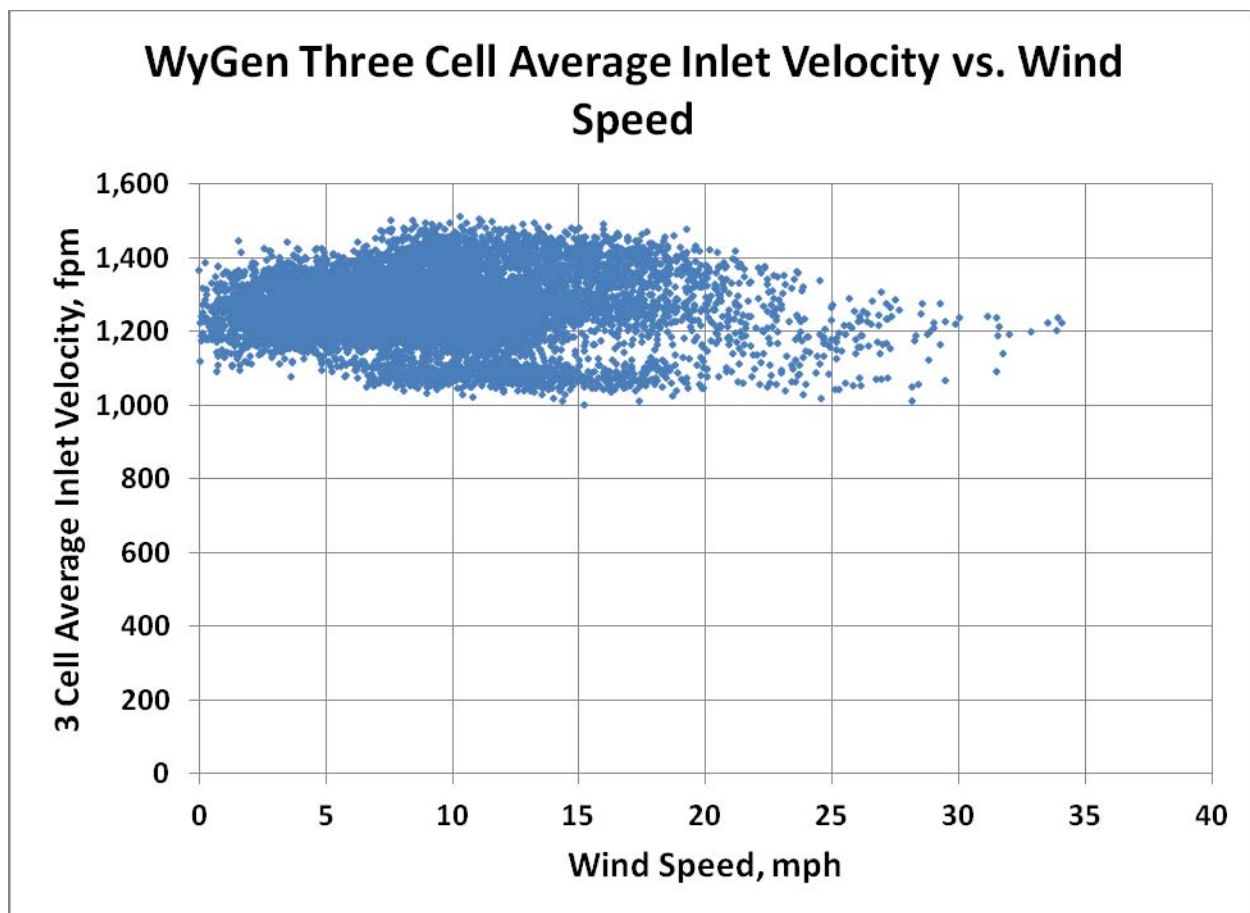


Figure 7-75. Wygen Three Cell inlet Velocity vs. Wind Speed—Entire Test Period.

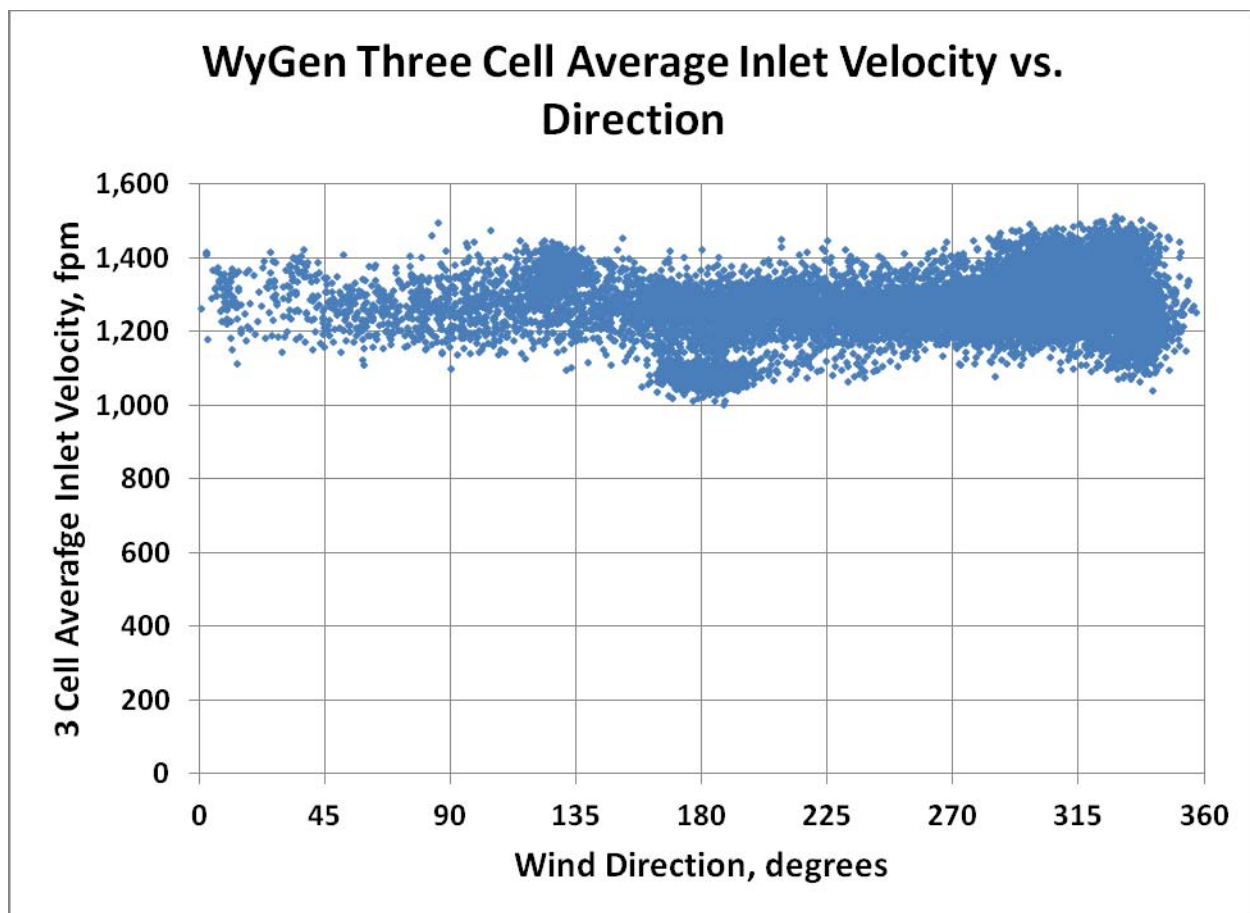


Figure 7-76. Wygen Three Cell inlet Velocity vs. Wind Direction—Entire Test Period.

CHAPTER 8:

Summary and Conclusions

8.1 Summary

8.1.1 Motivation

Dry cooling systems using air-cooled condensers (ACCs) are being selected more frequently in recent years as pressure mounts both for water conservation and for reduced water withdrawals for a variety of environmental concerns. As the use of dry cooling increases, the importance of understanding mechanisms affecting ACC performance becomes more critical to ensuring efficient operation of the individual plant and adequate power production for the network. The effect of wind on ACC performance is alleged to be the major challenge to successful ACC specification, design, and operation and is incompletely understood. With this in mind, field tests were undertaken with the objective of defining and quantifying the effects of wind on ACC performance and exploring means for mitigating the adverse wind effects.

8.1.2 Scope

Five western sites were selected for testing: the Bighorn Power Plant, El Dorado Energy Center, and the Apex Generating Station in Nevada; the Front Range Power Plant in Colorado; and Black Hills Power and Light's Wygen station in Wyoming. At each of the five sites, continuous measurements were made for several days of wind speed, wind direction, far-field ambient temperature, inlet air temperature at all ACC cells, inlet airflow at selected ACC cells, and plant operating variables of power output, steam flow, turbine backpressure, and fan status. At some sites additional measurements were made of fan static pressure rise in selected cells, fan motor currents on selected fans, and some additional interior air temperatures. The data were used to quantify the effects of wind on both hot air recirculation into cells and airflow reductions from degraded fan performance.

8.1.3 Data Analysis

The data were taken at one-minute intervals for 24 hours each day for the duration of testing. Only data obtained at times when all fans were running at full speed were used in the analysis.

The inlet air temperatures for all cells were averaged to define an ACC inlet temperature. Average recirculation was defined as the difference between the average inlet temperature and the minimum inlet temperature in any of the cells. Recirculation in individual cells was also determined, defined as the difference between the individual cell inlet temperature and the minimum cell inlet temperature. Average and maximum individual cell recirculation were plotted against wind speed and direction to look for trends or threshold effects. Additionally, recirculation was plotted against time for selected periods on the same axes as wind speed or direction to identify concurrent events. Also, cell inlet temperatures at a given time were arranged in simple array plots from the lowest to the highest to discern patterns of hot recirculated air distribution under the ACC.

Data on inlet air velocity, static pressure, and motor current were plotted against wind speed and direction and against time for selected periods on concurrent axes with wind speed or direction. The conclusions were all based on visual examination of the plotted results. No attempt was made to develop analytic correlations or to conduct statistical analyses of the data.

8.1.4 Results

Average recirculation for most conditions at all sites was less than 3°F. Occasional excursions to higher values of 4° to 8°F, with very infrequent instances as high as 10°F, were noted. Individual cell recirculation, primarily on edge and corner cells, of 10°F and above was not uncommon. The periods of high recirculation were typically associated not with sustained high wind speeds, but rather with short-lived periods of variable wind direction. Occasional instances of apparent high recirculation, based on the definition given above, were found to be associated with nighttime periods of very still air when temperature inversions were established and cold air near the ground was entrained into one or two cells, giving a spurious indication since the minimum temperature cell was used as the recirculation baseline.

Plotting all cell inlet temperatures at a given instant in ascending order gave insight into the mechanisms of recirculation. At light to moderate winds, the classical recirculation pattern showed relatively uniform inlet temperatures across the ACC with an abrupt increase in a few cells, usually downwind edge or corner cells. Many occasions of short-lived higher average recirculation were associated with a larger number of cells with high inlet temperatures, many of them interior cells and often clustered upwind. This occurred often after a shift in wind directions, suggesting that a previously established hot air exit plume was swept back toward and under the ACC. On those occasions of very still air but with indications of high recirculation, the pattern was the reverse of the classical recirculation pattern, with most of the ACC cells at a relatively uniform inlet temperature and one or two cells showing much lower inlet temperatures. These cells were usually directly above some obstruction under the ACC such as a motor control center, suggesting a diversion of slowly moving ground-level air induced by the ACC fans being turned upward by the obstruction and giving an anomalously low minimum cell inlet temperature.

Fan performance results are less easy to quantify or generalize. Measurements were typically taken at four locations under each fan being monitored. The average of the four velocities was used to estimate the total volume flow through the fan. Under low-wind conditions this was typically close to the design value. With increasing wind, cells near the edge of the ACC would sometimes show a reduction in average flow, occasionally up to 60% or 70%. At other times no significant reduction was observed under what appeared to very similar if not identical conditions.

Airflow across the inlet of a single fan usually, but again not always, showed the greatest reduction in the upwind region of the ACC. At times, at significant reduction was evident in one half of the fan inlet and little or no reduction in the other half. This is conceptually consistent with the idea that a separated flow region—beginning at the bottom of the windwall at the edge of the fan deck, extending in under the fan deck, and enveloping all or a portion of the inlet region of the nearest fan—is responsible for the reduction in airflow through the fan.

This is different from the more conventional interpretation that a strong crosswind disrupts the preferred uniform parallel flow normal to the fan inlet plane, which is the basis for most fan design calculations.

Static pressure and motor current results were sufficiently erratic and inconsistent that measurements were discontinued after the first test site, and any generalized interpretation of their significance is not attempted at this time.

8.2 Conclusions

The results of the field tests led to the following conclusions:

- Wind speeds above 20 miles per hour have been shown to result in turbine backpressures 1.5 to 2 in Hga higher than those which would be predicted from ACC design curves at a given steam flow and ambient temperature and which are achieved in practice under low-wind conditions.
- Recirculation is always present to some degree.
- Recirculation is highest at intermediate wind speeds and during times of variable wind direction.
- Recirculation occurs intermittently during time of variable winds, typically just after a shift in wind direction.
- Recirculation diminishes at the higher wind speeds, apparently as the hot exit air plume is blown clear of the ACC inlet area.
- The effect, typically an average recirculation of less than 3°F, is not sufficient to account for the overall reduction in ACC performance as inferred from the increase in turbine exhaust pressure under high-wind conditions.
- The major mechanism accounting for the observed effect of wind on ACC performance appears to be a reduction in airflow through the ACC resulting from fan performance being degraded by wind.
- The pattern of reduction in inlet air velocity across fan inlet areas suggests the likelihood of a separated region from the edge of the ACC enveloping the fan inlet.
- Precise correlation of wind conditions and measured ACC conditions with turbine backpressure is difficult to establish. This is due in part to plant control actions related to output control, steam flow control, and duct burning at combined-cycle plants which occur concurrently with variations in ambient wind conditions. The actions introduce variations in turbine backpressure, often with some time delay, confounding the interpretation of the instantaneous measurements made during the tests.

8.3 Recommendations

In order to develop more confident quantitative relationships between ambient wind conditions and ACC performance, additional testing and analysis is required. This should include:

- Closely coordinated field tests and CFD (computational fluid dynamics) modeling at a single well-described and completely instrumented operating ACC.
- More detailed measurements of both far- and near-field velocity and temperature fields.
- More extensive measurements of inlet velocity and static pressure on a greater number of cells.
- More precise and extensive measurements of static pressure underneath the ACC.
- More careful attention to pressure and velocity fields close to wind barriers under the ACC.
- More detailed and precise inclusion of plant operation and control actions into the interpretation of the data in order to normalize the simple ACC performance data of turbine exhaust pressure, steam flow, and ambient temperature against wind conditions.
- Careful calibration of detailed CFD models with field data in order to apply the models with confidence to the assessment of design modifications for the mitigating wind effects on fan performance.

8.4 Benefits to California

The state's burgeoning population demands more water, even as global warming effects on the Sierra snowpack make supply less secure. Under these conditions, air-cooled condensers become an increasingly attractive choice for power plant cooling. By advancing knowledge of wind effects on ACC performance—thereby enabling design modifications to mitigate those effects—this research promotes wider use of air-cooled condensers. In turn, dry cooling technology offers dramatic water savings over traditional wet cooling, along with greater freedom to build new power plants to meet growing demand and to site the new units closer to population centers.

References

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Glossary

acfm: Actual cubic feet per minute, a measure of airflow.

fpm: Feet per minute, a measure of airflow.

HRSG: Heat recovery steam generator. In a combined-cycle power station, hot exhaust from a gas turbine is fed to the HRSG, which makes use of the heat to generate steam to drive a steam turbine.

in H₂O: Inches of water, a measure of pressure, typically used in measurement of fan performance.

in Hg: Inches of mercury absolute; a measure of pressure typically used for turbine exhaust pressure.

MCC: Motor control center. Depending on location, this part of a power plant may block airflow around the ACC.

RTD probe: Resistance Temperature Detector; Used in this study to measure the inlet air temperature in individual ACC cells